PHYSICS TEXTBOOK PREPARED BY

MOEUN SOKMEAN

PHYSICS TEXTBOOK







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-រូបវិទ្យាថ្នាក់វិទ្យាល័យ ថ្នាក់ទី១០ ១១ ១២

-សម្រាយបញ្ជាក់រូបមន្តអគ្គិសនី និងម៉ាញេទិច

-មេរៀនសង្ខេប ថ្នាក់អនុវិទ្យាល័យថ្នាក់ទី៧ ៨ ៩

ALL PHYSICS LESSONS PREPARED BY MOEUN SOKMEAN

How to Derive a formula in physics

FOR GRADE 10 11 12

PHYSICS IGCSE

RUPP

4

SUMMARIZE

ALL LESSON

PHYSICS FORMULA

NEW EDITION 2022

Mathematics Review

Operation	Rule	Example
Multiplying Fraction	$\left(\frac{a}{b}\right)\left(\frac{c}{d}\right)=\frac{ac}{bd}$	$\left(\frac{2}{3}\right)\left(\frac{4}{5}\right)=\frac{(2)(4)}{(3)(5)}=\frac{8}{15}$
Division of Fraction	$\frac{\left(\frac{a}{b}\right)}{\left(\frac{c}{d}\right)}=\frac{ad}{bc}$	$\frac{\left(\frac{2}{3}\right)}{\left(\frac{4}{5}\right)}=\frac{(2)(5)}{(3)(4)}=\frac{10}{12}=\frac{5}{6}$
Addition and Subtracting Fraction	$\frac{a}{b}\pm\frac{c}{d}=\frac{ad\pm bc}{bd}$	$\frac{2}{3}-\frac{4}{5}=\frac{(2)(5)-(3)(4)}{(3)(5)}=-\frac{2}{15}$

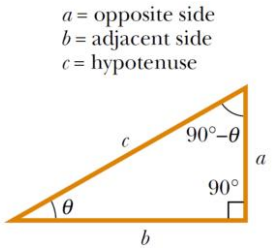
Scientific Notation

Positive Power	Rule	Example
10^0 $10^1=10$ $10^2=10\times10=100$ $10^3=10\times10\times10=1000$ $10^4=10\times10\times10\times10=10000$ $10^5=10\times10\times10\times10\times10=100000$	$\frac{1}{10^n}=10^{-n}$	$\frac{1}{10^{-3}}=10^{-(-3)}=10^3$
	$10^n\times10^m=10^{n+m}$	$10^3\times10^{-8}=10^{(3)+(-8)}=10^{-5}$
	$\frac{10^n}{10^m}=10^{n-m}$ $\left(10^n\right)^m=10^{nm}$	$\frac{10^{-3}}{10^9}=10^{(-3)-(9)}=10^{-12}$ $\left(10^{-3}\right)^4=10^{(-3)(4)}=10^{-12}$
Negative Power	**Rule**	**Example**
$10^{-1}=\frac{1}{10}=0.1$	$10^{1/n}=\sqrt[n]{10}$	$10^{1/2}=\sqrt[2]{10}$
	$10^{m/n}=\sqrt[n]{10^m}$	$10^{4/3}=\sqrt[3]{10^4}$
$10^{-2}=\frac{1}{100}=0.01$ $10^{-3}=\frac{1}{1000}=0.001$ $10^{-4}=\frac{1}{10000}=0.0001$ $10^{-5}=\frac{1}{100000}=0.00001$	$\left(a\times10^n\right)\pm\left(b\times10^n\right)=\left(a\pm b\right)\times10^n$ **EX:** $\left(5\times10^3\right)+\left(2\times10^3\right)=\left(5+2\right)\times10^3=7\times10^3$ **EX:** $\left(-3\times10^{-5}\right)+\left(7\times10^{-5}\right)=\left(-3+7\right)\times10^{-5}$	

Muliple and submultiple prefixes in the SI system.

	Base 10	English Word	Symbol
Prefix	10^{12}	tera	**T**
	10^9	giga	**G**
	10^6	mega	**M**
	10^3	kilo	**k**
	10^2	hecto	**h**
	10^1	deca	**da**
	1	-	**-**
Prefix	10^{-1}	deci	**d**
	10^{-2}	centi	**c**
	10^{-3}	milli	**m**
	10^{-6}	micro	**μ**
	10^{-9}	nano	**N**
	10^{-12}		

Trigonometry



$$\sin\theta=\frac{\text{opposite side}}{\text{hypotenuse}}=\frac{a}{c}$$

$$\cos\theta=\frac{\text{adjacent side}}{\text{hypotenuse}}=\frac{b}{c}$$

$$\tan\theta=\frac{\text{opposite side}}{\text{adjacent side}}=\frac{a}{b}$$

Chapter1 Mechanics
Lesson1 Linear Motion

I. Vector

i. Definition:

A **vector quantity** is any physical quantity that has both magnitude and direction (e.g., velocity, force, displacement).

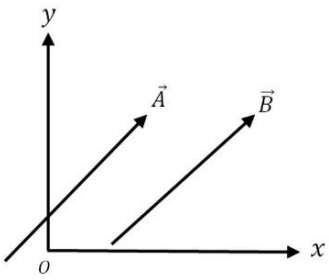
A **scalar quantity** is any physical quantity that has magnitude only, without direction (e.g., mass, time, temperature).

II. Two-Vector Addition

i. Definition:

Two vectors $\vec{A}$ and $\vec{B}$ are said to be **equal** if they have the **same magnitude (length)** and the **same direction**.

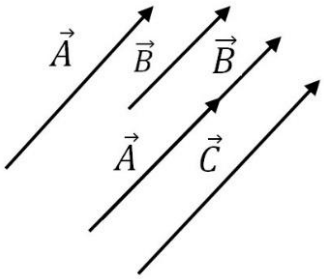
This can be written as: $\vec{A} = \vec{B}$ or simply $A = B$ $\left(\vec{A} \uparrow\uparrow \vec{B}\right)$



III. Vector Addition

When two vectors $\vec{A}$ and $\vec{B}$ have the same direction, their resultant vector $\vec{C}$ is $\vec{C} = \vec{A} + \vec{B}$. To construct vector $\vec{C}$, vector $\vec{B}$ is translated (without changing its direction or length) and placed such that its tail is at the head of vector $\vec{A}$.

The resulting vector $\vec{C}$ is drawn from the tail of $\vec{C}$ to the head of $\vec{B}$, and its magnitude is $C = A + B$

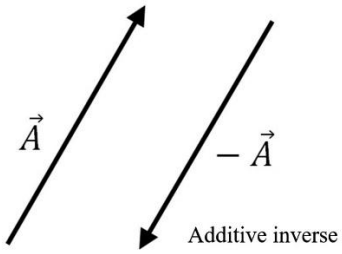


IV. Opposite Vector Addition

A vector opposite to vector $\vec{A}$ is a vector that has the same magnitude as $\vec{A}$, but points in the opposite direction. This opposite vector is denoted as $-\vec{A}$.

The sum of a vector and its opposite is the zero vector:

$$\vec{A} + \left(-\vec{A} = 0\right)$$



This means that when two vectors with equal magnitude and opposite direction are added, their net effect cancels out.

V. Vector Product

Vectors $\vec{A}$ and $\vec{B}$ can be multiplied using different methods to produce either a scalar quantity or a vector quantity.

i. Scalar Quantity Formation

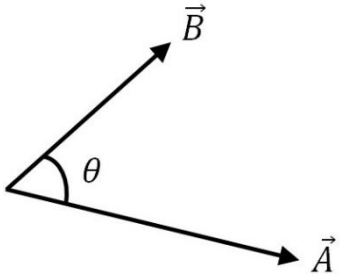
To form a **scalar quantity**, the **dot product** (also called the **scalar product**) of two vectors $\vec{A}$ and $\vec{B}$ is used.

It is denoted by: $\vec{A} \cdot \vec{B}$

The formula is: $\vec{A} \cdot \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right| \cos\theta$

Where, $\vec{A}$ and $\vec{B}$ are **vectors**.

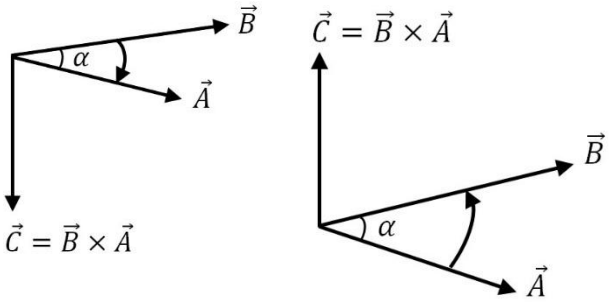
$\left|\vec{A}\right|\left|\vec{B}\right|$ are the **magnitudes of vectors**.



θ is the **angle between vectors** (in radians or degrees).

ii. Vector Quantity Formation

To form a **vector quantity**, the **cross product** (also called the **vector product**) of two vectors $\vec{A}$ and $\vec{B}$ is used.



It is denoted by: $\vec{A} \times \vec{B}$

The formula is: $\vec{A} \times \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right| \sin\alpha$

Where, $\vec{A}$ and $\vec{B}$ are **vectors**.

$\left|\vec{A}\right|\left|\vec{B}\right|$ are the **magnitudes of vectors**.

α is the **angle between vectors** (in radians or degrees).

VI. Coordinate System of a Vector

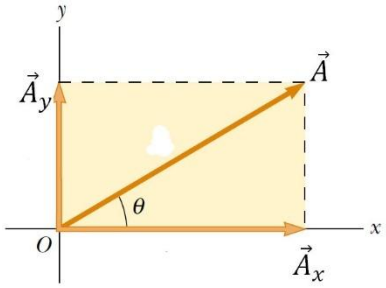
Suppose we have a vector $\vec{A}$ that lies in the xy-plane and forms an angle θ with the x-axis $\overrightarrow{ox}$.

The vector $\vec{A}$ can be resolved into two perpendicular components along the x- and y-axes:

$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$

Since $\overrightarrow{ox} \perp \overrightarrow{oy}$, we can use the Pythagorean theorem to find the magnitude of vector $\vec{A}$:

$$A = \sqrt{A_x^2 + A_y^2}$$



The components of vector $\vec{A}$ are:

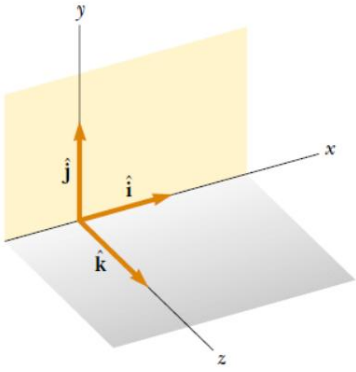
$$A_x = A\cos\theta, A_y = A\sin\theta$$

However, $\tan\theta = \dfrac{A_y}{A_x}$

This implies: $\theta = \tan^{-1}\left(\dfrac{A_y}{A_x}\right)$

Hence, the angle θ that vector $\vec{A}$ makes with the x-axis can be calculated as: $\theta = \tan^{-1}\left(\dfrac{A_y}{A_x}\right)$

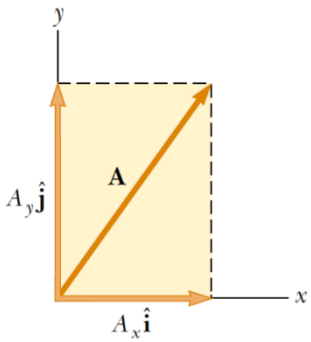
• Vector quantities often are expressed in terms of unit vectors. A unit vector is a dimensionless vector having a magnitude of exactly 1. Unit vectors are used to specify a given direction and have no other physical significance. They are used solely as a convenience in describing a direction in space. We shall use the symbols $\hat{i}, \hat{j}, \text{and } k$ to represent unit vectors pointing in the positive x, y, and z directions, respectively.



Consider a vector $\vec{A}$ lying in the xy plane forms an angle θ with the x-axis $\overrightarrow{ox}$.

A vector $\vec{A}$ lying in the xy-plane can be expressed using its components along the x- and y-axes:

$$\vec{A}_x = A_x\hat{i} \text{ and } \vec{A}_y = A_y\hat{j}$$



Unit vectors have magnitude 1: $\left|\hat{i}\right| = \left|\hat{j}\right| = \left|k\right| = 1$

Therefore, the vector $\vec{A}$ can be written as:

$$\vec{A} = A_x\hat{i} + A_y\hat{j}$$

The **position vector** $\vec{r}$ of a point in the xy-plane is written as: $\vec{r} = x\hat{i} + y\hat{j}$

Where, A_x is the **component of vector** $\vec{A}$ along the x-axis.

A_y is the **component of vector** $\vec{A}$ along the y-axis.

r is the position vector measured in (m).

x, y, z are coordinates/position components measured in (m).

$\hat{i}, \hat{j}, k$ are unit vectors (dimensionless).

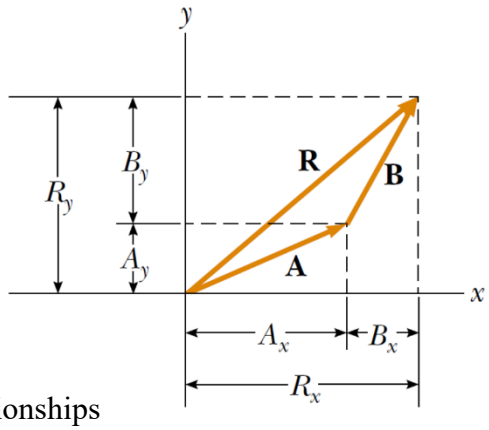
Note:

Unit vectors have magnitude 1, so $\left|\hat{i}\right| = \left|\hat{j}\right| = \left|k\right| = 1$, but $\hat{i} \neq \hat{j} \neq k$ because they point in different directions.

• Vector Addition and Subtraction in Coordinate System

Suppose we wish to add vector $\vec{B}$ to vector $\vec{A}$, where vector $\vec{B}$ has components $\vec{B}_x$ and $\vec{B}_y$. All we do is add the x and y components separately. The resultant vector $\vec{R}$ is therefore

$$\vec{R} = \vec{A} + \vec{B}$$
$$\vec{R} = \left(A_x + B_x\right)\hat{i} + \left(A_y + B_y\right)\hat{j}$$
$$\vec{R} = R_x\hat{i} + R_y\hat{j}$$



The components of the resultant vector are

$$R = \sqrt{R_x^2 + R_y^2}$$

This implies $$R = \sqrt{\left(A_x + B_x\right)^2 + \left(A_y + B_y\right)^2}$$

The angle it makes with the x axis from its components, using the relationships

$$\tan\theta = \frac{R_y}{R_x} = \frac{A_y + B_y}{A_x + B_x}$$

If $\vec{A}$ and $\vec{B}$ both have x, y, and z components, we express them in the form

$$\vec{A} = \vec{A}_x\hat{i} + \vec{A}_y\hat{j} + \vec{A}_z k \text{ and } \vec{A} = \vec{B}_x\hat{i} + \vec{B}_y\hat{j} + \vec{B}_z k$$

The sum of $\vec{A}$ and $\vec{B}$ is

$$\vec{R} = \vec{A} + \vec{B} = \left(A_x + B_x\right)\hat{i} + \left(A_y + B_y\right)\hat{j} + \left(A_z + B_z\right)k$$

Let, $R_x = A_x + B_x, R_y = A_y + B_y, R_z = A_z + B_z$

This implies $R = \sqrt{R_x^2 + R_y^2 + R_z^2}$

Hence, the magnitude of the vector is $R = \sqrt{R_x^2 + R_y^2 + R_z^2}$.

The angle θ that R makes with the x axis is found from the expression

$$\cos\theta_x = \frac{R_x}{R}$$

Hence, angle θ that R makes with the x axis is $\cos\theta_x = \frac{R_x}{R}$.

Where, A_x, A_y, A_z are **components of vector $\vec{A}$ in x, y, z directions** along the x-axis.

B_x, B_y, B_z are **components of vector $\vec{B}$ in x, y, z directions** along the y-axis.

R_x, R_y, R_z are **components of vector $\vec{R}$ in x, y, z directions** along the y-axis.

R is **magnitude of the resultant vector** measured in (m).

θ_x is the **angle between $\vec{R}$ and the x-axis** measured in $(\mathrm{rad\ or}^\circ)$.

VII. Displacement

i. Definition: **Displacement** of an object depends only on its initial and final positions, regardless of the path taken.

Example: An object moves along the x-axis from position x' to position x in the positive direction.

- If the object starts at point A and moves to point B, the displacement is the vector $\overline{AB}$.



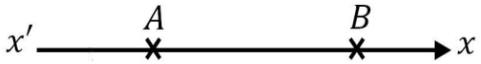
- If the object starts at A, moves to B, and then returns back to A, the total displacement is zero, since the initial and final positions are the same.

VIII. Distance

i. Definition: Distance is the total length of the path traveled by an object, regardless of the direction of motion.

Example: An object moves along the x-axis from position x' to position x in the positive direction.

- If the object starts at point A and moves to point B, the total distance traveled is $x = AB$.



- If the object starts at A, moves to B, and then returns back to A, the total distance traveled is $x = AB + BA = 2AB$.

IX. Speed and Velocity

i. Speed

Speed describes how fast an object is moving, regardless of its direction. It is defined as the ratio of the total distance traveled to the time taken.

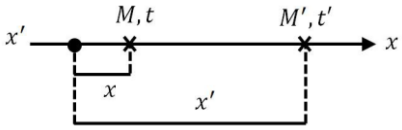
Mathematically, speed v is given by: $v = \dfrac{d}{t}$

Where, v is the **speed** measured in $(\mathrm{m/s})$.

d is the **total distance traveled** measured in (m).

t is the **time interval taken** measured in (s).

ii.Average Velocity



At time t, an object is located at point M, which has an **abscissa** x.

At a later time t', the object is at point M', which has an **abscissa** x'.

During the time interval $\Delta t = t' - t$, the object moves from point M to point M'.

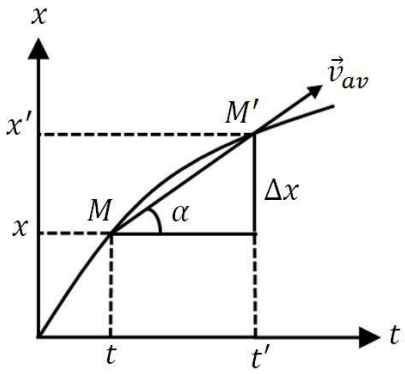
The displacement is $\Delta x = x' - x$, which represents the change in position along the x-axis.

The average velocity is written as

$$v_{av} = \frac{\Delta x}{\Delta t}$$

From the position-time graph,

$$\tan\alpha = \frac{\Delta x}{\Delta t} = v_{av}$$



Hence, the average velocity is the slope of the line MM'.

Average velocity has the following characteristics:

- Point of application: M
- Direction: along the line MM'
- Direction is from M to M'
- Magnitude (or intensity): $v_{av} = \dfrac{\Delta x}{\Delta t}$

Where, v_{av} is the **average velocity** measured in $(\mathrm{m/s})$.

Δx is the **displacement** measured in (m).

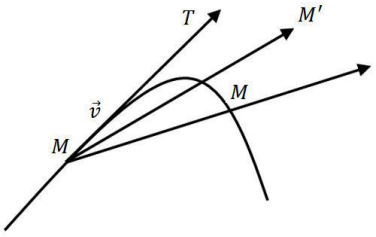
Δt is the **time interval** measured in (s).

iii.Instantaneous Velocity

Instantaneous velocity is the limit of the average velocity as Δt approaches zero and Δx approaches zero.

They write: $v = \lim\limits_{\Delta t \to 0} v_{av} = \lim\limits_{\Delta t \to 0} \dfrac{\Delta x}{\Delta t} = \dfrac{dx}{dt} = x'$

By trigonometry, when point M' approaches M, the line MM' intercepts the tangent line MT, and the average velocity $\vec{v}_{av}$ becomes the instantaneous velocity $\vec{v}$.



Average velocity has the following characteristics:

- Point of application: M
- Direction: along the tangent line the MT'

- Direction: same as the direction of motion
- Magnitude (or intensity): $v = \dfrac{\Delta x}{\Delta t}$

Where, v is the **instantaneous velocity** measured in $(\mathrm{m/s})$.

Δx is the **displacement** measured in (m).

Δt is the **time interval** measured in (s).

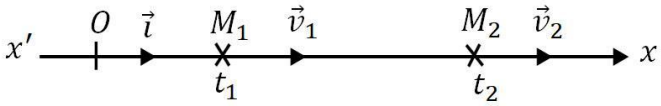
X. Acceleration

i. Average Acceleration

Assume that at instant t_1, an object has velocity v_1, and at instant t_2, the object has velocity v_2.

The time interval is $\Delta t = t_2 - t_1$

The change in velocity is $\Delta v = v_2 - v_1$



The average acceleration is then defined as

$$a_{av} = \frac{\Delta v}{\Delta t}$$

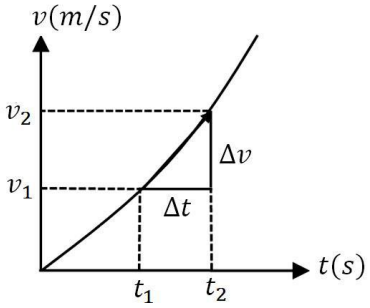
Where, a_{av} is the **average acceleration** measured in $(\mathrm{m/s^2})$.

Δv is the **change in velocity** measured in $(\mathrm{m/s})$.

Δt is the **time interval** measured in (s).

ii. Instantaneous Acceleration

Instantaneous acceleration is the limit of average acceleration as Δt approaches zero.



They derive: $a = \lim\limits_{\Delta t \to 0} a_{av} = \lim\limits_{\Delta t \to 0} \dfrac{\Delta v}{\Delta t} = \dfrac{dv}{dt} = \ddot{x}$

We know average acceleration is: $a_{av} = \dfrac{\Delta v}{\Delta t}$

If Δt approaches zero, then Δv also approaches zero, and the average acceleration becomes the instantaneous acceleration.

Hence, acceleration has the following characteristics:

- Point of application: at the position of the object
- Direction:
  - Along the positive direction of the path if $a > 0$
  - Along the negative direction of the path if $a < 0$

∗ To study the type of linear motion, we examine the acceleration a as follows:

- If $a = 0$, the motion is **uniform linear motion** (constant velocity).
- If $a > 0$, the motion is **accelerating motion.**
- If $a < 0$, the motion is **decelerated motion**
- If $v = 0$and remains constant, the object **stays at rest** (no motion).

XI. Uniform Linear Motion

An object is said to move in **uniform linear motion** when it moves along a straight line with constant velocity.

From this, we get: $v = \text{constant}$

Since the velocity does not change with time, the acceleration is:

$$a = \frac{dv}{dt} = 0$$

Therefore, in uniform linear motion, $a = 0$.

i. Time–Position Equation

We start with the definition of velocity:

$$v = \frac{dx}{dt}$$

Rewriting gives: $dx = vdt$

Integrating both sides:

$$\int_{x_0}^{x} dx = \int_{t_0}^{t} vdt$$

Assuming constant velocity v, the integration yields:

$$x - x_0 = v\left(t - t_0\right)$$

Therefore, the time–position equation is $x = x_0 + v\left(t - t_0\right)$

Where,

x is the **final position of the object** measured in (m).

x_0 is the **initial position of the object at time *t*0** measured in (m).

v is the **constant velocity of the object** measured in (m/s).

t is the **final time** measured in (s).

t_0 is the **initial time** measured in (s).

I. Non-Uniform Linear Motion

i. Definition: An object is said to be in **non-uniform linear motion** if it moves along a straight line with **changing velocity**.

If the velocity changes at a constant rate, the object is said to be moving with **uniform (constant) acceleration**.

In this case, we have: $a = \text{constant}.$

ii. Velocity Equation (Under Constant Acceleration)

We start from the definition of acceleration:

$$a = \frac{dv}{dt} \text{ or } dv = adt$$

Integrating both sides:

$$\int_{v_0}^{v} dv = \int_{t_0}^{t} adt$$

Assuming **constant acceleration** a, we get:

$$v - v_0 = a\left(t - t_0\right)$$

Therefore, the velocity equation becomes:

$$v = v_0 + a\left(t - t_0\right)$$

If the motion starts at $t_0 = 0$ and $v_0 = 0$:

$$v = at$$

Also, the **average velocity** over the time interval is $v_{av} = \dfrac{v + v_0}{2}$

Where, v is the **final velocity** measured in $(\mathrm{m/s})$.

v_0 is the **initial velocity at time t_0** measured in $(\mathrm{m/s})$.

v_{av} is the **average velocity of the object** measured in $(\mathrm{m/s})$.

a is the **uniform acceleration** measured in $(\mathrm{m/s^2})$.

t is the **final time** measured in (s).

t_0 is the **initial time** measured in (s).

iii. Time-Position Equation (Abscissa Equation)

We begin with the definition of velocity:

$$v = \frac{dx}{dt} \text{ or } dx = vdt$$

Since velocity under constant acceleration is:

$$v = v_0 + a\left(t - t_0\right)$$

We substitute into the position equation:

$$\frac{dx}{dt} = v_0 + a\left(t - t_0\right) \Rightarrow dx = \left[v_0 + a\left(t - t_0\right)\right]$$

Now integrate both sides:

$$\int_{x_0}^{x} dx = \int_{t_0}^{t}\left[v_0 + a\left(t - t_0\right)\right]dt$$

This gives: $x - x_0 = v_0\left(t - t_0\right) + \dfrac{1}{2}a\left(t - t_0\right)^2$

If the object starts at time $t_0 = 0$ and from rest $v_0 = 0$:

$$x = \frac{1}{2}at^2$$

Where, x is the **final position of the object** measured in (m).

x_0 is the **initial position of the object at time t_0** measured in (m).

v is the **constant velocity of the object** measured in $(\mathrm{m/s})$.

t is the **final time** measured in (s).

t_0 is the **initial time** measured in (s).

iv. Independent Time-Equation

Starting from the velocity equation:

$$v = v_0 + at \Rightarrow t = \frac{v - v_0}{a}$$

Substitute into the **abscissa (position) equation**:

$$x - x_0 = v_0 t + \frac{1}{2} a t^2$$

Substitute $t = \dfrac{v - v_0}{a}$ into (1):

$$x - x_0 = v_0 \left(\frac{v - v_0}{a} \right) + \frac{1}{2} a \left(\frac{v - v_0}{a} \right)^2$$

Simplify the right-hand side:

$$x - x_0 = \frac{v_0 \left(v - v_0 \right)}{a} + \frac{1}{2} \cdot \frac{\left(v - v_0 \right)^2}{a}$$

Multiply both sides by $2a$:

$$2a\left(x - x_0 \right) = 2v_0 \left(v - v_0 \right) + \left(v - v_0 \right)^2$$

Simplify: $2a\left(x - x_0 \right) = v^2 - v_0^2$

Thus, we get the **independent-time equation** $v^2 - v_0^2 = 2a\left(x - x_0 \right)$.

Where, x is the **final position of the object** measured in (m).

x_0 is the **initial position of the object at time t_0** measured in (m).

v is the **final velocity** measured in $(\mathrm{m/s})$.

v_0 is the **initial velocity at time t_0** measured in $(\mathrm{m/s})$.

a is the **uniform acceleration** measured in $(\mathrm{m/s^2})$.

v. Free Fall Motion

In free fall, the object moves under the influence of gravity alone. The acceleration becomes $a = g = 9.8\,\mathrm{m/s^2}$, directed downward.

∗ Equations of motion in free fall:

Position–Time Equation:

$$y = y_0 + v_0 t + \frac{1}{2} g t^2$$

Velocity–Time Equation:

$$v = v_0 + gt$$

Independent-Time Equation:

$$v^2 - v_0^2 = 2g\left(y - y_0 \right)$$

Where, y is the **final vertical position of the object** measured in (m).

y_0 is the **initial vertical position of the object at time t_0** measured in (m).

v is the **final velocity** measured in $(\mathrm{m/s})$.

v_0 is the **initial velocity at time t_0** measured in $(\mathrm{m/s})$.

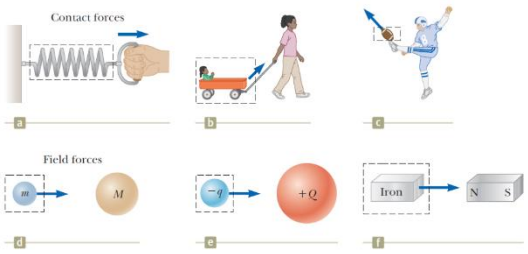
$g = 9.8$ is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

Chapter1 Mechanics
Lesson2 Newton's Law of Motion

I. Force

i. Definition: Force is a cause that makes an object tend to start moving, stop moving, deform, or change its direction of motion.

II. Newton's First Law (Principle of Inertia)



If an object is not acted upon by any force, or if the **net force** acting on it is zero, then:

- If the object was initially at rest, it will remain at rest forever.
- If the object was already in motion, it will continue to move with **uniform linear motion** (constant velocity).

Since the acceleration $a = 0$, the velocity remains constant $(v = \text{constant})$.

III. Newton's Second Law (Principle of Dynamics)

The net force acting on an object is equal to the product of the object's mass and its acceleration. Mathematically, this is expressed as:

$$\Sigma \vec{F} = m\vec{a}$$

Where, ΣF is the **net force acting on the object** measured in (N).

m is the **mass of the object** measured in (kg).

a is the **acceleration of the object** measured in (m/s^2).

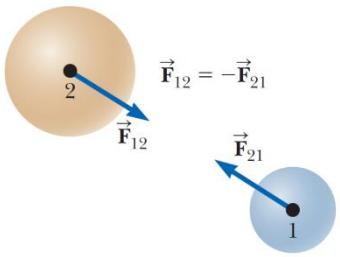
IV. Newton's Third Law (Principle of Reaction)

Statement:

For every action, there is an equal and opposite reaction.

If object m_1 exerts a force $\vec{F}_{12}$ on object m_2, then object m_2 exerts a force $\vec{F}_{21}$ on object m_1 such that:

$$\vec{F}_{12} = -\vec{F}_{21}$$



Characteristics of the Action–Reaction Force Pair:

- They act on **different objects** (different points of application).
- They act in **opposite directions** (face each other).
- They are **equal in magnitude**:

$$F_{12} = F_{21}$$

Where, F_{12} is the **force exerted by object 1 on object 2** measured in (N).

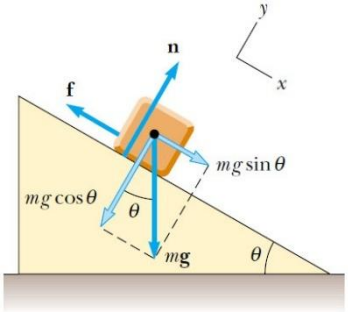
F_{21} is the **force exerted by object 2 on object 1** measured in (N).

V. Friction Force

When an object rests on an **inclined surface**, there is a **frictional force** acts due to contact between the object and the surface.

∗ **Characteristics of the Frictional Force:**

- **Point of application**: At the contact surface between the object and the inclined plane.
- **Direction**: Parallel to the surface (along the incline).
- **Direction**: Opposite to the direction of motion or the tendency of motion.



- **Magnitude**: Increases from 0 up μN.

$$f_r = \mu N = \mu mg\cos\theta$$

Where, f_r is the **frictional force** measured in (N).

N is the **normal force (perpendicular to the surface)** measured in (N).

$g = 9.8$ is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

μ is the **coefficient of friction (static or kinetic)** unitless.

m is the **mass of the object** measured in (kg).

θ is the **angle of the inclined plane (to horizontal)** measured in $(^\circ)$.

i. Static Friction

Static friction acts **when an object is at rest** and **prevents** it from starting to move.
It can be written as:

$$f_{s\max} = \mu N$$

In this case, static friction has a **maximum value**.
So we get: $f_{s\max} \le \mu N$

Hence, static friction is $f_{s\max} \le \mu N$

ii. Kinetic Friction

Kinetic friction acts when an object is already in motion across a surface.
It can be written as:

$$f_{s\max} = f_K = \mu N$$

VI. Object Slides on an Inclined Plane (Frictionless)



(a)

- **Acceleration of the object sliding on an inclined plane**

By **Newton's Second Law**, we consider forces acting on the object:

$$\Sigma\vec{F}_x + \Sigma\vec{F}_y = m\vec{a}$$

Along the incline (x-axis):

$$F_x = mg\sin\theta = ma_x$$

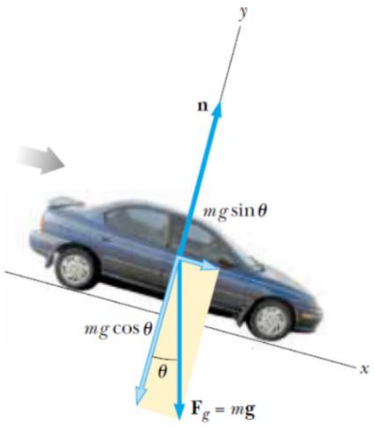
Perpendicular to the incline (y-axis):

$$F_y = N - mg\cos\theta = ma_y$$

Since there is **no motion perpendicular** to the plane,

$$a_y = 0 \Rightarrow N = mg\cos\theta$$

So, the **acceleration along the inclined plane** is $a_x = g\sin\theta$



(b)

- **Normal Force or Reaction Force**

Normal Force $N = mg\cos\theta$

So, the normal force is $N = mg\cos\theta$

- **Time Duration When Object Slides Down the Inclined Plane**

Using the equation of motion along the incline:

$$d = \frac{1}{2}a_x t^2$$

Where: $a_x = g\sin\theta$

This implies $d = \frac{1}{2}g\sin\theta t^2$

Solving for time t: $t = \sqrt{\frac{2d}{g\sin\theta}}$

• **Velocity of an Object Sliding Down a Frictionless Inclined Plane**

Using the **time-independent equation of motion**:

$$v_{xf}^2 - v_{xi}^2 = 2a_x d$$

Since the object starts from rest:

$$v_{xi} = 0$$

This implies $v_{xf}^2 = 2a_x d \Rightarrow v_{xf} = \sqrt{2a_x d}$

Now substitute $a_x = g\sin\theta$

$$v_{xf} = \sqrt{2gd\sin\theta}$$

Hence, the velocity of an object sliding down a frictionless inclined plane is $v_{xf} = \sqrt{2gd\sin\theta}$

• Equation of Motion Along the Inclined Plane (x-axis)

$$d = \frac{1}{2}a_x t^2 = \frac{1}{2}g\sin\theta t^2$$

Hence, displacement **along the inclined plane** is $d = \frac{1}{2}a_x t^2 = \frac{1}{2}g\sin\theta t^2$

Where, d is the displacement **along the inclined plane** measured in (m).

t is the **time taken to slide down the plane** measured in (s).

$g = 9.8$ is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

a_x is the **acceleration of the object along the inclined plane** measured in $(\mathrm{m/s^2})$.

θ is the **angle of the inclined plane (to horizontal)** measured in $(^\circ)$.

VII. The Atwood's Machine

An Atwood's Machine consists of **two masses** m_1 and m_2 connected by a light, inextensible string over a frictionless pulley.

Applying Newton's Second Law:

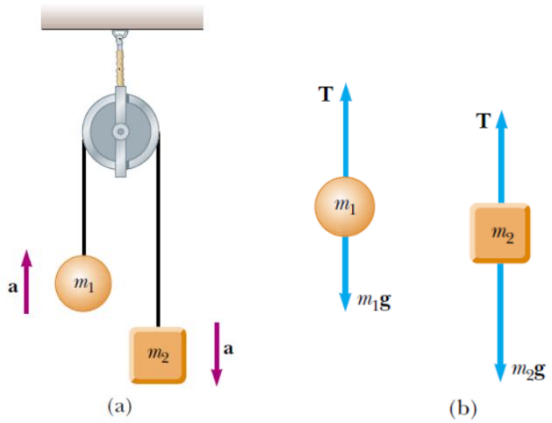
• For mass m_1: $\Sigma F_y = T - m_1 g = m_1 a \quad (1)$

• For mass m_2: $\Sigma F_y = m_2 g - T = m_2 a \quad (2)$

Add Equation (1) and (2):

$$T - m_1 g + m_2 g - T = m_1 a + m_2 a$$

$$(m_2 - m_1)g = (m_1 + m_2)a$$



Solving for time a: $a = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) g \quad (3)$

Substituting (3) into (1) to find Tension:

$$T = m_1 g + m_1 a = m_1 g + m_1 \left[\frac{\left(m_2 - m_1\right)}{\left(m_1 + m_2\right)} g\right]$$

Expanding for time T: $T = \left(\frac{2 m_1 m_2}{m_1 + m_2}\right) g$

Hence, the acceleration and tension are $a = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) g$, $T = \left(\frac{2 m_1 m_2}{m_1 + m_2}\right) g$.

Where, a is the **acceleration of the objects** measured in $(\mathrm{m/s^2})$.

T is the **tension in the string** measured in (N).

$g = 9.8$ is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

m_1 is the **mass of the first object** measured in (kg).

m_2 is the **mass of the second object** measured in (kg).

VIII. Free Falling

An object is said to be in **free fall** when it is acted upon **only by its weight**, meaning **the force of gravity** is the only force acting on it (air resistance is neglected).

In this case, the acceleration of the object is:

$$\vec{a} = \vec{g}$$

Since velocity is changing due to the constant gravitational acceleration, **free fall is a type of non-uniform motion** (uniform acceleration).

The position-time equation for an object in free fall is:

$$y = \frac{1}{2} g t^2 + v_0 t + y_0$$

Where, a is the **acceleration of the objects** measured in $(\mathrm{m/s^2})$.

y is the **final vertical position** measured in (m).

y_0 is the **initial vertical position** measured in $(\mathrm{m/s^2})$.

$g = 9.8$ is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

v_0 is the **initial vertical velocity** measured in $(\mathrm{m/s})$.

t is the **time** measured in (s).

Chapter1 Mechanics
Lesson3 Work-Energy and Power

I. Production of Constant Force

i. Scalar quantity and Vector quantity

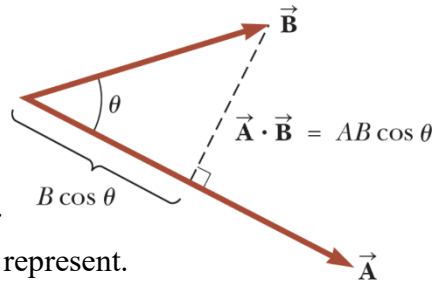
A. The Scalar Product of Two Vectors

When two vectors $\vec{A}$ and $\vec{B}$ form an angle θ between them, the **scalar product** (also called **dot product**) of these vectors is defined as:

$$P = \vec{A} \cdot \vec{B}$$

Using the definition of the dot product, this becomes:

$$P = AB\cos\theta$$



Where, $\vec{A}, \vec{B}$ are **vectors** depend on what those vectors represent.

P is the **scalar product** depending on what those vectors represent.

A is the **force** measured in (N).

B is the **displacement** measured in (m).

θ is the **angle** between the two vectors measured in $(°)$.

B. The Vector Product of Two Vectors

The **vector (cross) product** of two vectors $\vec{A}$ and $\vec{B}$ is a **vector quantity** given by:

$$\vec{P} = \vec{A} \times \vec{B}$$

The resulting vector $\vec{P}$ has the following characteristics:

- **Point of application**: At point O (the origin or a reference point)
- **Direction**: Perpendicular to both $\vec{A}$ and $\vec{B}$
- **Direction determination**: Follows the **right-hand rule**:
  - Point your **thumb** in the direction of $\vec{A}$
  - Point your **index finger** in the direction of $\vec{B}$
  - Your **middle finger** (perpendicular to the palm) points in the direction of $\vec{P}$
- **Magnitude** of the vector product:

$$P = AB\sin\theta$$

Where, $\vec{A}, \vec{B}$ are **vectors** depend on what those vectors represent.

P is the **vector product** depending on what those vectors represent.

A is the **force** measured in (N).

B is the **displacement** measured in (m).

θ is the **angle** between the two vectors measured in $(°)$.

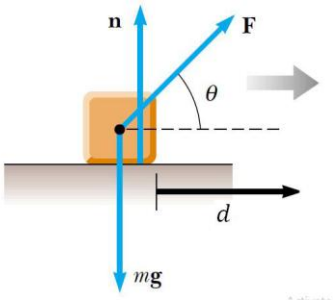
II. Work Done by a Constant Force

Assume an object experiences a **constant force** $\vec{F}$ and moves from point A to point B, being displaced by a distance $d =$ AB (displacement).

Definition of Work:

The work W done by the force is given by the **scalar product** of force and displacement: $W = \vec{F} \cdot \vec{d}$

Hence, work done by a constant force is $W = Fd\cos\theta$



Where, W is **work done** measured in (J).

F is the **magnitude of the force** measured in (N).

d is the **displacement** measured in (m).

θ is the **angle** between the two vectors $\vec{F}$ and $\vec{d}$ measured in $(^\circ)$.

Note: Cases based on the angle θ:

- $0^\circ \le \theta \le 90^\circ \Rightarrow \cos\theta > 0 \Rightarrow W > 0$: Work is positive. Force helps the motion.
- $\theta = 0^\circ \Rightarrow \cos 0^\circ = 1 \Rightarrow W = F \cdot d > 0$: Force and displacement are in same direction.
- $\theta = 90^\circ \Rightarrow \cos 90^\circ = 0 \Rightarrow W = Fd\cos\theta = 0$: Force is perpendicular to displacement.
- $90^\circ < \theta < 180^\circ \Rightarrow \cos\theta < 1 \Rightarrow W = Fd\cos\theta < 0$: Work is negative. Force opposes motion.
- $\theta = 180^\circ \Rightarrow \cos 180^\circ = -1 \Rightarrow W = Fd < 0$: Work is opposing.

III. Kinetic Energy and the Work-Energy Theorem

For a point mass m moving with velocity v, the **kinetic energy** is given by:

$$K = \frac{1}{2}mv^2$$

Where, K is the **kinetic energy** measured in (J).

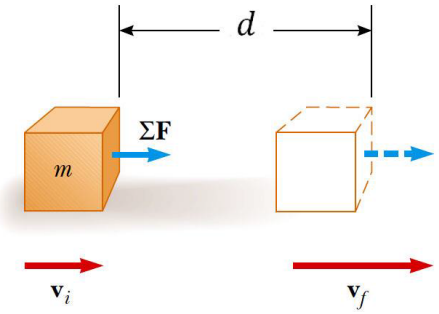
m is the **mass of the object** measured in (kg).

v is the **velocity** measured in $(\mathrm{m/s})$.

Note: Kinetic energy is the energy that an object possesses due to its motion.

i. The Work-Energy Theorem

Assume that an object experiences a force F and moves from point A with velocity v_i to point B with velocity v_f, over a displacement $d =$ AB.



The work done by the force F is:

$$W_{AB}(F) = \vec{F} \cdot \vec{d} = Fd\cos\theta$$

Where θ is the angle between force and displacement. If $\theta = 0^\circ$, then:

$$W_{AB}(F) = Fd$$

From kinematics, displacement is:

$$d = v_{av} \cdot t = \frac{v_i + v_f}{2} \cdot t$$

And from Newton's Second Law: $F = ma$

Since: $v_f = v_i + at \Rightarrow a = \dfrac{v_f - v_i}{t}$

Now substitute a and d into the work equation:

$$W_{AB}(F) = ma \cdot \left(\frac{v_f + v_i}{2} \cdot t \right)$$

$$W_{AB}(F) = m \cdot \frac{v_f - v_i}{t} \cdot \frac{v_f + v_i}{2} \cdot t$$

Simplify: $W_{AB}(F) = \dfrac{1}{2} m\left(v_f^2 - v_i^2\right)$

So, $W_{AB}(F) = \Delta K$. The **net work done** by the force equals the **change in kinetic energy** of the object.

Where, $K_i = \dfrac{1}{2} m v_i^2$ is the **kinetic energy at point A** measured in (J).

$K_f = \dfrac{1}{2} m v_f^2$ is the **kinetic energy at point B** measured in (J).

W_{AB} is the **work done** measured in (J).

IV. Power

Power is defined as the rate at which work is done or energy is transferred.

i. Average Power

If a force does work ΔW over a time interval Δt, then the **average power** is given by:

$$P_{av} = \frac{\Delta W}{\Delta t}$$

Where, P_{av} is the **power** measured in (W).

ΔW is the **work done** measured in (J).

Δt is the **time** measured in (s).

ii. Instantaneous Power

Instantaneous power is defined as the rate at which work is done at a specific instant of time. It is the **limit of average power** as the time interval $\Delta t \to 0$:

$$P = \lim_{\Delta t \to 0} P_{av} = \lim_{\Delta t \to 0} \frac{\Delta W}{\Delta t} = \frac{dW}{dt}$$

Since work $dW = \vec{F} \cdot \vec{dx}$, and velocity $\vec{v} = \dfrac{d\vec{x}}{dt}$, we can write:

$$P = \frac{dW}{dt} = \vec{F} \cdot \vec{v}$$

Hence, the instantaneous power is $P = Fv\cos\theta$.

Where, P is the **instantaneous power** measured in (W).

F is the **force** measured in (N).

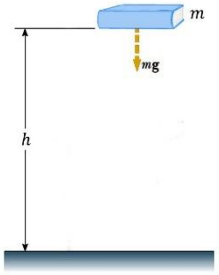
v is the **velocity** measured in $(\mathrm{m/s})$.

θ is the **angle between the force and velocity** measured in $(^\circ)$.

Note: $1\mathrm{kW} = 1000\,\mathrm{W}, 1\mathrm{MW} = 10^6\,\mathrm{W}, 1\mathrm{GW} = 10^9\,\mathrm{W}$

V. Potential Energy

Potential energy is the energy stored in an object due to its position in a force field, typically a gravitational field. The value of potential energy depends on the choice of a **reference level**, also called the **zero level** of potential energy.



$$U_g = mgh$$

Where, U_g is the **gravitational potential energy** measured in (J).

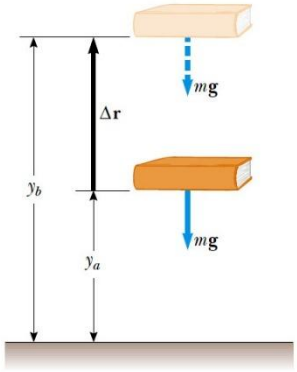
m is the **mass of the object** measured in (kg).

h is the **height above the reference level** measured in (m).

$g = 9.8$ is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

VI. Potential Energy of a System

Consider a book falling under the influence of gravity from position y_b to y_a. The gravitational force acting on the book is:



$$\vec{F}_g = -mg\hat{j}$$

The displacement of the book is:

$$\overrightarrow{\Delta r} = \left(y_a - y_b\right)\hat{j}$$

The **work done by gravity** is:

$$W = \vec{F}_g \cdot \overrightarrow{\Delta r} = -mg\hat{j} \cdot \left(y_a - y_b\right)\hat{j} = -mg\left(y_a - y_b\right)$$

Which simplifies to:

$$W = mg\left(y_b - y_a\right)$$

Since gravitational work equals the negative change in gravitational potential energy:

$$W = -\Delta U_g = mg\left(y_b - y_a\right) \quad or \quad \Delta U_g = U_{g,a} - U_{g,b} = mg\left(y_a - y_b\right)$$

If an object moves in both **horizontal and vertical** directions, displacement is:

$$\overrightarrow{\Delta r} = \left(x_b - x_a\right)\hat{i} + \left(y_b - y_a\right)\hat{j}$$

The gravitational force still acts **vertically**, so:

$$W = \vec{F}_g \cdot \overrightarrow{\Delta r} = -mg\hat{j} \cdot \left[\left(x_b - x_a\right)\hat{i} + \left(y_b - y_a\right)\hat{j}\right]$$

Since $\hat{i} \cdot \hat{j} = 0$, only the vertical component contributes:

$$W = -mg\left(y_b - y_a\right) = mg\left(y_b - y_a\right)$$

Hence, **work done by gravity** is $W = mg\left(y_b - y_a\right)$

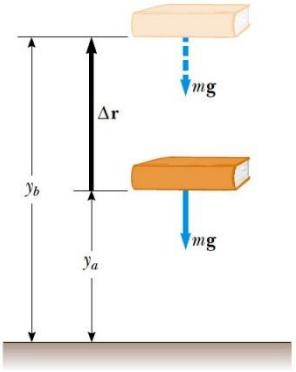
VII. Conservation of Mechanical Energy of an Isolated System

Consider a book falling under the influence of gravity from position y_b to y_a. The gravitational force acting on the book is:

$$\vec{F}_g = -mg\hat{j}$$

The displacement of the book is:

$$\overrightarrow{\Delta r} = \left(y_a - y_b\right)\hat{j}$$



The **work done by gravity** is:

$$W = \vec{F}_g \cdot \overrightarrow{\Delta r} = -mg\hat{j} \cdot \left(y_a - y_b\right)\hat{j} = -mg\left(y_a - y_b\right)$$

Which simplifies to:

$$W = mg\left(y_b - y_a\right)$$

Work–Kinetic Energy Theorem and Conservation of Mechanical Energy

$$W = \Delta K = K_f - K_i$$

In the case of gravitational force, the work done is also related to the change in gravitational potential energy:

$$\Delta K = -\Delta U_g \Rightarrow \Delta K + \Delta U_g = 0$$

This implies: $\left(K_f - K_i\right) + \left(U_f - U_i\right) = 0 \Rightarrow K_f + U_f = K_i + U_i$

Or, written in terms of **mechanical energy**: $E_{mech} = K + U$

Then, $\frac{1}{2}mv_f^2 + mgy_f = \frac{1}{2}mv_i^2 + mgy_i$

Where, W is the **work** measured in (J).

K is the **kinetic energy** measured in (J).

U_g is the **potential energy** measured in (J).

E_{mech} is the t**otal mechanical energy** measured in (J).

m is the **mass** measured in (kg).

v is the **velocity** measured in $(\mathrm{m/s})$.

y is the **height above the reference level** measured in (m).

$g = 9.8$ is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

VIII. Mechanical Energy

In the absence of non-conservative forces (like friction or air resistance), the mechanical energy of a system is **conserved**:

$$E_{MA} = E_{MB} \Rightarrow K_A + U_A = K_B + U_B$$

Where, $K_A = \frac{1}{2}mv_A^2$ is the **kinetic energy at point A** measured in (J).

$K_B = \frac{1}{2}mv_B^2$ is the **kinetic energy at point B** measured in (J).

$U_A = mgy_A$ is the **potential energy at point A** measured in (J).

$U_B = mgy_B$ is the **potential energy at point B** measured in (J).

I. Definition of Fluid

In physics, a **fluid** is any substance that can **flow**. By a more general definition, a **fluid** is a material that continuously deforms (changes shape) under the action of a **shear stress**, no matter how small the applied force is. Fluids include **liquids** (e.g., water, oil) and **gases** (e.g., air, steam).

i. Properties of Fluid

A fluid can flow. A fluid takes the shape of the object or container that holds it.

ii. Types of Fluids

Fluids are classified into two types:

- **Compressible fluids**
- **Incompressible fluids**

Note: In physics, a fluid that **cannot be compressed** is referred to as an **incompressible fluid**.

II. Atmospheric Pressure and Gas Pressure

i. Gas Pressure

Definition: Pressure is defined as the **force exerted per unit area**.

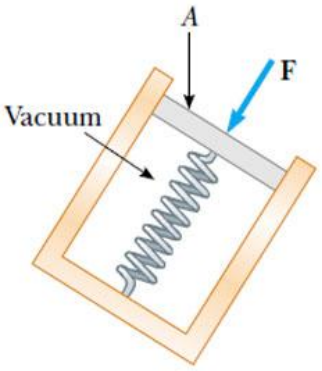
The formula for pressure is:

$$P = \frac{F}{A}$$

Where, P is the **pressure** measured in (Pa).

F is the **force** measured in (N).

A is the **area** measured in (m^2).



ii. Atmospheric Pressure

Definition: Atmospheric pressure is the pressure exerted by the weight of the Earth's atmosphere on a surface.

$$P_{atm} = \frac{mg}{A}$$

At sea level, the standard atmospheric pressure is:

$$P_{atm} = 1.013 \times 10^5\ \mathrm{Pa}$$

Note: $1\mathrm{Pa} = 1\mathrm{N}/\mathrm{m}^2$, $1\ \mathrm{atm} = 760\ \mathrm{mmHg} = 1.013\ \mathrm{bar}$

iii. Existence of Atmospheric Pressure

A well-known experiment demonstrates the existence of atmospheric pressure:

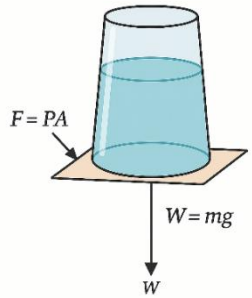
When a glass filled with water is covered with a sheet of paper and then inverted (or slightly inclined), the water does not spill out. This phenomenon occurs due to the **atmospheric pressure** acting perpendicular to the surface of the paper, which holds the water in place.

In the first figure, the water inside the glass is subjected to two forces:

- The **gravitational force** (its weight, $W = \mathrm{m}g$), acting downward.
- The **force due to atmospheric pressure** $F = PA$, acting upward on the paper.

According to the **condition of equilibrium**:

$$F = W$$



Substitute the values:

$$PA = mg$$

Solve for pressure: $P = \dfrac{mg}{A}$

Hence, the atmospheric pressure is $P = \dfrac{mg}{A}$

Where, P is the **pressure** measured in (Pa).

F is the **force** measured in (N).

A is the **area** measured in (m^2).

m is the **mass of the water** measured in (kg).

iv. Barometer

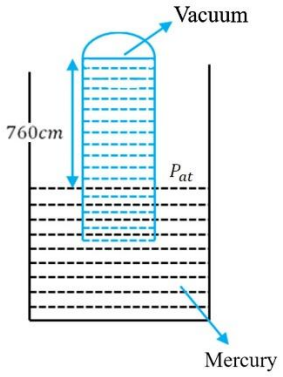
Definition: A **barometer** is a device used to measure **atmospheric pressure**.

Description:

It consists of a container filled with mercury and a long glass tube (about 1–2 meters in length) that is completely filled with mercury and sealed at one end. When the tube is inverted and the open end is submerged into the container of mercury (after removing the sealing cock), mercury falls slightly and a vacuum form at the top of the tube. The height of the mercury column remains at $h = 76\ \mathrm{cm}$ under normal atmospheric pressure.

Since points M and N lie on the same horizontal level (one inside the tube and one in the container), the pressure at both points must be equal:

$$P_M = P_N = P_{atm}$$



But pressure at point M is due to the mercury column:

$$P_M = \rho g h$$

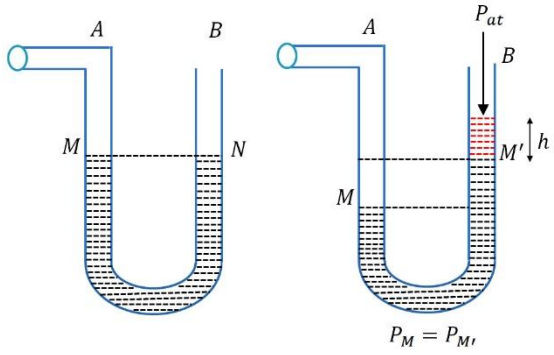
Where: $\rho = 13.6 \times 10^3\ \mathrm{kg/m^3}, g = 9.8\ \mathrm{m/s^2}, h = 76\ \mathrm{cm} = 0.76\ \mathrm{m}$

Substituting: $P_{atm} = 1.013 \times 10^3 \times 9.8 \times 76 \times 10^{-2}$

So, the atmospheric pressure is $P_{atm} = 1.013 \times 10^5\ \mathrm{Pa}$

v. Manometer

Definition: A **manometer** is a device used to measure the **pressure of a gas**. It typically consists of a **U-shaped tube** filled with **mercury** (or another liquid).



- Initially, **both branches** of the U-tube are open to the atmosphere. The mercury levels on both sides are equal, indicating that the pressure on both sides is the same (atmospheric pressure).
- When **gas is introduced** into **branch A**, it exerts additional pressure, causing the mercury level in branch A to **drop**, while the mercury level in branch B **rises**.
- We draw a **dotted horizontal line** from point **M** (on side A) to point **M′** (on side B), which lie on the **same horizontal level**.

Since fluid pressure at the same horizontal level in a continuous fluid is equal:

$$P_M = P_{M'}$$

At point M′ (open to the atmosphere and below a mercury column of height h):

$$P_{M'} = P_{atm} + \rho gh$$

Therefore, the gas pressure in branch A is:

$$P_{M'} = P_M = P_{M'} = P_{atm} + \rho gh$$

Where, $P_{M'}$ is the **gas pressure** measured in (Pa).

ρ is the **density of mercury** measured in $(\mathrm{kg/m^3})$.

h is the **difference in mercury levels between the two branches** measured in (m).

$g = 9.8$ is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

P_{atm} is the **atmospheric pressure** measured in (Pa).

III. Pressure Depends on Depth

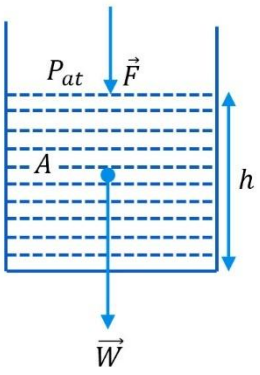
In a container filled with a liquid of **density ρ**, the **pressure at the base** depends on both the **atmospheric pressure** and the **pressure due to the liquid column** (which increases with depth).

Force due to atmospheric pressure:

$$F = P_{atm} A$$

Weight of the liquid column:

$$W = mg = \rho Vg = \rho ghA$$



Since both forces act vertically downward and are parallel:

$$F' = F + W = P_{atm} \cdot A + \rho ghA$$

$$F' = \left(P_{atm} + \rho gh\right) A$$

Solving for P: $$P = \frac{F'}{A} = \frac{\left(P_{atm} + \rho gh\right) A}{A}$$

So, the **total pressure** at depth h is $P = P_{atm} + \rho gh$. This formula shows that **pressure increases with depth** in a liquid.

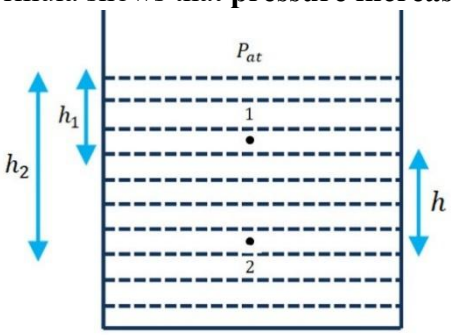
i. Hydraulic Principle (Pascal's Principle)

We consider a **stationary liquid** in a container.

The pressure at point 1 and point 2 is:

$$P_1 = P_{atm} + \rho gh_1$$

$$P_2 = P_{atm} + \rho gh_2$$



Subtracting: $P_2 - P_1 = \rho g\left(h_2 - h_1\right)$

Rewriting: $P_2 = P_2 + \rho g\left(h_2 - h_1\right)$

So, the pressure at point 2 is $P_2 = P_2 + \rho gh$.

Where, P_2 is the **pressure at point 2, located at depth h_2** measured in (Pa).

P_1 is the **pressure at point 1, located at depth h_1** measured in (Pa).

P_{atm} is the **atmospheric pressure** measured in (Pa).

ρ is the **density of mercury** measured in $(\mathrm{kg/m^3})$.

h is the **difference in depth causing the difference in pressure** measured in (m).

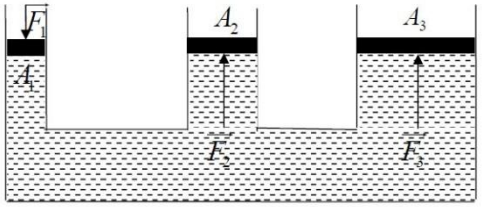
$g = 9.8$ is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

ii. Hydraulic Principle (Pascal's Principle)

A container has a **common base** and is filled with a liquid. It has **three branches**, each with a **piston** that has cross-sectional areas A_1, A_2, and A_3. They applied a force F_1 on **piston 1**.

The pressure experienced by this piston is:

$$P = \frac{F_1}{A_1}$$



According to Pascal's Principle, the **pressure is transmitted equally** through the liquid. So, the **pressure on all pistons remains the same**:

$$P_1 = P_2 = P_3$$

So: $$\frac{F_1}{A_1} = \frac{F_2}{A_2} = \frac{F_3}{A_3}$$

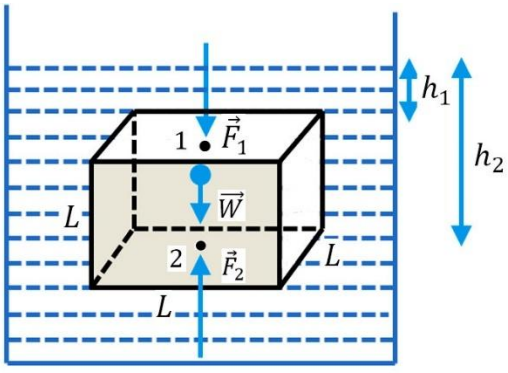
This shows that the **applied force is directly proportional to the cross-sectional area** of each piston.

IV. The Principle of Equilibrium Applied to Buoyant Force

i. Archimedes' Principle

A **rigid cube** with edge length L is **submerged** in a container filled with a liquid of **density ρ**. In the liquid, the rigid body experiences **forces from all directions**. Due to symmetry, the **horizontal forces cancel out**. But the **vertical forces do not**, and their sum results in a **net upward force** (buoyant force):

$$\vec{F}_b = \vec{F}_1 + \vec{F}_2$$



Since $\vec{F}_1 \uparrow\uparrow \vec{F}_2$

This implies $F_b = F_2 - F_1$

Now, $F_2 = P_2 A$ where $P_2 = P_{atm} + \rho gh_2$

$$\Rightarrow F_2 = \left(P_{atm} + \rho gh_2\right)A$$

And, $F_1 = P_1 A$ where $P_1 = P_{atm} + \rho gh_1$

$$\Rightarrow F_1 = \left(P_{atm} + \rho gh_1\right)A$$

Then the net force is:

$$F_b = \left(P_{atm} + \rho g h_2\right) A - \left(P_{atm} + \rho g h_1\right) A$$

$$F_b = \rho g \left(h_2 - h_1\right) A$$

Since $h_2 - h_1 = L$, we get:

$$F_b = \rho g A L$$

But $A = L^2$, so the volume of the cube is $V = L^3$

So, the buoyancy force is $F_b = \rho V g$

Archimedes' Principle states that a rigid body submerged in a stationary liquid experience a force exerted by the liquid, directed upward. This force is applied at the **center of buoyancy** and is called the **buoyant force**.

Where, F_b is the **buoyant force** measured in (N).

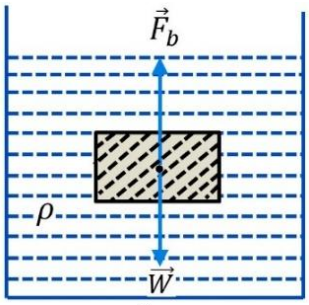
ρ is the **density of the liquid** measured in $(\mathrm{kg/m^3})$.

V is the **volume of the object submerged in the fluid** measured in $(\mathrm{m^3})$.

$g = 9.8$ is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

V. The Floated Body

Consider a rigid body placed in a liquid of density ρ. The body experiences two forces such as its **weight**, W, acting downward and a **buoyant force**, F_b, acting upward.



The behavior of the object depends on the relationship between W and F_b:

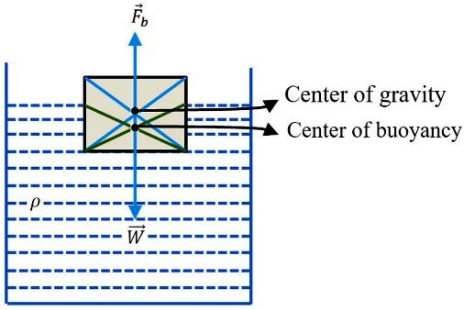
- If $W > F_b$, the object **sinks** and moves downward.
- If $W = F_b$, the object is **fully submerged** and remains in **equilibrium.**
- If $W < F_b$, the object **floats** on the surface of the liquid.

According to the equilibrium condition:

$$\Sigma F = 0 \Rightarrow W = F_b$$

But: $W = mg = \rho V g$ (object's weight)

$F_b = \rho' V' g$ (bouyancy force from displaced fluid)



So, under equilibrium:

$$\rho V g = \rho' V' g$$

Cancelling g: $\rho V = \rho' V'$

So, $\rho V = \rho' V'$ is the **condition of floating equilibrium** (or **floating condition**) for a body in a fluid.

Where, ρ is the **density of the object** measured in $(\mathrm{kg/m^3})$.

ρ' is the **density of the liquid** measured in $(\mathrm{kg/m^3})$.

V is the **volume of the object** measured in $(\mathrm{m^3})$.

V' is the **volume of the liquid displaced** measured in $(\mathrm{m^3})$.

VI. The Density of a Rigid Body and a Liquid

The **relative density** (also called **specific gravity**) of a rigid body or liquid is defined as the **ratio of its mass to the mass of an equal volume of water**.

It is written as: $d = \frac{m}{m_e}$

Where, d is the **relative density.**

m is the **mass of the rigid body or liquid** measured in (kg).

m_e is the **mass of water that occupies the same volume** measured in (kg).

The **density** of a rigid body or liquid is given by:

$$\rho = \frac{m}{V}$$

Where, ρ is the **density of a rigid body or liquid** measured in (kg/m^3).

m is the **mass of the rigid body or liquid** measured in (kg).

V is the **volume of the object or liquid** measured in (m^3).

The **density of water** is:

$$\rho_e = \frac{m_e}{V_e} \text{ or } m_e = \rho_e V_e$$

Since $V = V_e$, we can substitute:

$$d = \frac{m}{m_e} = \frac{\rho V}{\rho_e V_e} = \frac{\rho}{\rho_e}$$

Hence, $d = \frac{\rho}{\rho_e}$

This equation shows that **relative density is the ratio of the density of a substance to the density of water**.

Where, d is the **relative density.**

ρ is the **density of a rigid body or liquid** measured in (kg/m^3).

ρ_e is the **density of water** measured in (kg/m^3).

■ **Other Liquids (Different from Water)**

The relative density of a liquid other than water is defined as:

$$d' = \frac{\rho'}{\rho_e} \text{ or } \rho' = d'\rho_e$$

Substituting into the buoyancy equilibrium:

$$d\rho_e V = d'\rho_e V'$$

Cancelling ρ_e: $dV = d'V'$

■ **Volume Weight (Specific Weight) – Rigid Body (General Case)**

Volume weight (also called **specific weight**) is the **weight per unit volume**:

$$\overline{W} = \frac{W}{V} = \frac{mg}{V}$$

So, $\overline{W} = \rho g \Rightarrow \rho = \dfrac{\overline{W}}{g}$

- **Volume Weight-Liquid**

For a liquid: $\overline{W}' = \rho g \Rightarrow \rho' = \dfrac{\overline{W}'}{g}$

From Equation (1): $\dfrac{\overline{W}}{g} \cdot V = \dfrac{\overline{W}'}{g} \cdot V'$

Cancelling g: $\overline{W}\,V = \overline{W}'V'$

- **Floating Condition**

$$W = F_b$$

$$\rho V g = \rho' V' g \Rightarrow \rho V = \rho' V' \Rightarrow dV = d'V' \Rightarrow \overline{W}V = \overline{W}'V'$$

Quantity	Symbol	Equation
Relative Density	$d = \dfrac{\rho}{\rho_e}$	Unitless
Density	$\rho = \dfrac{m}{V}$	$(\mathrm{kg/m^3})$.
Volume Weight	$\overline{W} = \rho g$	$(\mathrm{N/m^3})$.
Floating Condition	$\rho V = \rho' V'$	Object floats

Chapter2 Thermodynamics
Lesson1 Temperature and Heat

I. Temperature and Heat

i. Definition: **Temperature** is a measure of how hot or cold an object is. **Heat** is the amount of energy transferred from a hotter object to a colder object.

II. Temperature Scales

Temperature scales are classified into three main types:

- Celsius scale (°C), also known as Satapheap
- Kelvin scale (K), also known as the absolute scale
- Fahrenheit scale (°F)

*** Relationships between temperature scales:**

- **Celsius to Kelvin:**

$$T = t + 273$$

Where, T is the **temperature in Kelvin** measured in (K).

t is the **temperature in Celsius** measured in (°C).

- **Celsius to Fahrenheit:**

$$F = \frac{9}{5}t + 32$$

Where, F is the **temperature in Fahrenheit** measured in (°F).

t is the **temperature in Celsius** measured in (°C).

III. Calculation of Temperature in Satapheap Scale (Celsius)

In a mercury thermometer, the volume of mercury changes with temperature. Therefore, the temperature can be measured by observing the change in the length of the mercury column, assuming the cross-sectional area of the capillary tube remains constant.

Let:

- l_0 be the level of mercury when the thermometer is placed in pure melting ice (0 °C).
- l_{100} be the level of mercury when placed in pure steam at boiling point (100 °C).
- l_t be the level of mercury when placed at an unknown temperature t

By proportion, the temperature t in Celsius can be calculated using the formula:

$$t = \frac{l_t - l_0}{l_{100} - l_0} \times 100\ ^\circ\text{C}$$

IV. Heat Energy

When an object with mass m and specific heat capacity c absorbs or releases heat, its temperature changes from an initial temperature T_1 to a final temperature T_2 (or vice versa).

Let t_1 and t_2 be the initial and final temperatures in °C.

But since: $T_2 = t_2 + 273$ and $T_1 = t_1 + 273$

we substitute: $Q = mc\left(t_2 + 273 - t_2 - 273\right) = mc\left(t_2 - t_1\right)$

So: $Q = mc\Delta t$

This heat transfer is calculated using the formula:

$$Q = mc\Delta t$$

Where, Q is the **heat energy** measured in (J).

m is the **mass of the object** measured in (kg).

c is the **specific heat capacity** measured in $(\mathrm{J/kg.°C})$.

Δt is the **change in temperature** measured in $(°\mathrm{C})$.

Note: The temperature difference is the same whether measured in Celsius or Kelvin. The change in temperature $\Delta \mathrm{T} = \Delta t = 1\ °\mathrm{C}$.

V. Heat Capacity

i. Definition: Heat capacity is the amount of heat energy absorbed or released by an object in order to increase or decrease its temperature by $1\ \mathrm{K}$ or $1°\mathrm{C}$.

Heat capacity $C = \dfrac{Q}{\Delta T}$

Where, C is the **heat capacity** measured in $(\mathrm{J/K\ or\ J/°C})$.

Q is the **heat energy** measured in (J).

Δt is the **change in temperature** measured in $(°\mathrm{C})$.

VI. Specific Heat Capacity

i. Definition: Specific heat capacity is the amount of heat energy absorbed or released by an object with a mass of $1\ \mathrm{kg}$ in order to increase its temperature by $1°\mathrm{C}$ or $1\ \mathrm{K}$.

They can write: $c = \dfrac{C}{m} = \dfrac{Q}{m\Delta t}$

So, they get: $Q = mc\Delta t$

Where, Q is the **heat energy** measured in (J).

m is the **mass of the object** measured in (kg).

c is the **specific heat capacity** measured in $(\mathrm{J/kg.°C})$.

***Special Cases:**

- In the case where the mass of an object is $m = 1\ \mathrm{kg}$ and the temperature change $\Delta T = T_2 - T_1 = 1\ \mathrm{K}$.

$$Q = c$$

- In the case where the mass of an object is $m = 1\ \mathrm{kg}$ and the temperature change $\Delta T = T_2 - T_1 = 1\ °\mathrm{C}$.

$$Q = c$$

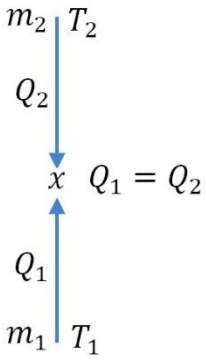
***Note:** Specific heat capacity of water: $c = 4200\ \mathrm{J/kg.^oC} \approx 4190\ \mathrm{J/kg.^oC}$.

VII. Law of Heat Exchange

They have two different temperatures are

When two objects at different temperatures come into contact. The first object has a higher temperature than the second. The object with the higher temperature **releases** heat to the object with the lower temperature. The amount of heat **released**

by the hot object is equal to the amount of heat **absorbed by the cold object**.



Heat released by the hot object:

$$Q_2 = m_2 c\left(t_2 - x\right)$$

Heat absorbed by the cold object:

$$Q_1 = m_1 c\left(x - t_1\right)$$

By the principle of heat exchange:

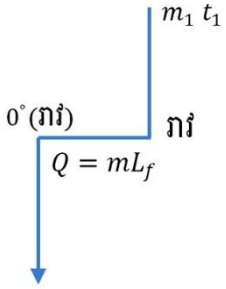
$$Q_1 = Q_2$$

So: $m_1 c\left(x - t_1\right) = m_2 c\left(t_2 - x\right)$

Where, x is the **equilibrium temperature (final temperature of both objects)** measured in $(\,^\circ\mathrm{C})$.

VIII. Latan Heat of Futon

i. Definition: Latent heat of fusion is the amount of heat energy required to melt an object with mass m at its melting temperature, **without any change in temperature**.



They write: $Q_f = mL_f$

Where, Q_f is the **latent heat of fusion** measured in (J).

m is the **mass of the object** measured in (kg).

L_f is the **specific latent heat of fusion** measured in $(\mathrm{J/kg})$.

*** Special Cases:**

- **Latent heat of fusion** is the amount of heat energy needed to melt $1\ \mathrm{kg}$ of a substance from solid to liquid at its melting point without changing temperature.
- **Latent heat of fusion of ice** is the energy required to melt $1\ \mathrm{kg}$ of ice at $0\ ^\circ\mathrm{C}$ into water without a change in temperature.

$$L_f = 3.34 \times 10^5\ \mathrm{J/kg}$$

IX. Latan Heat of Evaporation

i. Definition: Latent heat of evaporation (or vaporization) is the amount of heat energy required to change **1 kg** of a substance from liquid to vapor **at its boiling point**, without any change in temperature.
It is the transformation of a liquid into gas at the **boiling temperature and constant pressure**.
This can be written as:

$$Q_V = mL_V$$

Where, Q_V is the **latent heat of evaporation** measured in (J).

m is the **mass of the object** measured in (kg).

L_V is the **specific latent heat of evaporation** measured in $(\mathrm{J/kg})$.

***Note:** $1\ \mathrm{cal} = 4.18\ \mathrm{J}$

X. Melting, Boiling, Freezing, Steam, and Vaporization

i. Definition

- **Melting** is the process of changing a substance from **solid to liquid**.
- **Freezing (Clotting)** is the process of changing a substance from **liquid to solid**.
- **Condensation** is the process of changing **steam into liquid**.

■ **Boiling** is a rapid vaporization process where a liquid changes to gas, forming bubbles throughout the liquid and rising to the surface.

■ **Vaporization** is the process of changing from **liquid to gas without boiling**, usually occurring **only at the surface** of the liquid.

∗ Comparison Between Vaporization and Boiling

Feature	Boiling	Vaporization
Occurs at	Constant (boiling) temperature	Any temperature below boiling point
Process speed	Fast	Slow
Where it happens	Throughout the liquid	Only at the surface
Bubble formation	Yes, bubbles form inside and rise	No bubbles form
Temperature	Constant during boiling	Temperature can change
Heat source	Supplied by an external heat source	Supplied by the environment

Chapter2 Thermodynamics
Lesson2 Kinetic Theory of Molecules

I. Kinetic Theory of Molecules

The **kinetic theory of molecules** explains the properties of matter by describing the motion and interaction of particles (molecules or atoms).

Matter is classified into **three states** such as **solid**, **liquid** and **gas.**

The separation between particles in each state defines its physical properties. Particles cannot separate freely unless energy is added; they remain bound in a way that keeps the basic characteristics of the matter.

The reason **solid** has a fixed volume and shape is because its molecules are **closely packed**, **strongly bound**, and vibrate only slightly around fixed positions (equilibrium points).

The reason **gas** has no fixed volume or shape is because gas molecules are **widely separated**, **move freely and rapidly in all directions**, and experience almost **no attractive force** between them.

∗ **Arrangement of Molecules in the Three States of Matter:**

- **Solid**: Molecules are **tightly bound** and packed **very close together**.
- **Liquid**: Molecules are **less tightly bound**, form **clusters**, and are more separated than in solids.
- **Gas**: Molecules are **far apart**, move **randomly**, and have **no definite volume or shape**.

∗ **Properties of Solids:**

- Fixed **shape** and **volume**
- Strong molecular binding
- Require **strong force** to change shape
- High density
- Cannot be compressed

∗ **Properties of Liquids:**

- Irregular shape (takes the shape of the container)
- Definite volume
- Cannot be compressed easily

∗ **Properties of Gases:**

- Shape and volume can **change freely**
- Low density
- Can be compressed

Chapter3 Electricity and Magnetism
Lesson1 Electrostatics

I. Electric Charges

Electric charge exists in **two types**:

- **Positive electric charge** $(+)$
- **Negative electric charge** $(-)$

It is assumed that when a **chopstick is rubbed with a silk cloth**, it becomes **positively charged**, and when it is **rubbed with fur**, it becomes **negatively charged**.

∗ **Attraction and Repulsion Between Charges**

- **Like charges** (same sign) **repel** each other.
- **Unlike charges** (opposite signs) **attract** each other.

II. Electron Theory

All matter is made up of **molecules**, and each molecule is composed of **atoms**. Each **atom** consists of a **positively charged nucleus** at the center and **negatively charged electrons** that orbit around the nucleus continuously.

- In general, if the **total algebraic sum of all charges** in an atom is **zero**, the atom is said to be **electrically neutral** (as shown in **Figure A**).
- If an atom **loses one or more electrons**, the total charge becomes **positive**, and the atom becomes a **positive ion (cation)** (as shown in **Figure B**). Conversely, if an atom **gains extra electrons**, the total charge becomes **negative**, and the atom becomes a **negative ion (anion)** (as shown in **Figure C**).

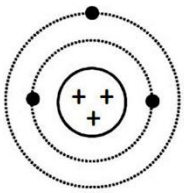


Fig (a)

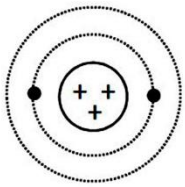


Fig (b)

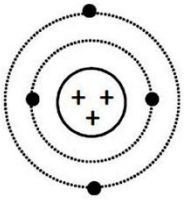


Fig (c)

The **electron** is a fundamental particle that carries a **negative electric charge**, denoted by $-e = -1.6 \times 10^{-19}$ C and its mass is $m_e = 9.11 \times 10^{-31}$ kg. The **electric charge** q of an object is the product of the number of charges and the fundamental charge. It is expressed as:

$$q = n|e|$$

Where, q is the **charge of the object** measured in (C).

n is the **number of charges (electrons or protons)**.

$|e|$ is the **magnitude of the fundamental charge** measured in 1.6×10^{-19} C.

A **point charge** is an electric charge that is considered to be **concentrated at a single point** in space. This assumption is valid when the size (dimension) of the charged object is **very small compared to the distance** over which the electric field or force is being studied.

III. Electrification

i. **Definition:** Electrification is the process by which an object **gains or loses electrons**, resulting in an **electric charge**. There are **three main types of electrification** such as **electrification by friction, electrification by contact** and **electrification by induction.**

- **Electrification by friction** is the process where **two objects are rubbed together**, causing electrons to be transferred from one object to the other, resulting in both objects becoming electrically charged.
- **Electrification by contact** occurs when a **charged object is brought into contact with a neutral object**, transferring some of the charge and causing the neutral object to become charged as well.
- **Electrification by induction** happens when a **charged object is brought near a neutral object without touching it**. The electric field of the charged object causes a **redistribution of charges** within the neutral object, which results in induced electrification.

IV. Electroscope

i. **Definition:** An **electroscope** is a device used to **detect whether an object is electrically charged or not**.

- For a **uniformly conductive object**, electric charge is **evenly distributed** across the surface.
- For a **conductive object with curved surfaces**, the **electric charge tends to concentrate at sharp or pointed areas**, such as edges or tips.
- The term **"no electric charge"** means the object contains **equal amounts of positive and negative charges**, so it is electrically **neutral**.
- The **phenomenon of the electroscope** is applied in various technologies and devices such as **lightning rods** and **Van de Graaff generators**.
- In everyday life, the **principle of the electroscope** is also used in **removing smoke and dust** from gases in **ore processing plants** and **electrostatic spray painting**.

Chapter3 Electricity and Magnetism
Lesson2 Direction Current and Magnetism

I. Direction Current

i. Definition: **Direct current (DC)** is the flow of **electric charge** through a conducting wire in a **specific, constant direction**.

- The relationship between electric charge, current, and time is given by:

$$Q = It \quad \text{or} \quad I = \frac{Q}{t}$$

Where, Q is the **amount of electric charge passing through the cross-sectional area** measured in (C).

I is the **electric current** measured in (A).

t is the **time taken** measured in (s).

- The amount of electric charge can also be calculated as:

$$Q = n|e|$$

Where, Q is the **charge of the object** measured in (C).

n is the **number of electrons passing through the area**.

$|e|$ is the **magnitude of charge of a single electron** measured in $1.6 \times 10^{-19}\ \mathrm{C}$.

*** Electric Current in Different Materials:**

- In **metals**, electric current is the flow of **free electrons**.
- In **electrolytes**, electric current is the flow of **positive and negative ions**.
- In **n-type semiconductors**, current is mainly due to the movement of **excess electrons**.
- In **p-type semiconductors**, current is mainly due to the movement of **holes** (which behave like **positive charges**).

II. Measure of Electric Current

i. Illustration of Ammeter

Definition: An **ammeter** is a device used to **measure the electric current** flowing through a circuit.

To accurately calculate the electric current shown by the ammeter, the following terms are used:

n_0: total number of calibration divisions (scale marks)

n: number of divisions pointed to by the needle

cal: calibration value per division (in amperes)

I: electric current (in amperes)

Formula for Ammeter Reading:

$$I = cal\,\frac{n}{n_0}$$

Where, I is the **measured current** measured in (A).

cal is the **calibration value per division**.

n/n_0 is the **fraction of the scale used.**

ii. Illustration of Voltmeter

Definition:

A **voltmeter** is a device used to **measure voltage** or the **electric potential difference** between two points in a circuit.

To calculate the voltage using the voltmeter reading, we consider:

n_0: total number of calibration divisions (scale marks)

n: number of divisions pointed to by the needle

cal: calibration value per division (in amperes)

V: electric potential difference (in voltage)

Formula for Voltmeter Reading:

$$V = cal\frac{n}{n_0}$$

Where, V is the **measured voltage** measured in (V).

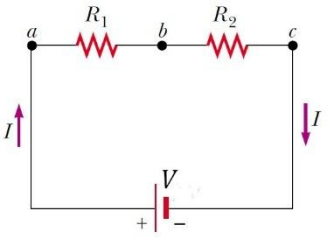
cal is the **calibration value per division**.

n_0 is the **full-scale division**.

n is the **needle position**.

iii. Series Combination of Resistors

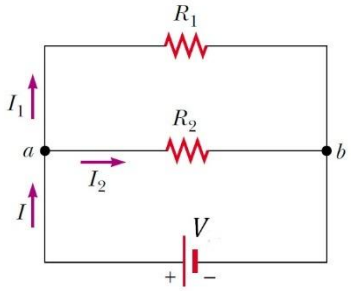
When resistors are connected **in series**, they are connected **end-to-end** so that the same current flows through each resistor. The total resistance is the **sum** of individual resistances.



- **Current**: $I = I_1 = I_2 = ... = I_n$
- **Voltage**: $V = V_1 + V_2 + ... + V_n$
- **Resistance**: $R = R_1 + R_2 + ... + R_n$

iv. Parallel Combination of Resistors

When resistors are connected **in parallel**, all resistors share the **same voltage** across them, and the total current is the **sum** of the individual currents through each resistor.



- **Current**: $I = I_1 + I_2 + ... + I_n$
- **Voltage**: $V = V_1 = V_2 = ... = V_n$
- **Resistance**: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + ... + \frac{1}{R_n}$

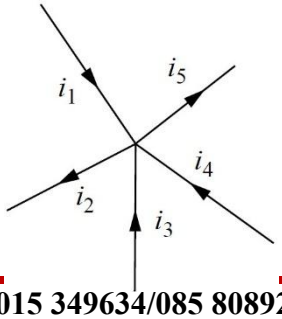
v. Voltage Addition (Loop Rule)

In a closed loop, the total voltage is the sum of the individual potential differences across each part of the loop.

They can write: $V_{ac} = V_{ab} + V_{bc} = -V_{ca}$

Note: (Voltage gains and drops are added algebraically around the loop.)

vi. Node's Law (Kirchhoff's Current Law - KCL)



At any junction (node), the total current entering equals the total current leaving.

They write: $\Sigma I_{in} = \Sigma I_{out}$

(This law expresses conservation of electric charge.)

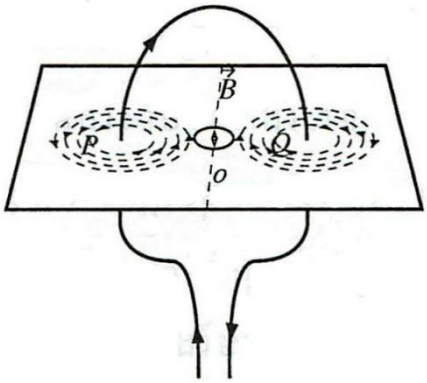
III. Magnetism

i. Magnetic Field Due to a Straight Current-Carrying Wire

When a current flows through a long straight conductor, it produces a magnetic field around it in concentric circles.

The magnetic field at a distance d from a straight, long current-carrying wire is given by the formula:

$$B = \frac{\mu_0 I}{2\pi d}$$



Where, B is the **magnetic field** measured in (T).

$\mu_0 = 4\pi\times 10^{-7}\ \mathrm{Tm/A}$ is the **permeability of free space.**

I is the **current in the wire** measured in (A).

d is the **perpendicular distance from the wire to the point of measurement** measured in (m).

ii. Magnetic Field Due to a Circular Loop of Wire

A wire is bent into a circular shape and penetrates a cardboard at two points, ***P*** and ***Q***. When an electric current is passed through the wire, a magnetic field is generated around it.

The magnetic field at the center ***O*** of the circular loop is given by the formula:

$$B = \frac{\mu_0 I}{2R}$$

Where, B is the **magnetic field** measured in (T).

$\mu_0 = 4\pi\times 10^{-7}\ \mathrm{Tm/A}$ is the **permeability of free space.**

I is the **current in the wire** measured in (A).

R is the **radius of the circular loop** measured in (m).

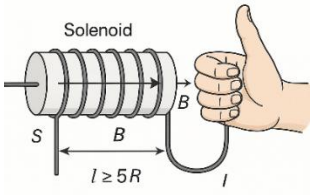
To observe the direction of the magnetic field at the center point ***O***, a compass is placed at ***O***. The north pole of the compass needle will point in the direction of the magnetic field. Alternatively, the direction can be determined using the **right-hand rule** or **compass S–N face method**.

iii. Magnetic Field Due to a Solenoid

A **solenoid** is a coil made by winding wire uniformly in a spiral around a cylindrical tube, where the length of the cylinder is greater than or equal to 5 times its radius $(l \geq 5R)$. When an electric current passes through the solenoid, it produces a magnetic field ***B*** inside the solenoid.

The magnetic field inside the solenoid is given by the formula:

$$B = \frac{\mu_0 NI}{l}$$



Where, B is the **magnetic field** measured in (T).

$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{Tm/A}$ is the **permeability of free space.**

I is the **current in the wire** measured in (A).

l is the **length of the solenoid** measured in (m).

N is the **radius of the circular loop** measured in (turns).

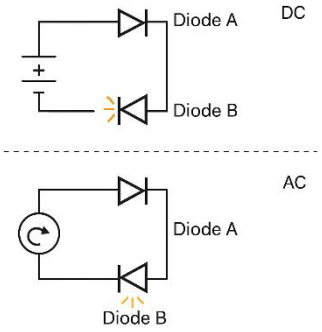
The **direction** of the magnetic field inside the solenoid can be determined using the **S–N (South–North) method** or the **right-hand grip rule**: curl your fingers in the direction of current flow, and your outstretched thumb will point in the direction of the magnetic field vector $\boldsymbol{B}$.

■ **Note:** The **magnetic spectrum** refers to the pattern or arrangement of **magnetic field lines** surrounding a magnet or current-carrying wire.

Chapter3 Electricity and Magnetism
Lesson3 Alternating Current

I. Nature of Alternating Current

i. The Difference Between Direct Current (DC) and Alternating Current (AC)



To compare the characteristics of DC and AC, an experiment is conducted using two diodes (A and B) connected in parallel but in opposite directions.

The following observations are made:

- When both ends of the diode system are connected to a **DC power source** (such as a battery), only **diode B emits light**, and **diode A does not emit light**. This indicates that **electric current flows in only one direction**.
- When the same setup is connected to the **dynamo of a vehicle** (which produces alternating current), the behavior is different:
  - As the dynamo slowly rotates, they observe that **the diodes alternately emit and stop emitting light**.
  - This shows that **electric current from the dynamo changes direction continuously** rather than flowing in one constant direction.

ii. The Simple DC and AC Generators

To produce alternating current (AC) in a power station, a **turbine** is used to rotate a **magnet (rotor)** around a coil of wire. This rotation induces an electromotive force (emf) in the wire due to **electromagnetic induction**. This is the basic principle behind both **DC (direct current)** and **AC (alternating current)** generators.

- In a **DC generator**, a **commutator** is used to ensure that current flows in one direction.
- In an **AC generator**, the current naturally alternates direction as the magnetic field changes polarity with rotation.

Relationship Between Frequency and Period

$$f = \frac{1}{T}$$

Where, f is the **frequency** measured in (Hz).

T is the **period** measured in (s).

Angular Frequency

$$\omega = \frac{v}{r}$$

Alternatively, angular frequency can also be expressed as:

$$\omega = 2\pi f$$

Where, ω is the **angular frequency** measured in (rad/s).

v is the **linear speed of the rotating object** measured in (m/s).

r is the **radius of rotation** measured in (m).

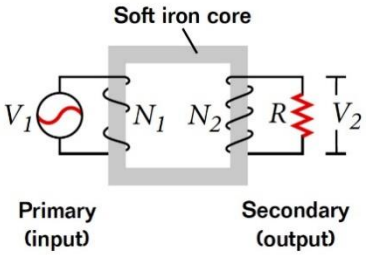
II. Function of Transformers

i. Construction and Combination of a Transformer

A transformer consists of coils of wire wrapped around a soft iron core. The core is often laminated made of thin iron sheets arranged parallel to each other and separated by insulation to reduce energy loss due to heating (known as eddy currents). There are two main coils, the primary coil is connected to the input alternating current (AC) source and the secondary coil is connected to the output circuit.

The transformer operates on the principle of electromagnetic induction, where a changing magnetic field in the primary coil induces a voltage in the secondary coil.

A transformer is used to increase or decrease the voltage of alternating current (AC) without changing the total power (neglecting losses).



Types of Transformers:

- A **Step-Up** Transformer (called **Soukvolter**) increases voltage from low to high.
- A **Step-Down** Transformer (called **Souvolter**) decreases voltage from high to low.

ii. Working Principle of a Transformer

A **transformer** works on the principle of **mutual inductance**, where electrical energy is transferred from the **primary coil** to the **secondary coil** through a **changing magnetic field**.

When an **alternating current (AC)** flows through the **primary coil**, it produces a **changing magnetic field** in the iron core. This changing magnetic field induces an **electromotive force (emf)** in the **secondary coil**.

- In a **step-up transformer**, the voltage in the secondary coil is **greater** than in the primary coil.
- In a **step-down transformer**, the voltage in the secondary coil is **less** than in the primary coil.

Transformer Equation:

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}$$

Where, V_S is the **voltage in the secondary coil** measured in (V).

V_P is the **voltage in the primary coil** measured in (V).

N_S is the **number of turns in the secondary coil** measured in (turns).

N_P is the **number of turns in the primary coil** measured in (turns).

Types Based on Turn Ratio:

- If $\frac{N_S}{N_P} > 1$: Step-up transformer (transformateur élévateur).
- If $\frac{N_S}{N_P} < 1$: Step-down transformer (transformateur abaisseur).
- If $\frac{N_S}{N_P} = 1$: Isolation transformer (transformateur d'isolement) used to isolate two circuits without changing voltage.

iii. Power Transmission in Transformers

Transformers are based on the principle of **conservation of energy**. Ideally, the **electrical power** in the primary coil is equal to the power in the secondary coil (neglecting losses):

$$P_S = P_P \Rightarrow V_S I_S = V_P I_P$$

Since power is defined as:

$$P = VI$$

We also use the transformer turn ratio:

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}$$

Combining this with the conservation of energy:

$$\frac{V_S}{V_P} = \frac{N_S}{N_P} = \frac{I_P}{I_S}$$

Where, V_S is the **voltage in the secondary coil** measured in (V).

V_P is the **voltage in the primary coil** measured in (V).

N_S is the **number of turns in the secondary coil** measured in (turns).

N_P is the **number of turns in the primary coil** measured in (turns).

I_S is the **current in the secondary coil** measured in (A).

I_P is the **current in the primary coil** measured in (A).

III. Electrical Power Transmission

When electrical power P_e is transmitted through a network with resistance R, and the system is supplied with voltage V, the **current** in the network is:

$$I = \frac{P_e}{V}$$

Where, I is the **electric current** measured in (A).

P_e is the **electrical power** measured in (W).

V is the **voltage** measured in (V).

Power Loss in the Network

Due to resistance in the transmission lines, some electrical energy is lost as **heat**. This **power loss**, called **Joule loss**, is given by:

$$P_J = RI^2 = R \times \left(\frac{P_e}{V}\right)^2$$

This shows that **higher voltage** results in **lower current**, which **reduces power loss** during transmission.

Where, P_J is the **power loss** measured in (W).

R is the **resistance** measured in (Ω).

V is the **voltage** measured in (V).

Energy Balance in Transformers

The total electrical power provided by the primary coil is divided into two parts:

$$P_{e1} = P_{e2} + P_J$$

Where, P_{e1} is the **input electrical power from the primary coil** measured in (W).

P_{e2} is the **output electrical power to the secondary coil** measured in (W).

P_{J} is the **power lost as heat in the transmission line** measured in (W).

Also: $P_{e1} = V_P I_P ; P_{e2} = V_S I_S ; P_J = R I_S^2$

Efficiency of a Transformer

The **efficiency** of a transformer is defined as the ratio of output power to input power:

$$eff = Rd = \frac{P_{e2}}{P_{e1}} \times 100\% = \frac{V_S I_S}{V_P I_P} \times 100\%$$

Where, $eff = Rd \leq 1$ is the **efficiency** measured in $(\%)$.

An ideal transformer has $eff = Rd = 100\%$, meaning:

$$\frac{P_{e2}}{P_{e1}} = \frac{V_S I_S}{V_P I_P} = 1$$

Also related by the transformer equation:

$$\frac{V_S}{V_P} = \frac{N_S}{N_P} = \frac{I_P}{I_S}$$

∗ Methods to Reduce Power Loss (Heat)

To minimize energy loss in transmission:

- **Reduce resistance *R***: Use transmission wires with a **large cross-sectional area** (thick wires).
- **Reduce current *I***:

Use a **step-up transformer** to increase voltage before transmission.
Higher voltage → lower current → less heat loss.

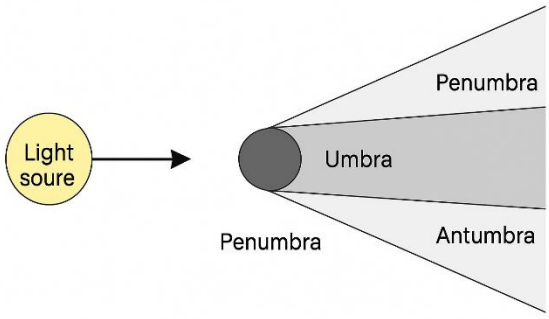
Chapter4 Optics
Lesson1 Nature and Rectilinear Propagation of Light

I. Rectilinear Propagation of Light

In a **uniform**, **transparent**, and **isotropic** medium (a medium where light travels equally in all directions), **light travels in straight lines**. This principle is known as the **Rectilinear Propagation of Light**.

II. Umbra, Penumbra, and Antumbra

Consider a point source of light and an opaque disk placed between the light source and a screen (E).

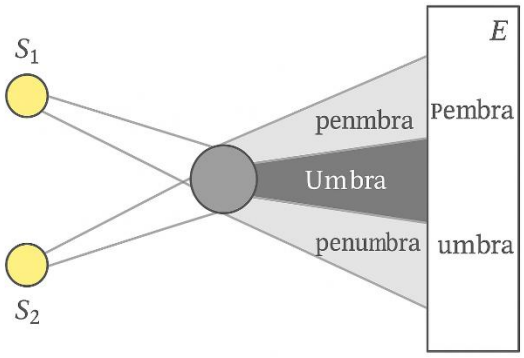


When light from the source is blocked by the opaque object, it creates shadows on the screen. These shadows are divided into different regions:

- On the edges of the shadow, where some light is partially blocked, the region is called the **penumbra.**
- At the central bottom part of the shadow, where all light is completely blocked, the region is called the **umbra.**
- On the screen (E), beyond the umbra and surrounded by the penumbra, there can be a region where the entire light source is visible around the object but does not directly illuminate that spot. This region is called the **antumbra**.

III. Umbra and Penumbra from Extended Light Sources

Consider two extended light sources, S_1 and S_2 , emitting light toward an opaque object. A screen (E) is placed behind the object to observe the shadow patterns.



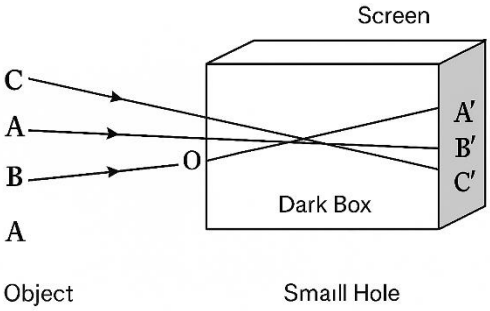
Due to the extended nature of the sources, the shadow formed consists of two main regions:

∗Near the opaque object, two regions are observed:

- Umbra: The region where light from both sources is completely blocked.
- Penumbra: The region where light from only one source reaches, while the other is blocked.
- On the screen (E), the shadow also consists of umbra is a completely dark region formed where neither light source illuminates and penumbra is a partially lit region formed where only one of the sources illuminates.

IV. Pinhole Camera

A rectangular room or box is used to construct a pinhole camera. On the front side, a small hole (O) is made, and on the back side, a rough or translucent screen is placed to receive the image.



When objects A, B, and C are placed in front of the hole (O), light rays from each point of the objects pass through the small opening. On the back screen, images A', B', and C' are formed.

Chapter4 Optics
Lesson2 Laws of Reflection and Refraction of Light

I. Definition

A **plane mirror** is a **flat, smooth surface** that reflects light **uniformly**. **Examples** of plane mirrors include the surface of still water, a polished tabletop, and a glass **plane mirror**.

II. Properties of a Plane Mirror

An object ***A*** is placed in front of a **plane mirror (M)**, and an **observer** stands in front of the mirror. The observer looks into the mirror and sees the **image A'** of the object.

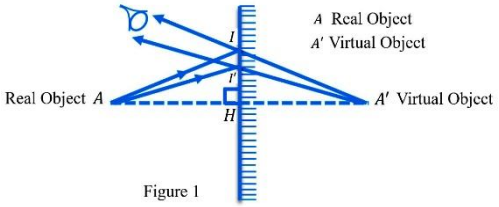


Figure 1

Since **light travels in straight lines**, the reflected light appears to come from point **A'** behind the mirror. However, in reality, the light **originates from the object A**, strikes the mirror, and is **reflected** into the observer's eyes.

III. Objects and Images

- An **object** is defined as the point where **incident rays originate** or appear to diverge from.
- A **real object** occurs when light rays are **actually diverging** from a point.
- A **virtual object** occurs when light rays are **converging** toward a point before reaching an optical surface.
- An **image** is formed at the point where **reflected or refracted rays intersect** (or appear to intersect).
- A **real image** is formed when **reflected or refracted rays actually converge**.
- A **virtual image** is formed when **reflected or refracted rays appear to converge** (but do not actually meet).

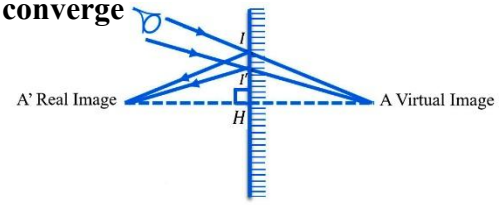


Figure 2

Examples (based on your figures):

- In the **first picture**: A is a **virtual image** formed by a plane mirror.
- In the **second picture**: A′ is a **real image** formed by a virtual object.

Important Note on Plane Mirrors:

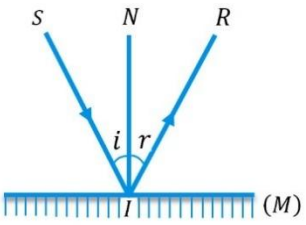
- A **plane mirror** usually forms a **virtual image** from a **real object**. However, under certain conditions, a **real image** can be formed from a **virtual object**.

Summary from the pictures:

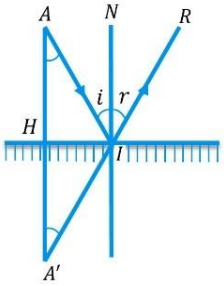
- In the **first picture**: The plane mirror forms a **virtual image A'** from a **real object A**.
- In the **second picture**: The plane mirror forms a **real image A'** from a **virtual object A**.

IV. Reflection of Light

A light ray ***SI*** strikes a **plane mirror** at point **I** and is reflected in the direction ***IR***.



- ***SI*** is the **incident ray**
- ***IR*** is the **reflected ray**
- ***I*** is the **point of incidence** (striking point)
- ***IN*** is the **normal** (a line perpendicular to the mirror surface at point I)
- $\angle i$ is the **angle of incidence** (between incident ray and normal)
- $\angle r$ is the **angle of reflection** (between reflected ray and normal)



∗Laws of Reflection (also known as Descartes' Laws):

- The **incident ray**, the **reflected ray**, and the **normal** all lie in the **same plane**.
- The **angle of incidence** $\angle i$ is **equal** to the **angle of reflection** $\angle r$.

∗ Construction Method for Image in a Plane Mirror

To construct the image of an object **A** in a **plane mirror**:

- Draw a **normal** line from point **H** on the mirror, **perpendicular** to the mirror surface. Measure and mark point **A'** such that: $AH = HA'$.
- Connect point **A'** to the **point of incidence** I on the mirror. The line IR will be the **reflected ray** seen by the observer.

This method shows how the **image A'** appears behind the mirror and how reflected rays reach the observer's eye.

V. Rotation of the Reflected Ray

A light ray SI is shined onto a **plane mirror** and strikes it at point I with an **angle of incidence** i. This ray is reflected in the direction IR, forming a **reflected angle** r equal to the incident angle.

$$i = r$$

Now, keeping the **incident ray** SI **fixed**, the **plane mirror** M is **rotated** around the point I by an angle α.

As a result, the reflected ray changes direction from IR to IR'.

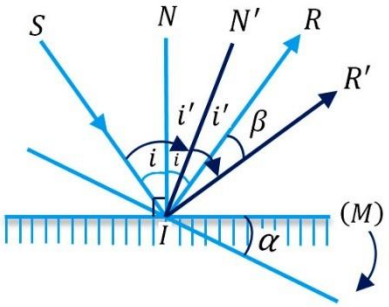
Let the angle between the **original reflected ray** IR and the **new reflected ray** IR' be β, such that:

$$\beta = \angle RIR'$$

From geometry:

The angle between the **incident ray** SI and the **new normal** N' after rotation is:

$$\beta = 2\angle SIN'$$



The original reflection angle:

$$\angle SIR = 2\angle SIN$$

Therefore, the change in direction of the reflected ray is:

$$\angle RIR' = 2\left(\angle SIN' - \angle SIN\right) = 2\angle NIN'$$

Since the mirror is rotated by angle α, then:

$$\angle NIN' = \alpha'$$

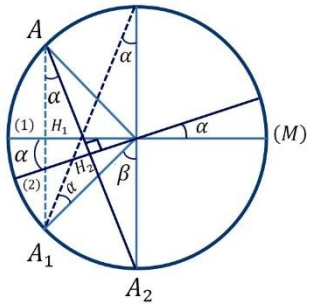
Thus: $\alpha' = 2\alpha$

When a plane mirror is rotated by an angle α, the reflected ray rotates by an angle $\beta = 2\alpha$.

VI. Rotation of the Reflected Ray

They have a **plane mirror** and an **object** A. For the initial position of the **plane mirror** M, the mirror gives an image A_1, which is **symmetric** to object A with respect to the mirror surface. Then, they **rotate the mirror** around a fixed point I by an angle α.

In this new position, the **plane mirror** gives an image A_2, which is **symmetric**



to object A with respect to the **new orientation** of the mirror.

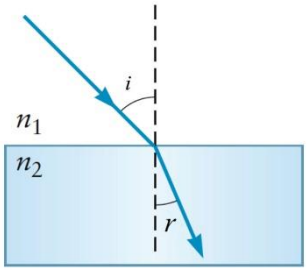
So, the image shifts from A_1 **to** A_2 due to the rotation of the mirror.

At each position, the image is always **symmetrical to the object** with respect to the mirror's surface.

VII. Refraction of Light

A light ray SI is shined onto the **boundary** between two **transparent materials** with **different refractive indices**. At the **point of incidence,** I, the ray is **refracted** as it enters the second medium.

- The light ray SI is the **incident ray**.
- At point I, it changes direction and continues as IR, the **refracted ray**.
- The line IN is the **normal**, drawn **perpendicular to the boundary surface** at the point of incidence.
- The angle between SI **and** IN is the **angle of incidence** i.
- The angle between IR **and** IN is the **angle of refraction** r.
- Plane (P) is the **incident plane**, which contains the incident ray and the normal.
- Plane (N) is the **refracted plane**, which contains the refracted ray and the normal.



VIII. Descartes' (Snell's) Laws of Refraction

They shine a ray of light incident of the boundary I incident ray is refracted if they increase the angle of incidence then the angle of refraction is also increase. A ray of light is shined onto the **boundary** I between two transparent materials. The **incident ray** is refracted when it enters the second medium. As the **angle of incidence** increases, the **angle of refraction** also increases.

Descartes' (Snell's) Laws of Refraction:

- **First Law**: The **incident ray**, the **refracted ray**, and the **normal** all lie in the **same plane**.
- **Second Law**: The **ratio of the sine of the angle of incidence to the sine of the angle of refraction** is a constant.

$$\frac{\sin i_1}{\sin r_1} = \frac{\sin i_2}{\sin r_2} = \ldots = \frac{\sin i_n}{\sin r_n} = \ldots = \text{constant}$$

Important Notes:

In the case where **incident ray** SI is refracted as **ray IR**, we write:

$$\frac{\sin i}{\sin r} = n$$

This constant n is the **relative refractive index**:

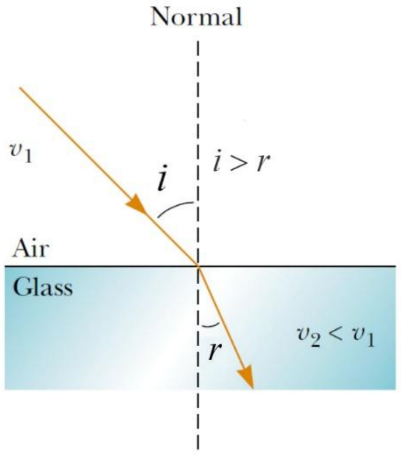
$$n = \frac{n_2}{n_1}$$

Thus, the **general form of Snell's Law** is:

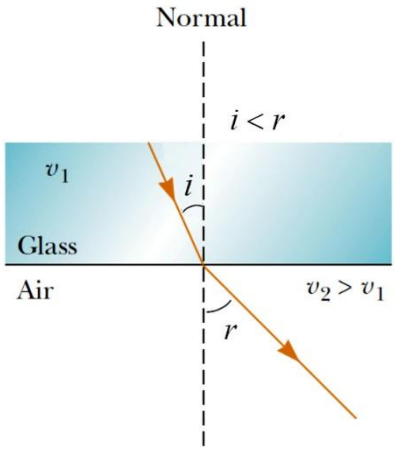
$$n_1 \sin i = n_2 \sin r$$

Refraction Conditions:

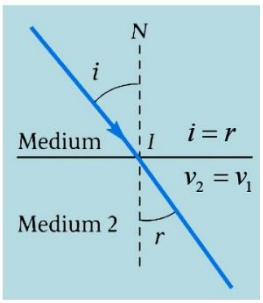
- If $\frac{n_2}{n_1} > 1 \Rightarrow n_2 > n_1 \Rightarrow \frac{\sin i}{\sin r} > 1 \Rightarrow \sin i > \sin r \Rightarrow i > r$:
  The light ray is refracted toward the normal.
- If $\frac{n_2}{n_1} < 1 \Rightarrow n_2 < n_1 \Rightarrow \frac{\sin i}{\sin r} < 1 \Rightarrow \sin i < \sin r \Rightarrow i < r$:
  The light ray is refracted away from the normal.
- If $\frac{n_2}{n_1} = 1 \Rightarrow n_2 = n_1 \Rightarrow \frac{\sin i}{\sin r} = 1 \Rightarrow \sin i = \sin r \Rightarrow i = r$:
  The light travel in a straight line without refraction.



(a)



(b)



(c)

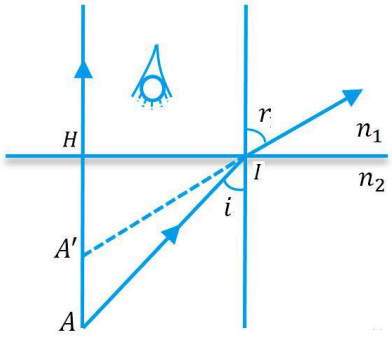
Chapter4 Optics
Lesson3 Double Plane and Slab

I. Definition

A **double plane** (or **optical slab**) is a system composed of **two transparent media** separated by a boundary. Examples include **water**, **oil**, and **fuel**.

II. Object in a Denser Medium

Consider two transparent media with refractive indices n_1 and n_2, where $n_2 > n_1$. An object $\boldsymbol{A}$ is placed in the **denser medium** n_2, and an observer located in the **less dense medium** n_1 looks at the object through the boundary.



By the **law of refraction** at point I:

$$n_2 \sin i = n_1 \sin r \quad (1)$$

In triangles $\triangle \perp HIA$ and $\triangle \perp HIA'$:

$$\tan i = \frac{HA}{HI} \text{ and } \tan r = \frac{HA'}{HI}$$

So: $$\frac{\tan i}{\tan r} = \frac{HA'}{HA}$$

Assuming small angles (i.e., light rays close to the normal), we approximate:

$$\tan i \approx \sin i, \tan r \approx \sin r \Rightarrow \frac{\sin i}{\sin r} = \frac{HA'}{HA}$$

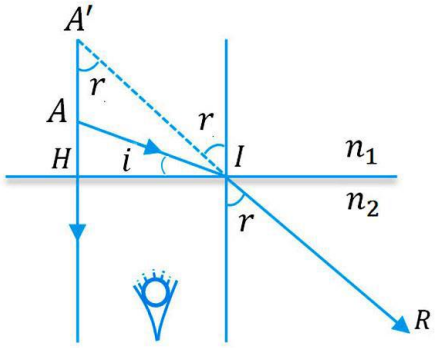
From equation (1): $\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$

Therefore: $$\frac{HA'}{HA} = \frac{n_2}{n_1}$$

Since $\frac{n_2}{n_1} > 1$, the ratio $\frac{HA'}{HA} > 1$, which means the **image $\boldsymbol{A}'$** appears **farther** from the boundary than the real object $\boldsymbol{A}$.

III. Object in a Less Dense Medium

Consider two transparent media with refractive indices n_1 and n_2, where $n_2 < n_1$. An object **A** is placed in the **less dense medium** n_1, and an observer located in the **denser medium** n_2 looks at the object through the boundary.



By the **law of refraction** at point I:

By the **law of refraction** at point I:

$$n_2 \sin i = n_1 \sin r \quad (1)$$

In triangles $\triangle \perp HIA$ and $\triangle \perp HIA'$:

$$\tan i = \frac{HA}{HI} \text{ and } \tan r = \frac{HA'}{HI}$$

So: $$\frac{\tan i}{\tan r} = \frac{HA'}{HA}$$

Assuming small angles (i.e., light rays close to the normal), we approximate:

$$\tan i \approx \sin i, \tan r \approx \sin r \Rightarrow \frac{\sin i}{\sin r} = \frac{HA'}{HA}$$

From equation (1): $\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$

Therefore: $$\frac{HA'}{HA} = \frac{n_2}{n_1}$$

Since $\frac{n_2}{n_1} > 1$, the ratio $\frac{HA'}{HA} > 1$, which means the **image** A' appears **farther** from the boundary than the real object A.

IV. A Slab

Definition: A **slab** is a transparent material bounded by two parallel surfaces (also called **double planes**).
Example: A sheet of transparent glass or plastic.

V. Light Passing Through a Slab

A thick slab of transparent material with thickness e and refractive index n is placed in **air**. A light beam SI is incident on the first surface of the slab at an angle of incidence i. The light is **refracted** at point I, then travels inside the slab along the path II', and finally emerges from the second surface at point I', continuing in the direction $I'R$ back into air.

By the Law of Refraction at points I and I':

At the first surface (air to slab):

$$\sin i = n \sin r \quad (1)$$

At the second surface (slab to air):

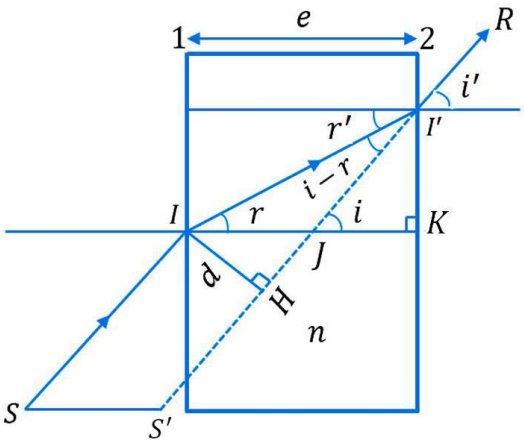
$$n \sin r' = \sin i \quad (2)$$

Since the slab surfaces are parallel, the angles of refraction inside the slab are equal:

$$i = r$$

Substitute into (1) and (2):

$$\sin i = \sin i' \Rightarrow i = i'$$



The **emergent ray** $I'R$ is **parallel** to the **incident ray** SI.

VI. Distance d (Lateral Displacement or Translational Radius)

When a light beam SI passes through a slab and emerges as $I'R$, it is displaced sideways by a distance $IH = d$, called the **lateral displacement** or **translational radius**.

In triangle **Δ⊥II′H**: Using the sine function:

$$\sin\left(i-r\right)=\frac{IH}{II'}$$

This gives: $IH = II'\sin\left(i-r\right)$ (1)

In triangle **Δ⊥II′J**: Using the cosine function:

$$\cos r=\frac{IJ}{II'}$$

This gives: $II'=\frac{IJ}{\cos r}=\frac{e}{\cos r}$ $\left(2\right)$ $\left(\text{since } IJ = e, \text{ the slab thickness}\right)$

Substitute (2) into (1):

$$IH=\frac{e}{\cos r}\sin\left(i-r\right)$$

So, lateral displace is $d=\frac{e\sin\left(i-r\right)}{\cos r}$.

Where, d is the **lateral displacement** measured in (m).

e is the **slab thickness** measured in (m).

i is the **angle of incidence** measured in $(°)$.

r is the **angle of refraction** measured in $(°)$.

VII. Lateral shift

A light beam enters a parallel-sided slab at point ***S*** and emerges at point ***S′***. The segment ***SS′*** is called the **lateral shift** of the light ray through the slab.

From the geometry, the quadrilateral ***IJSS′*** forms a **parallelogram**, so:

$$SS' = IJ \text{ but } IJ = IK - JK \quad (1)$$

In triangle **Δ⊥II′K**, we have:

$$\tan r=\frac{I'K}{JK}$$

In triangle **ΔJKI′**, we have:

$$\tan i=\frac{I'K}{JK}$$

From these: $\frac{\tan r}{\tan i}=\frac{JK}{IK}\Rightarrow JK=IK\cdot\frac{\tan r}{\tan i}$

Substituting into equation (1):

$$SS'=IK-IK\cdot\frac{\tan r}{\tan i}=IK\left(1-\frac{\tan r}{\tan i}\right)\ \left(2\right)$$

From Snell's Law: $\frac{\sin i}{\sin r} = n \Rightarrow \frac{\tan r}{\tan i} \approx \frac{\sin r}{\sin i} = \frac{1}{n}$ (for small angles)

Substitute into equation (2):

$$IJ = IK\left(1 - \frac{1}{n}\right)$$

Since $IK =$ e (thickness of the slab), and $IJ =$ SS', we conclude we conclude:

$$SS = e\left(1 - \frac{1}{n}\right)$$

Where, SS' is the **lateral shift** measured in (m).

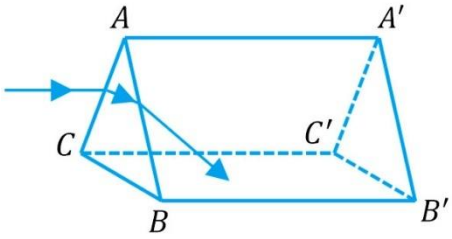
e is the **thickness of slabs** measured in (m).

n is the **refractive index of slab** dimensionless.

Chapter4 Optics
Lesson3 Prism

I. Definition

A **prism** is a transparent optical medium bounded by **two non-parallel plane surfaces** through which light can pass.

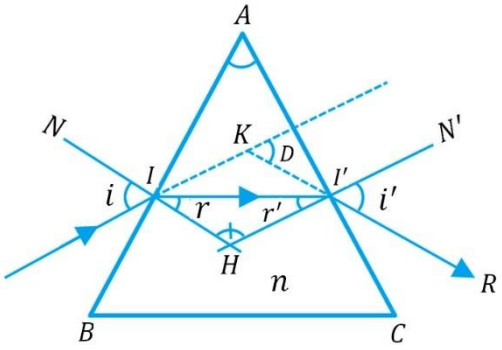


- AA' is called the **edge** of the prism.
- △ABC represents the **cross-sectional area** of the prism.
- The rectangle CC′BB′ is considered the **base** of the prism.
- The faces BB′ and AA′C′C are the **parallel surfaces** of the prism.
- The angle **A**, formed between the two refracting surfaces, is known as the **prism angle** or **apex angle** (also called the **vertex angle**).

II. Study Light Rays Passing Through a Prism

Consider a prism with a triangular cross-section △ABC, **apex angle A**, and **refractive index** n.

A light ray SI is incident on the first face AB of the prism at point I, making an **angle of incidence** i. The ray is **refracted** inside the prism along the path II', and then **emerges** from the second face AC at point I' into the air in the direction $I'R$, making an **angle of emergence** i'.



By the law of refraction (Snell's Law):

- At the first surface (air to prism):

$$\sin i = n \sin r \quad (1)$$

- At the second surface (prism to air):

$$\sin i' = n \sin r' \quad (2)$$

- The apex angle A of the prism is related to the internal angles by:

$$A = r + r'$$

∗Angle Convention Notes:

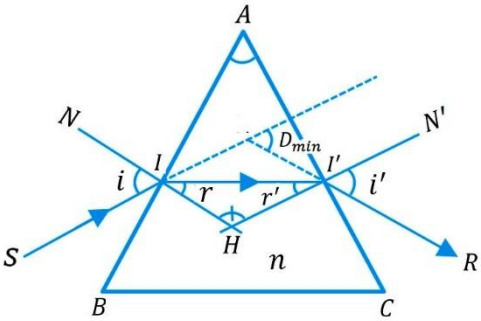
- Angles i and i' are considered **positive** when measured **below the normal line** (toward the inside of the prism) and **negative** if above the normal line.
- Angles r and r' are **positive** when measured **above the normal line** (within the prism) and **negative** if below the normal line.

III. Angle of Deviation

When a light ray passes through a prism, it changes direction. The total change in direction is called the **angle of deviation**, denoted by **D**. It is the angle between the direction of the **incident ray (SI)** and the direction of the **emergent ray** ($I'R$).

From triangle △IKI′, the angle of deviation is given by:

$$D = (i - r) + (i' - r')$$



Since the **apex angle** A of the prism satisfies:

$$A = r + r'$$

We can simplify the deviation angle:

$$D = i + i' - (r - r') = i + i' - A$$

∗Minimum Angle of Deviation

The deviation angle D is not constant it changes with the angle of incidence i. However, there exists a **minimum deviation angle**, denoted as D_{min}, which occurs when the light ray passes symmetrically through the prism. In this case:

$$i = i' \text{ and } r = r'$$

Thus: $$A = r + r = 2r \Rightarrow r = \frac{A}{2}$$

And: $$D_{min} = 2i - A$$

IV. Refractive Index of a Prism

Snell's law applies at point I, we get:

$$\sin i = n \sin r$$

We consider the case when the **deviated angle** D is **minimum**, so:

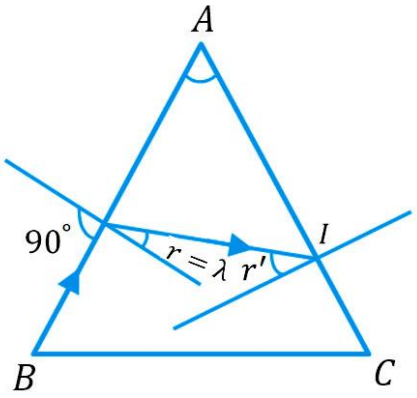
$$D = D_{min} \text{ when } i = i'$$

Since the ray travels symmetrically in the prism, we get:

$$r = r' \text{ and } A = r + r' = 2r \Rightarrow r = \frac{A}{2}$$

Also, from the deviated angle formula:

$$D = i + i' - A \Rightarrow D_{min} = 2i - A \Rightarrow i = \frac{D_{min} + A}{2}$$



Now substitute into Snell's law:

$$\sin i = \sin\left(\frac{D_{min} + A}{2}\right), \sin r = \sin\left(\frac{A}{2}\right)$$

Hence, the refractive index of the prism is:

$$n = \frac{\sin\left(\frac{D_{min} + A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Where, A is the **Apex angle of the prism** measured in (°).

D_{min} is the **minimum deviation angle** measured in (°).

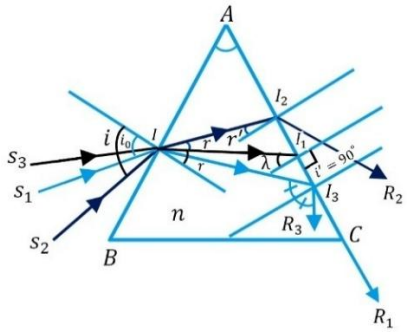
n is the **refractive index of slab** dimensionless.

V. Condition for Light Ray Leaving from a Prism

i. Condition Related to Prism Angle

Let us analyze when a light ray can leave or be totally internally reflected within a prism based on the **prism angle A** and a **critical angle λ**:

∗ Case 1: Prism Angle $A > 2\lambda$



Assume a light ray SI is incident on face AB with an incidence angle $i = 90°$.

At point I, the refracted angle is:

$$r = \lambda$$

Since the apex angle is:

$$A = r + r' \Rightarrow r' = A - r = A - \lambda$$

Now, because $A > 2\lambda$, we have:

$$r' = A - \lambda > 2\lambda - \lambda = \lambda$$

Thus, $r' > \lambda$, and **total internal reflection occurs** at face **AC** → **light cannot escape the prism**.

∗ Case 2: Prism Angle $A < 2\lambda$

Again, using $A = r + r'$, and $r = \lambda$, we get:

$$r' = A - \lambda < 2\lambda - \lambda = \lambda$$

This means $r' \leq \lambda$, so **light may escape the prism**.

∗ Critical Incidence Angle i_0

■ When $r' = \lambda$, the **emergent ray** travels **parallel to face AC**, and the **corresponding angle of incidence** at point I is called the **critical incidence angle i_0**.

■ Applying **Snell's law** at point I: $\sin i_0 = n \sin(A - \lambda)$

Condition	Result
$i > i_0$	$r' < \lambda$: Light escapes prism
$i = i_0$	$r' = \lambda$: Ray emerges parallel
$i < i_0$	$r' > \lambda$: Total internal reflection

Chapter1 **Mechanics**
Lesson1 **Motion in a Plane**

I. Kinematic of a point of matter

i. Definition: Kinematics is a branch of mechanics that studies the motion of objects without considering the causes of that motion.

II. Linear motion

i. Definition: An object is said to be in linear motion when it moves along a straight-line path.

ii. Position Vector

An object moves along the x-axis from x' to x. At an instant t, it is located at point M. The position vector of point M at time t is given by:

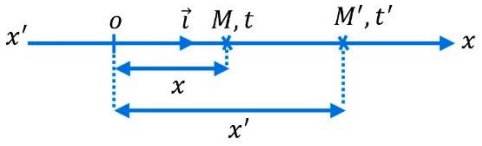


$$\overrightarrow{OM} = \overrightarrow{OM} \cdot \vec{i} = x\vec{i}$$

Magnitude: $\overline{OM} = x(t)$ and $x(t)$ is called a **position-time function.**

iii. Velocity

A. Average speed



At time t, an object is located at point M, which has an **abscissa** x.

At a later time t', the object is at point M', which has an **abscissa** x'.

During the time interval $\Delta t = t' - t$, the object moves from point M to point M'.

The displacement is $\overline{OM'} = x' - x = \Delta x$, which represents the change in position along the x-axis.

Hence, the average speed $v_{av} = \dfrac{\Delta x}{\Delta t}$

Where, v_{av} is the **average speed** measured in $(\mathrm{m/s})$.

Δx is the **displacement** measured in (m).

Δt is the **time interval** measured in (s).

B. Instantaneous velocity

The **instantaneous velocity** is defined as the limit of the average velocity as the time interval approaches zero:

When $\Delta t \to 0$, the ratio $\dfrac{\Delta x}{\Delta t}$, which represents the average velocity, becomes the **instantaneous velocity**.

This can be written as $v = \lim\limits_{\Delta t \to 0} \dfrac{\Delta x}{\Delta t}$

$$v = \frac{dx}{dt} = x'$$

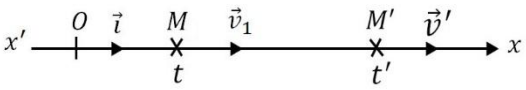
Hence, the instantaneous velocity is $v = x'\vec{i} = \dfrac{dx}{dt}\vec{i}$

i. Acceleration

A. Average acceleration

Assume that at instant t, an object at point M has velocity $\vec{v}$, and at instant t', the object at point M' has velocity $\vec{v}'$. During the time interval $\Delta t = t' - t$, the change in velocity is $\overrightarrow{\Delta v} = \vec{v}' - \vec{v}$.

This can be written as $\vec{a}_{av} = \dfrac{\overrightarrow{\Delta v}}{\Delta t}$



Hence, the average acceleration is $a_{av} = \dfrac{\Delta v}{\Delta t}$

Where, a_{av} is the **average acceleration** measured in $(\mathrm{m/s^2})$.

Δv is the **change in velocity** measured in $(\mathrm{m/s})$.

Δt is the **time interval** measured in (s).

B. Instantaneous acceleration

Instantaneous acceleration is defined as the limit of the **average acceleration** as the time interval approaches zero:

$$\vec{a} = \lim_{\Delta t \to 0} \vec{a}_{av} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t}$$

$$\vec{a} = \frac{d}{dt}\left(\frac{d\vec{x}}{dt}\right) = \frac{d^2\vec{x}}{dt^2} = \vec{x}'' = x''\vec{i}$$

Hence, the instantaneous acceleration is $\vec{a} = \dfrac{d\vec{v}}{dt} = \dfrac{d^2\vec{x}}{dt^2} = x''\vec{i}$

III. Motion in space

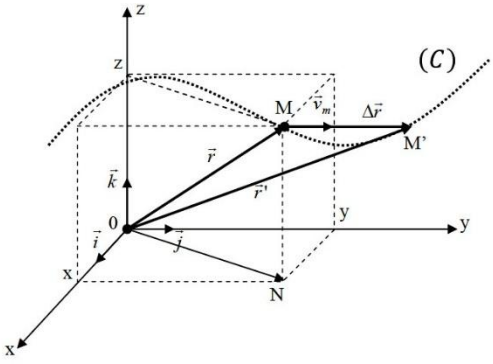
First method

Motion in space refers to the motion of an object in **three-dimensional space**, where its position changes over time along the x, y, and z-axes. The object's position at point M at any time t can be described by a **position vector**:

$$\overrightarrow{OM} = \vec{r}(t) = x\vec{i} + y\vec{j} + z\vec{k}$$

In magnitude $OM^2 = x^2 + y^2 + z^2$

Hence, the magnitude of a position vector is $OM = \sqrt{x^2 + y^2 + z^2}$



Second method

However, $\overrightarrow{OM} = \overrightarrow{ON} + \overrightarrow{NM}$

But, $\overrightarrow{ON} = \overrightarrow{OB} + \overrightarrow{OA}$

And, $|OA| = x,\ |OB| = y,\ |OC| = z$

Since, $\overrightarrow{OA} = |OA|\vec{i} = x\vec{i}, \overrightarrow{OB} = |OB|\vec{j} = y\vec{j},\ \overrightarrow{OC} = |OC|\vec{k} = z\vec{k}$

Hence, $\overrightarrow{OM} = \vec{r}(t) = x\vec{i} + y\vec{j} + z\vec{k}$

In magnitude $|\overrightarrow{OM}| = \overrightarrow{OM}$

Equivalent $(\overrightarrow{OM})^2 = (x\vec{i} + y\vec{j} + z\vec{k})^2$

$(\overrightarrow{OM})^2 = (x\vec{i})^2 + (y\vec{j})^2 + (z\vec{k})^2 + 2xy\vec{i}\vec{j} + 2xz\vec{i}\vec{k} + 2yz\vec{j}\vec{k}$

Since, $\vec{i}^2 = \vec{j}^2 = \vec{k}^2 = i^2 = j^2 = k^2 = 1$

But, $\vec{i}\cdot\vec{j} = \vec{j}\cdot\vec{k} = \vec{i}\cdot\vec{k} = 0$

Imply $|OM|^2 = x^2 + y^2 + z^2$

Hence, the magnitude of a position vector is $OM = \sqrt{x^2 + y^2 + z^2}$

Third method

In $\Delta \perp$ MON $|OM|^2 = |ON|^2 + |MN|^2$

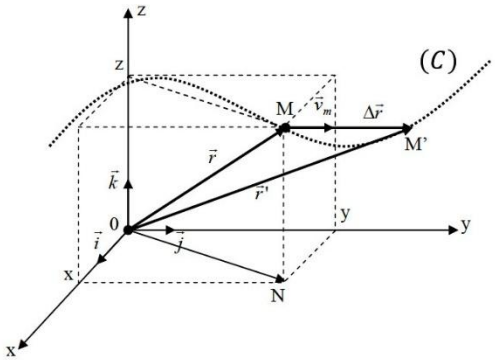
In $\Delta \perp$ OAN $|ON|^2 = |OA|^2 + |OB|^2$

But, $|NM|^2 = |OC|^2$

Imply $|OM|^2 = |OA|^2 + |OB|^2 + |OC|^2$

$|OM|^2 = x^2 + y^2 + z^2$

Hence, the magnitude of a position vector is $OM = \sqrt{x^2 + y^2 + z^2}$



i. Velocity

A. Average velocity

At time t, an object is at point M with position vector $\vec{r}$.

At a later time t', the object is at point M' with position vector $\vec{r}'$.

The time interval is $\Delta t = t' - t$, the displacement vector of the object during this time is

$\overrightarrow{MM'} = \overrightarrow{OM'} - \overrightarrow{OM} = \vec{r}' - \vec{r} = \overrightarrow{\Delta r}$

Hence, the average velocity is $\vec{v}_{av} = \dfrac{\overrightarrow{\Delta r}}{\Delta t}$

B. Instantaneous velocity

Instantaneous velocity is defined as the **limit of the average velocity** as the time interval Δt approaches zero.

We get, $\vec{v} = \lim\limits_{\Delta t \to 0} \vec{v}_{av} = \lim\limits_{\Delta t \to 0} \dfrac{\Delta\vec{r}}{\Delta t} = \dfrac{d\vec{r}}{dt}$

Where, $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

Imply $\vec{v} = \dfrac{d}{dt}\left(x\vec{i} + y\vec{j} + z\vec{k}\right)$

Since, $\vec{i}; \vec{k}; \vec{k} = \text{constant}$

$\vec{v} = \dfrac{dx}{dt}\vec{i} + \dfrac{dy}{dt}\vec{j} + \dfrac{dz}{dt}\vec{k}$

And, $v_x = \frac{dx}{dt}; v_y = \frac{dy}{dt}; v_z = \frac{dz}{dt}$

Hence, instantaneous velocity is $\vec{v} = v_x\vec{i} + v_y\vec{j} + v_z\vec{k}$

However, $\vec{v} = \begin{pmatrix} \vec{v_x} \\ \vec{v_y} \\ \vec{v_z} \end{pmatrix} \begin{pmatrix} \vec{i} \\ \vec{j} \\ \vec{k} \end{pmatrix}$

In magnitude $v^2 = v_x^2 + v_y^2 + v_z^2$

Hence, instantaneous velocity is $v = \sqrt{v_x^2 + v_y^2 + v_z^2}$

ii. Acceleration

A. Average acceleration

Assume that at instant t, an object at point M has velocity $\vec{v}$, and at instant t', the object at point M' has velocity $\vec{v}'$. During the time interval $\Delta t = t' - t$, the change in velocity is $\overrightarrow{\Delta v} = \vec{v}' - \vec{v}$.

This can be written as $\vec{a}_{av} = \frac{\overrightarrow{\Delta v}}{\Delta t}$

Hence, the average acceleration is $a_{av} = \frac{\Delta v}{\Delta t}$

Where, a_{av} is the **average acceleration** measured in $(\mathrm{m/s^2})$.

Δv is the **change in velocity** measured in $(\mathrm{m/s})$.

Δt is the **time interval** measured in (s).

B. Instantaneous acceleration

Instantaneous acceleration is defined as the limit of the **average acceleration** as the time interval approaches zero:

$$\vec{a} = \lim_{\Delta t \to 0} \vec{a}_{av} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

Since, $\vec{v} = \vec{v_x}\vec{i} + \vec{v_y}\vec{j} + \vec{v_z}\vec{k}$

Imply $\vec{a} = \frac{d}{dt}\left(\vec{v_x}\vec{i} + \vec{v_y}\vec{j} + \vec{v_z}\vec{k}\right)$

$$\vec{a} = \frac{dv_x}{dt}\vec{i} + \frac{dv_y}{dt}\vec{j} + \frac{dv_z}{dt}\vec{k}$$

And, $a_x = \frac{dv_x}{dt}, a_y = \frac{dv_y}{dt}, a_z = \frac{dv_z}{dt}$

Hence, $\vec{a} = a_x\vec{i} + a_y\vec{j} + a_z\vec{k}$

Where, $a_x; a_y; a_z$ components of acceleration $\vec{a}$

In magnitude $a^2 = a_x^2 + a_y^2 + a_z^2$

Hence, the instantaneous acceleration is $a = \sqrt{a_x^2 + a_y^2 + a_z^2}$

IV. Curvilinear motion

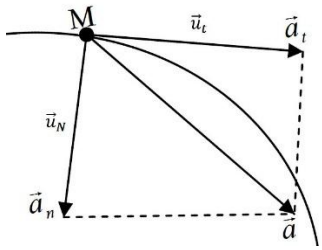
An object moves along a curved path C. At an instant t, the object is at point M. The **acceleration** of the object at that point is the **time rate of change of its velocity vector**, which can be expressed as

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\vec{v}(t + \Delta t) - \vec{v}(t)}{\Delta t}$$

$$\vec{a} = \vec{a}_T + \vec{a}_N$$

But, $\vec{a}_T \perp \vec{a}_N$

We get, $a^2 = a_T^2 + a_N^2$



Hence, the acceleration of the object is $a = \sqrt{a_T^2 + a_N^2}$

Where, $a_T = \dfrac{dv}{dt}$ is the **tangential acceleration** measured in $(\mathrm{m/s^2})$.

$a_N = \dfrac{v^2}{r}$ is the **normal (or centripetal) acceleration** measured in $(\mathrm{m/s^2})$.

$R = \rho$ is the **radius of curvature** measured in (m).

V. Linear motion

i. Definition: An object is said to be in linear motion if it moves along a straight line with constant velocity.

Assuming the object moves with constant velocity $\vec{v}$ along the $x'x$-axis, we obtain:

Acceleration $a = \dfrac{dv}{dt}$

But, $v = \text{constant} \Rightarrow a = 0$

Instantaneous velocity $v = \dfrac{dx}{dt} \Rightarrow dx = vdt$

$$\int dx = \int vdt$$

$$x + c_1 = vt + c_2$$

Since, $c_2 - c_1 = c$

We get, $x = vt + c_2 - c_1$

$$x = vt + c$$

If $t = 0 \Rightarrow x_0 = c$

Hence, **time-position equation** is $x = x_0 + vt$

VI. Uniformly accelerated linear motion or constant acceleration motion

i. Definition: An object moves with constant acceleration if it travels along a straight line and its velocity changes uniformly over time.

Consider an object moving along the $x'x$-axis with a non-uniform velocity under constant acceleration.

Acceleration $a = \dfrac{dv}{dt} \Rightarrow dv = adt$

$$\int_{v_0}^{v} dv = \int_{t_0}^{t} a dt$$

$$v - v_0 = a\left(t - t_0\right)$$

Hence, the equation of velocity for uniformly accelerated linear motion is $v = v_0 + a\left(t - t_0\right)$

At $t_0 = 0$ s $\quad v = v_0 + at$

Where, v_0 is the **initial velocity at time t_0** measured in $(\mathrm{m/s})$.

v is the **final velocity at time t** measured in $(\mathrm{m/s})$.

a is the **constant acceleration** measured in $(\mathrm{m/s^2})$.

$t - t_0$ is called **time elapsed** measured in (s).

ii. Time-position equation of motion

Instantaneous velocity $v = \dfrac{dx}{dt} \Rightarrow dx = vdt$

$$\int_{x_0}^{x} dx = \int_{t_0}^{t} \left[v_0 + a\left(t - t_0\right)\right] dt$$

$$x - x_0 = \int_{t_0}^{t} v_0 dt + \int_{t_0}^{t} a\left(t - t_0\right) dt$$

Let, $u = t - t_0 \Rightarrow du = dt$

Imply $\int u dv = \dfrac{u^2}{2} = \dfrac{\left(t - t_0\right)^2}{2}$

We get, $x - x_0 = v_0\left(t - t_0\right) + \dfrac{1}{2} a\left(t - t_0\right)^2$

At instant, $t_0 = 0, x_0 = 0, v_0 = 0$

Hence, time position equation of motion with constant acceleration is $x = \dfrac{1}{2} at^2$

* Alternative kinematic equation with constant acceleration motion

If an object moves with **constant acceleration**, starting from point O with initial velocity v_0 and reaching point A with velocity v, then the **average velocity** v_{av} during the motion from O to A is:

$$v_{av} = \frac{v_0 + v}{2}$$

Time position equation $\quad x = \left(\dfrac{v_0 + v}{2}\right) t$

Hence, time position equation of motion with constant acceleration is $x = \left(\dfrac{v_0 + v}{2}\right) t$

iii. Time-Independent Equation of Motion

We have $a = \frac{dv}{dt}$ and $v = \frac{dx}{dt}$

Equivalent $\frac{a}{v} = \frac{dv}{dx} \Rightarrow vdv = adx$

Imply $\int_{v_0}^{v} vdv = \int_{x_0}^{x} adx$

$$\frac{1}{2}\left(v^2 - v_0^2\right) = a\left(t - t_0\right)$$

Hence, time-independent equation of motion is $v^2 - v_0^2 = 2a\left(t - t_0\right)$

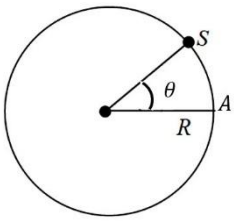
Where, v_0 is the **initial velocity at time t_0** measured in $(\mathrm{m/s})$.

v is the **final velocity at time t** measured in $(\mathrm{m/s})$.

a is the **constant acceleration** measured in $(\mathrm{m/s^2})$.

$t - t_0$ is called **time elapsed** measured in (s).

VII. Non-Circular motion



i. Definition: Circular motion is the movement of an object along a circular path, where its distance from a fixed central point (the radius) stays constant.

ii. Angle or displacement

An object moves along a **circular path** with **center** O and **radius** R.

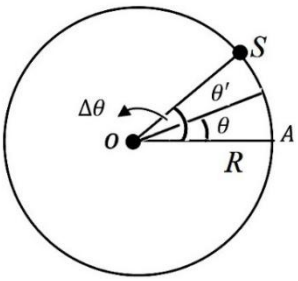
It starts from point A at instant t_0 and reaches point B at instant t.

The angle θ (t) (in radians) is the **angular displacement** swept out by the radius R during this time. The **arc length** or **linear position** x of the object along the circular path from point A is given by:

$$x = AS = R\theta$$

iii. Angular frequency

A. Average angular frequency



An object moves along a circular path, starting from point A at instant t_0,

reaching point B at time t, and then reaching point M at time t'.

In the time interval $\Delta t = t' - t$, the **angular displacement** is given by: $\Delta\theta = \theta' - \theta$

Hence, the average angular frequency is $\omega_{av} = \frac{\Delta\theta}{\Delta t}$

Where, ω_{av} is the **average angular velocity** measured in $(\mathrm{rad/s})$.

$\Delta\theta$ is the **angular displacement** measured in (rad).

Δt is the **time interval during which the displacement occurs** measured in (s).

B. Instantaneous angular frequency

Instantaneous angular frequency is defined as the limit of the average angular velocity as the time interval $\Delta t \to 0$.

Instantaneous angular frequency $\omega = \lim_{\Delta t \to 0} \omega_{av} = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt} = \theta'$

Hence, the instantaneous angular frequency is $\omega = \frac{d\theta}{dt} = \theta'$

iv. Angular acceleration

A. Average angular acceleration

Assume that at time t, the object has an angular velocity ω, and at time t', it has an angular velocity ω'.

Over the time interval $\Delta t = t' - t$, the change in angular velocity is $\Delta\omega = \omega' - \omega$.

Hence, the average angular acceleration is $\alpha_{av} = \frac{\Delta\omega}{\Delta t}$

Where, α_{av} is the **average angular acceleration** measured in $(\mathrm{rad/s^2})$.

$\Delta\omega$ is the **change in angular velocity** measured in $(\mathrm{rad/s})$.

Δt is the **time interval during which the angular velocity occurs** measured in (s).

B. Instantaneous angular acceleration

Instantaneous angular acceleration is defined as the limit of the average angular acceleration as the time interval $\Delta t \to 0$.

Instantaneous angular frequency $\alpha = \lim_{\Delta t \to 0} \alpha_{av} = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt} = \frac{d}{dt}\left(\frac{d\theta}{dt}\right) = \frac{d^2\theta}{dt^2} = \theta''$

Hence, the instantaneous angular acceleration is $\alpha = \frac{d^2\omega}{dt^2} = \alpha''$

VIII. Uniform Circular motion

i. Definition: An object is said to undergo uniform circular motion if it moves along a circular path with constant speed. At instant t, the object is at point A. The **position** of the object at instant t is:

$$s = OA = R\theta_{rad}$$

However, $v = R\omega$

But, $\omega = \frac{v}{r} = \text{constant}$

Hence, the angular acceleration is $\alpha = \frac{d\omega}{dt} = 0$

ii. Time-position equation of motion

We have $\omega = \frac{d\theta}{dt}$

Equivalent $d\theta = \omega dt$

Imply $\int_{\theta_0}^{\theta} d\theta = \omega \int_{t_0}^{t} dt$

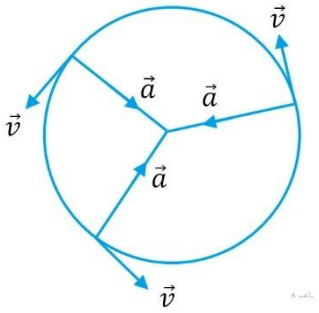
$$\theta - \theta_0 = \omega(t - t_0)$$

$$\theta = \theta_0 + \omega(t - t_0)$$

Incase $t_0 = 0, \theta_0 = 0$

Hence, time-position equation of motion is $\theta = \omega t$

iii. Centripetal acceleration



We have $\vec{a} = \vec{a}_T + \vec{a}_N$

Since, $a_T = \dfrac{dv}{dt} = 0 \ (v = \text{constant})$

Imply $\vec{a} = \vec{a}_N$

Equivalent $a = a_N = \dfrac{v^2}{r} = \dfrac{\left(r\omega\right)^2}{r} = r\omega^2$

Hence, centripetal acceleration is $a_N = \dfrac{v^2}{r} = r\omega^2$

IX. Non-uniform Circular motion

i. Definition: An object moves with **non-uniform circular motion** when its angular acceleration is **not constant**.

ii. Angular acceleration

We have $\alpha = \dfrac{d\omega}{dt} \Rightarrow d\omega = \alpha dt$

Imply $\displaystyle\int_{\omega_0}^{\omega} d\omega = \alpha \int_{t_0}^{t} dt$

$$\omega - \omega_0 = \alpha\left(t - t_0\right)$$

$$\omega = \omega_0 + \alpha\left(t - t_0\right)$$

At instant $t_0 = 0, \theta_0 = 0, \omega_0 = 0$

Hence, the angular acceleration is $\omega = \alpha t$

iii. Angular displacement equation of motion

We have $\omega = \dfrac{d\theta}{dt} \Rightarrow d\theta = \omega dt$

Imply $\displaystyle\int_{\theta_0}^{\theta} d\theta = \int_{t_0}^{t} \omega dt$

$$\theta - \theta_0 = \int_{t_0}^{t} \left[\omega_0 + \alpha\left(t - t_0\right)\right] dt$$

$$\theta = \theta_0 + \omega\left(t - t_0\right) + \frac{1}{2}\alpha\left(t - t_0\right)^2$$

At instant $t_0 = 0, \theta_0 = 0, \omega_0 = 0$

Hence, $\theta = \dfrac{1}{2}\alpha t^2$ is **angular displacement equation** for **rotational motion with constant angular acceleration**.

iv. Angular Equation of Motion (Time-Independent Form)

We have $\alpha = \frac{d\omega}{dt}$ and $\omega = \frac{d\theta}{dt}$

Divide both side $\frac{\alpha}{\omega} = \frac{d\omega}{d\theta} \Rightarrow \omega d\omega = \alpha d\theta$

Imply $\int_{\omega_0}^{\omega} \omega d\omega = \int_{t_0}^{t} \alpha d\theta$

$$\frac{1}{2}\left(\omega^2 - \omega_0^2\right) = \alpha\left(t - t_0\right)$$

Hence, time-independent angular equation of motion is $\omega^2 - \omega_0^2 = 2\alpha\left(t - t_0\right)$

Where, ω_0 is the **initial angular velocity at time t**$_0$ measured in (rad/s).

ω is the **final angular velocity at time t** measured in (rad/s).

α is the **angular acceleration** measured in (rad/s^2).

t - t_0 is called **time elapsed** measured in (s).

X. Projectile Motion

An object that is thrown or launched from a point into the atmosphere is called a **projectile**. In this study, we consider the case where the object is influenced only by gravity, neglecting air resistance and the buoyant force of the air.

i. Motion Through Each Axis

An object is launched from a point O with an initial velocity v_0 at an angle α to the horizontal axis $(\overrightarrow{ox})$.

At instant $t = 0, x_0 = 0, y_0 = 0$

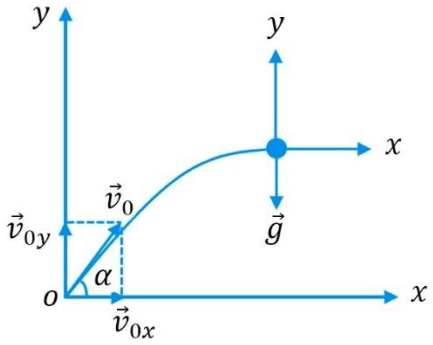
According to the second law of Newton

We have $\Sigma\vec{F} = m\vec{a}$

An object is experienced only by gravity

$\vec{F}_g = m\vec{g}$

Imply $m\vec{a} = m\vec{g} \Leftrightarrow \vec{a} = \vec{g}$ (1)



- Horizontal axis $(\overrightarrow{ox})$

Acceleration $a_x = 0 \Rightarrow v_x = \text{constant}$. Therefore, the object moves with **uniform linear motion**.

We get, $v_x = v_{0x} = v_0 \cos\alpha$

Time position equation of motion $x = x_0 + vt$

We get, $x = v_0 \cos\alpha$ (2)

Hence, time position equation of motion is $x = v_0 \cos\alpha$.

- Vertical axis $(\overrightarrow{oy})$

Acceleration $a_y = -g$. Therefore, then the object undergoes **non-uniform linear motion**.

Initial velocity $v_{0y} = v_0 \sin\alpha$

Final velocity $v_y = v_{0y} + a_y t$

Equivalent $v_y = v_0 \sin\alpha - gt$

Time position equation of motion $y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$

$$y = v_0 \sin\alpha t - \frac{1}{2}gt^2 \quad (3)$$

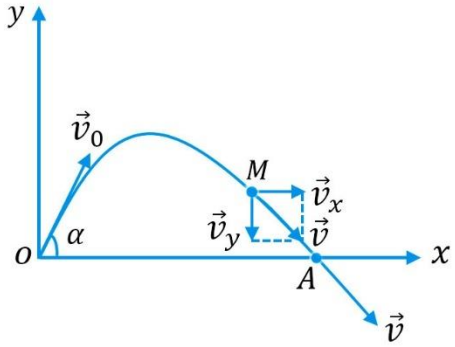
Hence, time position equation of motion is $y = v_0 \sin\alpha t - \frac{1}{2}gt^2$

ii. Trajectory equation

The **trajectory equation** is a mathematical expression that describes the **path** of a projectile in two-dimensional motion under the influence of gravity, **without involving time explicitly**.

It is written in the form: $y = f(x)$

From equation (1) $t = \dfrac{x}{v_0 \cos\alpha}$



Substitute equation (1) into (2): $y = v_0 \sin\alpha\left(\dfrac{x}{v_0 \cos\alpha}\right) - \dfrac{1}{2}g\left(\dfrac{x}{v_0 \cos\alpha}\right)^2$

$$y = x\tan\alpha - \frac{1}{2}g\frac{x^2}{v_0^2 \cos^2\alpha}$$

Hence, the trajectory equation is $y = x\tan\alpha - \dfrac{gx^2}{2v_0^2 \cos^2\alpha}$

iii. Range of trajectory

The **range** is the **horizontal distance** traveled by a projectile from its launch point until it lands back on the same vertical level (usually the ground). Assume that a projectile crosses the horizontal $(\overrightarrow{ox})$ axis at point A.

We get, $A(x = \overline{OA}; y = 0)$

We have, $y = x\tan\alpha - \dfrac{gx^2}{2v_0^2 \cos^2\alpha}$

Equivalent $0 = x\tan\alpha - \dfrac{gx^2}{2v_0^2 \cos^2\alpha}$

$$\frac{gx}{2v_0^2 \cos^2\alpha} = \tan\alpha$$

$$x = \frac{\sin\alpha}{\cos\alpha} \cdot \frac{2v_0^2 \cos^2\alpha}{g} = \frac{2v_0^2 \sin\alpha \cos\alpha}{g}$$

Hence, the range of projectile is $x = \dfrac{v_0^2 \sin 2\alpha}{g}$

Range of projectile $x = \overline{OA}$ is directly proportional to $\sin 2\alpha$. To achieve the maximum range of a projectile, the condition is $\sin 2\alpha = 1$.

We get, $\sin 2\alpha = \sin \frac{\pi}{2}$

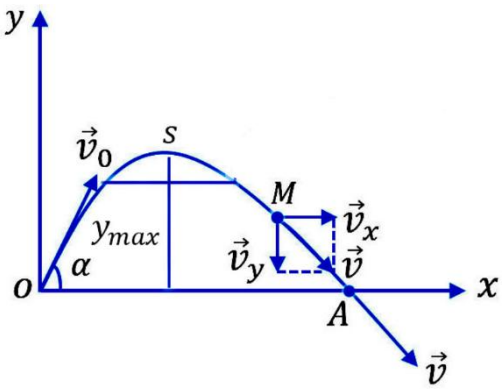
$2\alpha = \frac{\pi}{2} \Rightarrow \alpha = \frac{\pi}{4} = 45^\circ$

Imply $x_{\max} = \frac{v_0^2}{g}$

Hence, the maximum range of a projectile is $x_{\max} = \frac{v_0^2}{g}$

iv. Instantaneous velocity

Assume that at instant t, an object at point M has a velocity $\vec{v}$. This velocity can be resolved into two components: $\vec{v} = \vec{v}_x + \vec{v}_y$



Since, $\vec{v}_x \perp \vec{v}_y$

We get, $v^2 = v_x^2 + v_y^2$

And $v_x = v_{0x} = v_0 \cos\alpha$

$v_y = v_{0y} = v_0 \sin\alpha$

At instant $t_0 = 0, v_{0y} = v_0 \sin\alpha$

We get, $a_y = -g$

Imply $v_y = v_0 \sin\alpha - gt$

$v^2 = (v_0 \cos\alpha)^2 + (v_0 \sin\alpha - gt)^2$

$v^2 = v_0^2 \cos^2\alpha + v_0^2 \sin^2\alpha + g^2t^2 - 2gtv_0 \sin\alpha$

$v^2 = v_0^2(\cos^2\alpha + v_0^2 \sin^2\alpha) - 2g(-\frac{1}{2}gt^2 + v_0 \sin\alpha t)$

We get, $v^2 = v_0^2 - 2gy$

Hence, the instantaneous velocity is $v^2 - v_0^2 = \pm 2gy$

Noted: Take the positive sign $(+)$ when the acceleration due to gravity $\vec{g} \uparrow\uparrow \overrightarrow{oy}$.

Take the negative sign $(-)$ when the acceleration due to gravity $\vec{g} \uparrow\downarrow \overrightarrow{oy}$.

v. Maximum Height

We have $\vec{v} = \vec{v}_x + \vec{v}_y$

At the peak (highest) point of the projectile's trajectory, the vertical component of velocity $v_y = 0$.

Imply $\vec{v} = \vec{v}_x$

And $v = v_x = v_0 \cos\alpha$

Previous equation $\left(v_0 \cos\alpha\right)^2 - v_0^2 = -2gy_{max}$

$$v_0^2 \cos^2\alpha - v_0^2 = -2gy_{max}$$

$$v_0^2\left(\cos^2\alpha - 1\right) = -2gy_{max}$$

$$-v_0^2\left(1 - \cos^2\alpha\right) = -2gy_{max}$$

From formula $1 - \cos^2\alpha = \sin^2\alpha$

We get, $v_0^2 \sin^2\alpha = 2gy_{max}$

Hence, the maximum height of the trajectory is $y_{max} = \dfrac{v_0^2 \sin^2\alpha}{2g}$

Chapter1 Mechanics
Lesson2 Application of Newton law and Rotational Motion

I. Newton Laws

i. First Law of Newton or Principle of Inertia

If an object experiences no net external force (i.e., the sum of all forces acting on it is zero), then:

- If it was initially at rest, it will remain at rest.
- If it was in motion, it will continue moving in a straight line at constant speed (uniform linear motion).

This can be written as $\Sigma\vec{F} = \vec{0}$

Acceleration $\vec{a} = \vec{0} \Rightarrow \vec{v} = \text{constant}$

- If $\vec{v} = 0$, the object will remain at rest.
- If $\vec{v} \neq 0$, the object will continue moving in a straight line.

ii. Second Law of Newton or Fundamental Law of Dynamics

According to fundamental law of dynamics

We have $\vec{F} = \dfrac{d\vec{p}}{dt}$

Since, $\vec{p} = m\vec{v}$

Imply $\vec{F} = \dfrac{d(m\vec{v})}{dt}$

$\vec{F} = m\dfrac{d\vec{v}}{dt} = m\vec{a}$

Hence, the force acts upon on the object is $\vec{F} = m\vec{a}$

Where, $\vec{a}$ is the absolute acceleration.

But, $\vec{a} = \vec{a}_r + \vec{a}_e + \vec{a}_c$

Since, $\vec{a}_c = 2\vec{\omega} \times \vec{v}_r$

Imply $\vec{F} = m\vec{a}_r + m\vec{a}_e + m\vec{a}_c$

$\vec{F} - m\vec{a}_e - m\vec{a}_c = m\vec{a}_r$

However, $\vec{F}_{ie} = m\vec{a}_e$ is inertial force in a non-inertial (accelerating) reference frame.

$\vec{F}_{ic} = -m\vec{a}_c$ is Coriolis inertial force in a non-inertial (rotating) reference frame.

Hence, the fundamental law of dynamics for inertial reference frame is $\vec{F} + \vec{F}_{ie} + \vec{F}_{ic} = m\vec{a}_r$

iii. Third Law of Newton or Reaction Law

If body A exerts a force on body B, then body B simultaneously exerts a force of equal magnitude and opposite direction on body A.

According to third law of Newton

We get, $\vec{F}_{AB} = -\vec{F}_{BA}$

Hence, the third law of Newton is $\vec{F}_{AB} = -\vec{F}_{BA}$

II. Newton Laws Application

i. Non-Uniform Circular Motion

A. A cone pendulum

A **cone pendulum** is a **mass attached to a string or rod** that moves in a **horizontal circular path**, causing the string to sweep out the shape of a **cone**.

When an axis (Δ) rotates with angular velocity $\vec{\omega}$, a string or rod suspended from it deflects and makes a constant angle α with the vertical.

• Applying Newton's Second Law in a Reference Frame

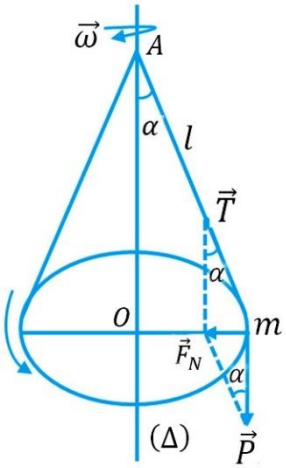
We get, $\vec{P} + \vec{T} = m\vec{a}$

But, $\vec{a} = \vec{a}_T + \vec{a}_N$

Since, $\vec{a}_T = \dfrac{d\vec{v}}{dt} = 0,\ \vec{v} = \text{constant}$

We get, $\vec{a} = \vec{a}_N$

Hence, the expression of Newton's Second Law in a cone pendulum is $\vec{P} + \vec{T} = m\vec{a}_N = \vec{F}_N$



However, $\tan\alpha = \dfrac{F_N}{P} = \dfrac{ma_N}{mg}$

And, $a_N = \dfrac{v^2}{r} = \dfrac{(r\omega)^2}{r} = r\omega^2$

Imply $\tan\alpha = \dfrac{(r\omega)^2}{g}$

In $\Delta \perp OAm$ $\sin\alpha = \dfrac{r}{l} \Rightarrow r = l\sin\alpha$

We get, $\dfrac{\sin\alpha}{\cos\alpha} = \dfrac{l\sin\alpha\omega^2}{g}$

$\dfrac{1}{\cos\alpha} = \dfrac{l\omega^2}{g}$

Hence, $\cos\alpha = \dfrac{g}{l\omega^2}$

• Applying Newton's Second Law in a Non-Reference Frame

We get, $\vec{F} + \vec{F}_{ie} + \vec{F}_{ic} = m\vec{a}_r$

Since, $\vec{F}_{ie} = -m\vec{a}_e$ and $\vec{F}_{ic} = -m\vec{a}_c$

But, $\vec{a}_c = 2\omega \times \vec{v}_r$

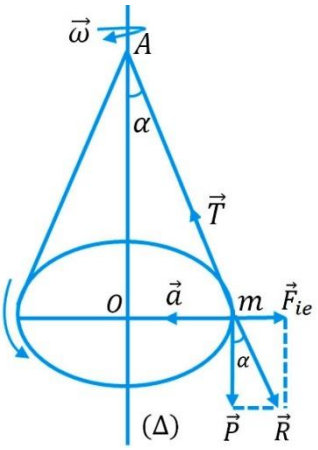
With respect to the observer, the pendulum is at rest.

We get, $\vec{v}_r = 0; \vec{a}_r = \vec{0}$

Therefore, $\vec{a}_c = 0$

Imply $\vec{F}_{ic} = \vec{0}$

We get, $\vec{F} + \vec{F}_{ie} = \vec{0}$



But, $\vec{F} = \vec{P} + \vec{T}$

$\vec{P} + \vec{T} + \vec{F}_{ie} = \vec{0}$

$m\vec{a} + \vec{F}_{ie} = \vec{0}$

Hence, the expression of Newton's Second Law in a cone pendulum is $\vec{F}_{ie} = -m\vec{a}$

* Inertial force $\vec{F}_{ie}$ have

- The direction same as $\vec{a}$
- The direction opposite from $\vec{a}$
- The magnitude: $F_{ie} = ma$

The equation can be written as $\vec{P} + \vec{T} + \vec{F}_{ie} = \vec{0}$

Let, $\vec{R} = \vec{P} + \vec{F}_{ie}$

$\vec{R} + \vec{T} = \vec{0}$

Imply, $\vec{R} = -\vec{T}$

However, $\tan\alpha = \frac{F_{ie}}{P} = \frac{ma_N}{mg} = \frac{a_N}{g}$

$\tan\alpha = \frac{(r\omega)^2}{g}$

But, $r = l\sin\alpha$

Imply, $\frac{\sin\alpha}{\cos\alpha} = \frac{l\sin\alpha \cdot \omega^2}{g}$

$\frac{1}{\cos\alpha} = \frac{l\omega^2}{g}$

Hence, $\cos\alpha = \frac{g}{l\omega^2}$

In order for the string to deviate from the vertical position, the angle α must be greater than 0.

We get, $\cos\alpha < 1$

$\frac{g}{l\omega^2} < 1 \Rightarrow \omega^2 > \frac{g}{l}$

Imply, $\omega > \sqrt{\frac{g}{l}}$

But, $\omega_0 > \sqrt{\frac{g}{l}}$

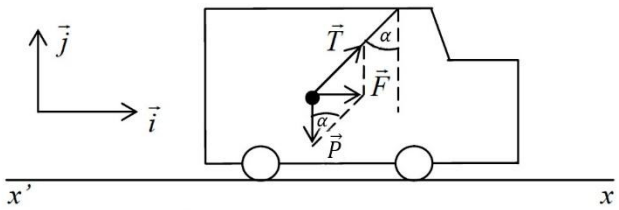
Hence, $\omega > \omega_0$

B. Linear Motion

A vehicle is accelerating along a straight road with acceleration $\vec{a}$. An observer inside the vehicle notices that a pendulum suspended from the ceiling deflects by an angle α from the vertical due to the horizontal acceleration.

• Applying Newton's Second Law in a Reference Frame

The pendulum is subjected to two forces: the **gravitational force (weight)** acting vertically downward and the **tension** in the string acting along the direction of the string.



This can be written as $\vec{P}+\vec{T}=m\vec{a}$

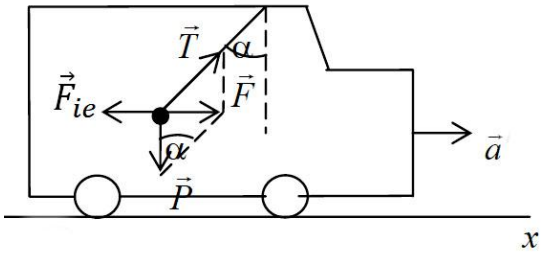
But, $\tan\alpha=\dfrac{F}{P}=\dfrac{ma}{mg}$

Hence, $\tan\alpha=\dfrac{a}{g}$

• Applying Newton's Second Law in a Non-Reference Frame

This can be written as $\vec{F}+\vec{F}_{ie}+\vec{F}_{ic}=m\vec{a}_r$

With respect to the observer, the pendulum is at rest.



We get, $\vec{v}_r=0;\vec{a}_r=\vec{0}$

But, $\vec{F}_{ic}=-m\vec{a}_c$

Since, $\vec{a}_c=2\omega\times\vec{v}_r$

Therefore, $\vec{a}_c=0$

Imply $\vec{F}_{ic}=\vec{0}$

We get, $\vec{F}+\vec{F}_{ie}=\vec{0}$

And, $\vec{F}=\vec{P}+\vec{T}=m\vec{a}$

Equivalent $\vec{P}+\vec{T}+\vec{F}_{ie}=\vec{0}$

$m\vec{a}+\vec{F}_{ie}=\vec{0}$

We get, $\vec{F}_{ie}=-m\vec{a}$

This can be written as $\vec{F}+\vec{F}_{ie}+\vec{T}=\vec{0}$

Let, $\vec{R}=\vec{P}+\vec{F}_{ie}$

We get, $\vec{R}+\vec{T}=\vec{0}\Leftrightarrow\vec{R}=-\vec{T}$

$R=T$

However, $\tan\alpha=\dfrac{a}{g}$

III. Torque and Angular Frequency

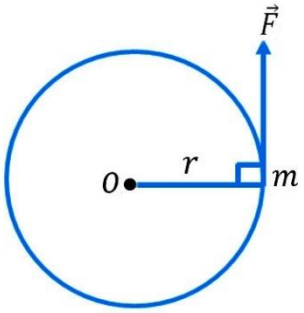
A particle of mass m moves in a circular path of radius r around a fixed center O, under the action of a force F. This force produces a **torque** about point O, which affects the particle's **angular motion**.

Applying Newton's Second Law

We get, $\vec{F}=m\vec{a}\Leftrightarrow F=ma\Rightarrow a=\dfrac{F}{m}$

But, $\vec{a}=\vec{a}_T+\vec{a}_N$

And, $a=a_T=\dfrac{dv}{dt}=\dfrac{d}{dt}(r\omega)$



We get, $a = r\dfrac{d\omega}{dt} = r\alpha$; α is angular acceleration

$$r\alpha = \frac{F}{m}$$

Multiply both sides by r

We get, $r^2\alpha = \dfrac{Fr}{m}$

But, $\vec{\tau} = \vec{F} \times \vec{r} \Rightarrow \tau = Fr\sin\left(\vec{r}, \vec{F}\right)$

$$\tau = Fr$$

Imply $r^2\alpha = \dfrac{\tau}{m}$

Hence, $\tau = mr^2\alpha = I\alpha$

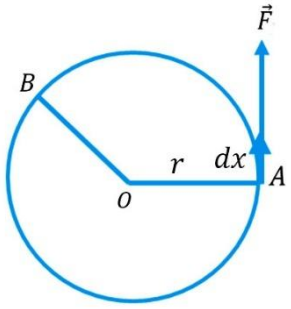
Where, τ is the **net torque** measured in $(\mathrm{N.m})$.

$I = mr^2$ is the **moment of inertia relative to (Δ) axis** measured in $(\mathrm{kg.m^2})$.

α is the **angular acceleration** measured in $(\mathrm{rad/s})$.

IV. Work and Power in Rotational Motion

Assume that a particle moves from point A to point B under the action of a force $\vec{F}$. To simplify the calculation, the curved path $(\mathrm{arc\ AB})$ can be divided into infinitesimally small straight segments dx, allowing it to be treated as a straight-line segment locally.



An **infinitesimal amount of work** done $dW = Fdx$

Imply $\int_A^B dW = \int_0^l Fdx$

We get, $W_{AB} = Fl$

But, $l = r\theta$

Therefore, $W_{AB} = \tau\theta$

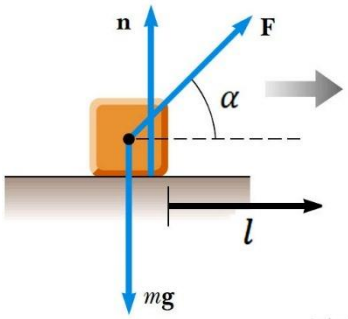
Where, W_{AB} is the **work done from point *A* to *B*** measured in (J).

τ is the **net torque** measured in $(\mathrm{N.m})$.

θ is the **angular displacement** measured in (rad).

V. Work Done in Translational Motion

When an object is acted upon by a force $\vec{F}$ at an angle α with the horizontal and moves a distance l along the horizontal, the **work done by the force** is given by:



$$W_{AB} = Fl\cos\alpha \quad \alpha = \left(\vec{F}, \vec{l}\right)$$

Hence, the **work done by the force** is $W_{AB} = Fl\cos\alpha$

Where, W_{AB} is the **work done from point A to B** measured in (J).

F is the **magnitude of the applied force** measured in (N).

l is the **displacement of the object** measured in (m).

α is the **angle between the force and the direction of motion** measured in $(^\circ)$.

VI. Power

Power is the rate at which **work** is done or **energy** is transferred over time.

i. Average power

If a force does an amount of work ΔW during a time interval Δt, then the **average power** is defined as:

$$P_{av} = \frac{\Delta W}{\Delta t}$$

Where, P_{av} is the **average power** measured in (W).

ΔW is the **work done** measured in $(\mathrm{N.m})$.

Δt is the **time interval** measured in (s).

ii. Instantaneous power

Instantaneous power is defined as the **limit** of the average power when the time interval Δt approaches zero:

$$P = \lim_{\Delta t \to 0} P_{av} = \lim_{\Delta t \to 0} \frac{\Delta W}{\Delta t} = \frac{dW}{dt}$$

Hence, instantaneous power is $P = \dfrac{dW}{dt}$

This means instantaneous power measures how quickly work is being done **at a precise moment**.

iii. Power in Rotational Motion

Power in rotational motion measures how quickly work is done by a torque when an object rotates.

Work in rotational motion $W = \tau\theta$

Since, $$P = \frac{dW}{dt}$$

We get, $$P = \frac{d}{dt}\left(\tau\theta\right)$$

$$P = \tau\frac{d\theta}{dt}$$

Imply $$P = \tau\omega$$

Hence, power in rotational motion is $P = \tau\omega$.

Where, P is the **power in rotational motion** measured in (W).

τ is the **net torque** measured in $(\mathrm{N.m})$.

ω is the **angular velocity** measured in $(\mathrm{rad/s})$.

iv. Power in Translational Motion

Power in translational motion measures how fast a force does work when moving an object in a straight line.

Work in rotational motion $W = \vec{F} \cdot \vec{l}$

Since, $$P = \frac{dW}{dt}$$

We get, $P = \frac{d}{dt}\left(\vec{F} \cdot \vec{l}\right)$

$P = \vec{F}\frac{d\vec{l}}{dt}$

Imply $P = \vec{F} \cdot \vec{v}$

Hence, power in rotational motion is $P = Fv\cos\alpha$.

Where, P is the **power in translational motion** measured in (W).

v is the **velocity** measured in $(\mathrm{m/s})$.

F is the **applied force** measured in (N).

α is the angle between **velocity** and **force** measured in degrees $(°)$.

VII. Kinetic Energy in Rotational Motion

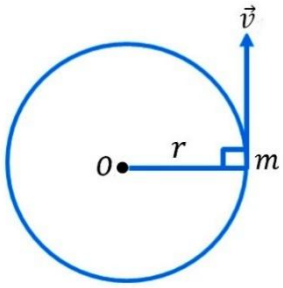
An object moves in a circular path of radius r around a center point O with a linear (tangential) velocity $\vec{v}$.

Kinetic energy $K = \frac{1}{2}mv^2$

Since, $v = r\omega$

We get, $K = \frac{1}{2}m\left(r\omega\right)^2$

$K = \frac{1}{2}mr^2\omega^2$



Hence, kinetic energy in rotational motion is $K = I\omega^2$

Where, K is the **Rotational kinetic energy** measured in (J).

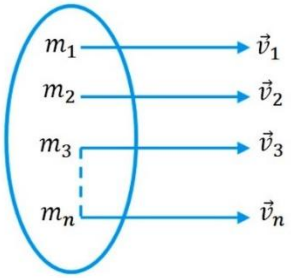
I is the **Moment of inertia** measured in $(\mathrm{kg.m^2})$.

ω is the **Angular velocity** measured in $(\mathrm{rad/s})$.

VIII. Kinetic Energy of an Object

An object of total mass mmm, composed of many point masses $m_1, m_2, m_3, ..., m_n$, where each point mass moves with velocity $\vec{v}_1, \vec{v}_2, \vec{v}_3, ..., \vec{v}_n$, respectively, has a total kinetic energy given by the sum of the kinetic energy of all its components:

$$K = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \frac{1}{2}m_3v_3^2 + ... + \frac{1}{2}m_nv_n^2$$



Hence, the kinetic energy of an object is $K = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \frac{1}{2}m_3v_3^2 + ... + \frac{1}{2}m_nv_n^2$.

Where, K is the **translational kinetic energy** measured in (J).

m is the **mass of an object** measured in (kg).

v is the **velocity** measured in $(\mathrm{m/s})$.

i. Kinetic Energy in Translational Motion

For translational motion, we get $v = v_1 = v_2 = v_3 = ... = v_n$

We have $K_t = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \frac{1}{2}m_3v_3^2 + ... + \frac{1}{2}m_nv_n^2$

Imply $K_t = \frac{1}{2}v\left(m_2 + m_2 + m_3 + ... + m_n\right)$

Let, $m = m_2 + m_2 + m_3 + ... + m_n$ is the mass of an object.

Hence, translational kinetic energy is $K_t = \frac{1}{2}mv^2$

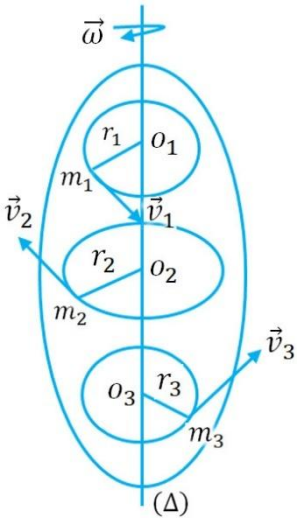
Where, K is the **translational kinetic energy** measured in (J).

m is the **total mass of an object** measured in (kg).

v is the **velocity** measured in $(\mathrm{m/s})$.

ii. Kinetic Energy in Rotational Motion

When a rigid body rotates about a fixed axis (Δ), every point in the body moves in a circle around that axis.



We have, $v_1 = r_1\omega, v_2 = r_2\omega, v_3 = r_3\omega, v_n = r_n\omega$

Imply $K_r = \frac{1}{2}m_1\left(r_1\omega\right)^2 + \frac{1}{2}m_2\left(r_2\omega\right)^2 + \frac{1}{2}m_3\left(r_3\omega\right)^2 + ... + \frac{1}{2}m_n\left(r_n\omega\right)^2$

$K_r = \frac{1}{2}m_1\left(r_1\omega\right)^2 + \frac{1}{2}\omega^2\left(m_2r_1^2 + m_2r_2^2 + m_3r_3^2 + ... + +m_nr_n^2\right)$

But, $I = m_2r_1^2 + m_2r_2^2 + m_3r_3^2 + ... + +m_nr_n^2$

The kinetic energy of such rotational motion is given by: $K_r = \frac{1}{2}I\omega^2$.

When an object undergoes **both rotational** and **translational motion** (such as a rolling wheel), its total kinetic energy is the **sum of translational** and **rotational kinetic energies**.

$$K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

Where, K is the **total kinetic energy** measured in (J).

I is the **moment of inertia of an object respected to axis (Δ)** measured in $(\mathrm{kg.m^2})$.

ω is the **angular velocity** measured in $(\mathrm{rad/s})$.

m is the **mass of an object** measured in (kg).

v is the **velocity** measured in $(\mathrm{m/s})$.

IX. Moment of Inertia

The moment of inertia $I = \sum_{i=1}^{n} m_i r_i^2$ is valid only for systems of **discrete point masses**. For objects with a **continuous mass distribution** (where the mass is spread out rather than separate), such as shown in the figure above, the moment of inertia must be expressed in **integral form**:

$$I = \int r^2 dm$$

Density of an object $\rho = \dfrac{dm}{dV} \Rightarrow dm = \rho dV$

Hence, moment of inertia is $I = \int r^2 dm = \int \rho r^2 dV$

Where, I is the **moment of inertia of an object respected to axis (Δ)** measured in (kg.m^2).

r is the **perpendicular distance from the axis of rotation (Δ)** measured in (m).

dm is **a tiny mass element of the object** measured in (kg).

i. Moment of Inertia of a Point Mass

The **moment of inertia of a point mass (or point of matter)** is a measure of how difficult it is to change its rotational motion about a specific axis. It depends on the mass and its distance from the axis of rotation.

Hence, moment of inertia of a point of mass is $I_{(\Delta)} = mr^2$

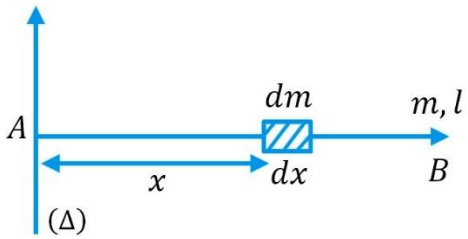
Where, I is the **moment of inertia of a point mass to axis (Δ)** measured in (kg.m^2).

r is the **perpendicular distance from the axis of rotation (Δ)** measured in (m).

m is **mass of the object** measured in (kg).

ii. A Uniform Bar

A bar $AB = l$ has mass m.



A. Relative to an axis (Δ) through one end

The moment of inertia of an infinitesimal mass element dm at a distance x from the axis (Δ) is $dI = x^2 dm$

Density $\rho = \dfrac{m}{l} = \dfrac{dm}{dx}$

Infinitesimal mass element: $dm = \dfrac{m}{l} dx$

We get, $dI = x^2 \dfrac{m}{l} dx$

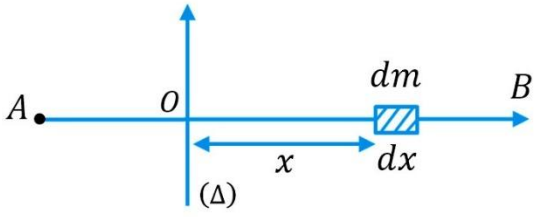
Integrate from 0 to l: $\int dI = \dfrac{m}{l} \int_0^l x^2 dm$

$$I = \frac{ml^3}{3l}$$

Hence, moment of inertia of a **uniform bar** rotating about its **end is** $I = \dfrac{1}{3} ml^2$

B. Relative to an axis (Δ) through mid-point

The moment of inertia of an infinitesimal mass element dm at a distance x from the axis (Δ) is $dI = x^2 dm$



Density $\rho = \dfrac{dm}{dx} = \dfrac{m/2}{l/2}$

Infinitesimal mass element: $dm = \frac{m}{l}dx$

We get, $dI = x^2 \frac{m}{l}dx$

Integrate from 0 to $\frac{l}{2}$: $\int dI = \frac{m}{l}\int_0^{\frac{l}{2}} x^2 dm$

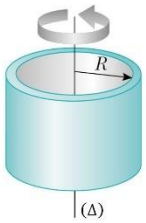
$$I = \frac{1}{24}ml^2$$

Total moment of inertia about axis (Δ) is $I_{(\Delta)} = 2I = 2\frac{1}{24}ml^2$

Hence, moment of inertia of a **uniform bar** rotating about its **center is** $I_{(\Delta)} = \frac{1}{12}ml^2$

iii. A hollow cylinder

A hollow cylinder has mass m and radius R, and rotates about a central axis (Δ) through its center O.

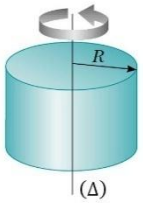


The **moment of inertia** relative to the axis (Δ) is: $I_{(\Delta)} = mR^2$

Hence, the moment of inertia of a hollow cylinder is $I_{(\Delta)} = mR^2$

iv. Disk or Cylinder Sphere

A disk has mass m and radius R, and rotates about a central axis (Δ) through its center O.



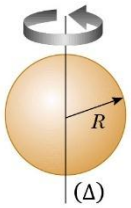
The **moment of inertia** relative to the axis (Δ) is: $I_{(\Delta)} = \frac{1}{2}mR^2$

Hence, the moment of inertia of a disk is $I_{(\Delta)} = \frac{1}{2}mR^2$

v. A Uniform Solid Sphere

A uniform solid sphere has mass m and radius R, and rotates about a central axis (Δ) through its center O.

The **moment of inertia** relative to the axis (Δ) is: $I_{(\Delta)} = \frac{2}{5}mR^2$

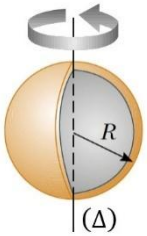


Hence, the moment of inertia of a uniform solid sphere is $I_{(\Delta)} = \frac{2}{5}mR^2$

vi. A Hollow Sphere

A hollow sphere has mass m and radius R, and rotates about a central axis (Δ) through its center O.

The **moment of inertia** relative to the axis (Δ) is: $I_{(\Delta)} = \frac{2}{3}mR^2$

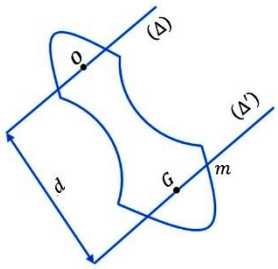


Hence, the moment of inertia of a hollow sphere is $I_{(\Delta)} = \frac{2}{3}mR^2$

vii. Huygens' Theorem (Parallel Axis Theorem)

Huygens' theorem states that:

The moment of inertia of an object about any axis (Δ) is equal to the moment of inertia of the object about a parallel axis (Δ') through its center of gravity, plus the product of the object's mass and the square of the perpendicular distance d between the two axes.



This can be written as $I_{(\Delta)} = I_{(\Delta')} + md^2$

Hence, moment of inertia of an object about axis (Δ) is $I_{(\Delta)} = I_{(\Delta')} + md^2$.

Where, $I_{(\Delta)}$ is the **moment of inertia of an object to axis (Δ)** measured in (kg.m^2).

$I_{(\Delta')} = I_{\text{cm}}$ is the **moment of inertia about the center of mass** measured in (kg.m^2).

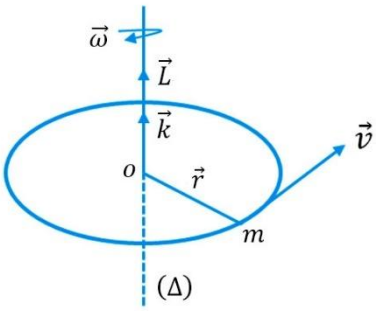
d is the **perpendicular distance between two axes** measured in (m).

m is **mass of the object** measured in (kg).

X. Angular Momentum

i. Definition: Angular momentum is a measure of the rotational motion of an object. It is the rotational analog of linear momentum.

A particle of mass mmm rotates about an axis (Δ) at a perpendicular distance $\vec{r}$ from the axis, moving with a tangential velocity $\vec{v}$.



Angular momentum $\vec{L} = \vec{r} \times \vec{p}$

Momentum $\vec{p} = m\vec{v}$

We get, $\vec{L} = \vec{r} \times m\vec{v}$

Hence, $\vec{L} = \vec{r} \times m\vec{v}$

∗ Angular momentum $\vec{L}$ has

- Direction perpendicular to an axis (Δ)
- Direction determined by right hand rule

→ Curl the fingers of your right hand in the direction of the object's rotation (the direction the object is spinning).

→ Your extended thumb then points in the direction of the angular momentum vector $\vec{L}$.

- Magnitude $L = rmv\sin\alpha \quad \alpha\left(\vec{r},\vec{v}\right) = 90^\circ$

We get, $L = rmv$

But, $v = r\omega$

Imply, $L = rmr\omega$

And, $I = mr^2$

We get, $L = I\omega$

$\vec{L} = I\vec{\omega}\vec{k}$

Hence, the angular momentum is $L = I\omega$

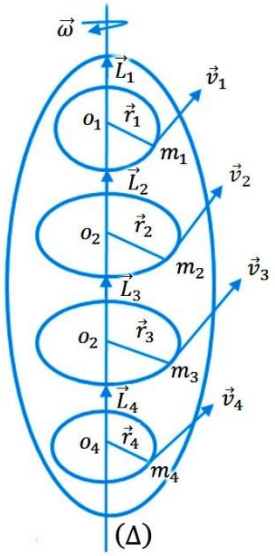
Where, L is the **angular momentum** measured in $(\text{kg.m}^2/\text{s})$.

I is the **moment of inertia about the axis of rotation** measured in (kg.m^2).

ω is the **angular velocity** measured in (rad/s).

ii. Angular momentum of an object

An object of total mass mmm, composed of point masses m_1, m_2, m_3 and m_4, rotates about an axis (Δ) with angular velocity $\vec{\omega}$. The angular momentum of the system is



$$\vec{L} = \vec{L}_1 + \vec{L}_2 + \vec{L}_3 + \vec{L}_4$$

We get, $$L = L_1 + L_2 + L_3 + L_4$$

But, $$\vec{L}_1 = \vec{r}_1 \times m_1\vec{v}_1 \Rightarrow L_1 = r_1 m_1 v_1 \sin\left(\vec{r}_1, \vec{v}_1\right)$$

$$L_1 = r_1 m_1 v_1$$

Since, $$v_1 = r_1\omega \Rightarrow L_1 = r_1 m_1 r_1 \omega = m_1 r_1^2 \omega$$

And, $$L_2 = m_2 r_2^2 \omega$$

$$L_3 = m_3 r_3^2 \omega$$

And, $$L_4 = m_4 r_4^2 \omega$$

We get, $$L = \left(m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + m_4 r_4^2\right)\omega$$

However, $I = \sum_{i=1}^{n} m_i r_i^2$ is the moment of inertia of an object relative to an axis (Δ).

Therefore, $$L = I\omega$$

Hence, the angular moment of an object is $L = I\omega$.

Where, L is the **angular momentum** measured in $(\text{kg.m}^2/\text{s})$.

I is the **moment of inertia about the axis of rotation** measured in (kg.m^2).

ω is the **angular velocity** measured in (rad/s).

iii. Angular Momentum Theorem (Second Law of Newton in Rotational Motion)

We have, $$\vec{L} = \vec{r} \times m\vec{v}$$

Imply $$\frac{d\vec{L}}{dt} = \frac{d}{dt}\left(\vec{r} \times m\vec{v}\right)$$

$$\frac{d\vec{L}}{dt} = \frac{d\vec{r}}{dt} \times m\vec{v} + \vec{r} \times \frac{d}{dt}\left(m\vec{v}\right)$$

$$\frac{d\vec{L}}{dt} = \vec{v} \times m\vec{v} + \vec{r} \times \vec{F}$$

Since, $\vec{v} \times m\vec{v} = 0$ normal to each other.

We get, $$\frac{d\vec{L}}{dt} = \vec{\tau}\left(\vec{F}\right) \quad (1)$$

However, $$\vec{L} = I\vec{\omega}\vec{k}$$

Therefore, $\dfrac{d\vec{L}}{dt} = \dfrac{d}{dt}\left(I\omega\vec{k}\right)$

$$\frac{d\vec{L}}{dt} = I\left(\frac{d\omega}{dt}\right)\vec{k}$$

We get, $\dfrac{d\vec{L}}{dt} = I\alpha\vec{k}\ (2)$

Compare equation (1) and (2)

We get, $\vec{\tau}\left(\vec{F}\right) = I\alpha\vec{k}$

Hence, $\vec{\tau}\left(\vec{F}\right) = I\alpha\vec{k}$ is called angular momentum theorem or second law of newton in rotational motion.

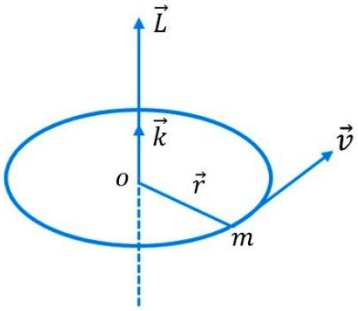
Where, α is the **angular acceleration** measured in $(\mathrm{rad/s^2})$.

I is the **moment of inertia** measured in $(\mathrm{kg.m^2})$.

τ is the **net torque** measured in $(\mathrm{N.m})$.

iv. Conservation of Angular Momentum

In order for $\vec{\tau}\left(\vec{F}\right) = \vec{0}$, unless $\vec{F} = \vec{0}$ and $\vec{r} \uparrow\uparrow \vec{F}$



We get, $\dfrac{d\vec{L}}{dt} = \vec{0}$

Imply $\vec{L} = \overrightarrow{\text{constant}}$ mass moves in a plane.

$L = \text{constant}$

Hence, the conservation of angular momentum is $I_1\omega_1 = I_2\omega_2$

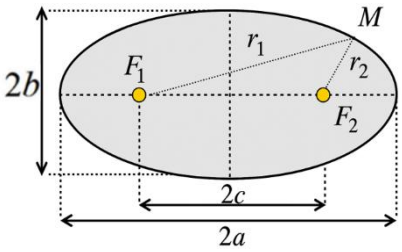
I. Kepler's Law

Kepler's laws are classified into three laws.

• **Kepler's first law** states that planets move in elliptical orbits, with the Sun at one focus. An ellipse is defined as the set of all points such that the sum of the distance from each point to two foci is a constant.

$$MF_1 + MF_2 = 2a = \text{constant}$$

$$r_1 + r_2 = 2a = \text{constant}$$



Where, F_1 and F_2 **are the fixed points of focus** measured in (m).

$2a$ is the **length of major axis** measured in (m).

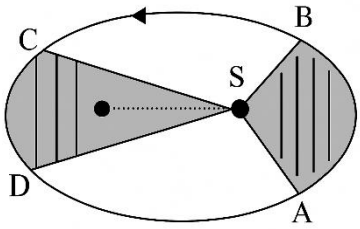
$2b$ is the **length of minor axis** measured in (m).

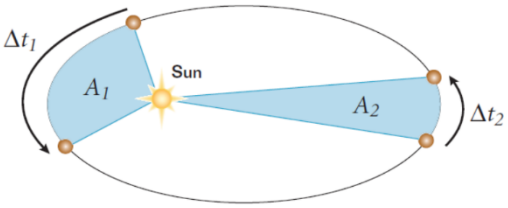
$2c$ is the **distance between two focuses of an ellipse** measured in (m).

$c^2 = a^2 - b^2$, Where eccentricity of ellipse: $e = \mathrm{c}/a$

If $e = 0$ $(0 < e < 1)$ then ellipse becomes a circle.

• **Kepler's second law** states that an imaginary line from the Sun to a planet sweeps out equal areas in equal times. If the planets travelled by AB or CD in equal duration, the area $\mathrm{SAB} = \mathrm{SDA}$.





The **area of an ellipse** $S = \pi ab$

Where, S is the **area of an ellipse** measured in (m^2).

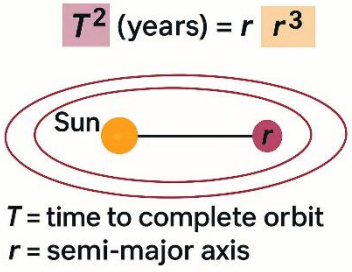
a is the **semi-major axis** measured in (m).

b is the **semi-minor axis** measured in (m).

The rate of area swept out (areal velocity) is constant, and that constant is often denoted a

$$\frac{dS}{dt} = \frac{c}{2};\ c \text{ is the areal constant.}$$

• **Kepler's third law** states that the square of the ratio of the periods of any two planets is equal to the cube of the ratio of their distances from the Sun.



A planet orbiting the Sun is kept in orbit by the gravitational pull from the Sun. The gravitational force provides the necessary centripetal force for circular motion:

$$\frac{GM_{\text{Sun}}M_{\text{Planet}}}{r^2} = \frac{M_{\text{Planet}}v^2}{r}$$

$$v = \frac{2\pi r}{T}$$

$$T^2 = \left(\frac{4\pi^2}{GM_{\text{Sun}}}\right)r^3 = K_S r^3$$

Where, T is the **orbital period of a planet** measured in (s).

K_s is the independent of the mass of the planet, and so is valid for any planet.

r is the **average orbital radius or semi-major axis** measured in (m).

However, **Kepler's Third Law** states that the **square of the orbital period** of a planet is **proportional to the cube of the semi-major axis** of its orbit.

Law of gravitation and centripetal force for a planet orbiting a star:

$$\frac{GmM}{r^2} = \frac{mv^2}{r}$$

The period is the time to complete one orbit:

$$T = \frac{2\pi r}{v} = 2\pi\sqrt{\frac{r^3}{GM}} \Rightarrow T^2 = \frac{4\pi^2}{GM}r^3$$

Thus, $$\frac{T^2}{r^3} = \frac{4\pi^2}{GM} = \text{constant}$$

For two planets A and B: $$\left(\frac{T_A}{T_B}\right)^2 = \left(\frac{r_A}{r_B}\right)^3$$

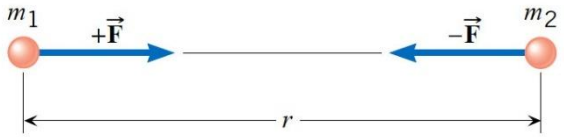
Where, T_{A} is the **orbital period of a planet *A*** measured in (s).

T_{B} is the **orbital period of a planet *B*** measured in (s).

r_{A} is the **semi-major axis of planet *A*'s orbit** measured in (m).

r_{B} is the **semi-major axis of planet *B*'s orbit** measured in (m).

II. Newton's Law of Universal Gravitation



Every particle in the universe attracts every other particle with a force that is:

- Directly proportional to the product of their masses,
- And inversely proportional to the square of the distance between their centers.

We get, $$\vec{F}_{12} = -\vec{F}_{21} = -G\frac{m_1 m_2}{r^2}\vec{i}$$

Hence, the gravitation force between two masses is $F_{12} = F_{21} = G\dfrac{m_1 m_2}{r^2}$

Where, $F_{12} = F_{21}$ is the **gravitational force two objects** measured in (N).

m_1 and m_2 are **masses of the two objects** measured in (kg).

$G = 6.67 \times 10^{-11}\ \mathrm{N.m^2/kg^2}$ is the **universal gravitational constant.**

r is the **distance between the centers of the two masses** measured in (m).

III. Gravitational Field

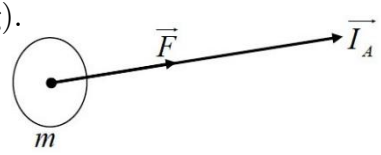
The **gravitational field** at a point in space is the **gravitational force per unit mass** experienced by a small test mass placed at that point.

This can be written as $g = \dfrac{F}{m}$

Where, g or I is the **gravitational field (vector)** measured in $(\mathrm{N/kg})$.

F is the **gravitational force on the mass** measured in (N).

m is the **mass of the test object** measured in (kg).



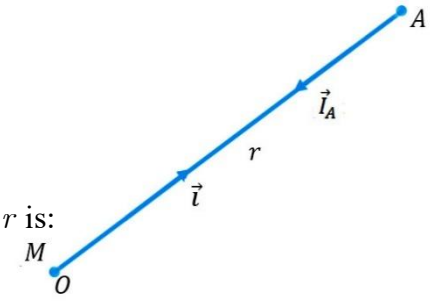
IV. Gravitational Field Due to a Mass M

Using $F = mg$, and substituting the expression for F:

$$g = \frac{F}{m} = G\frac{M}{r^2}$$

This means the gravitational field strength due to a mass M at a distance r is:

$$g = G\frac{M}{r^2}$$



Where, g or I is the **gravitational field (vector)** measured in $(\mathrm{N/kg})$.

$G = 6.67 \times 10^{-11}\ \mathrm{N.m^2/kg^2}$ is the **universal gravitational constant.**

M is the **mass of the large object creating the gravitational field** measured in (kg).

r is the **distance between the center of the mass M and the point where you are measuring the field or force** measured in (m).

V. Gravitational Field of Earth or Acceleration Due to Gravity

i. Gravitational Field of Earth

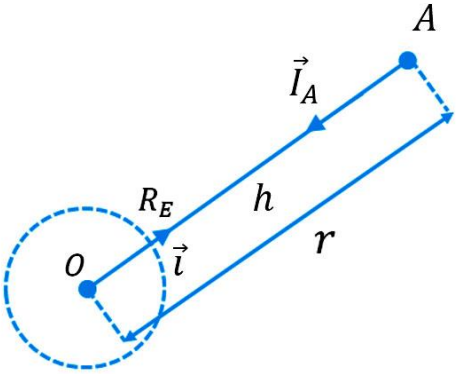
Near the surface of the Earth, the **gravitational field** is called **acceleration due to gravity** and is denoted by g. It represents the acceleration that any object experiences due to Earth's gravity.

We have, $\vec{g} = -G\dfrac{M_E}{r^2}\vec{i}$

Since, $r = R_E + h$

We get, $g = G\dfrac{M_E}{(R_E + h)^2}$

Hence, the acceleration due to gravity is $g = G\dfrac{M_E}{(R_E + h)^2}$



If $h = 0$, that means point ***A*** is located **on the surface of the Earth**.

Therefore, $g = G\dfrac{M_E}{R_E^2}$

Where, g or I is the **gravitational field (vector)** measured in $(\mathrm{N/kg})$.

$G = 6.67 \times 10^{-11}\ \mathrm{N.m^2/kg^2}$ is the **universal gravitational constant.**

$M_E = 5.97 \times 10^{24}\ \mathrm{kg}$ is the **mass of the Earth** measured in (kg).

$R_E = 6400\ \mathrm{km}$ is the **radius of the Earth** measured in (m).

ii. Gravitational Field Strength of Earth

An object of mass m experiences a weight

$$F_g = mg$$

The gravitational force creates the weight, so

$$F = F_g = mg$$

Gravity g depends on the height and latitude.

Therefore, $g = \dfrac{GM_E}{\left(R_E + h\right)^2}$ (1)

At height *h* from the Earth's surface, when $h = 0$,

we get: $g_0 = \dfrac{GM_E}{R_E^2}$ (2)

Divide the first equation by the second:

$$\frac{g}{g_0} = \frac{GM_E \big/ \left(R_E + h\right)^2}{GM_E / R_E^2}$$

Cancel out GM_E from numerator and denominator:

$$\frac{g}{g_0} = \frac{R_E^2}{\left(R_E + h\right)^2}$$

Now multiply both sides by g_0:

$$g = g_0 \frac{R_E^2}{\left(R_E + h\right)^2}$$

Hence, $g = g_0 \dfrac{R_E^2}{\left(R_E + h\right)^2}$ This formula tells you the **acceleration due to gravity at height *h*** above the Earth's surface, based on the known surface gravity g_0.

Note: The radius of the Earth is slightly longer at the equator and shorter at the poles, since the Earth is not a perfect sphere but an **oblate spheroid** (slightly flattened at the poles and bulging at the equator).

As a result, the gravitational field strength is **stronger at the poles** than at the equator: $g_p > g_e$

Where, g_p is gravitational acceleration at the **pole** measured in $(\mathrm{N/kg})$.

g_e is gravitational acceleration at the **equator** measured in $(\mathrm{N/kg})$.

g is gravity at any other height or location measured in $(\mathrm{m/s^2})$.

g_0 is standard gravity at Earth's surface (reference point) measured in $(\mathrm{m/s^2})$.

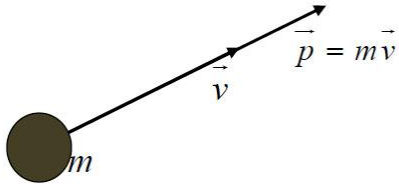
$R_E = 6400\ \mathrm{km}$ is the **radius of the Earth** measured in (m).

Chapter1 Mechanics
Lesson4 Impulse and Momentum

I. Linear Momentum

i. Definition: The **momentum** of a point mass at an instant is the product of its **mass** m and its **velocity** v at that instant.

This can be written as $p = mv$



Where, p is momentum (a vector quantity) measured in $(\mathrm{kg.m/s})$.

m is mass of the object measured in (kg).

v is velocity of the object measured in $(\mathrm{m/s})$.

If a **rigid body** is composed of a distribution of masses $m_1,\ m_2,\ m_3,\ldots,m_n$ and moves with a common velocity v, then the **total momentum** of the body is:

$$\vec{p}_t = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \ldots + \vec{p}_n$$
$$\vec{p}_t = m_1\vec{v} + m_2\vec{v} + m_3\vec{v} + \ldots + m_n\vec{v}$$
$$\vec{p}_t = \left(m_1 + m_2 + m_3 + \ldots + m_n\right)\vec{v}$$

Denoted $M = m_1 + m_2 + m_3 + \ldots + m_n$

We get, $\vec{p}_t = M\vec{v}$

Hence, the **total momentum** of the body is: $p_t = Mv$

Where, p_t is **total linear momentum** measured in $(\mathrm{kg.m/s})$.

$M = \sum_{i=1}^{n} m_i$ is **total mass** of the rigid body measured in (kg).

v is **common velocity** of the whole body measured in $(\mathrm{m/s})$.

II. Linear Momentum and Impulse

i. Definition: **Impulse** is the **change in linear momentum** of an object caused by an **external force** acting over a **time interval**.

Vector expression $\vec{I} = \vec{F}\Delta t = \Delta\vec{p}$

Hence, linear momentum and impulse theorem is $I = F\Delta t = \Delta p$.

Where, I is **impulse (vector)** measured in $(\mathrm{N.s})$.

$\Delta p = m\Delta v$ is **change in linear momentum** measured in $(\mathrm{kg.m/s})$.

F is **average external force** measured in (N).

Δt is **time duration of the force** measured in (s).

III. Conservation of Linear Momentum

For a system, if the **net external force** F_{ext} acting on it is zero, the system is called an **isolated system**. In an **isolated system**, the linear momentum does not change. That is:

$$F\Delta t = \Delta p = 0$$

Since, $F_{ext} = 0$

Hence, Δp This means the **total linear momentum of an isolate system is conserved**.

IV. Collision

If two masses m_1 and m_2 **collide** with each other over a time interval Δt, then **each mass experiences an impulse** due to the **force exerted by the other**.

$$I_1 = \Delta p_1 = F_1 \Delta t \text{ and } I_2 = \Delta p_2 = F_2 \Delta t$$

Since, $\Delta p_1 = \Delta p_2$

$$m_1 v_1' - m_1 v_1 = -(m_2 v_2' - m_2 v_2)$$

$$m_1 v_1' - m_1 v_1 = m_2 v_2 - m_2 v_2'$$

$$m_1 v_1' + m_2 v_2' = m_1 v_1 + m_2 v_2$$

Hence, $\Sigma \vec{p}_f = \Sigma \vec{p}_i$

Where, v_1' and m_1 is **the velocity and mass after collision** measured in $(\mathrm{m/s})$ and (kg).

v_2' and m_2 is **the velocity and mass after collision** measured in $(\mathrm{m/s})$ and (kg).

v_1 and m_1 is **the velocity and mass before collision** measured in $(\mathrm{m/s})$ and (kg).

v_2 and m_2 is **the velocity and mass before collision** measured in $(\mathrm{m/s})$ and (kg).

i. Inelastic Collision

In an **inelastic collision**, the following characteristics apply:

- **Objects stick together after collision** (for a **perfectly** inelastic collision).
- The **total linear momentum** is conserved: $p_i = p_f$
- **Kinetic energy is not conserved.** Some kinetic energy is transformed into internal energy (heat, sound, deformation, etc.).

Change in Kinetic Energy: $\Delta K = K_f - K_i$

Note: This value is **negative** in inelastic collisions, indicating a **loss of mechanical energy**.

Coefficient of Restitution (e):

The **coefficient of restitution** is a measure of how "bouncy" the collision is. It's defined as:

$$e = \left| \frac{v_f - V_f}{v_i - V_i} \right|$$

Where, v_f and V_f are **the initial velocities of the two objects after collision** measured in $(\mathrm{m/s})$.

v_i and V_i are **the initial velocities of the two objects before collision** measured in $(\mathrm{m/s})$.

$e = 0$ is **perfectly inelastic (objects stick together).**

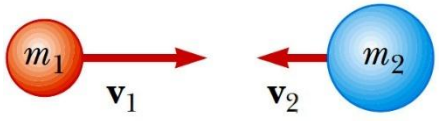
$0 < e < 1$: inelastic.

$e = 1$ is **perfectly elastic (kinetic energy conserved).**

ii. Elastic Collision

A **collision is said to be elastic** if the following conditions are met:

- **Objects move separately** after the collision (they do not stick together).
- The **total momentum** before and after the collision is conserved.



$$p_i = p_f$$

• The **total kinetic energy** before and after the collision is also conserved:

$$K_i = K_f$$

Kinetic Energy Conservation: $\frac{1}{2}m_1v_{1i}^2 + \frac{1}{2}m_2v_{2i}^2 = \frac{1}{2}m_1v_{1f}^2 + \frac{1}{2}m_2v_{2f}^2$

Note: In an elastic collision, the **relative speed of approach** before the collision equals the **relative speed of separation** after the collision, but in the **opposite direction**:

$$v_{1i} - v_{2i} = -\left(v_{1f} - v_{2f}\right)$$

Hence, $v_{1i} - v_{2i} = -\left(v_{1f} - v_{2f}\right)$ This equation is very useful when solving elastic collision problems algebraically.

V. Recoil of Gun and Missile Propulsion

• **The recoil of gun:** a bullet in the system has mass m_1 and the gun has mass m_2, then we get:

Momentum before shooting: $m_1v_1 + m_2v_2 = 0$

Momentum after shooting: $m_1v_1' + m_2v_2' = m_1v_1 + m_2v_2 = 0$

$$m_1v_1' + m_2v_2' = 0$$

This implies $v_1' = -\frac{m_2}{m_1}v_2'$

Hence, the recoil velocity of the gun is $v_1' = -\frac{m_2}{m_1}v_2'$

Where, v_1' is the **velocity of the bullet after firing** measured in $(\mathrm{m/s})$.

v_2' is the **recoil velocity of the gun after firing** measured in $(\mathrm{m/s})$.

m_1 is the **mass of the bullet** measured in (kg).

m_2 is the **mass of the gun** measured in (kg).

• **Missile Propulsion:** If, at an instant t, the missile moves with velocity $\vec{v}$, then the missile has momentum $\vec{p} = m\vec{v}$. If the missile burns a small amount of mass Δm, then the remaining part of the missile has mass m - Δm. The missile is accelerated forward by a velocity change $\overrightarrow{\Delta v}$, while the ejected mass Δm moves with an exhaust velocity $\vec{v_e}$.

(Mass ejected backward with exhaust velocity)

$$v_{\Delta m} = v - v_e$$

((Remaining missile moves forward with increased velocity)

$$v_{m-\Delta m} = v + \Delta v$$

By Conservation of Momentum:

$$\Delta v = \Delta m\left(v - v_e\right) + \left(m - \Delta m\right)\left(v + \Delta v\right)$$

Expanding and simplifying:

$$mv = \Delta m v - \Delta m v_e + mv + m\Delta v - v\Delta m - \Delta m \Delta v$$

Since, $\Delta m \Delta v$ is very small, it can be neglected:

$$m\Delta v - \Delta m v_e = \Delta m \Delta v \approx 0$$

So we get: $m\Delta v = \Delta m v_e$

Alternatively, $\mu = \dfrac{\Delta m}{\Delta t}$ is the **rate of mass ejection** (mass flow rate).

Then: $\Delta m = \mu \Delta t$

Substitute into the momentum relation:

$$m\Delta v = \mu \Delta t \times v_e$$

Divide both sides by Δt

$$\frac{\Delta v}{\Delta t} = \frac{\mu \times v_e}{m} = a$$

Since $F = ma$

we get: $F = m \times \dfrac{\mu \times v_e}{m}$

Therefore: $F = \mu \times v_e$

Hence, $F = \mu \times v_e$

Where, F is the **thrust force** measured in (N).

$\mu = dm/dt$ is the **mass ejection rate** measured in $(\mathrm{kg/s})$.

v_e is the **exhaust velocity** measured in $(\mathrm{m/s})$.

Δm is the **change in mass** measured in (kg).

Δt is the **time interval** measured in (s).

v, Δv is the **velocity / velocity change** measured in $(\mathrm{m/s})$.

a is the **acceleration** measured in $(\mathrm{m/s^2})$.

m is the **mass of the missile** measured in (kg).

p is the **momentum measured** in $(\mathrm{kg.\ m/s})$.

Chapter1 Mechanics
Lesson5 Conservation of Energy

I. Kinetic Energy and the Work–Kinetic Energy Theorem

i. Kinetic Energy

An object with mass m moving at velocity v possesses **kinetic energy** given by the formula:

$$K = \frac{1}{2}mv^2$$

Where, K is the **kinetic energy** measured in (J).

m is the **mass of the object** measured in (kg).

v is the **velocity of the object** measured in $(\mathrm{m/s})$.

ii. Work–Kinetic Energy Theorem

Assume an object of mass m is acted upon by a **constant net force** F, which is **in the same direction** as the displacement d.

The work done by the net force is:

$$W = Fd \quad (1)$$

From the kinematic equation (independent of time), we have:

$$v_2^2 - v_1^2 = 2ad$$

Solving for d: $$d = \frac{v_2^2 - v_1^2}{2a} \quad (2)$$

Now substitute this value of d into equation (1):

$$W = F \cdot \frac{v_2^2 - v_1^2}{2a}$$

Since $F = ma$, substitute into the equation:

$$W = ma \cdot \frac{v_2^2 - v_1^2}{2a}$$

$$W = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

Hence, $W = \Delta K = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$. This is the **Work–Kinetic Energy Theorem**, which states: The net work done on an object is equal to the change in its kinetic energy.

Where, ΔK is the **change in kinetic energy** measured in (J).

W is the **work done on an object** measured in (J).

m is the **mass of the object** measured in (kg).

v_1 is the **initial velocity of the object** measured in $(\mathrm{m/s})$.

v_2 is the **final velocity of the object** measured in $(\mathrm{m/s})$.

II. Potential Energy

The **gravitational potential energy** of an object is given by:

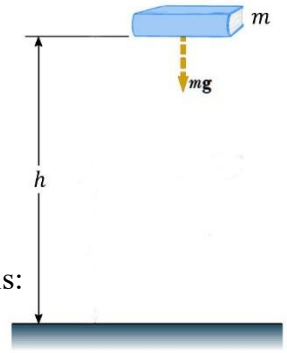
$$U_g = mgh$$

Where, U_g is the **gravitational potential energy** measured in (J).

m is the **mass of the object** measured in (kg).

g is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

h is the **height above the reference level** measured in (m).



- **Work Done by Gravitational Force**

When an object moves vertically by a height h, the work done by the **force of gravity** is:

$$W_g = -\Delta U$$

Where, ΔU is the **change in gravitational potential energy** measured in (J).

W_g is the **work done by the weight** measured in (J).

The negative sign indicates that gravity does **positive work** when an object moves **downward** (potential energy decreases) and **negative work** when moving **upward** (potential energy increases).

- **Characteristic of Gravitational Work**

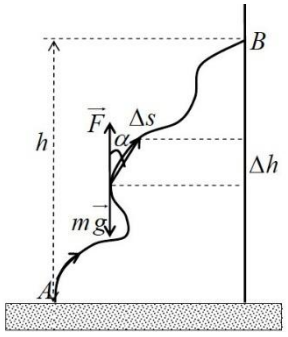
The **work done by gravity** depends **only on the change in height**, not on the path taken.

Therefore, if an object moves from point A to point B vertically:

$$W_{AB} = mg\left(h_A - h_B\right) = -\Delta U$$

Or if you define h as final minus initial:

$$W_g = mg\left(h_f - h_i\right) = -\Delta U \Rightarrow W_g = -\Delta U$$



Where, ΔU is the **change in gravitational potential energy** measured in (J).

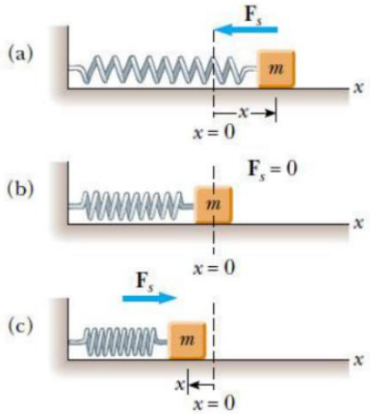
W_g is the **work done by the weight** measured in (J).

h is the **height above the reference level** measured in (m).

III. Elastic Potential Energy

Consider a spring: Whenever it is compressed or stretched, it has potential energy because it does work on the object.

In order for a spring to be stretched or compressed by a distance x, a force proportional to x is needed $F = kx$. The spring also exerts a restoring force in the opposite direction $F = -kx$. The work done by the compressing force is $W = Fx$ and the stretching (or compressing) force increases gradually from 0 to kx, we take the **average force** as $F = \frac{1}{2}kx$.



Hence, the work done to compress or stretch the spring is $W = \left(\frac{1}{2}kx\right)x = \frac{1}{2}kx^2$

Thus, elastic potential energy stored in a spring is $U_s = \frac{1}{2}kx^2$

Where, U_s is the **elastic (spring) potential energy** measured in (J).

k is the **spring constant** measured in $(\mathrm{N/m})$.

x is the **compression or extension from equilibrium** measured in (m).

IV. Conservation of Mechanical Energy

Mechanical energy of an object or a system is the sum of its **kinetic energy** and **potential energy**.

This can be written as: $E = K + U$

Where, E is the **total mechanical energy** measured in (J).

K is the **kinetic energy** measured in (J).

U is the **potential energy** measured in (J).

Conservation of Mechanical Energy

In a system where only conservative forces act (like gravity or spring force), the mechanical energy of the system is conserved.

This means: $E = K + U = \text{constant}$

Or equivalently: $E_i = E_f$

Which expands to: $K_i + U_i = K_f + U_f$

So, the total mechanical energy remains unchanged: $E = K + U = \text{constant}$

V. Application of Conservation of Energy

i. Free Fall Motion

When an object is in free fall, mechanical energy is conserved (assuming no air resistance).

From conservation of energy:

$$K_i + U_i = K_f + U_f$$

$$\frac{1}{2}mv_0^2 + mgh = \frac{1}{2}mv_f^2 + mgy$$

Canceling out mass m and rearranging:

$$v_f^2 = v_0^2 + 2g\left(h - y\right)$$

So, the final velocity is $v_f = \sqrt{v_0^2 + 2g\left(h - y\right)}$

Where, v_f is the **final velocity** measured in $(\mathrm{m/s})$.

v_0 is the **initial velocity** measured in $(\mathrm{m/s})$.

$g = 9.8$ is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

h is the **initial height** measured in (m).

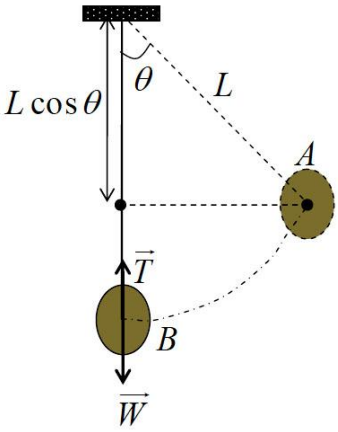
y is the **final height** measured in (m).

ii. Pendulum Motion

Velocity of a Pendulum at Point B

Take the **potential energy reference** at the lowest point O.

∗ At **point A** (the highest position):



- The kinetic energy is: $K_A = \frac{1}{2}mv_A^2 = 0 \quad \left(\text{since}, v_A = 0\right)$
- The potential energy is $U_g = -mgL\cos\theta$

∗ At **point B** (the lowest point):

- The kinetic energy is: $K_B = \frac{1}{2}mv_B^2$
- The potential energy is $U_g = -mgL$

From the **conservation of mechanical energy**: $K_A + U_A = K_B + U_B$

Substitute the values: $0 - mgL\cos\theta = \frac{1}{2}mv_B^2 - mgL$

Rearranging: $mgL\left(1-\cos\theta\right) = \frac{1}{2}mv_B^2$

Solving for v_B: $v_B = \sqrt{2gL\left(1-\cos\theta\right)}$

Hence, the velocity at point B is

$$v_B = \sqrt{2gL\left(1-\cos\theta\right)}$$

∗ **General Case: Instantaneous Velocity at Any Position**

At any point along the swing:

$$v = \sqrt{2gL\left(\cos\alpha - \cos\theta\right)}$$

Velocity at the Lowest Point (Equilibrium Position)

This occurs when $\theta = 0$, so:

$$v_{\max} = \sqrt{2gL\left(1-\cos\alpha\right)}$$

∗ **Tension in the String at Point B**

Use the dynamics of **uniform circular motion** at the lowest point:

Net force toward the center:

$$\Sigma F = T - W = ma = \frac{mv_B^2}{L}$$

Substitute $v_B^2 = 2gL\left(1-\cos\theta\right)$

$$T - mg = \frac{m\left[2gL\left(1-\cos\theta\right)\right]}{L}$$

$$T = mg + 2mg\left(1-\cos\theta\right)$$

Hence, tension in the string at point B is $T = mg\left(3-2\cos\theta\right)$

Where, v_B is the **velocity at the lowest point** measured in (m/s).

v_A is the **velocity at the highest point** measured in $(\mathrm{m/s})$.

$g = 9.8$ is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

L is the **length of the string of a pendulum** measured in (m).

θ is the **angle** measured in degrees $(°)$.

α is the **angle at any point along the swing** measured in degrees$(°)$.

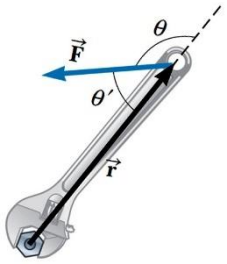
T is the **tension in the string** measured in (N).

m is the **mass of the sphere** measured (kg).

Chapter1 Mechanics
Lesson6 Equilibrium and Elasticity

I. Torque and Vector Quantity

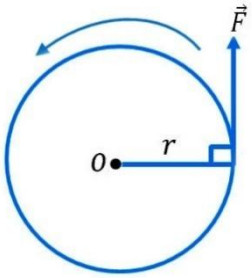
i. Torque



Assume you want to tighten a nut by using a wrench in the **clockwise direction**. If you apply the force **close to the nut**, it is **harder to turn**. However, the nut can be turned more **easily** if you apply the force **farther away** from the nut along the wrench. This is similar to **pushing a door** it is easier to push the door open if you apply force **far from the hinge (pivot point)** than near it.

ii. Definition

Torque is the turning effect of a force applied **tangentially** to a circle of radius ***r***, with the center located on the **axis of rotation**.



It is determined by the formula: $\tau = Fr$

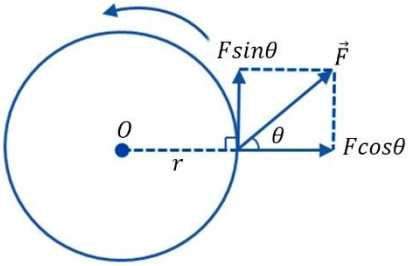
Where, τ is the **torque** measured in $(\mathrm{N.m})$.

F is the **force applied tangentially** measured in (N).

r is the **distance from the axis of rotation** measured in (m).

iii. Moment of Force

If a force is applied at an **angle θ** to a line that passes through the **center of a circle (axis of rotation)**, the force can be resolved into **two components**, such a **tangential component** $F\cos\theta$ and a **normal (perpendicular) component** $F\sin\theta$. Only the **normal component** $F\sin\theta$ contributes to the **rotational motion** it causes the object to **turn**.



The tangential component $F\cos\theta$ does **not** produce any rotation about the axis.

By definition, the **moment of rotational force (torque)** is given by:

$$\vec{\tau} = \vec{F} \times \vec{r}$$

Hence, the magnitude of moment of force is $\tau = Fr\sin\theta$

Sign Convention for Moment of Force (Torque):

- If $\tau > 0$, the moment causes **anticlockwise (counterclockwise)** rotation.
- If $\tau < 0$, the moment causes **clockwise** rotation.

Where, τ is the **torque** measured in $(\mathrm{N.m})$.

F is the **force applied tangentially** measured in (N).

r is the **distance from the axis of rotation** measured in (m).

θ is the **angle between force and the lever arm** measured in degrees $(°)$.

II. Equilibrium Conditions

For an object to be in **complete equilibrium**, two conditions must be satisfied:

- **First Condition: Translational Equilibrium**

If the **net force** acting on an object is **zero**, then the object does not accelerate in any direction. This means the **sum of all forces** in both horizontal and vertical directions must be zero.

$$\Sigma F_x = 0 \text{ and } \Sigma F_y = 0$$

This ensures the object remains at rest or moves with constant velocity (no linear acceleration).

- **Second Condition: Rotational Equilibrium**

If the object is in equilibrium, the **net torque** (moment of force) acting on it must also be **zero**. This prevents the object from rotating or changing its rotational motion.

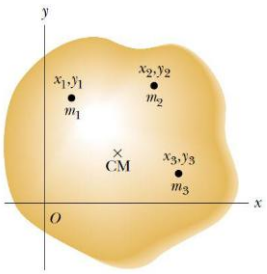
$$\Sigma \tau = 0$$

These two conditions together ensure that the object is in **static or dynamic equilibrium**, meaning it is stable and not accelerating either linearly or rotationally.

III. Center of mass or Barycenter Theorem

i. Definition: The **center of mass** is the point at which the system is in **equilibrium** under a uniform gravitational field. In general, the center of mass of a system is determined by the distribution of **multiple masses**.

- In the case the object is located on a horizontal axis:



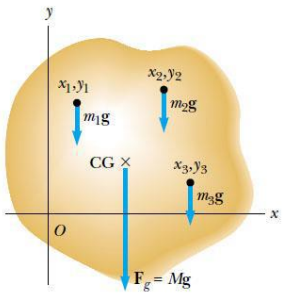
$$x_{cm} = \frac{m_1x_1 + m_2x_2 + \ldots + m_nx_n}{m_1 + m_2 + m_2} = \frac{\sum_{i=1}^{n} m_ix_i}{\sum_{i=1}^{n} m_i}$$

Where, x_{cm} is the ***x*-coordinate of the center of mass** measured in (m).

m_i is the **mass of each object** measured in (kg).

x_i is the **position of each object along the x-axis** measured in (m)

- In the case the object is located on a vertical axis:



$$y_{cm} = \frac{m_1y_1 + m_2y_2 + \ldots + m_ny_n}{m_1 + m_2 + m_2} = \frac{\sum_{i=1}^{n} m_iy_i}{\sum_{i=1}^{n} m_i}$$

Where, y_{cm} is the ***y*-coordinate of the center of mass** measured in (m).

m_i is the **mass of each object** measured in (kg).

y_i is the **position of each object along the *y*-axis** measured in (m).

IV. Elasticity Characteristics of a Rigid Body

To study the statics of objects under the action of forces, follow these steps:

- Draw the diagram of the system.
- **Isolate the object** to be analyzed. Draw a **free-body diagram** showing all the forces acting on the object. Test the direction of each force if your choice is incorrect, it will lead to a **negative value** in calculation, indicating that your assumed direction was opposite to the actual one.
- **Create a coordinate system** and find the **net force** along the two axes. Then apply the **first condition of equilibrium**:

$$\Sigma F_x = 0 \text{ and } \Sigma F_y = 0$$

- **Choose a principal axis** for calculating torque acting on the object. Remember, the origin for torque calculation can be **any point**, but it is best to choose a point that **simplifies the calculation**. Then apply the **second condition of equilibrium**:

$$\Sigma \tau = 0$$

- Formula for Modulus of Elasticity: $\text{Modulus of Elasticity} = \dfrac{\text{Stress}}{\text{strain}}$

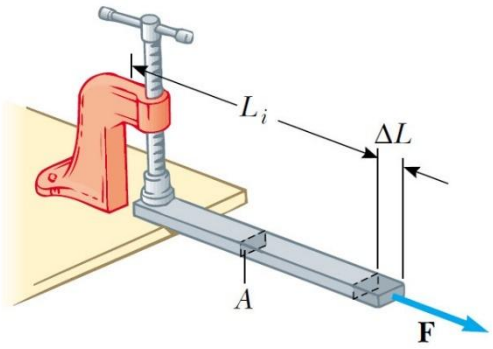
Stress is the **external force acting on an object per unit area**.
Strain is the **measure of deformation**, defined as the **ratio** of change in dimension to the original dimension (it has **no units**).

∗ Types of Deformation:
Young's Modulus: Measures the resistance of an object to **change in length** under tension or compression.
Shear Modulus: Measures the resistance of an object to **shearing deformation**, where one-layer slides over another.
Bulk Modulus: Measures the resistance of a **solid or fluid** to **change in volume** under uniform pressure.

i. Young's Modulus: Elasticity in Longitudinal Direction

Young's Modulus is a measure of the **elasticity of a material** when it is subjected to a force **along its length** (longitudinal direction), such as in stretching or compressing a rod or wire.

It is defined by the formula: $$Y = \frac{F/A}{\Delta L/L_i}$$



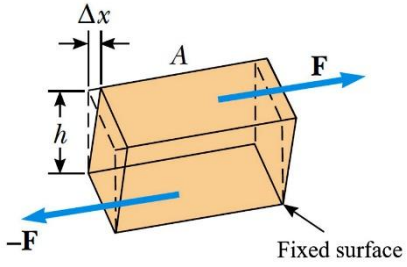
Where,
Y is the **Young's modulus** measured in (Pa).
F is the **applied force** measured in (N).
A is the **cross-sectional area** measured in (m^2).
ΔL is the **change in length** measured in (m).
L_i is the **original length** measured in (m).

Notes:
- Young's modulus describes how **stiff** or **stretch-resistant** a material is.
- Higher Y means the material is **less elastic** and more **rigid**.

ii. Shear Modulus: Elasticity in Shape Deformation

Shear Modulus (also called the **modulus of rigidity**) measures a material's ability to **resist shape deformation** when subjected to **shearing forces** forces that cause one layer of the material to slide over another.

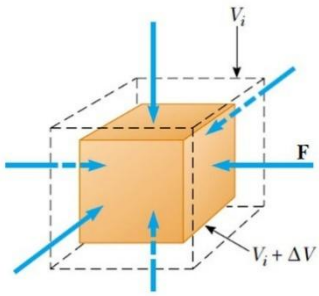
It is defined by the formula: $$S = \frac{F/A}{\Delta x/h}$$



Where,
S is the **Shear modulus** measured in (Pa).
F is the **tangential force applied** measured in (N).
A is the **area over which the force is applied** measured in (m^2).
Δx is the **horizontal displacement** measured in (m).
h is the **height or thickness of the object measured** in (m).

Notes:
- Shear modulus describes how a material resists **distortion of shape** (without changing volume).
- Larger values of S mean greater **resistance to shear deformation**.

iii. Bulk Modulus: Elasticity in Volume Deformation



Bulk Modulus measures a material's **resistance to uniform compression**, that is, how much it resists a **change in volume** when subjected to an external pressure applied from all directions (such as in fluids or solids under pressure).

It is defined by the formula: $B = -\dfrac{\Delta P}{\Delta V / V_i} = -\dfrac{\Delta F / A}{\Delta V / V_i}$

Where, B is the **Bulk modulus** measured in (Pa).

ΔP is the **change in pressure** measured in (Pa).

ΔF is the **change in applied force** measured in (N).

A is the **area over which the force is applied** measured in (m^2).

ΔV is the **change in volume** measured in (m^3).

V_i is the **original volume measured** in (m^3).

Notes:

- Bulk modulus quantifies the material's resistance to **volume compression**.
- Higher B means the material or fluid is **less compressible**.

I. Definition: **Fluid** is a type of material that **can flow** and **takes the shape of its container**. Examples of fluids include **liquids** and **gases**.

i. Properties and Types of Fluids

- A **fluid takes the shape of the container** that holds it.
- Fluids can be classified in **two different ways** such as **compressible fluids** and **incompressible fluids**. However, fluids are also classified into two other types: **ideal fluid** and **real fluid**.
- **Ideal Fluid:** An ideal fluid is a theoretical fluid with no internal resistance (no viscosity). Its molecules do not stick to each other and can slide freely past one another. Example: *Pure water (in theory)* is often treated as an ideal fluid in calculations.
- **A real fluid** is what we encounter in real life. It has **viscosity**, meaning the molecules **stick to each other** and **resist motion**. Examples include: oil, honey, air, and even water under real conditions.

II. Volume Flow Rate

i. Definition: **Volume flow rate** is the amount of fluid that flows across a **tube** or **pipe per unit time**.

Assume that during a time interval Δt, a volume ΔV of fluid flows through the tube.

Then, the volume flow rate is given by:

$$Q_V = \frac{\Delta V}{\Delta t}$$

Where, Q_V is the **volume flow rate** measured in $(\mathrm{m}^3/\mathrm{s})$.

ΔV is the **volume of fluid that flows** measured in (m^3).

Δt is the **time interval** measured in (s).

Suppose a fluid flows through a tube of **length** l and **cross-sectional area** A, with a **constant velocity** v, over a **time interval** t.

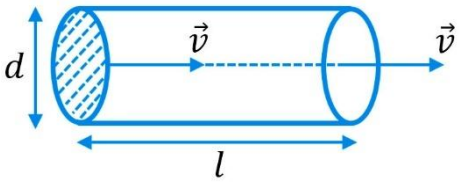
By definition $Q_V = \frac{V}{t}$

Since, volume $V = Al$ and $l = vt$

This implies $V = Avt$

Substitute $Q_V = \frac{Avt}{t}$

Hence, $Q_V = Av$



Where, Q_V is the **volume flow rate** measured in $(\mathrm{m}^3/\mathrm{s})$.

A is the **cross-sectional area of the tube** measured in (m^2).

v is the **flow velocity** measured in (m/s).

III. Mass Flow Rate

i. Definition: **Mass flow rate** is the amount of **fluid mass** that flows through a **tube or pipe per unit time**.

If, during a time interval Δt, a mass Δm of fluid flows, then the **mass flow rate** is:

$$Q_m = \frac{\Delta m}{\Delta t}$$

***Relationship Between Volume Flow Rate and Mass Flow Rate:**

The density of a fluid is defined as:

$$\rho = \frac{\Delta m}{\Delta V}$$

If we express this in terms of flow rates:

$$\rho = \frac{Q_m}{Q_V}$$

Hence, mass flow rate is $Q_m = \rho Q_V$

Where, Q_m is the **mass flow rate** measured in $(\mathrm{kg/s})$.

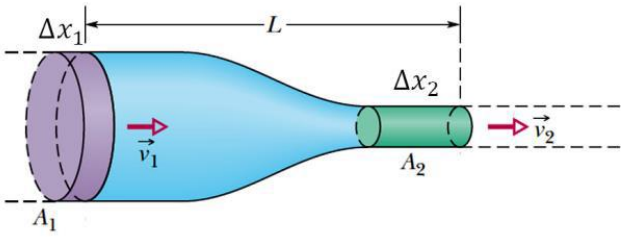
Q_V is the **volume flow rate** measured in $(\mathrm{m^3/s})$.

ρ is the **fluid density** measured in $(\mathrm{kg/m^3})$.

IV. Equation of Continuity

Suppose a tube has a **changing cross-sectional area**, and fluid with **density** ρ flows through it.

- At cross-sectional area A_1, the fluid flows with **velocity** $\vec{v}_1$ over a distance Δx_1 during a time interval t.
- At cross-sectional area A_2, the fluid flows with **velocity** $\vec{v}_2$ over a distance Δx_2 during the same time interval t.



- The **mass** of fluid flowing through area A_1 is:

$$\Delta m_1 = \rho_1 \cdot \Delta V_1$$

Since the **volume** of fluid that flows is:

$$\Delta V_1 = A_1 \cdot \Delta x_1$$

And $\Delta x_1 = v_1 \cdot t$, we get:

$$\Delta V_1 = A_1 v_1 t$$

Substituting this into the mass equation:

$$\Delta m_1 = \rho_1 A_1 v_1 t$$

- The **mass** of fluid flowing through area A_2 is:

$$\Delta m_2 = \rho_2 \cdot \Delta V_2$$

Since the **volume** of fluid that flows is:

$$\Delta V_2 = A_2 \cdot \Delta x_2$$

And $\Delta x_2 = v_2 \cdot t$, we get:

$$\Delta V_2 = A_2 v_2 t$$

Substituting this into the mass equation:

$$\Delta m_2 = \rho_2 A_2 v_2 t$$

By conservation of mass

$$\Delta m_1 = \Delta m_2$$

Since it is the **same fluid**, the **density** is constant: $\rho = \rho_1 = \rho_2$

Using the earlier result:

$$\Delta m_1 = \rho_1 A_1 v_1 t \text{ and } \Delta m_2 = \rho_2 A_2 v_2 t$$

So, $$\rho A_1 v_1 t = \rho A_2 v_2 t$$

Canceling ρ and t from both sides (since they are constant):

Hence, the equation of continuity is $A_1 v_1 = A_2 v_2$.

Note: This continuity equation, velocity is inversely proportional to cross-sectional area, assuming the fluid is incompressible and flow is steady.

Where, A_1 is the **cross-sectional area at section 1** measured in (m^2).

A_2 is the **cross-sectional area at section 2** measured in (m^2).

v_1 is the **fluid velocity at section 1** measured in $(\mathrm{m/s})$.

v_2 is the **fluid velocity at section 2** measured in $(\mathrm{m/s})$.

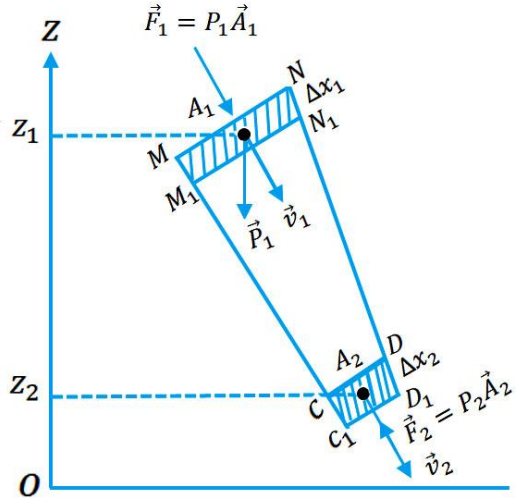
V. Bernoulli's Equation

We consider an inclined tube with varying cross-sectional area. A fluid of density ρ flows through it.

- At time t, at section A_1, the fluid flows with velocity $\vec{v}_1$ over a small distance Δx_1.
- At the same instant t, at section A_2, the fluid flows with velocity $\vec{v}_2$ over a small distance Δx_2

∗Apply the work-energy theorem

$$\Delta K = W(F_1) + W(F_2) + W(P)$$



∗Work done by pressure forces

- At section A_1: $W(F_1) = F_1 \Delta x = P_1 A_1 \Delta x = P_1 \Delta V$
- At section A_2: $W(F_2) = -F_2 \Delta x = -P_2 A_2 \Delta x_2 = -P_2 \Delta V$

∗Work done by gravity

- The work done by gravity is

$$W(P) = \Delta m g (z_1 - z_2)$$

But since $\Delta m = \rho \Delta V$, this becomes:

$$W(P) = \rho \Delta V g (z_1 - z_2)$$

∗ Change in kinetic energy

$$\Delta K = \frac{1}{2}\Delta m v_2^2 - \frac{1}{2}\Delta m v_1^2 = \frac{1}{2}\rho \Delta V v_2^2 - \frac{1}{2}\rho \Delta V v_1^2$$

∗ Combine all terms

$$\frac{1}{2}\rho \Delta V v_2^2 - \frac{1}{2}\rho \Delta V v_1^2 = P_1 \Delta V - P_2 \Delta V + \rho \Delta V g (z_1 - z_2)$$

Divide both sides by ΔV

$$\frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2 = P_1 - P_2 + \rho g\left(z_1 - z_2\right)$$

Rearrange: $$P_1 + \frac{1}{2}\rho v_1^2 + \rho g z_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g z_2$$

Hence, Bernoulli's equation is $P + \frac{1}{2}\rho v^2 + \rho g z = \text{constant}$.

Where, P is the **pressure** measured in (Pa).

ρ is the **density of the fluid** measured in (kg/m^3).

v is the **velocity of the fluid** measured in (m/s).

$g = 9.8$ is the **acceleration due to gravity** measured in (m/s^2).

z is the **height (elevation) from reference level** measured in (m).

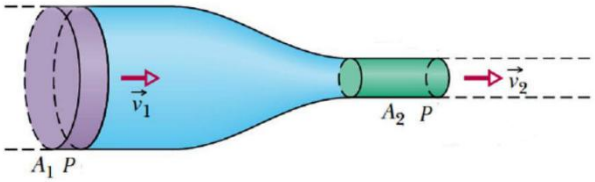
*** Case: Horizontal Tube (Constant Height)**

In a horizontal tube, the elevation at both ends is the same:

$$z_1 = z_2$$

So, Bernoulli's equation simplifies to:

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$



Hence, Bernoulli's equation simplifies to $P + \frac{1}{2}\rho v^2 = \text{constant}$.

VI. Torricelli's Equation

Consider a container filled with a fluid of density ρ, and the height of the fluid column is h. The container is open to the atmosphere at the top. At the base of the container, there is a small hole with a cross-sectional area much smaller than the cross-sectional area of the container.

We apply Bernoulli's equation between point 1 and 2:

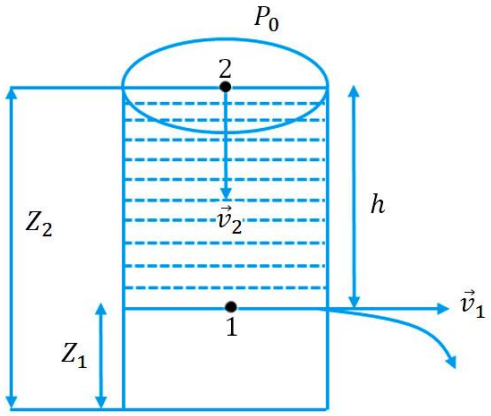
$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g z_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g z_2$$

Let's define $P_1 = P_2 = P_0$ (both open to atmosphere)

This implies $$\frac{1}{2}\rho v_1^2 + \rho g z_1 = \frac{1}{2}\rho v_2^2 + \rho g z_2$$

$$\frac{1}{2}\left(v_1^2 - v_2^2\right) = g\left(z_2 - z_1\right)$$

$$\frac{1}{2}\left(v_1^2 - v_2^2\right) = gh$$



We apply the **continuity equation** between point (1) and (2):

$$A_1 v_1 = A_2 v_2 \Rightarrow \frac{A_1}{A_2} = \frac{v_2}{v_1}$$

Since, $A_2 \gg A_1$, then $\frac{A_2}{A_1} \approx 0$, so : $\frac{v_2}{v_1} = 0 \Rightarrow v_1 = 0$

Plug into previous equation:

$$v_2^2 + 0 = 2gh$$

Hence, exit speed of the fluid is $v_2 = \sqrt{2gh}$

Where, v_2 is the **exit speed of the fluid** measured in $(\mathrm{m/s})$.

$g = 9.8$ is the **acceleration due to gravity** measured in $(\mathrm{m/s^2})$.

h is the **height of the fluid above the hole** measured in (m).

The velocity of fluid flow is not constant; it is **proportional to the square root of the height** of the fluid above the opening. This relationship is known as **Torricelli's Theorem**. It states that the **speed of a fluid flowing out of a small hole at the bottom of a container** does **not depend on the density** of the fluid. The speed is the same as the speed an object would have if it were in **free fall from the same height *h***, assuming it starts from rest.

We know that the **pressure at depth *h*** in a fluid is given by:

$$P = \rho gh$$

Solving for h: $h = \frac{P}{\rho g}$

Substitute this into **Torricelli's equation**:

$$v_2 = \sqrt{2gh} \Leftrightarrow v_2^2 = 2gh$$

Now replace h with $\frac{P}{\rho g}$:

$$v_2^2 = 2g\frac{P}{\rho g} = \frac{2P}{\rho}$$

Hence, velocity of the fluid as it exits is $v_2 = \sqrt{\frac{2P}{\rho}}$

Where, v_2 is the **exit speed of the fluid** measured in $(\mathrm{m/s})$.

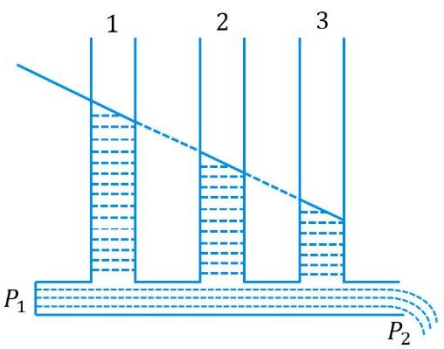
P is the **pressure due to the fluid column** measured in (Pa).

ρ is the **density of the fluid** measured in $(\mathrm{kg/m^3})$.

VII. Pressure Loss (Head Loss)

They install a tube or pipe in series with a common base.

- When a **real fluid** is placed into the tube, it is observed that the **molecules interact with each other**, causing them to **stick together** and generate **friction**. Due to this **friction**, part of the fluid's energy is lost and **transformed into heat**.
- It is observed that the **fluid levels in three vertical tubes** connected along the pipe **decreases linearly**. This observation allows them to determine that the **pressure drops linearly** from the first tube to the last. This **decrease in "losing charge"** or **pressure loss**.



It is denoted by: $\Delta P = P_2 - P_1$

Where, ΔP is the **pressure loss (or loss of head)** measured in (Pa).

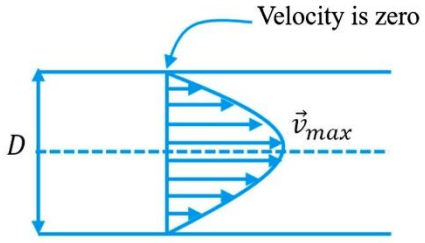
P_1 is the **pressure at the starting point** measured in (Pa).

P_2 is the **pressure at the ending point** measured in (Pa).

VIII. Poiseuille's Law

i. Poiseuille's Observation

Consider a canal, pipe, or lake where fluid is flowing:



- For an **ideal fluid** (with no internal friction), the **velocity at all points across the cross-section** of the canal is **almost equal**.
- For a **real fluid**, the **velocity is not uniform** across the cross-section of a canal.
- **At the walls** (sides) of the canal or pipe, the **fluid velocity is zero** due to adhesion and friction with the boundary (no-slip condition). **Moving inward** from the walls, the **fluid velocity increases**.
- **At the centerline (axis)** of the canal, the **velocity reaches its maximum** value, v_{max}.

The **average velocity** of the fluid is given by:

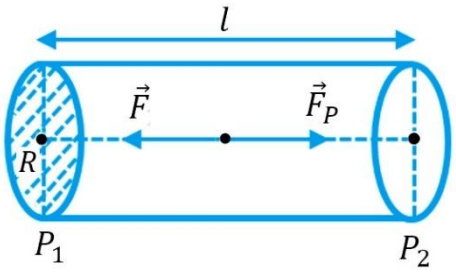
$$v_{av} = \frac{v_0 + v_{max}}{2}$$

Where, v_{av} is the **average fluid speed across the cross-section** measured in $(\mathrm{m/s})$.

$v_0 = 0$ is the **velocity at the wall** measured in $(\mathrm{m/s})$.

v_{max} is the **maximum velocity at the center** measured in $(\mathrm{m/s})$.

Hence, the **average velocity** of the fluid is $v_{av} = \frac{v_{max}}{2}$



- When we examine the **velocity distribution across the diameter** of a tube (pipe or canal), it forms a **parabolic curve**. The **vertex** of this parabola corresponds to the **maximum fluid velocity**, located at the center (axis) of the tube. This **variation in velocity** occurs due to **internal friction** (viscosity) between adjacent fluid layers.
- At a point inside the fluid, there are two forces acting in opposite directions $\vec{F}_P$ due to the pressure difference between the two ends of the tube, $\Delta P = P_2 - P_1$, pushing the fluid forward and $\vec{F}$ due to internal friction (viscous drag), opposing the motion. This viscous force is proportional the **length** of the tube l, the **fluid's velocity** v, and the **fluid's viscosity** η and is **inversely proportional** to the **cross-sectional area** A of the tube.

From balancing the forces, we get the **Poiseuille equation**:

$$P_2 - P_1 = \frac{8\pi\eta v l}{A}$$

Since volume flow rate is defined as:

$$Q_V = Av \quad \left(\text{where } A = \pi R^2\right)$$

Substitute for v: $v = \dfrac{(P_1 - P_2)A}{8\pi\eta l}$

Then the volume flow rate becomes:

$$Q_V = A \cdot \frac{(P_1 - P_2)A}{8\pi\eta l} = \frac{(P_1 - P_2)A^2}{8\pi\eta l}$$

Substitute $A = \pi R^2$:

$$Q_V = \frac{(P_1 - P_2)(\pi R^2)^2}{8\pi\eta l} = \frac{(P_1 - P_2)\pi R^4}{8\eta l}$$

Hence, the volume flow rate is $Q_V = \dfrac{(P_1 - P_2)\pi R^4}{8\eta l}$

Where,

Q_V is the **volume flow rate** measured in $(\mathrm{m}^3/\mathrm{s})$.

$P_1 - P_2$ is the **pressure difference across the tube** measured in (Pa).

R is the **radius of the tube** measured in (m).

η is the **dynamic viscosity of the fluid** measured in $(\mathrm{Pa.s})$.

l is the **length of the tube** measured in (m).

Chapter2 Thermodynamics
Lesson1 Temperature and Heat

Temperature of a substance is a measure of the average kinetic energy of its molecules.

I. First Law of Thermodynamic

The Zeroth Law of Thermodynamics states that if two objects, A and B, are each in thermal equilibrium with a third object, C, then A and B are also in thermal equilibrium with each other.

We get, $T_A = T_B$

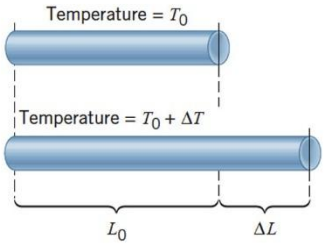
And, $T_B = T_C$

Hence, $T_A = T_C$

ii. Solid Expansion

A. Linear Expansion

Linear expansion $\Delta L = \alpha L_0 \Delta T$



Where, L_0 is the original length of the solid measured in (m).

α is the coefficient of linear expansion measured in $(1/°\mathrm{C} \text{ or } 1/\mathrm{K})$.

ΔT is the temperature change measured in $(°\mathrm{C})$.

ΔL is the change in length of the solid measured in (m).

B. Area Expansion

Total area $(l + \Delta l)(w + \Delta w) = A + \Delta A$

$$(l + \alpha L \Delta T)(w + \alpha w_0 \Delta T) = A + \Delta A$$

$$lw + l\alpha w\Delta T + W\alpha l\Delta T + lw\alpha^2\Delta T^2 = A + \Delta A$$

$$A + A\alpha\Delta T + \alpha A\Delta T + A\alpha^2\Delta T + A\alpha^2\Delta T = A + \Delta A$$

$$\Delta A = A\left[2\alpha\Delta T + (\alpha\Delta T)^2\right]$$

Since, $\alpha = 10^{-6} \Rightarrow (\alpha\Delta T)^2$ small compared to $2\alpha\Delta T$

Hence, $\Delta A = 2\alpha A_0 \Delta T = \eta A_0 \Delta T$

Where, A_0 is the original area of the solid measured in (m^2).

η is the coefficient of area expansion measured in $(1/°\mathrm{C} \text{ or } 1/\mathrm{K})$.

ΔT is the temperature change measured in $(°\mathrm{C})$.

ΔA is the change in area of the solid measured in (m).

C. Volume Expansion

Volume expansion $\Delta V = 3\alpha V_0 \Delta T = \beta V_0 \Delta T$

Where, V_0 is the original volume of the solid measured in (m^3).

β is the coefficient of volume expansion measured in $(1/°\mathrm{C} \text{ or } 1/\mathrm{K})$.

ΔT is the temperature change measured in $(°\mathrm{C})$.

ΔV is the change in volume of the solid measured in (m^3).

iii. Liquid Expansion

Liquid expansion $\Delta V = V - V_0 = \beta V_0 \Delta T$

Where, V_0 is the original volume of the solid measured in (m^3).

β is the coefficient of volume expansion measured in $(1/°\mathrm{C} \text{ or } 1/\mathrm{K})$.

ΔT is the temperature change measured in $(°\mathrm{C})$.

ΔV is the change in volume of the solid measured in (m^3).

And, $V = V_0\left(1 + \beta T\right)$

Where, V_0 is the volume of liquid at $0\,°\mathrm{C}$ measured in (m^3).

iii. Gases Expansion

Gases expansion $\Delta V = \gamma V_0 \Delta T$

Hence, $\gamma = \dfrac{\Delta V}{V_0} \cdot \dfrac{1}{\Delta T}$

Where, V_0 is the original volume of gases measured in (m^3).

γ is the coefficient of volume expansion measured in $(1/°\mathrm{C} \text{ or } 1/\mathrm{K})$.

ΔT is the temperature change measured in $(°\mathrm{C})$.

ΔV is the change in volume of gases measured in (m^3).

And, $V = V_0\left(1 + \gamma T\right)$

Where, $\gamma = \dfrac{1}{273\,°\mathrm{C}}$ is the under constant pressure.

iv. Elasticity Deformation

Elastic deformation (stress) is directly proportional to the applied external force that causes a change in shape.

This can be written as $\sigma = \dfrac{F}{A}$

Where, σ is stress measured in (Pa).

F is the force applied measured in (N).

A is cross-sectional area over which the force is applied measured in (m^2).

Strain is the ratio of the change in length to the original length of a material.

This can be written as $\tau = \dfrac{|\Delta L|}{L} = \dfrac{F}{A}$

Where, τ is shear stress measured in (Pa).

L is the original length of the material measured in (m).

ΔL is the change in length (elongation or compression) of the material measured in (m).

F is the force applied measured in (N).

A is cross-sectional area over which the force is applied measured in (m^2).

$\Delta L/L$ is strain.

Young's Modulus is a measure of a material's **stiffness** or **resistance** to being stretched or compressed.

This can be written as $Y = \dfrac{F/A}{\Delta L/L} = \dfrac{\sigma}{\tau}$

Where, Y is Young's Modulus measured in (Pa).

$\sigma = \text{stress} = \dfrac{F}{A}$ is force per unit area measured in (Pa).

$\tau = \text{strain} = \dfrac{\Delta L}{L}$ (relative change in length).

Note: A **high** Young's modulus = the material is **very stiff** (e.g., steel)
A **low** Young's modulus = the material is **flexible** or **elastic** (e.g., rubber)

II. Heat Energy

This can be written as $Q = mc\Delta t = C\Delta t$

Where, Q is heat energy measured in (J).

m is mass of matters measured in (kg).

Δt is the change in temperature measured in $(^\circ\mathrm{C})$.

c is specific heat capacity measured in $(\mathrm{J/kg.^\circ C})$.

C is heat capacity measured in $(\mathrm{J/kg})$.

Noted: $1\,\mathrm{J} = 0.24\,\mathrm{cal} \Rightarrow 1\,\mathrm{cal} = 2.4\,\mathrm{J}$

III. Laten Heat of Fusion and Evaporation

• **Latent Heat of Fusion** is the amount of heat energy required to change a substance from a solid to a liquid (melting) or from a liquid to a solid (freezing) without changing its temperature.

This can be written as $Q = mL_f$

Where, Q is heat energy measured in (J).

m is mass of matters measured in (kg).

L_f is specific latent heat of fusion measured in $(\mathrm{J/kg})$.

• **Latent Heat of Vaporization** is the amount of heat energy required to change a substance from a liquid to a gas (vaporization or boiling) or from a gas to a liquid (condensation) without changing its temperature.

This can be written as $Q = mL_v$

Where, Q is heat energy measured in (J).

m is mass of matters measured in (kg).

L_v is specific latent heat of vaporization measured in $(\mathrm{J/kg})$.

IV. Heat Transfer

Heat transfer is the movement of thermal energy from a region of higher temperature to a region of lower temperature.

Rate of heat transfer through a surface $H = \frac{\Delta Q}{\Delta t} = -kA\frac{\Delta T}{\Delta x}$

Where, $H = \frac{\Delta Q}{\Delta t}$ is rate of heat transfer measured in $(\mathrm{J/s})$.

A is cross-sectional area of the plate measured in $(\mathrm{m^2})$.

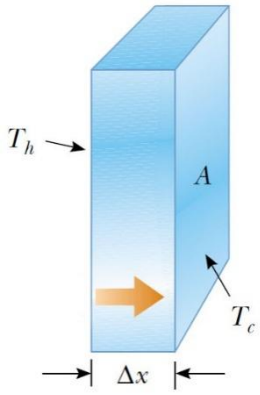
Δt is time taken for the heat to transfer measured in (s).

ΔT is temperature difference across the plate measured in $(°\mathrm{C})$.

Δx is thickness of the plate measured in (m).

ΔQ is amount of heat transferred measured in (J).

k is thermal conductivity of the material measured in $(\mathrm{W/m.K})$.



V. Radiation

Radiation is the transfer of heat energy through **electromagnetic waves**, typically **infrared radiation**, **without the need for a medium** (no solid, liquid, or gas is required).

Radiation $R = \frac{P}{A} = e\sigma T^4$

Where, R is Radiant energy flux or radiant heat intensity measured in $(\mathrm{W/m^2})$.

A is cross-sectional area of the plate measured in $(\mathrm{m^2})$.

P is Power radiated or total energy emitted per second measured in (W).

e is emissivity $(0 \le e \le 1)$.

T is absolute temperature measured in (K).

σ is Stefan–Boltzmann constant measured in $(\mathrm{W/m^2.K^4})$.

I. Ideal Gas equation

II. Boyle's Law (at Constant Temperature)

Boyle studied the relationship between pressure and volume of a fixed amount of gas at constant temperature. He stated that, for a given mass of gas at constant temperature, the product of its pressure and volume remains constant.

This can be written as $PV = P_1V_1 = P_2V_2 = \ldots = \text{constant}$

- If m is the mass and V_1 is volume under pressure P_1, then the density ρ_1 of the gas is given by

$$\rho_1 = \frac{m}{V_1} \Rightarrow V_1 = \frac{m}{\rho_1}$$

- If m is the mass and V_2 is volume under pressure P_2, then the density ρ_2 of the gas is given by

$$\rho_2 = \frac{m}{V_2} \Rightarrow V_2 = \frac{m}{\rho_2}$$

- Ratio $\dfrac{V_1}{V_2} = \dfrac{m/\rho_1}{m/\rho_2} \Leftrightarrow \dfrac{\rho_2}{\rho_1} = \dfrac{P_2}{P_1}$

Hence, $\dfrac{\rho_2}{\rho_1} = \dfrac{P_2}{P_1}$

However, P is the atmospheric pressure measured in (atm).

V is the volume of gas measured in (m^3).

We get, $P = P_0\left(1+\gamma t\right)$ and $V = V_0\left(1+\alpha t\right)$

Hence, $P = P_0\left(1+\gamma t\right)$

This shows that as temperature increases, pressure increases **linearly** (if volume is constant). This applies to **gases in sealed containers** where volume does not expand.

$$V = V_0\left(1+\alpha t\right)$$

This describes **thermal expansion** how a material's volume increases as temperature increases (common for liquids and solids).

Where, P is the atmospheric pressure at $\text{t}\ (°\text{C})$ measured in (atm).

P_0 is the atmospheric pressure at 0 °C measured in (atm).

t is the temperature measured in $(°\text{C})$.

γ is the coefficient of pressure expansion measured in $(1/°\text{C})$.

V is the volume of gas at $\text{t}\ (°\text{C})$ measured in (m^3).

V_0 is the volume of gas at 0 °C measured in (m^3).

α is the coefficient of volume expansion measured in $(1/°\text{C})$.

The coefficient of volume expansion of gas is $\alpha = \gamma = \dfrac{P-P_0}{P_0T} = \dfrac{V-V_0}{V_0T}$

ii. Charles's Law (at Constant Pressure)

Charles studied the behavior of gases during a temperature change by assuming that the pressure remains constant. He found that the volume of a gas changes directly with its temperature.

Charles's Law states that, for a fixed mass of gas at constant pressure, the volume of the gas is directly proportional to its absolute temperature (in kelvin).

This can be written as $\frac{V}{T} = \frac{V_1}{T_1} = \frac{V_2}{T_2} \ldots = \text{constant}$

Where, T is the temperature of gas measured in (K).

V is the volume of gas measured in (m^3).

iii. Gay-Lussac's Law (at Constant Volume)

Gay-Lussac studied the behavior of gases by assuming that the volume remains constant and that the pressure of a gas depends directly on its temperature.

Gay-Lussac's Law states that, for a fixed amount of gas at constant volume, the pressure of the gas is directly proportional to its absolute temperature.

This can be written as $\frac{P}{T} = \frac{P_1}{T_1} = \frac{P_2}{T_2} \ldots = \text{constant}$

Where, P is the pressure of gas measured in (atm).

V is the volume of gas measured in (m^3).

Gay-Lussac's Law states that, at constant volume, the pressure of a gas is directly proportional to its absolute temperature.

For combine the Gas Laws

- Boyle's Law (constant temperature):

$$P \propto \frac{1}{V} \Rightarrow PV = \text{constant (if T fixed)}$$

- Charles's Law (constant pressure):

$$V \propto T \Rightarrow \frac{V}{T} = \text{constant (if P fixed)}$$

- Gay-Lussac's Law (constant volume):

$$P \propto T \Rightarrow \frac{P}{T} = \text{constant (if V fixed)}$$

- From these, the general relationship for an ideal gas is: $PV \propto T$

Or introducing a constant nR: $PV = nRT$

Since, Number of moles n is related to the **mass** m of the gas and its **molar mass** M (also called molecular weight, often denoted):

$$n = \frac{m}{M}$$

This equation can be written as $PV = \frac{m}{M}RT$

Where, P is the pressure of gas measured in (Pa).

V is the volume of gas measured in (m^3).

M or μ is the molar mass or molecule mass of gas measured in $(\mathrm{kg/mol})$.

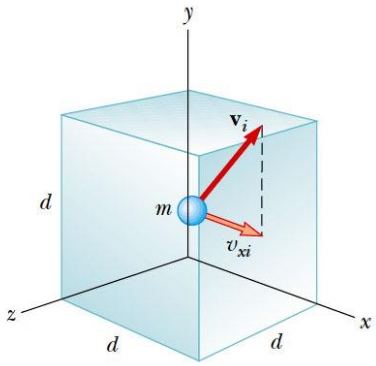
$R = 8.31\ \mathrm{J/mol.K}$ is the universal gas constant.

n is the number of moles of gas measured in (moles).

m is the mass of the gas measured in (kg).

III. Molecular Sample of an Ideal Gas

Since the collisions between molecules occur only along the x-axis, the conservation of momentum applies only in the x-direction and can be written as



We get, $\vec{p}_x = m\vec{v}_x$

Before collision: $p_x = mv_x$

After collision: $p'_x = -mv_x$

We get, $\Delta p_x = p'_x - p_x$

$\Delta p_x = mv_x + mv_x = 2mv_x$

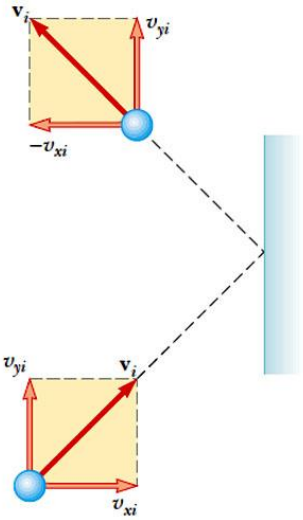
Impulse-Momentum Theorem:

We get, $F\Delta t = \Delta p_x$

Where, F is the force exerted by the molecule on the wall over a time interval.

Δt is the time interval.

For double collision of molecules



We get, $F = \dfrac{\Delta p_x}{\Delta t} = \dfrac{2mv_x}{2d/v_x} = \dfrac{mv_x^2}{d}$

Sum of average speed: $\overline{v^2} = \overline{v_x^2} + \overline{v_y^2} + \overline{v_z^2} \Rightarrow \overline{v^2} = 3\overline{v_x^2} \Rightarrow \overline{v_x^2} =$

Since the molecules move only along the x-axis

We get, $F = \dfrac{N}{3}\left(\dfrac{m\overline{v^2}}{d}\right)$

Since, $P = \dfrac{F}{A},\ A = d^2$

We get, $P = \dfrac{\dfrac{N}{3}\left(\dfrac{m\overline{v^2}}{d}\right)}{d^2} = \dfrac{1}{3}\left(\dfrac{N}{d^3}m\overline{v^2}\right) = \dfrac{1}{3}\left(\dfrac{N}{V}\right)m\overline{v^2} = \dfrac{2}{3}\left(\dfrac{N}{V}\right)\left(\dfrac{1}{2}m\overline{v^2}\right)$

Hence, $P = \dfrac{2}{3}\left(\dfrac{N}{V}\right)\left(\dfrac{1}{2}m\overline{v^2}\right)$

Where, F is the average force measured in (N).

$\bar{v}$ is the average speed measured in $(\mathrm{m/s})$.

d is the edge of the cube measured in (m^2).

m is the mass measured in (kg).

N is the number of molecules.

P is pressure of gas measured in (Pa).

Pressure of molecule: $P = \frac{2}{3}\left(\frac{N}{V}\right)\left(\frac{1}{2}m\overline{v^2}\right)$

We get, $PV = nRT = \frac{N}{N_A}RT = Nk_BT$

Hence, $k_B = \frac{R}{N_A} = 1.38 \times 10^{-23}\ \mathrm{J/K}$ is Boltzmann's constant.

Chapter2 **Thermodynamics**
Lesson3 **Periodic Motion**

I. Circular Motion

i. Angular frequency

Angular velocity is the rate of change of angular displacement with respect to time basically, how quickly an object spins or rotates.

Angular frequency $\omega = \dfrac{\theta}{t}$

Where, ω is the angular frequency measured in $(\mathrm{rad/s})$.

θ is angular displacement (angle turned), measured in (rad).

t is time taken for the rotation, measured in (s).

However, $\omega = 2\pi N$

Where, N is the number of revolutions per second $(\mathrm{rev/s})$.

ii. Period of Motion

The period T is the time taken for one complete cycle or revolution of a rotating object. It represents how long it takes the object to make one full turn.

The formula $T = \dfrac{2\pi}{\omega}$

Where, T is period of rotation the time taken for one complete revolution measured in in (s).

ω is the angular frequency measured in $(\mathrm{rad/s})$.

- Relationship between T and N

If an object completes N revolutions in time T, then: $T = \dfrac{1}{N}$

Where, N is the number of revolutions per second $(\mathrm{rev/s})$.

T is period of rotation the time taken for one complete revolution measured in (s).

II. Harmonic Motion

Harmonic motion is a type of periodic motion where an object moves back and forth about an equilibrium (central) position in a regular, repeating pattern.

General equation $y = A\sin\left(\omega t + \phi\right)$

Where, A is the amplitude of the oscillation measured in (m) or (cm).

ω is the angular frequency measured in $(\mathrm{rad/s})$.

t is the time taken measured in (s).

ϕ is the phase angle measured in (rad).

y is the maximum displacement measured in (m) or (cm).

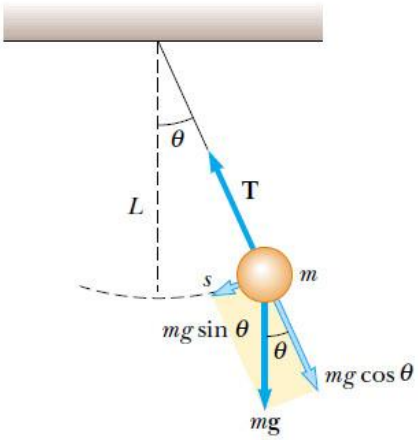
IV. Oscillation

Oscillation is the repetitive back-and-forth or periodic motion of an object around a central position or equilibrium point, often taking the form of sinusoidal motion.

i. Pendulum

A pendulum is a simple mechanical system consisting of a weight (called the bob) suspended from a fixed point by a massless string or rod that can swing freely back and forth under the influence of gravity. When

the bob is pulled from its equilibrium position by an angle θ_m and then released without any initial speed, it undergoes oscillations characterized by its own period and frequency, which are given by the following expressions:



Equation of harmonic motion: $\frac{d^2\theta}{dt^2} = \frac{g}{l}\sin\theta = 0$

For small angle $\sin\theta = \theta \left(\theta \text{ measured in radian}\right)$

General solution $\theta = \theta_m \cos\left(\omega t + \phi\right)$

Angular frequency $\omega^2 = \frac{g}{l} \Rightarrow \omega = \sqrt{\frac{g}{l}}$

Period $T = 2\pi\sqrt{\frac{l}{g}}$

Frequency $f = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$

Hence, $T = 2\pi\sqrt{\frac{l}{g}}$ and $f = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$

Where, T is period of rotation the time taken for one complete revolution measured in (s).

f is the frequency measured in (Hz).

$g = 9.8\ \text{N/kg}$ is the gravity.

• Conservation of energy

In a mechanical system like a pendulum oscillating under gravity (assuming no friction or air resistance), the total mechanical energy remains constant throughout the motion.

The conservation of energy for a pendulum can be expressed using mechanical energy as the sum of kinetic and potential energy remaining constant during its motion.

$$E = K + U = \text{constant}$$

Where, $K = \frac{1}{2}mv^2$ is the kinetic energy measured in (J).

$U = mgh$ is the gravitational potential energy measured in (J).

ii. Physical Pendulum Period

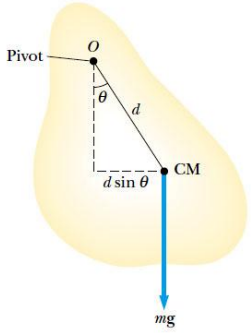
Equation of harmonic motion: $\frac{d^2\theta}{dt^2} = \frac{g}{l}\sin\theta = 0$

For small angle $\sin\theta = \theta \left(\theta \text{ measured in radian}\right)$

We get, $\frac{d^2\theta}{dt^2} = \frac{mgd}{I}\theta = 0$

General solution $\theta = \theta_m \cos\left(\omega t + \phi\right)$

Angular frequency $\omega^2 = \frac{mgd}{I} \Rightarrow \omega = \sqrt{\frac{mgd}{I}}$



Period $T = 2\pi\sqrt{\dfrac{I}{mgd}}$

Where, T is period of rotation the time taken for one complete revolution measured in (s).

I is moment of inertia about the pivot measured in $(\mathrm{kg.m^2})$.

m is the mass of the pendulum measured in (kg).

f is the frequency measured in (Hz).

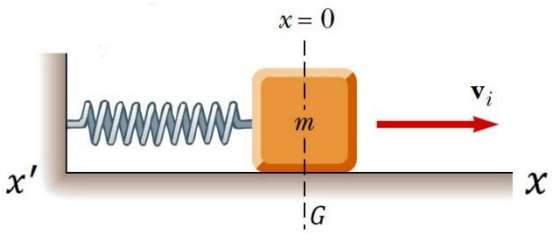
$g = 9.8\ \mathrm{N/kg}$ is the gravity.

d is distance from pivot to center of mass measured in (m).

iv. Horizontal Spring-Mass System

A horizontal spring-mass system consists of a mass m attached to a spring, moving frictionlessly along the x-axis. The spring has a spring constant k. When the mass is pulled from its equilibrium position and released, it undergoes oscillatory motion. The period and frequency of this motion are given by the following expressions:

Equation of harmonic motion: $\dfrac{d^2\theta}{dt^2} = \dfrac{k}{m}x = 0$



General solution $x = A\cos\left(\omega t + \phi\right)$

Angular frequency $\omega^2 = \dfrac{g}{l} \Rightarrow \omega = \sqrt{\dfrac{g}{l}}$

Period $T = 2\pi\sqrt{\dfrac{m}{k}}$

Frequency $f = \dfrac{1}{2\pi}\sqrt{\dfrac{k}{m}}$

Hence, $T = 2\pi\sqrt{\dfrac{m}{k}}$ and $f = \dfrac{1}{2\pi}\sqrt{\dfrac{k}{m}}$

Where, T is period of the oscillation measured in (s).

m is the mass attached to a spring measured in (kg).

f is the frequency measured in (Hz).

k is the spring constant measured in $(\mathrm{N/m})$.

Noted: If a spring experiences a force F, it causes a change in length Δl, according to Hooke's Law:

$$F = k\Delta l$$

Where, F is the applied force measured in (N).

Δl is the change in length measured in (m).

k is the spring constant measured in $(\mathrm{N/m})$.

- Conservation of energy

Assume that $x = \overline{OG}$ is the displacement of the spring-mass system from the equilibrium position at time t, and x_m is the maximum displacement (amplitude). Taking the spring's natural (undeformed) length as the reference, the elastic potential energy stored in the spring at any instant t is given by:

$$PE_s = \frac{1}{2}kx^2$$

Where, PE_s is the elastic potential energy stored in the spring measured in (J).

x is the extension or elongation measured in (m).

k is the spring constant measured in $(\mathrm{N/m})$.

Note: It is observed that when $x = 0$ (i.e., the spring is at its natural, undeformed length), the elastic potential energy PE_s is zero:

$$PE_s = \frac{1}{2}kx^2 = 0, \text{ When } x = 0$$

- The kinetic energy of a mass m moving with instantaneous speed v is given by:

$$KE = \frac{1}{2}mv^2$$

Where, KE is the kinetic energy measured in (J).

m is the mass attached to a spring measured in (kg).

v is the instantaneous speed measured in $(\mathrm{m/s})$.

- For undamped oscillations, the total mechanical energy of the system remains constant over time.

$$E = KE + PE_s = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \text{constant}$$

v. Vertical Spring-Mass System

We consider a vertical spring-mass system as shown on the right-hand side. Initially, the spring is at its natural (unstretched) length. When a mass m is attached, the spring extends by a distance x_0 to reach a new equilibrium position due to the balance between the spring force and gravitational force.

If the mass is then pulled further downward by a distance x from the equilibrium position and released, it undergoes vertical oscillations. The motion occurs symmetrically about the equilibrium point with an amplitude A, oscillating both above and below the center of gravity.

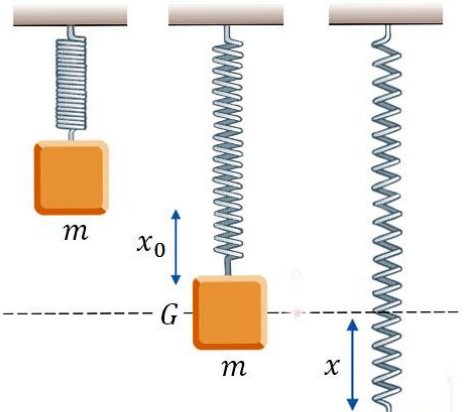
- Kinetic energy $KE = \frac{1}{2}mv^2$

Where, KE is the kinetic energy measured in (J).

m is the mass attached to a spring measured in (kg).

v is the speed of the block measured in $(\mathrm{m/s})$.

- Elastic potential energy $PE_s = \frac{1}{2}kx^2$



Where, PE_s is the elastic potential energy stored in the spring measured in (J).

x is the extension or elongation measured in (m).

k is the spring constant measured in $(\mathrm{N/m})$.

• Gravitational potential energy $PE_g = mgz$

Where, PE_g is the gravitational potential energy stored in the spring measured in (J).

h is the height of G compared to initial position measured in (m).

• The period of oscillation for a vertical spring-mass system is the same as that of a horizontal spring-mass system, and it is given by:

$$T = 2\pi\sqrt{\frac{m}{k}}$$

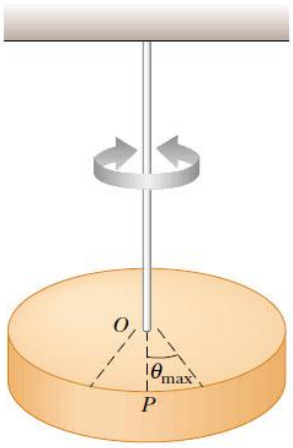
Where, T is period of the oscillation measured in (s).

m is the mass attached to a spring measured in (kg).

k is the spring constant measured in $(\mathrm{N/kg})$.

vi. Torsional Pendulum

Equation of harmonic motion: $\dfrac{d^2\theta}{dt^2} = \dfrac{k}{l}\sin\theta = 0$



For small angle $\sin\theta = \theta \left(\theta \text{ measured in radian}\right)$

We get, $\dfrac{d^2\theta}{dt^2} = \dfrac{k}{l}\theta = 0$

General solution $\theta = \theta_m \cos\left(\omega t + \phi\right)$

Angular frequency $\omega^2 = \dfrac{g}{I} \Rightarrow \omega = \sqrt{\dfrac{g}{I}}$

Period $T = 2\pi\sqrt{\dfrac{I}{k}}$

Frequency $f = \dfrac{1}{2\pi}\sqrt{\dfrac{k}{I}}$

Hence, $T = 2\pi\sqrt{\dfrac{I}{k}}$ and $f = \dfrac{1}{2\pi}\sqrt{\dfrac{k}{I}}$

Where, T is period of torsional pendulum measured in (s).

I is moment of inertia about the pivot measured in $(\mathrm{kg.m^2})$.

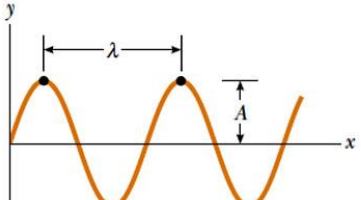
f is the frequency measured in (Hz).

k is the spring constant measured in $(\mathrm{N/m})$.

Chapter2 **Wave**
Lesson4 **Wave**

I. Wave speed of a wave

Wave speed is the rate at which a wave travels through a medium. It is given by the formula:



$$v = \lambda f = \frac{\omega}{k} = \frac{\lambda}{T}$$

Where, T is period of a wave measured in (s).

v is the wave speed measured in $(\mathrm{m/s})$.

f is the frequency of a wave measured in (Hz).

k is the wave number measured in $(\mathrm{rad/m})$.

λ is the wavelength measured in (m).

II. Wavelength

Wavelength λ is the distance a wave travels during one complete cycle of vibration.

$$\lambda = vT = \frac{2\pi}{k} = \frac{v}{f}$$

III. Period of a wave

The **period** of a wave is the time it takes for **one complete wave cycle** to pass a fixed point.

$$T = \frac{2\pi}{\omega} = \frac{1}{f} = \frac{2\pi}{vk}$$

V. Angular frequency

Angular frequency is a measure of how quickly an object or wave goes through its cycle, expressed in radians per second. It is commonly used in oscillations and wave motion.

$$\omega = 2\pi f$$

VI. Sinusoidal equation

Waves exhibit properties such as reflection, refraction, and diffraction.

- **Reflection**: When a wave bounces off a surface and changes direction, but stays in the same medium.

Example: Echo from a wall.

- **Refraction**: The bending of a wave as it passes from one medium into another with a different wave speed.

Example: A straw looking bent in water.

- **Diffraction**: The spreading of a wave as it passes through a narrow opening or around an obstacle.

Example: Sound bending around a corner.

The **wave equation for a string** describes how waves propagate along a stretched string (such as on a guitar or violin).

$$y = A\sin 2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right)$$

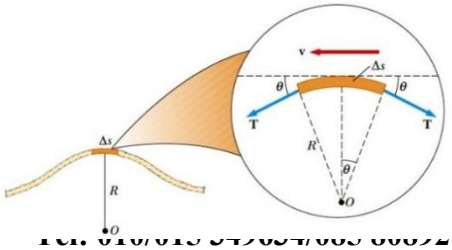
Where, A is the amplitude of a wave measured in (m) or (cm).

t is the time taken measured in (s).

T is period of a wave measured in (s).

λ is the wavelength measured in (m).

x is the position measured in (m).



Wave equation for a string $v = \sqrt{\frac{F_T}{\mu}}$

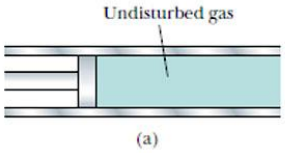
Where, F_T is the tension measured in (N).

$\mu = m/l$ is the density or mass per unit length measured in $(\mathrm{kg/m})$.

v is the wave speed on a string measured in $(\mathrm{m/s})$.

VII. Speed of sound in a solid

The **speed of sound in a solid** depends on the **elastic properties** and the **density** of the material. It is generally **faster in solids** than in liquids or gases because particles in solids are more tightly packed and interact more strongly.

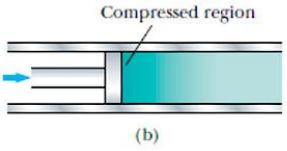


Speed of sound in a solid $v = \sqrt{\frac{Y}{\rho}}$

Where, Y is the **Young's modulus** of the material measured in $(\mathrm{Pa\ or\ N/m^2})$.

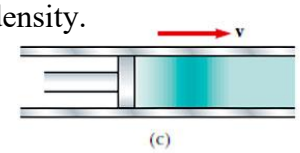
ρ is the **density** of the material (solid) measured in $(\mathrm{kg/m^3})$.

v is the **speed of sound** in a solid measured in $(\mathrm{m/s})$.



VIII. Speed of sound in a liquid

The **speed of sound in a liquid** depends on how compressible the liquid is and its density.

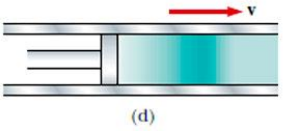


Speed of sound in a liquid $v = \sqrt{\frac{B}{\mu}}$

Where, B is the **bulk modulus** of the liquid measured in $(\mathrm{Pa\ or\ N/m^2})$.

μ is the **density** of a liquid measured in $(\mathrm{kg/m^3})$.

v is the **speed of sound** in a liquid measured in $(\mathrm{m/s})$.



IX. Speed of sound in a gas

The **speed of sound in a gas** depends on the gas's temperature, its molar mass, and the specific heat ratio.

Speed of sound in a gas $v = v_0\sqrt{\frac{T}{273}}$

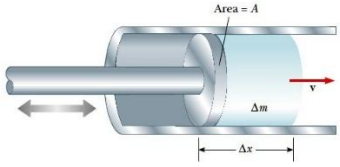
Where, T is the **absolute temperature** of a gas measured in (K).

v_0 is the **speed of sound** at $0°\mathrm{C}$ $(273\ \mathrm{K})$, **approximately 331 m/s.**

v is the **speed of sound** in a gas measured in $(\mathrm{m/s})$.

X. Intensity of sound

Sound intensity is the **power transmitted per unit area** by a sound wave. It measures how much energy the wave carries through a surface perpendicular to the direction of wave propagation.



Sound intensity $I = \frac{P}{A}$

Where, I is the **sound intensity** measured in $(\mathrm{W/m^2})$.

P is the **power of the sound wave** measured in (W).

A is the **area through which the sound energy passes** measured in $(\mathrm{m^2})$.

XI. Rate of Intensity of sound

The **rate of intensity of sound** usually refers to the **sound intensity level**, denoted by β, which expresses the intensity of a sound on a logarithmic scale measured in **decibels (dB)**.

Rate of intensity of sound $\beta = 10\log\left(\frac{I}{I_0}\right)$

Where, β is the **sound intensity level** measured in (dB).

I is the **intensity of the sound wave** measured in $(\mathrm{W/m^2})$.

$I_0 = 10^{-12}\,\mathrm{W/m^2}$ is the **reference intensity** (threshold of hearing).

XII. Doppler effect

The **Doppler Effect** is the **apparent change in the frequency or wavelength** of a wave (such as sound or light) **due to the relative motion** between the **source** of the wave and the **observer**.

i. Sound Source at Rest, Observer Moves Toward the Source

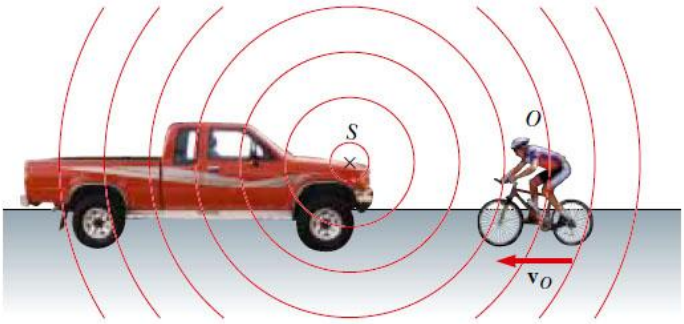
- The **sound source is stationary**.
- The **observer is moving toward** the source.
- As a result, the observer receives more wavefronts per second → **higher frequency heard**.

We get, $f' = \frac{v'}{\lambda}$

$f' = \frac{v + v_0}{\lambda}$

Equivalent $f' = \left(1 + \frac{v_0}{v}\right) f$

Hence, $f' = \left(\frac{v + v_0}{v}\right) f$



Where, f is the **source frequency** measured in (Hz).

f' is the **observed frequency** measured in (Hz).

v is the **speed of sound in air** measured in $(\mathrm{m/s})$.

v_0 is the **speed of observer (toward the source)** measured in $(\mathrm{m/s})$.

ii. Observer at Rest, Sound Source Moves outward the Source

- The **observer is stationary**.
- The **sound source is moving away** from the observer.
- This causes the sound waves to **spread out**, increasing wavelength → resulting in a **lower observed frequency**.

Hence, $f' = \left(\frac{v}{v + v_s}\right) f$

Where, f is the **actual frequency emitted by the source** measured in (Hz).

f' is the **observed frequency** measured in (Hz).

v is the **speed of sound in air** measured in $(\mathrm{m/s})$.

v_s is the **speed of the source (moving away from observer)** measured in $(\mathrm{m/s})$.

iii. Observer and Sound Source Moves toward each other

- The observer is moving toward the source.
- The source is moving toward the observer.
- As a result, the waves are compressed, and the observer moves into the waves, so the observed frequency increases.

Hence, $$f' = \left(\frac{v + v_s}{v - v_s}\right) f$$

Where, f is the **actual frequency emitted by the source** measured in (Hz).

f' is the **observed frequency** measured in (Hz).

v is the **speed of sound in air** measured in $(\mathrm{m/s})$.

v_s is the **speed of the source (moving away from observer)** measured in $(\mathrm{m/s})$.

iv. Observer and Sound Source Moves outward from each other

- The **observer is moving away** from the source.
- The **source is also moving away** from the observer.
- The wavefronts become **more spread out**, and the observer receives **fewer waves per second** → the **observed frequency decreases**.

Hence, $$f' = \left(\frac{v - v_0}{v + v_s}\right) f = \left(\frac{v \pm v_0}{v \mp v_s}\right) f$$

Where, f is the **actual frequency emitted by the source** measured in (Hz).

f' is the **observed frequency** measured in (Hz).

v is the **speed of sound in air** measured in $(\mathrm{m/s})$.

v_s is the **speed of the source (moving away from observer)** measured in $(\mathrm{m/s})$.

Note: if $+v_0$ and $-v_s$ means that source and observer move toward each other.

if $-v_0$ and $+v_s$ means that source and observer move away from each other.

Vocabulary

• **conductors:** Any material that permits electrons to readily move.
• **electric charge**
• **electron:** A particle that carries negative charge and resides in the center of an atom. Electrons attract protons and are in motion around the nucleus.
• **ion:** An atom that has gained or lost an electron. Ions do not have an equal number of positive and negative
charges.
• **insulators:** Materials that do not conduct electric current, also known as non-conductors.
• **negative charge**
• **neutron:** A neutral particle that has no electric charge. Neutrons reside in the center of an atom.
• **positive charge**
• **proton:** A particle that carries positive charge and resides in the center of an atom. Protons attract electrons.
• **static electricity:** An imbalance of one kind (positive or negative) of charge. It can be caused by friction between two materials or placing objects in contact.

I.Coulomb's Law

Electrostatic charging

Electrostatic charging is the process by which an object gains or loses electric charge, resulting in it becoming electrically charged. There are three types of electrostatic charging, such as **frictional charging, charging by conduction** and **charging by induction.**

II. Electric charge

Electric charge is classified into two types, such as positively charge and negatively electric charge.
Assumption: When a positively charged rod is brought near a neutral nylon foil, the foil will experience an induced charge separation. The negative charges in the foil are attracted to the rod, and this creates a net attractive force between the rod and the foil, causing the foil to be drawn toward the positively charged rod. The underlying assumptions of **opposite charges attracting**, **like charges repelling**, and **electrostatic induction** explain this behavior.

• **Like charges or similar charges** repel each other.
• **Unlike charges or opposite charges** attract each other.

Electric charge and mass of electron	Electric charge and mass of electron
Electric charge: $-e = -1.6 \times 10^{-19}$ C	Electric charge: $e = 1.6 \times 10^{-19}$ C
Mass: $m_e = 9.11 \times 10^{-31}$ kg	Mass: $m_p = 1.67 \times 10^{-27}$ kg

To calculate the **electric charge of the nucleus of an atom**, you need to consider the number of **protons** in the nucleus because the nucleus consists of protons and neutrons, and only protons carry a positive charge.

$$q_{necleus} = Ze$$

Where, q_{ucleus} is the **total electric charge** of the nucleus.

Z is the **atomic number**, which represents the number of protons in the nucleus.

e is the **elementary charge**, approximately $e = 1.6 \times 10^{-19}$ C.

III. Quantization of Charge:

Further, charge on an object can be increased or decreased in multiples of e. It is because, during the charging process an integral number of electrons can be transferred from one body to the other body. This is known as **quantization of charge** or discrete nature of charge.

$$q = \pm n|e|$$

Where, n is the number of electrons.

$e = 1.6 \times 10^{-19}$ C is the charge of electron measured in (C).

q is electric charge measured in (C).

The process of **charge redistribution** between two conductive spheres, which have charge q_A and q_B that touch each other and then separate. When two charged conductors come into contact, charge flows between them until they reach the same electric potential with charges $q_A^{'}$ and $q_B^{'}$ is given by

$$q_A^{'} = q_B^{'} = \frac{q_A + q_B}{2}$$

IV. Forces between Charges

It was observed in carefully conducted experiments with charged objects that they experience force when brought close (not touching) to each other. This force can be attractive or repulsive. **Like charges repel each other and unlike charges attract each other.** Figure 10.2 describes this schematically.

This is the reason for charging by induction as described in section 10.2 and Fig. 10.1.

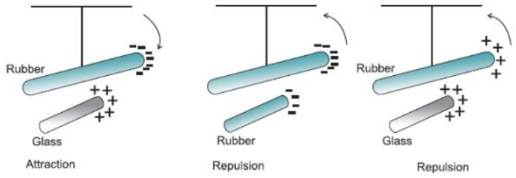


Fig. 10.2: Attractive and repulsive force.

Coulomb's law: The force between two charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between the two charges.

$$F = k\frac{|q_1||q_2|}{r^2}$$

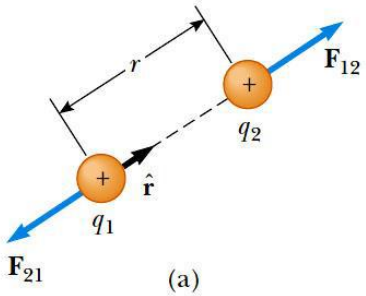
Where, F is force between two charges measured in (N).

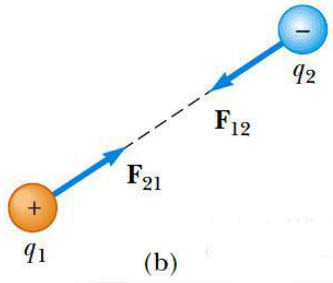
q_1 and q_2 are two charges measured in (C).

$k = 9 \times 10^9$ Nm^2/C^2 is a constant of proportionality called Coulomb's constant.

$\varepsilon = \varepsilon_0 \varepsilon_r$ is called permittivity of a medium.

$\varepsilon_r = \dfrac{\varepsilon}{\varepsilon_0}$ is the relative permittivity or dielectric constant of the medium and is denoted k.





r is the distance between the two charges measured in (m).

V. Principle of Superposition

The principle of superposition states that when a number of charges are interacting, the resultant force on a particular charge is given by the vector sum of the forces exerted by individual charges.

Consider a number of point charges $q_1,\ q_2,\ q_3\ \ldots$ kept at points $A_1,\ A_2,\ A_3\ \ldots$ as shown in Fig. 10.5. The force exerted on the charge q_1 by q_2 is $\vec{F}_{12}$. The value of $\vec{F}_{12}$ is calculated by ignoring the presence of other charges. Similarly, we find $\vec{F}_{13}$, $\vec{F}_{14}$ etc, one at a time, using the coulomb's law.

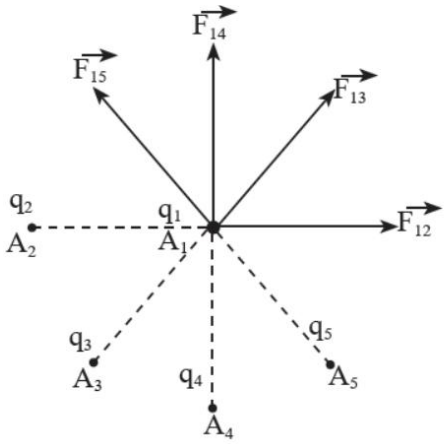


Fig. 10.5: Principle of superposition.

Total force $\vec{F}_1$ on charge q_1 is the vector sum of all such forces.

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{13} + \ldots + \vec{F}_{1N}$$

$$\vec{F}_1 = \frac{1}{4\pi\varepsilon_0}\left[\frac{|q_1||q_2|}{r_{12}^2}\hat{r}_{12} + \frac{|q_1||q_3|}{r_{13}^2}\hat{r}_{13} + \frac{|q_1||q_2|}{r_{14}^2}\hat{r}_{13} + \ldots + \frac{|q_1||q_n|}{r_{1n}^2}\hat{r}_{1N}\right]$$

The force $\vec{F}$ exerted by these charges on a test charge q_0 can be written using the summation notation Σ as follows,

$$\vec{F}_{\text{test}} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \ldots + \vec{F}_N = \sum_{n=1}^{N} F_N = \frac{1}{4\pi\varepsilon_0}\sum_{n=1}^{N}\frac{q_0 q_n}{r_{0n}^2}\hat{r}_{0n}$$

VI. Electric Field

Vocabulary

- electric field: The field that surrounds a charged particle.
- electric field line: A simple way of representing an electric field, also known as electric lines of force.
- test charge: Used to measure an electric field. The electric field is defined as the ratio of the force acting on the positive test charge, at some point in space, to the magnitude of the test charge.

The field that surrounds a charged particle is called **an electric field.**

The electric field is measured by observing the force on a small positive "test charge" placed within the field. (The test charge has to be small so it would have no effect upon the charges which have created the field we wish to measure.) The electric field is then defined as the ratio of the force acting on the positive test charge, at some point in space, to the magnitude of the test charge.

$$E = \frac{F}{|q_0|} = \frac{k|q||q_0|}{r^2|q_0|} = k\frac{|q|}{r^2}$$

Where, F is force between two charges measured in (N).

q_0 is a small test charge measured in (C).

$k = 9 \times 10^9\ \mathrm{Nm^2/C^2}$ is a constant of proportionality called Coulomb's constant.

r is the distance from a test charge measured in (m).

E is an electric field measured in $(\mathrm{N/C\ or\ V/m})$.

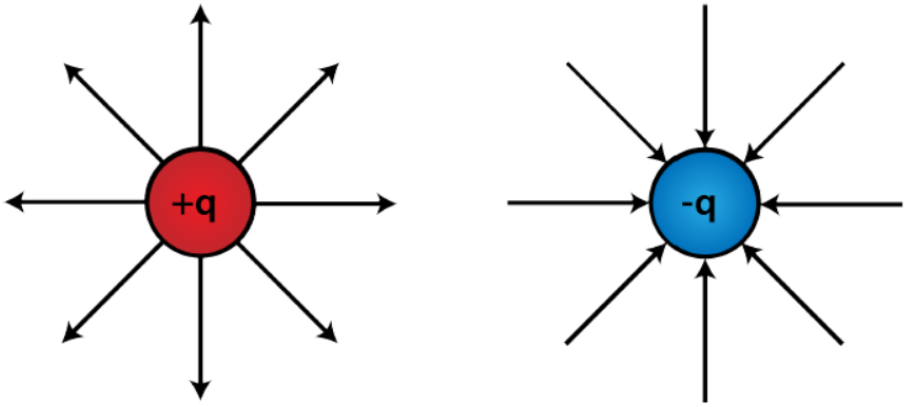
Note:

If $q > 0$: Outward electric field.

If $q < 0$: Inward electric field.

VII. Electric Field Lines for Isolated Positive and Negative Charges

The force experienced by a test charge in the presence of a positive charge is directed away from the positive charge, since like charges repel. The electric field lines would therefore be drawn pointing away from the positive charge. Similarly, the force experienced by the test charge in the presence of a negative charge is directed toward the negative charge, since unlike charges attract. The electric field lines would therefore be drawn pointing away from the negative charge.



The force F acting on a charge q in an electric field E is directly proportional to both the charge and the strength of the electric field:

$$F = |q|E$$

Noted:

- If $q > 0$: the electric force is in the same direction of electric field $\left(\vec{F} \uparrow\uparrow \vec{E}\right)$.
- If $q < 0$: the electric force is in the opposite direction of electric field $\left(\vec{F} \uparrow\downarrow \vec{E}\right)$.

✌ **How do you determine the location of a third charge between two charges so that the net electric force acting on it is zero?**

- **Case1:** If q_1 and q_2 have the same sign and $|q_1| > |q_2|$ thus, a third charge experienced by a zero net force should be placed between two charges, closer to the smaller magnitude charge q_2.

• **Case2:** If q_1 and q_2 have opposite sign and $|q_1| > |q_2|$ thus, a third charge experienced by a zero net force should be placed should be placed outside the region between the charges, closer to the charge q_2 with the smaller magnitude.

VIII. Uniform Fields

There is one special example you also need to know:

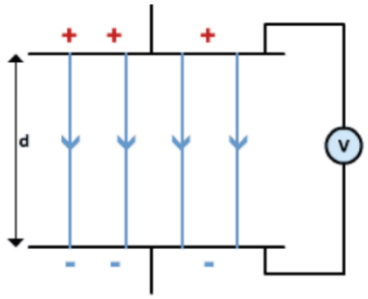


Fig 5.11

This is a uniform field. The field strength at any point in this field is: $E = \dfrac{V}{d}$

Where, E is a uniform electric field measured in $(\mathrm{N/C\ or\ V/m})$.

V is the potential difference between two plates measured in (V).

d is the distance separation the plates measured in (m).

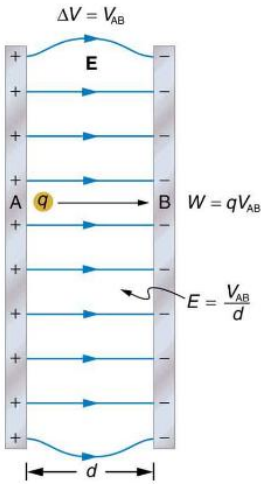
IV. A charged particle in electric field

✌ Case#1: Initial velocity is parallel to electric field

❍ If a charged particle is placed in an electric field and experiences a force with an acceleration a where $a > 0$, this means that the particle is accelerating under the influence of the **electric force** exerted by the electric field.

If the particle is experiencing an acceleration $a > 0$, the force on the particle must be greater than zero.

$$a = \frac{F}{m} = \frac{|q|E}{m}$$



❍ Determine the velocity exit the electric field at point B

• **Electric Potential Energy at Point A**: The electric potential energy of the charge q at point A is: $U_A = qV_A$.

• **Electric Potential Energy at Point B**: Similarly, the electric potential energy at point B is: $U_B = qV_B$

• **Change in Potential Energy**: The difference in potential energy as the particle moves from point A to point B is: $\Delta U = U_A - U_B = q\left(V_A - V_B\right) = qV_{AB}$

• **Kinetic Energy at Point B**: The work done on the particle by the electric field is converted into kinetic energy. If the particle starts from rest at point A, then the total change in energy will be equal to the particle's kinetic energy K_B at point B: $K_B = \frac{1}{2}mv_B^2, K_A = \frac{1}{2}mv_A^2$

• **Conservation of Energy**: The electric potential energy lost is equal to the kinetic energy gained, so:

$$qV_{AB} = \frac{1}{2}mv_B^2 \Rightarrow v_B = \sqrt{\frac{2qV_{AB}}{m} + v_A^2}$$

❍ Determine the time t taken for a charged particle to move between two plates in an electric field, we assume that the electric field accelerates the particle uniformly from plate A to plate B.

• **Use kinematic equations**: $v_B = at + v_A \Rightarrow t = \frac{v_B - v_A}{a}$

• To find the time taken, we use the kinematic equation for constant acceleration: $d = v_A t + \frac{1}{2}at^2$

Since the particle starts from rest at plate A (i.e., $v_A = 0$), the equation simplifies to: $d = \frac{1}{2}at^2$

Substituting $a = \frac{F}{m} = \frac{|q|E}{m}$ into the equation: $d = \frac{1}{2}\left(\frac{qV_{AB}}{md}\right)t^2 \Rightarrow t^2 = \frac{2md^2}{qV_{AB}}$

Hence, time t taken for a charged particle to move between two plates in an electric field is $t = \sqrt{\frac{2md^2}{qV_{AB}}}$

❍ **Equations of Motion in the Electric Field:**

• **Position as a Function of Time**: The position x (t) of the particle at any time t can be found using the following equation of motion: $x(t) = x_0 + v_0 t + \frac{1}{2}at^2$

Substituting $a = \frac{F}{m} = \frac{|q|E}{m}$ into the equation therefore, the equation of position as a function of time can be written as, $x(t) = x_0 + v_0 t + \frac{qV_{AB}}{2md}t^2$ is a quadratic relationship in time.

✌ **Case#2: Initial velocity is perpendicular electric field**

❍ A charged particle moves in electric field, a charged particle have:

• Horizontal-axis $\overrightarrow{ox}$: a charged particle moves with constant velocity:

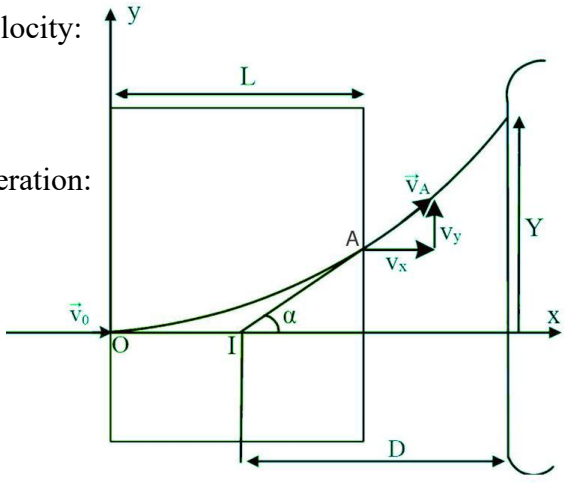
$$x = v_x t = v_0 t \quad (v_x = v_0) \therefore t = \frac{x}{v_0}$$

• Vertical-axis $\overrightarrow{oy}$: a charged particle moves with constant acceleration:

$$y = \frac{1}{2}at^2 + v_{0y}t = \frac{1}{2}at^2 \quad (v_{0y} = 0)$$

Since: $t = \frac{x}{v_0}, a = \frac{F}{m} = \frac{|q|E}{m}$

• Substitute t and a therefore, $y = \frac{|q|E}{2m}\left(\frac{x}{v_0}\right)^2 = \frac{|q|E}{2mv_0^2}x^2$



• Hence, $y = \dfrac{|q|E}{2mv_0^2}x^2$, $y = ax^2$ is called a **quadratic equation** and represents a **parabola** when plotted on a graph, where a is a constant.

❍ Determine the velocity exit the electric field at point A

We have: $\vec{v}_A = \vec{v}_x + \vec{v}_y \Leftrightarrow v_A^2 = v_x^2 + v_y^2 \Rightarrow v = \sqrt{v_x^2 + v_y^2}$

Since: $v_x = v_0$, $v_y = at = \dfrac{|q|E}{m} \times \dfrac{x}{v_0} = \dfrac{|q|El}{mv_0} \quad \therefore x = l$

Hence: $v_A = \sqrt{v_0^2 + \left(\dfrac{|q|El}{mv_0}\right)^2}$

❍ Determine the angle of electric deflection

• The angle of deflection α is the angle between the particle's final velocity vector and the original direction of motion (along the y-axis). This angle is given by:

$$\tan\alpha = \left(\frac{v_y}{v_x}\right) = \left(\frac{v_y}{v_0}\right)$$

Since: $v_y = at = \dfrac{|q|E}{m} \times \dfrac{l}{v_0} = \dfrac{|q|El}{mv_0}$

Hence: $\tan\alpha = \dfrac{|q|El}{mv_0^2}$

• Determine the displacement in the direction of the Electric Field:

In $\Delta \perp IA'H'$: $\tan\alpha = \dfrac{Y}{D} \Rightarrow Y = D\tan\alpha \quad \left(Y = A'H', D = IH'\right)$

In $\Delta \perp Av_Av_x$: $\tan\alpha = \dfrac{v_y}{v_x} = \dfrac{v_y}{v_0} \left(v_x = v_0\right)$

We get: $Y = D\dfrac{v_y}{v_x}$

Since: $v_y = at = \dfrac{|q|E}{m} \times \dfrac{l}{v_0}$

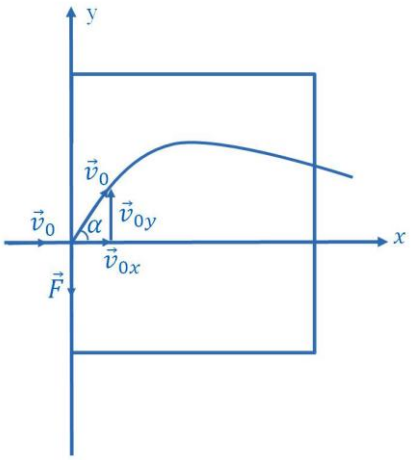
Hence: $Y = \dfrac{|q|El}{mv_0^2}D$

And: $E = \dfrac{V}{d}$

Therefore, $Y = \dfrac{|q|Dl}{mv_0^2d}V$ So, the distance of deflection is proportional to voltage and coefficient $\dfrac{|q|Dl}{mv_0^2d}$.

✌ Case#3: Initial velocity has direction created an angle with electric field.

When the **initial velocity** of a charged particle is at an **angle to the electric field**, the motion becomes a combination of uniform motion (perpendicular to the electric field) and accelerated motion (parallel to the electric field). The initial velocity has two components: one parallel to the electric field and one perpendicular to it.



❍ A charged particle moves in electric field, a charged particle have:

• Horizontal-axis $\overrightarrow{ox}$: a charged particle moves with constant velocity:

$$x = v_x t = v_0 t = v_0 \cos\alpha \times t \Rightarrow t = \frac{x}{v_0 \cos\alpha}$$

• Vertical-axis $\overrightarrow{oy}$: a charged particle moves with constant acceleration:

$$y = -\frac{1}{2}at^2 + v_{0y}t = -\frac{1}{2}at^2 + v_0 \sin\alpha \quad \left(v_{0y} = v_0 \sin\alpha\right)$$

Since, $t = \frac{x}{v_0 \cos\alpha}, a = \frac{F}{m} = \frac{|q|E}{m}$

• Substitute t and a therefore, $y = -\frac{|q|E}{2m}\left(\frac{x}{v_0 \cos\alpha}\right)^2 + v_0 \sin\alpha \frac{x}{v_0 \cos\alpha} = -\frac{|q|E}{2mv_0^2 \cos^2\alpha}x^2 + x\tan\alpha$

Hence, $y = -\frac{|q|E}{2mv_0^2 \cos^2\alpha}x^2 + x\tan\alpha$, $y = ax + b$ is represents a **quadratic equation** in terms of x.

✌ For a charged particle to completely leave an electric field

• For a charged particle to completely leave an electric field: $x = l$ and $y < OM$, we get

$$y = -\frac{|q|E}{2mv_0^2 \cos^2\alpha}x^2 + x\tan\alpha < OM$$

• For a charged particle touching at point M, unless: $x = l$ and $y = O'M$, we get

$$y = -\frac{|q|E}{2mv_0^2 \cos^2\alpha}l^2 + l\tan\alpha = O'M$$

• To keep a charged particle from leaving an electric field: $x = l$ and $y > O'M$, we get

$$y = -\frac{|q|E}{2mv_0^2 \cos^2\alpha}l^2 + l\tan\alpha > O'M \quad \left(x < l, y = O'M\right)$$

• For a charged particle completely leave an electric field at O unless, $x = l$ and $y = 0$, we get

$$y = -\frac{|q|E}{2mv_0^2 \cos^2\alpha}l^2 + l\tan\alpha = 0$$

IX. Electric potential

Electric potential

Electric field potential (V) is defined as the **work done in moving a unit charge through a distance (d) in an electric field.**

Electric field potential, $V = \dfrac{\text{Work done}}{\text{Charge}} = \dfrac{\text{Force}(\text{F}) \times \text{distance}(\text{m})}{\text{Charge}}$

The **work** done on a charge q as it moves between two points A and B in an electric field, and it's related to both the **electric field** E, the **distance** d, and the **potential difference** (voltage) V_{AB} between the two points.

$$W_{AB} = qEd = qV_{AB}$$

Where, W_{AB} is work done on the charge q when it moves a distance d through a uniform electric field E.

q is the charge of the particle being moved.

E is the magnitude of the uniform electric field.

d is the distance the charge moves in the direction of the electric field.

V_{AB} is the **electric potential difference** (voltage) between points A and B.

The electric potential V_A at point A in relation to the work done $W_{A\infty}$ in bringing a unit positive charge from a reference point (often at infinity) to that point A.

$$V_A = \frac{W_{A\infty}}{q}$$

Where, V_A is the electric potential at point.

$W_{A\infty}$ is the work done in moving a charge q from infinity (or a reference point where the electric potential is considered to be zero) to point A.

q is the charge being moved.

The **electric potential** V due to a point charge q at a distance r from that charge: $V = 9 \times 10^9 \times \dfrac{q}{r}$

Where, V is the electric potential at a distance r from the charge q.

q is the magnitude of the point charge creating the electric potential.

r is the distance from the charge q to the point where the electric potential V is being measured.

Electric potential difference: $V_{AB} = -V_{BA} = V_A - V_B = \Delta V$

$$V_{AB} = \frac{W_{AB}}{q}$$

Note: The relationship between electric potential and work done in an electric field.

- This indicates that the potential difference is equal in magnitude but opposite in sign when the points are switched. It shows the symmetric nature of electric potential.
- This expression defines the potential difference between points A and B. It states that the potential at point A minus the potential at point B gives the potential difference V_{AB}.
- This equation states that the potential difference V_{AB} is equal to the work done W_{AB} in moving a charge q from point B to point A.

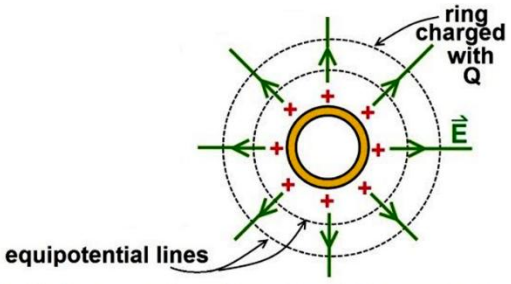
Electric potential energy: $U_{AB} = U_A - U_B = qV_{AB} = q\left(V_A - V_B\right) = qV_A - qV_B$

Note:

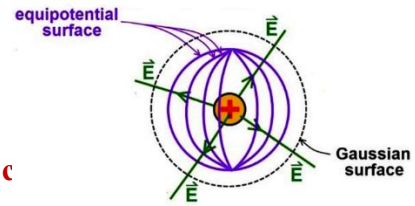
- $U_{AB} = U_A - U_B$: This equation defines the change in electric potential energy U_{AB} when a charge moves from point B to point A. Here, U_A is the electric potential energy at point A, and U_B is the electric potential energy at point B.
- $U_{AB} = qV_{AB}$: This shows that the change in electric potential energy U_{AB} is equal to the charge q multiplied by the potential difference V_{AB} between points A and B.
- $U_{AB} = q\left(V_A - V_B\right)$ This equation expresses the change in potential energy in terms of the potentials at points A and B. It indicates that the change in potential energy is dependent on the difference in electric potential at those points.
- $U_{AB} = qV_A - qV_B$: This represents the distribution of electric potential energy based on the individual potentials at points A and B.

X. Equipotential

- Equipotential lines always intersect electric lines perpendicular to the direction of the electric field.
- The voltage (potential) remains constant along the equipotential lines.



- It is the surface of region (like a sphere) on which the potential is constant.
- As with a Gaussian surface the electric field is perpendicular to the surface everywhere on the surface.



XII. Introduction to capacitors and c

Capacitor

A **capacitor** is a device that stores charges. It consists of two parallel metal plates held together and separated by an insulating medium called the *dielectric*. There are many types of capacitors depending on the dielectric used. The different type of capacitors includes, electrolytic capacitors, metal foil type capacitors, variable capacitors etc. In this chapter, we shall study the parallel plate capacitor.

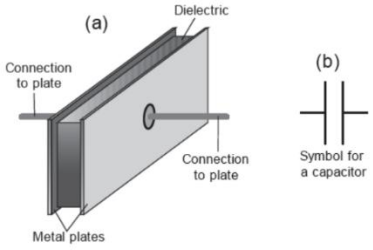


Fig. 5.18: A parallel capacitor

The ***capacitance C*** of a capacitor is defined as *the ratio of the charge stored on the plates to the potential difference between the plates.*

$$C = \frac{q}{V}$$

Where, C is the capacitance measured in (F).

q is the electric charge measured in (C).

V is the potential difference measured in (V).

2. Area of overlap of the plates

An increase in the effective area or the area of overlap of the plates increases the capacitance due to increased surface for holding charges (Fig. 5.27).

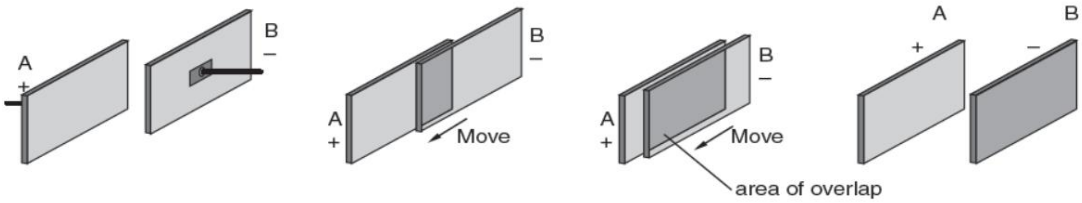


Fig. 5.27

Capacitance, C_1 is therefore proportional to the area of overlap i.e $C \alpha A$

3. Nature of the medium between the plates, called the dielectric

An insulating medium like mica, paraffin wax, ebonite, polythene or polystyrene sheet, waxed paper, glass can be introduced in the space between the plates of an air capacitor, covering the entire area between the plates. This material is called a *dielectric material.*

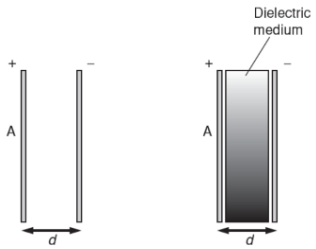


Fig. 5.28: Capacitance depends upon the dielectric

Introduction of a dielectric material de ce, V, between the plates and hence the capacitance of the capacitor increas ice with a dielectric material to the capacitance without a dielectric material is called the *dielectric constant* of the material.

Since $C \propto \frac{l}{A}$, C α A and C increases when a dielectric material is placed between the plates then,

$$C = \varepsilon_0 \frac{A}{d}$$

Where, where ε_0 is the permittivity of the dielectric material used.

$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0}$ is the relative permittivity or dielectric constant of the medium

A is the area of overlap plates measured in (m^2).

d is distance of separation measured in (m).

C is the capacitance measured in (F).

(a) Capacitors in parallel

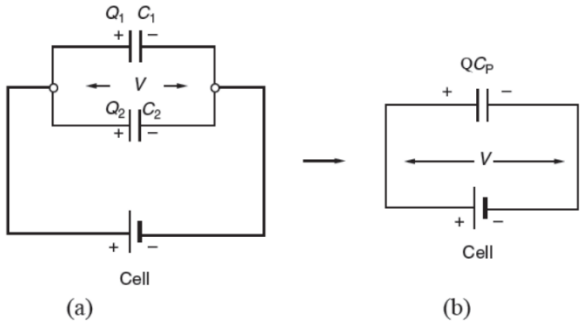


Fig. 5.32: Capacitors in parallel

In this connection, the potential difference, V, across the plates of each capacitor is the same.
The total charge Q of the effective capacitor, C_p, should be equal to the sum of the charge Q_1 and Q_2

$$Q = Q_1 + Q_2, \text{ Where, } Q = CV$$

Hence, $C_pV = C_1V + C_2V \Leftrightarrow C_pV = V\left(C_1 + C_2\right) \Rightarrow C_p = C_1 + C_2$

The effective capacitance C_p has increased and it is the sum of the individual capacitance. If there are N capacitors are connected in parallel therefore, the effective capacitance C_p can be written as

$$C_p = C_1 + C_2 + C_3 + \ldots + C_n$$

(b) Capacitors in series

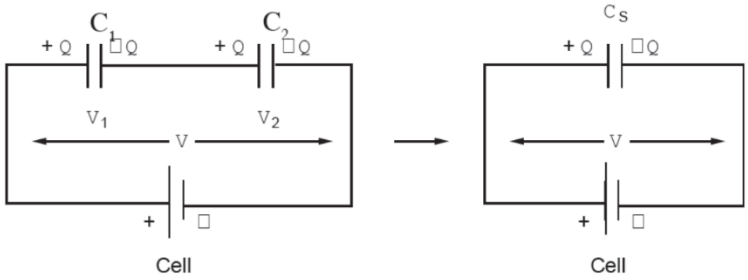


Fig. 5.33: Capacitors in series

In this connection, the charge Q on each plate of the capacitor is the same, but the total potential difference, V, is split as V_1 and V_2 across each capacitor.

$V = V_1 + V_2$, But charge $Q = CV \quad \therefore\ V = \dfrac{Q}{C}$

Hence, $\dfrac{Q}{C} = \dfrac{Q}{C_1} + \dfrac{Q}{C_2} \Leftrightarrow \dfrac{Q}{C} = Q\left(\dfrac{1}{C_1} + \dfrac{1}{C_2}\right) \Rightarrow \dfrac{1}{C} = \dfrac{1}{C_1} + \dfrac{1}{C_2}$

The effective capacitance C_S has decreased and is always less than the least value of capacitance. If there are N capacitors connected in series, therefore the capacitance C_S can be written as

$$\frac{1}{C_S} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \ldots + \frac{1}{C_n}$$

Dipoles RC:

- Time constant of RC circuit: $\tau = RC$
- Electric current of dipoles RC: $i = \frac{dq}{dt} = C\frac{dV}{dt}$
- Electric energy of capacitors: $E_C = \frac{1}{2}CV^2 = \frac{q^2}{2C}$

✌ **Calculating the minimum number of capacitors needed in a circuit and illustrating their combination involves understanding how capacitors can be arranged in series and parallel configurations.**

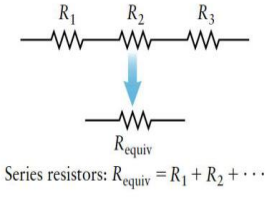
Assume that we have many identical capacitors, each with capacitance C_0 and effective capacitor C.

- If $C < C_0$, we need to add a new capacitor C_x to combine in series with C_0 then we find the capacitor C_x.
- If $C > C_0$, we need to add a new capacitor C_y to combine in parallel with C_0 then we find the capacitor C_y

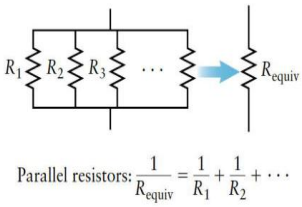
Electric current, Resistance and Electromotive force

I. Combination of resistance in series and in parallel

- Series combination: $R = R_1 + R_2 + R_3 + \ldots + R_n$
- Parallel combination: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots + \frac{1}{R_n}$



Series resistors: $R_{equiv} = R_1 + R_2 + \cdots$

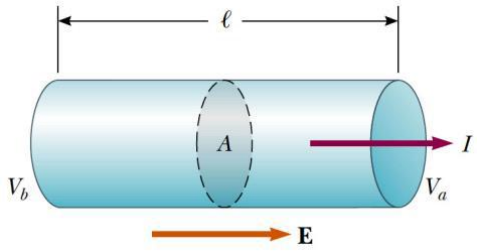


Parallel resistors: $\frac{1}{R_{equiv}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$

II. Resistance depends on temperature

- Whenever the temperature of a conductor is increased by $1^{\circ}C$ hence, resistance of a conductor increased by $\alpha = \frac{R_t - R_0}{R_0 t}$ where, α is called heat coefficient of resistance.

Hence, $R_t = R_0(1 + \alpha t)$.



III. Resistance of a wire

- Resistance of a wire: $R = \rho \frac{l}{A}$

Where, ρ is the resistivity of a wire measured in (Ωm).

$l = \pi DN$ is the length of a wire measured in (m).

N is the number of turns

D is the diameter of a inductor measured in (m).

d is the diameter of a wire measured in (m).

$A = \frac{\pi d^2}{4}$ is the cross-sectional area of a wire measured in (m^2).

IV. Power and Electric Energy

❍ Electromotive force of a source is the size defined by ratio work done and electric charge.

- We can write: $E = \frac{W}{q}$

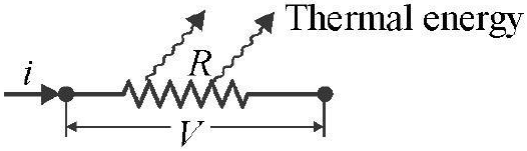
Where, W is the work done of moving electric charge measured in (J).

- Electrical power: $P_e = VI$
- Electric energy: $W = P_e t = VIt$



V. Ohm's conductor

- Voltage across two points of ohm's conductor: $V = RI$
- Electric power of ohm's conductor: $P_e = VI = RI^2 = \frac{V^2}{R}$
- Electric energy of ohm's conductor: $W_e = P_e t = VIt = RI^2 t$



- Efficiency of ohm's conductor: Ohm's conductor gives 100% efficiency.

VI. Electrical device (Motor and electrolysis)

The condition for electrical device to be operated unless: $V > E'$

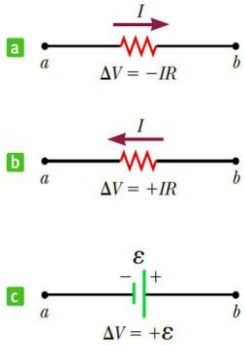
❍ Ohm's law for an electrical device: $V' = E' + r'I$



Where, E' is the reverse electromotive force of electrical device measured in (V).

r' is the internal resistance of the device measured in (Ω).

I is the current flowing through the device measured in (A).



❍ Electrical power for an electric device: $P_e = P_U + P_J$

$$P_e = E'I + r'I^2$$

Where, P_e is total electrical power supplied to the device measured in (W).

P_U is the useful (or mechanical) power output of the device measured in (W).

P_J is the power lost due to Joule heating measured in (W).

❍ Electrical energy for an electrical device: $W_e = W_U + W_J$

$$W_e = P_U t + P_J t$$

$$W_e = E'It + r'I^2 t$$

Where, $W_e = VIt$ is the total electrical energy supplied to the device measured in (J).

$W_U = E'It$ is the useful or mechanical energy output of the device measured in (J).

$W_J = r'I^2 t$ is the energy lost due to Joule heating measured in (J).

❍ Efficiency for an electrical device: $Rd = \dfrac{W_U}{W_e} = \dfrac{P_U}{P_e} = \dfrac{E'}{V}$

Noted: If electro motor is not operating or melt anode in electrolysis: $V = r'I$

I. DC electric source (Battery)

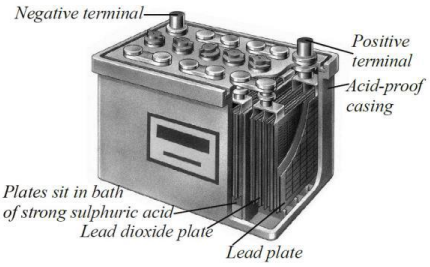
❍ Voltage across a DC electric source: $V = E - rI$

Where, E is the electromotive force of the source measured in (V).

V is the terminal voltage across the DC source measured in (V).

I is the current flowing through the source measured in (A).

r is the internal resistance of the source measured in (A).



- If $r = 0 \Rightarrow V = E =$ constant, battery is an ideal dc electrical source.
- If $I = 0 \Rightarrow V = E$ is the open circuit, there is no electric current across a circuit.
- If $V = 0 \Rightarrow I_{CC} = \dfrac{E}{r}$ is a short circuit.

❍ Electric power of a DC electric source: $P_g = P_e + P_J$

Where, $P_g = EI$ is the chemical or total power output of the electric source power measured in (W).

$P_e = VI$ is the electrical power provided by the electric source measured in (W).

$P_J = rI^2$ is the power lost as heat (Joule heating) in the circuit measured in (W).

❍ Electric energy of a DC electric source: $W_g = W_e + W_J$

Where, $W_g = EIt$ is the total electric energy of the electric source measured in (J).

$W_e = VIt$ is the useful electric energy of the electric source measured in (J).

$W_J = rI^2 t$ is the electrical energy lost as heat (Joule heating) in the circuit measured in (J).

❍ Efficiency of a DC electric source: $Rd = \frac{W_e}{W_g} = \frac{P_e}{P_g} = \frac{V}{E}$

II. General Ohm's Law

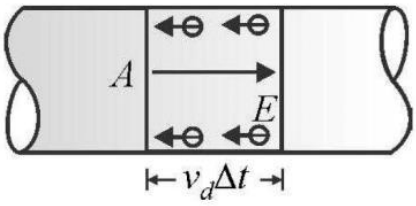
❍ Ohm's law: $E - E' = \Sigma IR$

❍ Battery's capacity

• When a battery charging: $V = E + rI$

• When a battery is operating: $V = E - rI$

❍ Electric current



• Electric current across an electrical wire: $I = \frac{q}{t}$

• Current density: $J = \frac{I}{A}$

• Relationship between current and drift velocity of electron: $I = nAve$, $e = 1.6 \times 10^{-19}$ C

Where, $n = \frac{N}{V}$ is the number of electrons per unit volume measured in $\left(atoms/m^3\right)$.

$n_{atom/m^3} = \frac{\rho}{m_0}$ is the number of atoms per unit volume measured in $\left(atoms/m^3\right)$.

$m_0 = \frac{M}{N_A}$ is the mass of each atom measured in $\left(kg\right)$.

N is the total number of electrons across a cross-sectional area of a conductor.

$V = AL$ is the volume of a conductor measured in $\left(m^3\right)$.

v is the drift velocity measured in $\left(m/s\right)$.

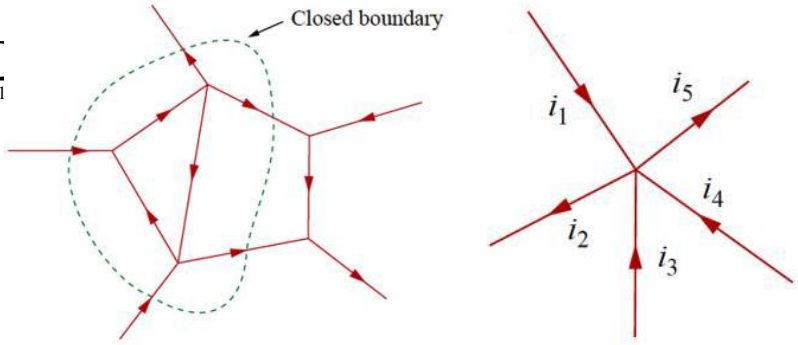
A is the cross-sectional area measured in $\left(m^2\right)$.

J is the current density measured in $\left(A/m^2\right)$.

III. Kirchhoff Law

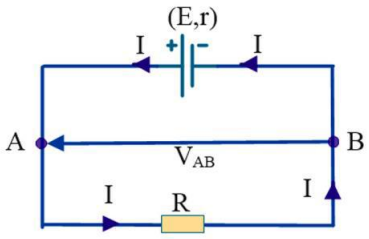
❍ **Kirchhoff's Current Law (KCL)**, also known as the **junction rule**, states that the total current entering a junction (or node) in an electrical circuit must equal the total current leaving the junction. Mathematically, it can be expressed as:

$$\sum_{i=1}^{n}$$



❍ Method how to determine the electric current across closed circuit (electrical circuit is given as follow)

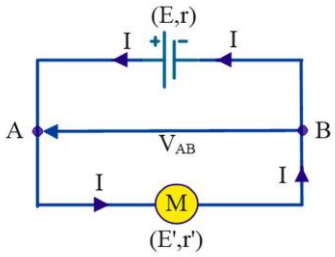
✌ Data given: An electrical circuit has a cell (E,r) connected by two ends and a resistor.



- Ohm's law for a cell: $V_{AB} = E - rI \quad (1)$
- Ohm's law for a resistor: $V_{AB} = RI \quad (2)$
- Comparing equation (1) and (2) thus, $E - rI = RI \Rightarrow RI + rI = E$

Hence, the electric current across closed circuit is $I = \dfrac{E}{R+r}$

✌ Data given: An electrical circuit has a cell (E,r) and an electric motor (E',r')

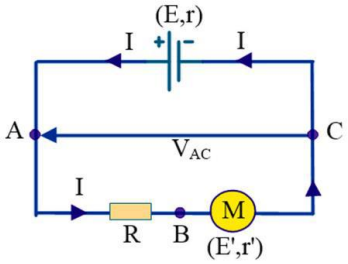


- Ohm's law for a cell: $V_{AB} = E - rI \quad (1)$
- Ohm's law for an electric motor: $V_{AB} = E' + r'I \quad (2)$
- Comparing equation (1) and (2) thus, $E - rI = E' + r'I$

$$E - E' = rI + r'I$$

Hence, the electric current across closed circuit is $I = \dfrac{E - E'}{r + r'}$

✌ Data given: An electrical circuit has a cell (E,r) and an electric motor (E',r') and resistor are connected in series as shown in figure.



- Ohm's law for a cell: $V_{AC} = E - rI \quad (1)$
- For a resistor and an electric motor: $V_{AC} = V_{AB} + V_{BC}$
- Ohm's law for a resistor: $V_{AB} = RI$
- Ohm's law for a motor: $V_{BC} = E' + r'I$

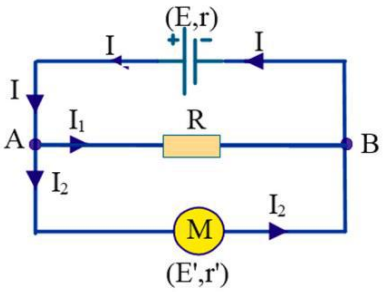
$$V_{AC} = RI + E' + r'I \quad (2)$$

- Comparing equation (1) and (2) thus, $E - rI = RI + E' + r'I \Leftrightarrow E - E' = RI + r'I + rI$

$$E - E' = \left(R + r' + r\right)I$$

Hence, the electric current across closed circuit is $I = \dfrac{E - E'}{R + r + r'}$

❍ Method how to determine the electric current I_1, I_2 and I_3 across closed circuit (electrical circuit is given as follow)



- Since, a motor and a resistor are connected in parallel: $I = I_1 + I_2 \quad (a)$
- Ohm's law for a cell: $V_{AB} = E - rI \Rightarrow I = \dfrac{E - V_{AB}}{r} \quad (1)$
- Ohm's law for a motor: $V_{AB} = E' + r'I_2 \Rightarrow I_2 = \dfrac{V_{AB} - E'}{r'} \quad (2)$
- Ohm's law for a resistor: $V_{AB} = RI_1 \Rightarrow I_1 = \dfrac{V_{AB}}{R} \quad (3)$

• Substitute (1), (2), and (3) into equation (a): $\frac{E-V_{AB}}{r}=\frac{V_{AB}}{R}+\frac{V_{AB}-E'}{r'}$

$$\frac{E}{r}-\frac{V_{AB}}{r}=\frac{V_{AB}}{R}+\frac{V_{AB}}{r'}-\frac{E'}{r'}\Rightarrow\frac{E}{r}+\frac{E}{r'}=\frac{V_{AB}}{R}+\frac{V_{AB}}{r'}+\frac{V_{AB}}{r}$$

$$E\left(\frac{1}{r}+\frac{1}{r'}\right)=V_{AB}\left(\frac{1}{R}+\frac{1}{r'}+\frac{1}{r}\right)$$

Hence, voltage across AB is $V_{AB}=\dfrac{\frac{E}{r}+\frac{E'}{r'}}{\frac{1}{R}+\frac{1}{r'}+\frac{1}{r}}$, substitute V_{AB} in equation (1), (2), and (3) to determine the electric current I_1, I_2 and I_3 across closed circuit.

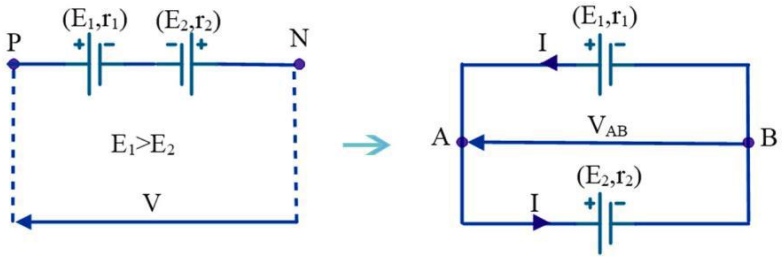
❍ Method how to determine the electric current across closed circuit (opposite combination of a cell)

Noted: If two cells of a source E_1 and E_2 combine opposite to each other (Where $E_1 > E_2$)

We get, E_1 is acting as an electric source: $V = E_1 - r'I$

E_2 is acting as an electrical component: $V = E_2 + r_2 I$

A. Two cells are opposite arranged as shown in figure.



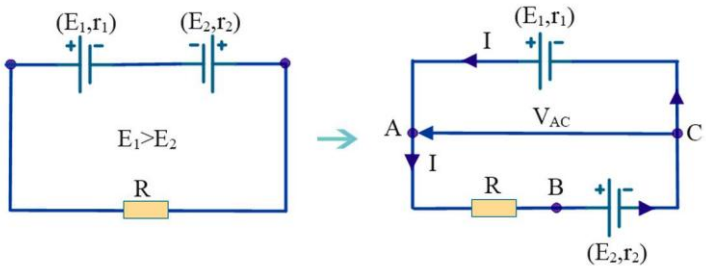
• Ohm's law of a cell: $V_{AB} = E_1 - r_1 I \quad (1)$

• Ohm's law for an electrical component: $V_{AB} = E_2 + r_2 I \quad (2)$

• Comparing equation (1) and (2) thus, $E_1 - r_1 I = E_2 + r_2 I \Rightarrow E_1 - E_2 = I\left(r_1 + r_2\right)$

Hence, the electric current across closed circuit is $I = \dfrac{E_1 - E_2}{r_1 + r_2}$

B. Two cells are opposite arranged and have a resistor as shown in figure



• Ohm's law of a cell: $V_{AC} = E_1 - r_1 I \quad (1)$

• By summation of voltage: $V_{AC} = V_{AB} + V_{BC}$

Since, $V_{AB} = RI, V_{BC} = E_2 + r_2 I$ $\left(E_2 \text{ is an electrical component}\right)$

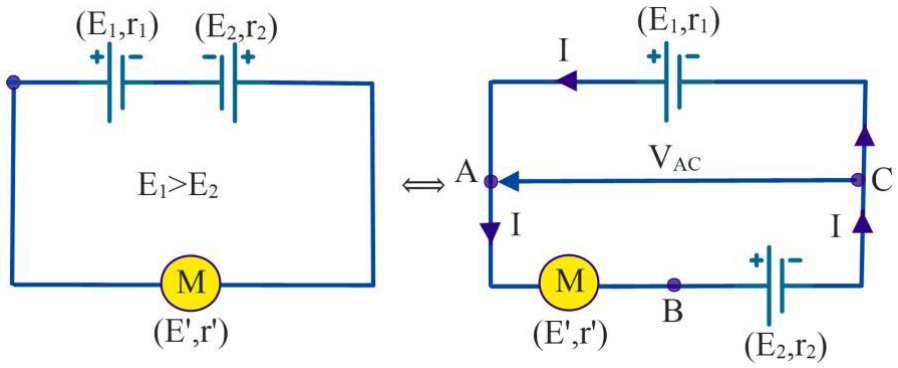
So, $V_{AC} = RI + E_2 + r_2 I$ (2)

- Comparing equation (1) and (2) thus, $E_1 - r_1 I = RI + E_2 + r_2 I$

$$E_1 - r_1 I = RI + E_2 + r_2 I \Rightarrow E_1 - E_2 = I\left(R + r_2 + r_1\right)$$

Hence, the electric current across closed circuit is $I = \dfrac{E_1 - E_2}{R + r_1 + r_2}$

C. **Two cells are opposite arranged and have a motor as shown in figure**



- Ohm's law of a cell: $V_{AC} = E_1 - r_1 I$ (1)
- By summation of voltage: $V_{AC} = V_{AB} + V_{BC}$

Since, $V_{AB} = E' + r' I, V_{BC} = E_2 + r_2 I$

So, $V_{AC} = E' + r' I + E_2 + r_2 I$ (2)

- Comparing equation (1) and (2) thus, $E_1 - r_1 I = E' + r' I + E_2 + r_2 I$

$$E_1 - \left(E' + E_2\right) = I\left(r' + r_1 + r_2\right)$$

Hence, the electric current across closed circuit is $I = \dfrac{E_1 - \left(E' + E_2\right)}{r' + r_1 + r_2}$

Chapter1 Thermal Physics
Lesson1 Kinetic Theory of ideal gases

I. Temperature Scales

❍ Relationship between temperature in Kelvin and Celsius is given by

$$T = t + 273$$

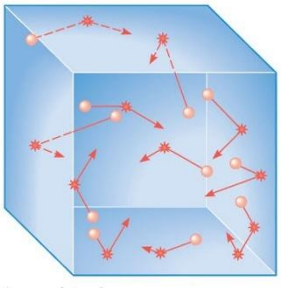
❍ Relationship between temperature in Celsius and Fahrenheit is given by

$$T_F = \frac{9}{5}T_C + 32$$

II. Kinetic theory of gasses

The kinetic theory of gases describes the microscopic properties of an ideal gas, based on several assumptions:

- All molecules in the gas are identical.
- The number of molecules in the gas is large, and the average separation between them is large compared with their dimensions.
- The molecules execute random motion and may change their direction of motion after every collision.
- The molecules make elastic collisions with each other and with the walls of
- Molecules don't exert force on each other except during a collision.



III. Pressure of an ideal gas theory.

The pressure of N molecules of an ideal gas contained in a volume V is: $P = \frac{2}{3}\left(\frac{N}{V}\right)\frac{1}{2}m_0(v^2)_{av}$

Where, P is the pressure of gas measured in (Pa).

V is the volume of a container measured in (m^3).

N is the number of molecules.

m_0 is the mass of one molecule measured in (kg).

$(v)_{av}$ is the average speed of molecule of an ideal gas measured in (m/s).

IV. Kinetic energy and temperature

Ideal Gases Equation

The pressure of an ideal gas depends on temperature T, number of molecules N and volume V of an ideal gas by experiment show that

- Pressure is proportional to the temperature: $P \propto T$
- Pressure is proportional to the number of molecules: $P \propto N$
- Pressure is inversely proportional to the volume of an ideal gas: $P \propto 1/V$

$$P \propto \frac{NT}{V} \Rightarrow P = k_B \frac{NT}{V} \Leftrightarrow PV = Nk_BT$$

Where, k_B is Boltzmann's constant: $k_B = 1.38 \times 10^{-23}\ \text{J/K}$

$$N = nN_A \quad \left(N_A = 6.022 \times 10^{23}\ \text{particles/mole}\right)$$

$$R = k_B N_A = 1.38 \times 10^{23}\ \text{J/K} \times 6.022 \times 10^{23}\ \text{particles/mole} = 8.31\ \text{J/mol.K}$$

Hence, an ideal gas equation can be written as: $PV = nRT$

IV. Kinetic energy and temperature

We have: $P = \frac{2}{3}\left(\frac{N}{V}\right)\frac{1}{2}m_0(v^2)_{av}$

Since, $K_{av} = \frac{1}{2}m_0(v^2)_{av}$

Hence, $P = \frac{2}{3}\left(\frac{N}{V}\right)K_{av} \Rightarrow K_{av} = \frac{3}{2}\frac{PV}{N} = \frac{3}{2}\frac{Nk_BT}{N} = \frac{3}{2}k_BT$

Total kinetic energy of N molecules: $K = NK_{av} = \frac{3}{2}Nk_BT = \frac{3}{2}nRT = \frac{3}{2}PV$

V. Root-mean-square speed of molecules v_{rms}

Average kinetic energy of each molecule is $K_{av} = \frac{3}{2}k_BT = \frac{1}{2}m_0(v^2)_{av} \Rightarrow (v^2)_{av} = \frac{3k_BT}{m_0}$

Root-mean-square speed: $v_{rms} = \sqrt{(v^2)_{av}} = \sqrt{\frac{3k_BT}{m_0}}$

Since, molar mass $M = m_0N_A \Rightarrow m_0 = \frac{M}{N_A}$

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

Using relation: $P = \frac{2}{3}\left(\frac{N}{V}\right)\left(\frac{1}{2}m_0(v^2)_{av}\right) = \frac{1}{3}\frac{Nm_0v_{rms}^2}{V} = \frac{1}{3}\frac{mr_{rms}^2}{V} = \frac{1}{3}\rho v_{rms}^2$

$$v_{rms} = \sqrt{\frac{3P}{\rho}}$$

Where, ρ is density measured in $\left(\text{kg}/\text{m}^3\right)$.

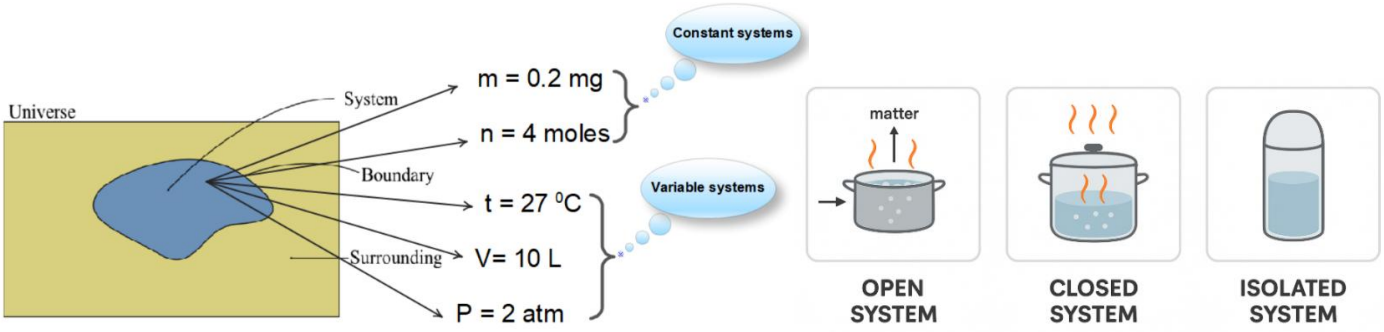
Review

- Number of moles: $n = \frac{m}{M} = \frac{N}{N_A} = \frac{V}{V_{mole}}$ $\quad \therefore V_{mole} = 22.4\ l/\text{mole at STP}$.
- Mass of one molecule: $m_0 = \frac{m}{N} = \frac{M}{N_A}$
- Total mass of ideal gases: $m = m_0 N$
- Normal temperature and pressure (NTP): $T = 20\,^{\circ}\text{C} = 293\ \text{K}, P = 1\text{atm} = 1\times10^5\text{Pa}$
- Standard temperature and pressure (STP): $T = 0\,^{\circ}\text{C} = 273\ \text{K}, P = 1\text{atm} = 1\times10^5\text{Pa}$
- Relation between pressure in Pascal and millimeter mercury: $760\ \text{mmHg} = 76\ \text{cmHg} = 1\times10^5\ \text{Pa}$
- Submultiple unit of volume: $1l = 1dm^3, 1\text{m}l = 1\times10^{-3}l = 1\times10^{-3}\text{dm}^3 = 1\times10^{-6}\text{m}^3$

Chapter1 Thermal Physics
Lesson 2 First Law of Thermodynamics

I. Thermodynamics System

❍ In thermodynamics, a **system** is an object or set of objects that we study in relation to its external environment. Everything outside of this boundary is called the **surroundings** or the **environment**. The system can exchange energy, matter, or both with its surroundings, depending on the type of system.



II. State of a system in thermodynamics

❍ In thermodynamics, the **state of a system** is a set of variables that describe its physical condition or properties at a given moment. The state of a system can be described by various **state variables** or **state functions**, such as pressure, volume, temperature, and composition. Furthermore, the state of a system is classified into two types: **constant systems** and **variable systems**.

* There are three main types of thermodynamic systems:

Non-Cyclic Process (Open System):

An open system or non-cyclic process can exchange both matter and energy (heat or work) with its surroundings. Hence, the initial state and final state of a system is not the same.

Example: A boiling pot of water without a lid. Heat and water vapor can escape into the surroundings.

Cyclic Process (Closed System):

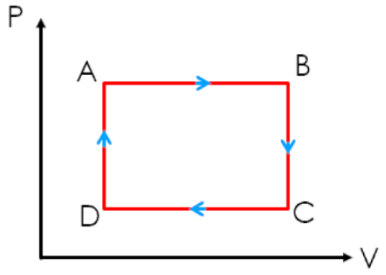
A **cyclic process** is the process in which the initial and final states of a system are the same.

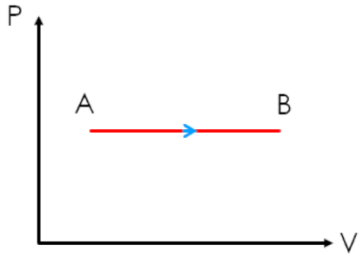
Example: A sealed pressure cooker. Heat can enter or leave, but the steam (matter) stays inside.

Isolated System:

An isolated system cannot exchange either matter or energy with its surroundings.

Example: An insulated thermos bottle with a hot liquid inside (though in practice, perfect isolation is difficult to achieve.



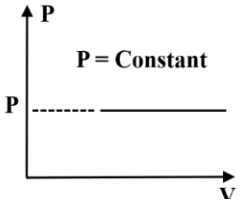


III. Thermodynamics process

❍ **A thermodynamic system** is a system that undergoes thermodynamic processes and changes its state from an initial state to a final state through various thermodynamic processes.

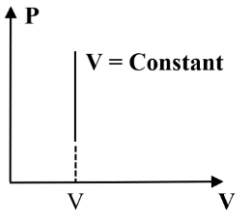
1. The equation of combined gas law is given by

$$PV = nRT \Leftrightarrow \frac{PV}{T} = nR = \text{constant} \Rightarrow \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$



2. Isobaric process is defined as a process that occurs at constant pressure.

$$\frac{V_1}{V_2} = \frac{V_2}{T_2}$$



3. Isochoric process is defined as a process that occurs at constant volume.

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

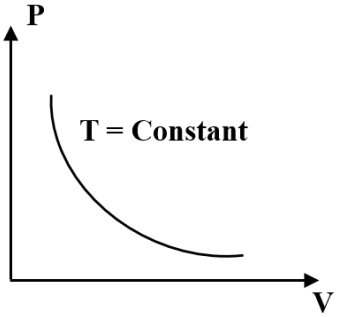
4. Isothermal process is defined as a process that occurs at constant temperature. Isothermal process, when a gas expands or compresses isothermally (constant temperature) thus

$$P_1V_1 = P_2V_2$$

Where, P_1 is initial pressure measured in (Pa).

P_2 is final pressure measured in (Pa).

V_1 is initial volume measured in (m^3).



V_2 is final volume measured in $\left(\mathrm{m}^3\right)$.

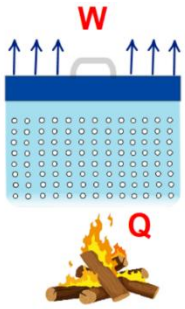
T_1 is initial temperature measured in (K).

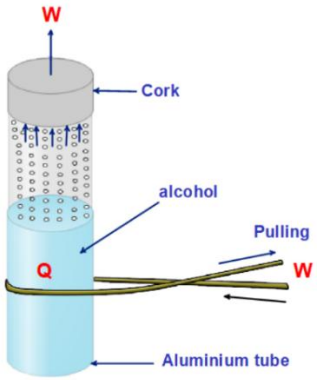
T_2 is final temperature measured in (K).

IV. First Law of thermodynamics

❍ **First law of thermodynamic:** In thermodynamic process, heat absorbed by a system is the sum between work done by the system and the change in internal energy.

$$Q = \Delta U + W \Leftrightarrow \Delta U = Q - W$$





∗**Noted:** Heat and work are equivalent to each other, meaning that heat can be used to perform work, and work can be converted into heat.

❍ **Sign Convention for first law thermodynamic**

Heat Added to the System (Q):

Positive (+) when heat is added to the system from the surroundings.

Negative (−) when heat is removed from the system (i.e., heat is lost to the surroundings).

Work Done by the System (W):

Positive (+) when the system does work on the surroundings (e.g., expansion work).

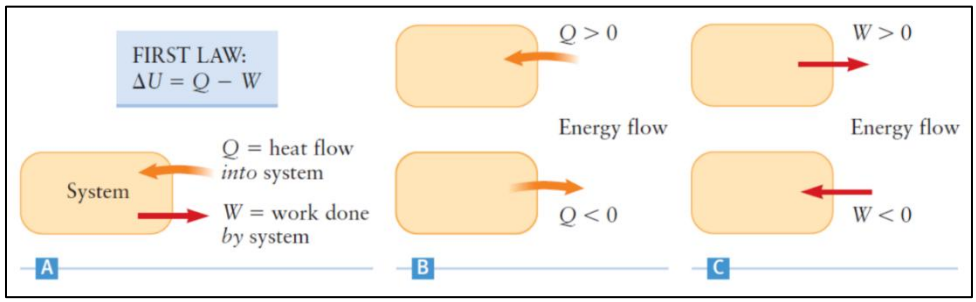
Negative (−) when work is done on the system by the surroundings (e.g., compression work).

Positive (ΔU> 0): When the internal energy of the system increases.

This can occur if the system absorbs heat (Q>0) and/or does less work on the surroundings (W is relatively small or negative).

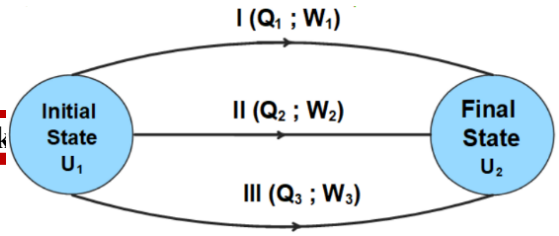
Negative (ΔU< 0): When the internal energy of the system decreases.

This can occur if the system loses heat (Q<0) and/or does more work on the surroundings (W is relatively large and positive).



❍ **Principle of initial and final states in thermodynamics**

The principle of initial and final states stated that when a system changes from an initial state to a final state by exchanging only heat Q and work W with its surroundings, the algebraic sum $(Q-W)$ depends only on the initial and final states of the system and not on the path taken during the process.



$$\Delta U = Q_1 - W_1 = Q_2 - W_2 = Q_3 - W_3 = U_2 - U_1 = \text{constant}$$

❍ Internal energy of first law of thermodynamic: Total kinetic energy of N molecules:

$$U = K = NK_{av} = \frac{3}{2}Nk_BT = \frac{3}{2}nRT = \frac{3}{2}PV$$

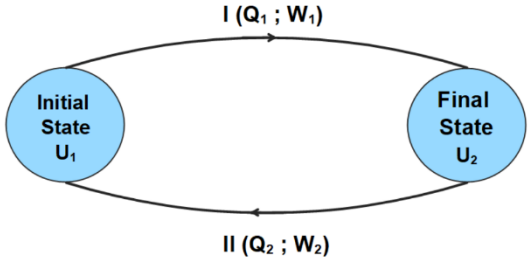
❍ The change in internal energy of an ideal gas

$$\Delta U = U_f - U_i = \frac{3}{2}Nk_B\left(T_f - T_i\right) = \frac{3}{2}Nk_B\Delta T = \frac{3}{2}nR\Delta T$$

❍ The **principle of equivalence** stated that when a system undergoes a cyclic process by exchanging only heat Q and work W with its surroundings.

- If the system does work ($W > 0$), it absorbs the amount of heat ($Q > 0$).
- If the system receives work ($W < 0$), it expels the amount of heat ($Q < 0$).

❍ If a system changes its state from (1) to (2) and return from state (2) to state (1) then we get,



- From state (1) to (2): $\Delta U = U_2 - U_1 = Q_1 - W_1$
- From state (2) to (1): $\Delta U_2 = U_1 - U_2 = Q_2 - W_2$
- Total internal energy in one complete one cycle process:

$$\Delta U = \Delta U_1 + \Delta U_2 = \left(U_2 - U_1\right) - \left(U_1 - U_2\right) = 0$$

$$\left(Q_1 - W_1\right) + \left(Q_2 - W_2\right) = 0$$

$$\left(Q_1 + Q_2\right) - \left(W_1 + W_2\right) = 0$$

Let, $Q = Q_1 + Q_2$ and $W = W_1 + W_2$, $Q = W$, $Q = W$

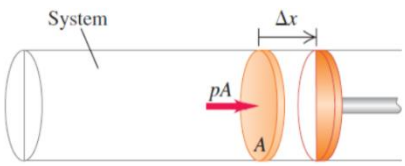
∗**Noted:** For a **closed system**, when the system undergoes a complete cycle, it exchanges only heat and work with the environment, but its internal energy returns to the initial state.

Work done in thermodynamic processes

❍ Isochoric process is the process that volume of the system in thermodynamic process is constant (V = constant)

$$W = P\Delta V = P\left(V_f - V_i\right) = 0 \quad \left(V_f = V_i = \text{constant}\right)$$

Work done by an ideal gas at constant pressure is



$W = P\Delta V = F\Delta x = F\left(x_f - x_i\right) \quad \therefore F = PA$

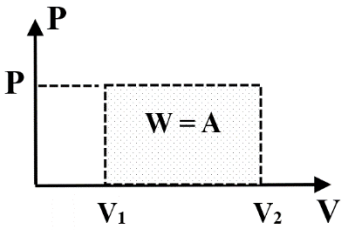
We get, $W = PA\left(x_f - x_i\right) = P\left(Ax_f - Ax_i\right) = P\left(V_f - V_i\right)$

Hence, $W = P\Delta V$

❍ **Work done at constant pressure**

Work done by an ideal gas is $W = P\Delta V$

Work done by an ideal gas is the area of rectangle with length and breadth P and ΔV.



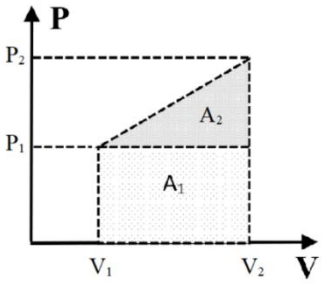
❍ **Work done at pressure increases linearly**

Work done by an ideal gas is $W = A_1 + A_2$

Area: $A_1 = P_1\left(V_2 - V_1\right) = P_1\Delta V_1$

Area: $A_2 = \frac{1}{2}\left(P_2 - P_1\right)\left(V_2 - V_1\right) = \frac{1}{2}\Delta P\Delta V$

Work done by an ideal gas: $W = P_1\Delta V + \frac{1}{2}\left(P_2 - P_1\right)\Delta V$



∗ **Noted:** In case, pressure equally increases is $W = P_{av}\Delta V \;\therefore P_{av} = \frac{P_1 + P_2}{2}$

❍ **Work done at constant temperature**

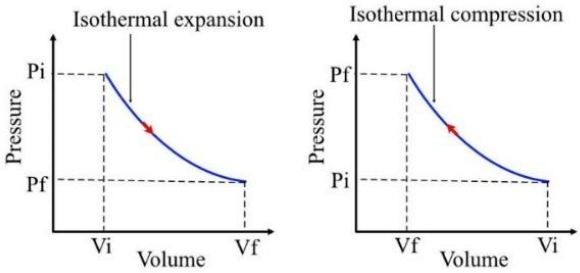
Work done by an ideal gas is $dW = PdV$

Using ideal gas equation: $PV = nRT \Rightarrow P = \frac{nRT}{V}$

$$\Rightarrow W = \int dW = \int_{V_1}^{V_2} PdV = \int_{V_1}^{V_2} \frac{nRT}{V}dV$$

$$\Leftrightarrow W = nRT\int_{V_1}^{V_2} \frac{dV}{V} = nRT\left(\ln V_2 - \ln V_1\right)$$

Hence, $W = nRT\ln\frac{V_2}{V_1}$



Isothermal processes

❍ **Heat transfer at Constant Volume Process (Isochoric):**

At constant volume, the gas does no work (since $\Delta V = 0$) and all the heat added goes into changing the internal energy.

The first law of thermodynamics: $\Delta U = Q_V$

Where Q_V is the heat added at constant volume.

For an ideal gas, the change in internal energy at constant volume is related to temperature:

$$\Delta U = nC_v\Delta T$$

Where: n is the number of moles,

C_v is the molar heat capacity at constant volume,

ΔT is the change in temperature.

❍ **Heat transfer at Constant Pressure Process (Isobaric):**

At constant pressure, the gas does work as it expands or contracts, so the heat added goes into both changing the internal energy and doing work.

The first law of thermodynamics: $Q_P = \Delta U + W$

Where Q_P is the heat added at constant pressure, and $W = P\Delta V$ is the work done by the gas.

For an ideal gas at constant pressure, the heat added is related to temperature as:

$$Q_P = nC_p\Delta T$$

Where: C_p is the molar heat capacity at constant pressure.

Since the change in internal energy ΔU is always given by $nC_v\Delta T$, the difference between Q_P and Q_V arises because of the work done during expansion:

$$Q_P = \Delta U + P\Delta V = nC_p\Delta T$$

The relation between C_p and C_V is given by:

$$C_p - C_v = R$$

Where R is the ideal gas constant. This reflects the additional energy required to do work at constant pressure.

Chapter1 Thermal Physics

Lesson3 Engine

I. Adiabatic process

An adiabatic process is a process that happened when there is no energy transferred into the system and out of the system so, heat $Q = 0$.

Apply first law of thermodynamic: $\Delta U = Q - W \quad (Q = 0)$

$$\Delta U = -W$$

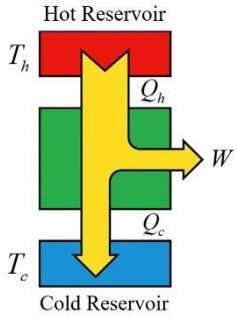
II. Carnot theory and maximum efficiency

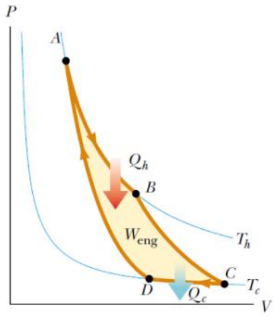
❍ In 1824, **Sadi Carnot** introduced a theoretical concept of engines that used heat to perform work, meaning heat energy was used to do work. His method of designing the machine depends on the reversibility of the cycle, known as the Carnot cycle.

❍ The **Carnot engine** is a theoretical or ideal engine due to its maximum heat efficiency. It absorbs heat energy from a hot reservoir at a high temperature to perform work, then transfers the remaining heat energy to a cold reservoir at a lower temperature.

❍ No real heat engine operating between two energy reservoirs can be more efficient than a Carnot engine operating between the same two reservoirs.

∗ **Diagram illustrating the operation of the Carnot cycle**





❍ The heat energy Q_h represents the amount of energy absorbed by the gas from the hot reservoir.

❍ The heat energy Q_c represents the amount of energy expelled by the gas to the cold reservoir.

❍ The **net work done (W)** by the Carnot engine over one complete cycle is the difference between the heat absorbed and the heat rejected.

$$W = Q_h - Q_c$$

The thermal efficiency of the engine is: $e = \frac{W}{Q_h} = \frac{Q_h - Q_c}{Q_h} = 1 - \frac{Q_c}{Q_h}$

For (Carnot cycle) ideal machine: $\frac{Q_c}{Q_h} = \frac{T_c}{T_h} \Rightarrow e = 1 - \frac{T_c}{T_h}$

Hence, the thermal efficiency of a Carnot engine is $e = 1 - \frac{T_c}{T_h} = 1 - \frac{Q_c}{Q_h} = \frac{W}{Q_h}$

Where, Q_h is the heat absorbed by hot reservoir (J), Q_c is the heat absorbed by cold reservoir (J).

T_h is heat at hot reservoir (J), T_c is heat at cold reservoir (J), e is the Carnot efficiency.

Noted:

❍ All engines that operate using heat involve two heat reservoirs with different temperatures (T_c and T_h). The thermal efficiency reaches its maximum value when the working fluid follows a reversible cycle.

❍ The maximum thermal efficiency does not depend on the type of working fluid or the specific process of the reversible cycle.

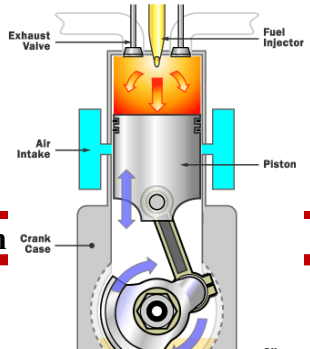
❍ The thermal efficiency depends on the temperatures of the hot reservoir (T_h) and the cold reservoir (T_c).

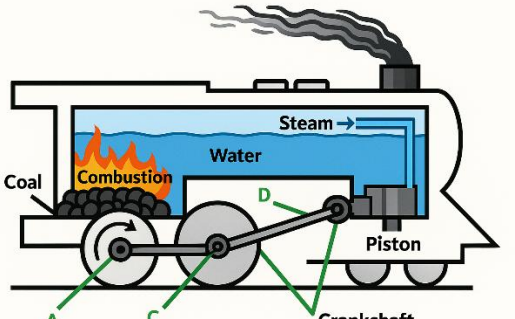
III. Gasoline and Diesel Engines

❍ Engines or motors that operate using heat are called heat engines or motors.

❍ Engines are classified into two types: internal combustion engines and external combustion engines.

- An **internal combustion engine (ICE)** is an engine in which the combustion chamber is the place where the combustion of fuel is used to perform work.
- An **external combustion engine (ECE)** is an engine in which the combustion chamber is outside the place where the combustion of fuel is used to perform work.



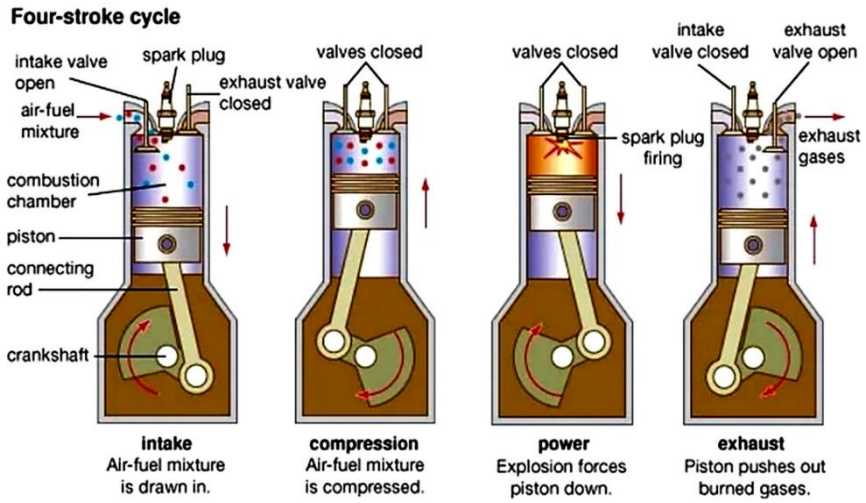


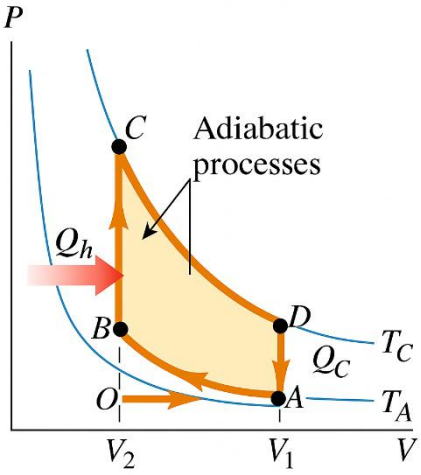
Internal combustion engine **External combustion engine**

❍ Internal combustion engines are classified into two types: spark-ignition engines (powered by a spark) and compression-ignition engines (such as diesel engines).

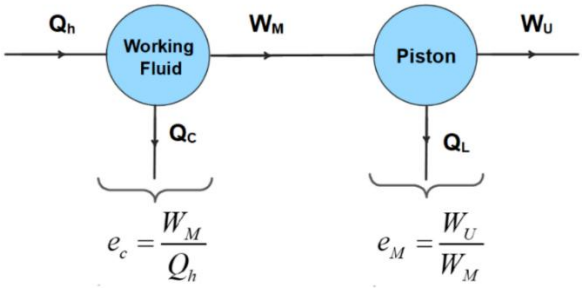
❍ A **four-stroke engine** is a type of internal combustion engine that completes a power cycle in four distinct strokes of the piston within the cylinder. The four strokes are:

- **Intake Stroke**: The intake valve opens, and the piston moves down, drawing a mixture of fuel and air into the cylinder.
- **Compression Stroke**: The intake valve closes, and the piston moves back up, compressing the fuel-air mixture.
- **Power Stroke**: The spark plug ignites the compressed fuel-air mixture, causing an explosion that forces the piston back down, producing power.
- **Exhaust Stroke**: The exhaust valve opens, and the piston moves back up, expelling the exhaust gases from the cylinder.





Noted: One complete cycle corresponds to four processes, which result in two rotations of the crankshaft, producing one power stroke and one intake process.



❍ Efficiency of four strokes engine

- Thermal efficiency: $e_c = \dfrac{W_M}{Q_h}$

  Where, W_M is mechanical work measured in (J), Q_h is the heat energy measured in (J).

- Mechanical efficiency: $e_M = \dfrac{W_U}{W_M}$

  Where, W_U is useful work measured in (J), W_M is the mechanical work measured in (J).

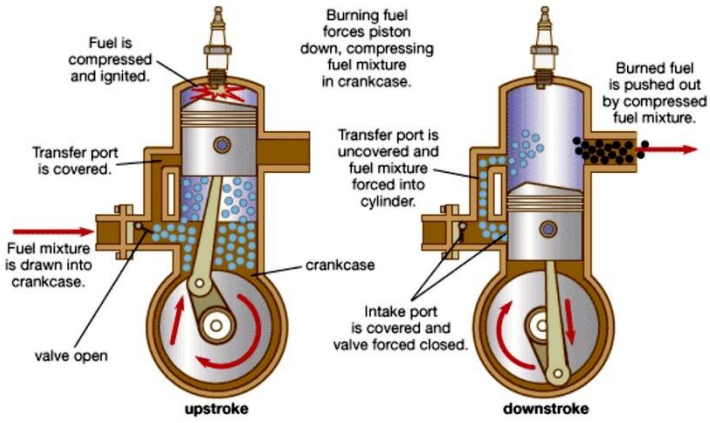
- Useful efficiency or overall efficiency: $e = \dfrac{W_U}{Q_h} = \dfrac{W_U}{W_M} \times \dfrac{W_M}{Q_h} = e_M \times e_c$

- If the air–fuel mixture is assumed to be an ideal gas, then the efficiency of the Otto cycle is

$$e = 1 - \frac{1}{\left(V_1/V_2\right)^{\gamma-1}}$$

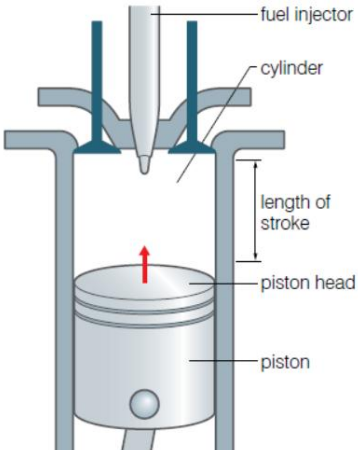
❍ A **two-stroke engine** is another type of internal combustion engine, and it completes a power cycle in just two strokes of the piston.

- **First Stroke (Compression and Exhaust)**: The piston compresses the fuel-air mixture as it moves up the cylinder, while simultaneously pushing the exhaust gases out.
- **Second Stroke (Power and Intake)**: The piston moves down, causing the compressed fuel-air mixture to ignite, which drives the piston down (power stroke). At the same time, a new fuel-air mixture is drawn into the cylinder.



IV. Diesel engine

❍ A **diesel engine** is a type of internal combustion engine that operates using **compression ignition**, where the fuel is ignited by the heat generated from compressing air in the cylinder, rather than using a spark plug like in a **gasoline engine**. Diesel engines are commonly used in vehicles, industrial machines, and heavy duty applications due to their fuel efficiency and durability.



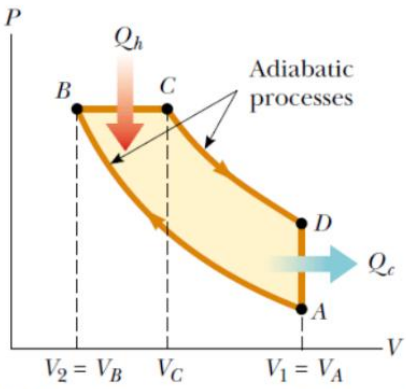


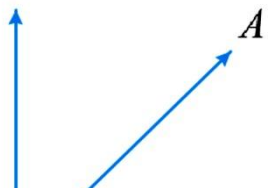
Figure 22.14 *PV* diagram for an ideal diesel engine.

Chapter2 Wave
Lesson1 Superposition of wave and standing wave

I. Superposition

❍ Superposition of waves is a fundamental principle in wave theory that states that when two or more waves overlap or travel through the same medium at the same time, the resulting wave is the sum of the individual waves. This principle applies to all types of waves, including sound waves, light waves, and water waves.

II. The initial condition of a wave medium



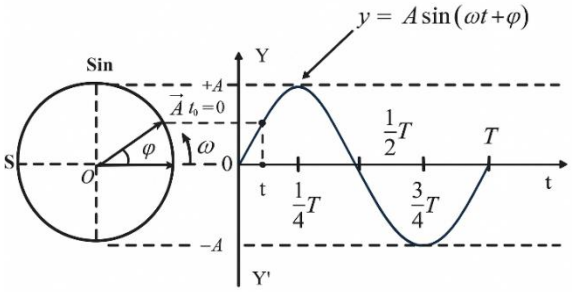
❍ The equation of a simple harmonic motion is given as: $y = A\sin\left(\omega t + \phi\right)$

Where:

A is the amplitude or maximum displacement (in meters),
ω is the angular velocity (in radians per second),
ϕ is the phase angle (in radians),
$\left(\omega t + \phi\right)$ is the phase of motion at any time t (in radians),
y is the displacement or maximum displacement of the particle at time t (in meters).

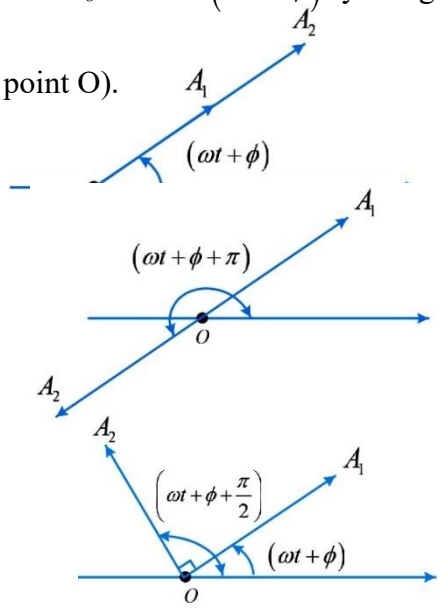
∗ The **Fresnel Method** is a technique used in periodic functions, such as simple harmonic motion. This method uses A as a vector amplitude or a vector Fresnel that rotates with an angular frequency ω at an instant equal to zero, with a phase angle φ, where φ is the angle between vector amplitude A and the x-axis.

$$y = A\sin\left(\omega t + \phi\right) \Rightarrow \vec{A}\begin{vmatrix}\omega \\ \phi = \left(\vec{A}, \overrightarrow{ox}\right)\end{vmatrix}$$



❍ A phasor is a vector that rotates in a plane. It represents a sinusoidal function $y = A\sin\left(\omega t + \phi\right)$ by using both magnitude and direction.

Origin: The object oscillates around its equilibrium point (denoted as point O).
Phase: Created an initial phase respect to ox-axis.
Direction: Outward.
Magnitude: Amplitude of a wave.



❍ **Special cases**

Equal Phases (in phase)
If the phases are equal, then the change in phase is given by:

$$\phi_1 = \phi_2 \Rightarrow \Delta\phi = \phi_2 - \phi_1 = 0$$

Phase Difference of π (in antiphase)
If the phases differ by π, then the change in phase is given by:

$$\Delta\phi = \phi_2 - \phi_1 = \pi$$

Phase Difference of π/2 (90 degrees out of phase)
If the phases differ by π/2, then the change in phase is given by:

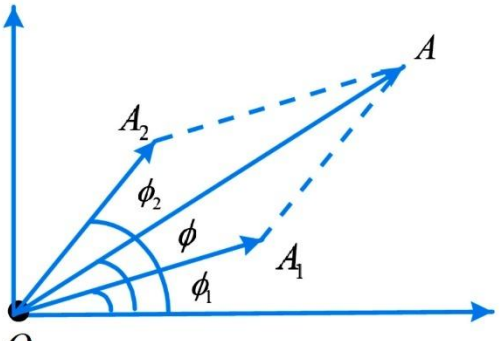
$$\Delta\phi = \phi_2 - \phi_1 = \frac{\pi}{2}$$

III. Superposition of sinusoidal

❍ When two waves overlap, their amplitudes combine, resulting in a new wave. This principle is known as the superposition of waves.

We have, $y_1 = A_1\sin\left(\omega t + \phi_1\right)$ and $y_2 = A_2\sin\left(\omega t + \phi_2\right)$

Using superposition: $y = y_1 + y_2 = A_1\sin\left(\omega t + \phi_1\right) + A_2\sin\left(\omega t + \phi_2\right)$



$$y = A\sin\left(\omega t + \phi\right)$$

❍ Amplitude and Phase

In case the same amplitude

$$y = y_1 + y_2 = A\sin\left(\omega t + \phi_1\right) + A\sin\left(\omega t + \phi_2\right)$$

$$y = A\left[\sin\left(\omega t + \phi_1\right) + \sin\left(\omega t + \phi_2\right)\right]$$

Using formula: $\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$

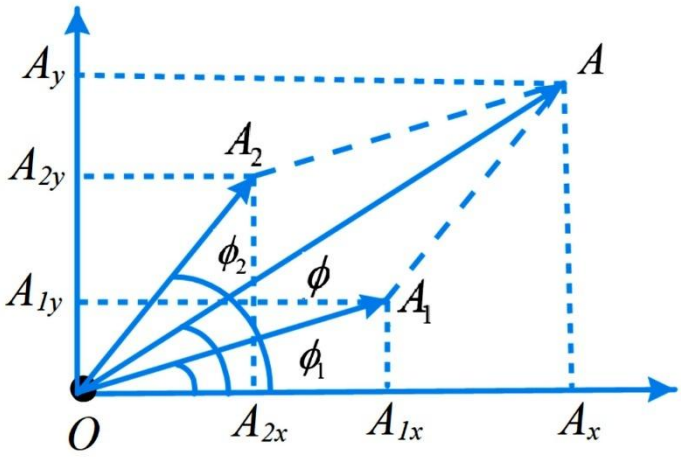
$$y = A\left[2\sin\left(\frac{(\omega t + \phi_1) + (\omega t + \phi_2)}{2}\right)\cos\left(\frac{(\omega t + \phi) - (\omega t + \phi_2)}{2}\right)\right]$$

$$y = 2A\cos\left(\frac{\phi_1 - \phi_2}{2}\right)\sin\left(\omega t + \frac{\phi_1 + \phi_2}{2}\right)$$

In case different amplitude

$$y = y_1 + y_2 = A_1\sin\left(\omega t + \phi_1\right) + A_2\sin\left(\omega t + \phi_2\right)$$

Superposition wave equation: $y = A\sin\left(\omega t + \phi\right)$



Amplitude A

$$\vec{A} = \vec{A}_1 + \vec{A}_2$$

$$A^2 = A_1^2 + A_1^2 + 2A_1A_2\cos\left(\phi_2 - \phi_1\right)$$

And $\vec{A} = \vec{A}_x + \vec{A}_y$

$$A_x = A_{1x} + A_{2x} = A_1\cos\phi_1 + A_2\cos\phi_2$$

$$A_y = A_{1y} + A_{2y} = A_1\sin\phi_1 + A_2\sin\phi_2$$

$$A = \sqrt{A_x^2 + A_y^2} = \sqrt{\left(A_1\cos\phi_1 + A_2\cos\phi_2\right)^2 + \left(A_2\sin\phi_1 + A_2\sin\phi_2\right)^2}$$

❍ **Special case**

If $(\phi_2 - \phi_1 = 0)$ then $A = A_1 + A_2$ and $\phi = \phi_1 = \phi_2$

If $\phi_2 - \phi_1 = \pi/2$ then $A = \sqrt{A_1^2 + A_2^2}$

$$\tan\varphi = \frac{A_1\sin\phi_1 + A_2\sin\phi_2}{A_1\cos\phi_1 + A_2\cos\phi_2}$$

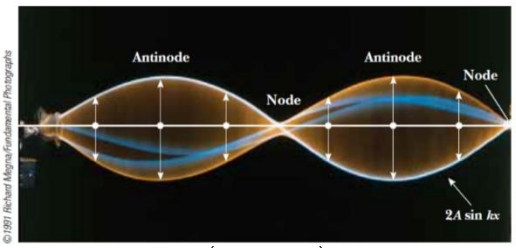
If $\phi_2 - \phi_1 = \pi$

If $A_1 > A_2$ then $A = A_1 - A_2$ and $\phi = \phi_1$

If $A_1 < A_2$ then $A = A_2 - A_1$ and $\phi = \phi_2$

IV. Standing Waves

❍ A standing wave is a wave that remains in a constant position, resulting from the interference of two waves traveling in opposite directions with the same frequency and amplitude. Unlike traveling waves, which move through a medium, standing waves oscillate in place.



Wave equation: $y_1 = A\sin\left(kx - \omega t\right)$ and $y = A\sin\left(kx + \omega t\right)$

The equations represent two waves traveling in opposite directions:
Wave 1: moving in the positive + x direction
Wave 2: moving in the negative - x direction
And, wave number: $k = 2\pi/\lambda$

Superposition wave: $y = y_1 + y_2 = A\left[\sin\left(kx - \omega t\right) + \sin\left(kx + \omega t\right)\right]$

Using formula: $\sin a \pm \sin b = \sin a \cos \pm \cos a \sin b$

$$y = A\left[\sin kx\cos\omega t - \cos kx\sin\omega t + \sin kx\cos\omega t + \cos kx\sin\omega t\right]$$

$$y = 2A\sin kx\cos\omega t$$

Nodes are the points of zero amplitude: $A' = 2A\sin kx = 0$

$$\Leftrightarrow \sin kx = 0 \Rightarrow kx = n\pi \quad (n = 0,1,2,\ldots)$$

$$\frac{2\pi}{\lambda}x = \frac{\pi}{2} + n\pi$$

$$\Rightarrow x = n\frac{\lambda}{2} \quad (n = 0,1,2,\ldots)$$

Hence, node positions are $x = n\dfrac{\lambda}{2} \quad (n = 0,1,2,\ldots)$

❍ The positions in the medium at which this maximum displacement occurs are called antinodes.

$$A' = 2A\sin kx = \pm 2A$$

$$\Leftrightarrow \sin kx = \pm 1 \quad \Rightarrow kx = \frac{\pi}{2} + n\pi \quad (n = 0,1,2,\ldots)$$

$$\frac{2\pi}{\lambda}x = \frac{\pi}{2} + n\pi$$

$$\Rightarrow x(2n+1)\frac{\lambda}{4} \quad (n = 0,1,2,\ldots)$$

Let, $n' = (2n+1)$ if $n = 0,1,2,\ldots$ then $n' = 1,3,5,\ldots$

$$\Rightarrow x = n'\frac{\lambda}{4} \quad (n' = 1,3,5,\ldots)$$

Hence, the antinode positions are: $x = n'\dfrac{\lambda}{4} \quad (n' = 1,3,5,\ldots)$

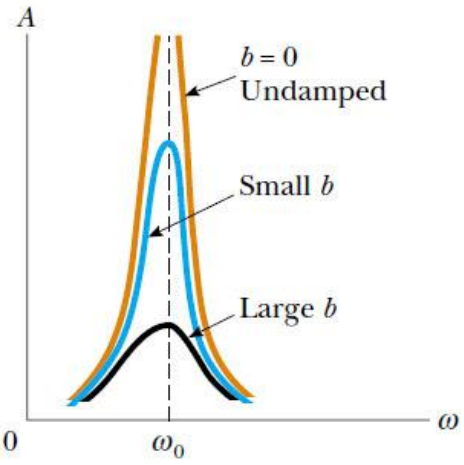
V. Resonance

❍ **Resonance** refers to the phenomenon where an object or system vibrates at a higher amplitude at certain frequencies. These frequencies, known as **resonant frequencies**, occur when an external force or energy source causes the object to vibrate in sync with its natural frequency.

1. Mechanical Resonance: In physical systems (like bridges or buildings), resonance occurs when external vibrations match the natural frequency of the structure, which can lead to large oscillations and even structural failure if unchecked. The famous collapse of the Tacoma Narrows Bridge in 1940 is an example of mechanical resonance.

2. Acoustic Resonance: This happens in musical instruments or audio systems when sound waves amplify at the natural frequency of the medium, enhancing certain tones. For instance, the body of a guitar resonates to amplify sound produced by the strings.

3. Electrical Resonance: In electrical circuits, resonance occurs when the inductive and capacitive reactance cancel each other out, allowing for the maximum flow of current. This principle is used in radio tuning, where the resonant frequency of the circuit matches the frequency of the radio signal.



❍ **Undamped Resonance (b = 0):**
- This is the sharpest peak (orange curve).
- There is no damping (resistance to motion), meaning the system oscillates at maximum amplitude when driven at the resonant frequency (ω_0).
- Since there is no energy loss, the resonance is extremely sharp and narrow, occurring at exactly ω_0
- This would lead to very high amplitudes at resonance.

❍ **Small Damping (b):**
- This represented by the blue curve.
- The system experiences a small amount of damping.
- The resonance peak is still relatively sharp, but the maximum amplitude is slightly reduced compared to the undamped case.
- The frequency response broadens slightly, meaning the system can resonate within a slightly wider range of frequency around ω_0.

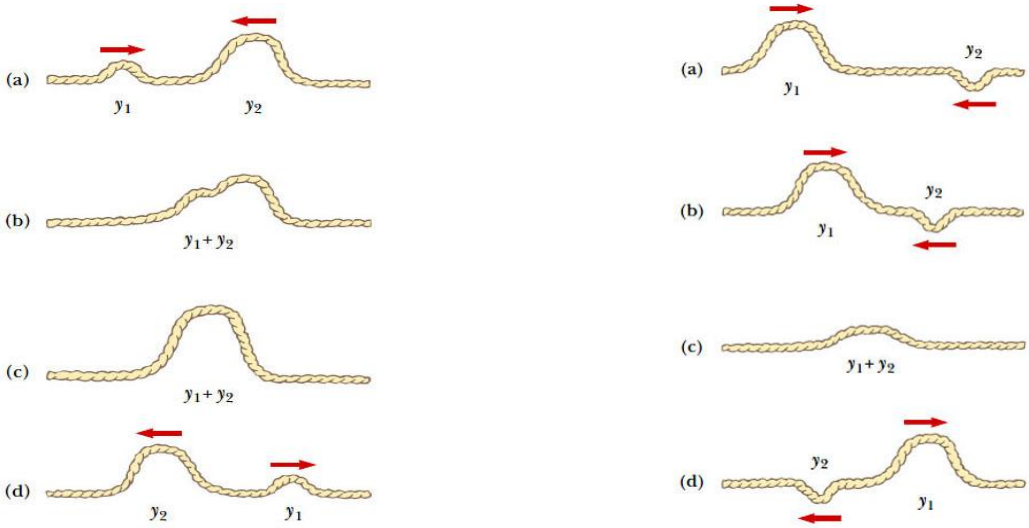
❍ **Large Damping (b):**
- This is shown by the black curve.
- The damping is significant is this case, leading to a much broader and lower resonance peak.
- The system does not respond strongly at the resonant frequency, and the amplitude is much smaller. The system can still resonate but the effects are less pronounced, and the resonance occurs over a much wider range of frequency.

Chapter2 Wave
Lesson2 Interference and Diffraction

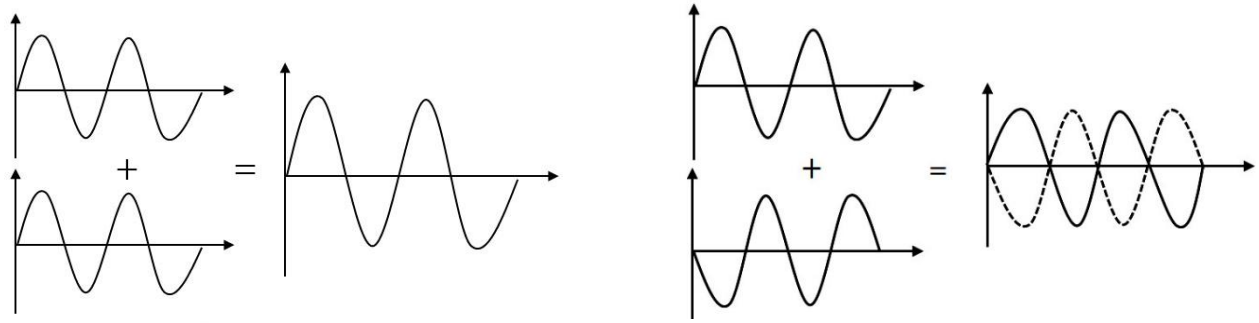
I. Interference

❍ **Interference** is a phenomenon that occurs when two or more waves overlap and combine, leading to a modification in the resulting wave pattern. Interference can happen with any type of waves—sound waves, light waves, water waves, or electromagnetic waves.

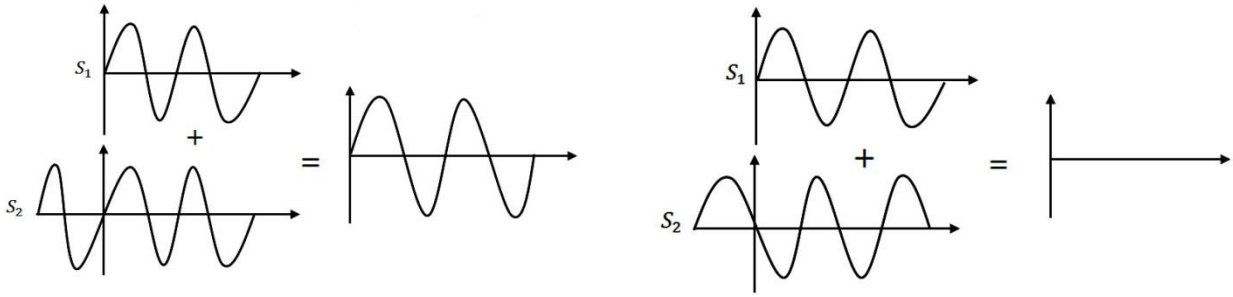
❍ **Mechanical interference**



Constructive interference **Destructive interference**



For two waves are in phase



❍ **Condition for Constructive and Destructive Interference:**

The condition for constructive interference is given by: $d_2 - d_1 = n\lambda$

The condition for destructive interference is given by: $d_2 - d_1 = (2n+1)\lambda/2$

Wave function: $y = A\sin(\omega t - kx)$

For point M that receives waves from two sources S_1 and S_2:

$$y_1 = A\sin(\omega t - kd_1) \text{ and } y_2 = A\sin(\omega t - kd_2)$$

Superposition of wave equation at point M is $y = y_1 + y_2 = A\left[\sin(\omega t - kd_1) + \sin(\omega t - kd_2)\right]$

Using formula: $\sin a + \sin b = 2\sin\frac{a+b}{2}\cos\frac{a-b}{2}$

Imply, $y = 2A\left[\sin\left(\frac{\omega t - kd_1 + \omega t - kd_2}{2}\right)\cos\left(\frac{\omega t - kd_1 - \omega t + kd_2}{2}\right)\right]$

$$y = 2A\cos\left(\frac{k(d_2 - d_1)}{2}\right)\sin\left(\omega t - \frac{k(d_2 + d_1)}{2}\right)$$

Since, wave number: $k = \frac{2\pi}{\lambda}$

We get, $y = 2A\cos\left(\pi\frac{d_2 - d_1}{\lambda}\right)\sin\left(\omega t - \pi\frac{d_2 + d_1}{\lambda}\right)$

Maximum amplitude of superposition wave: $A' = 2A\cos\left(\pi\frac{d_2 - d_1}{\lambda}\right) \quad (n = 0,1,2,3,\ldots)$

Maximum amplitude unless, $\cos\left(\pi\frac{d_2 - d_1}{\lambda}\right) = \pm 1$

$$\Rightarrow \pi\frac{d_2 - d_1}{\lambda} = n\pi$$

$$\Rightarrow d_2 - d_1 = n\lambda \quad (n = 0,1,2,3,\ldots)$$

Hence, path difference is $d_2 - d_1 = n\lambda \quad (n = 0,1,2,3,\ldots)$

Minimum amplitude of superposition wave: $A' = 0$

$$\Leftrightarrow 2A\cos\left(\pi\frac{d_2 - d_1}{\lambda}\right) = 0 \Rightarrow \pi\frac{d_2 - d_1}{\lambda} = \frac{\pi}{2} + n\pi = \left(\frac{1}{2} + n\right)\pi$$

$$\Rightarrow d_2 - d_1 = (2n+1)\frac{\lambda}{2} \quad (n = 0,1,2,3)$$

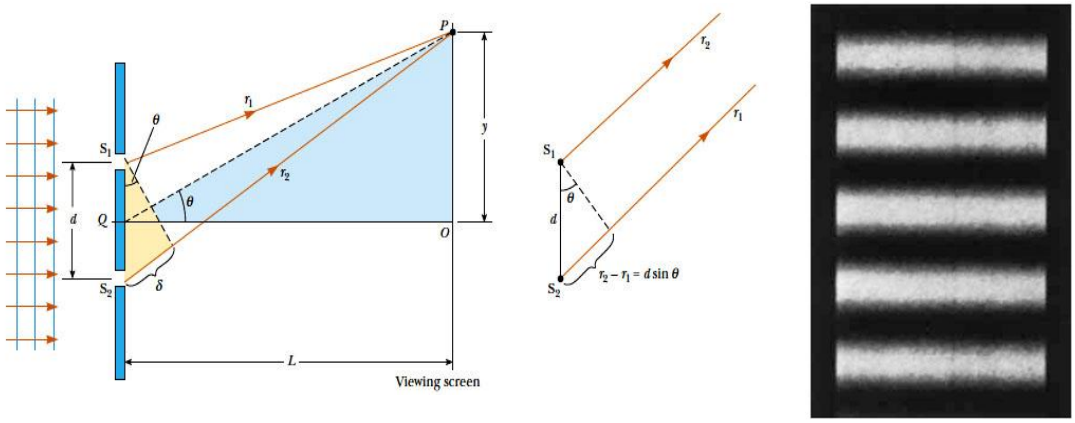
Hence, path difference is $d_2 - d_1 = (2n+1)\frac{\lambda}{2} \quad (n = 0,1,2,3)$

II. Interference of Light Waves

❍ Two sources are said to be coherent sources if they produce waves of same amplitude, same frequency having either no initial phase different or a constant phase difference.

❍ **YOUNG'S DOUBLE SLIT EXPERIMENT: Young's Double-Slit Experiment** is a famous experiment in physics that demonstrates the wave nature of light through interference patterns. Conducted by Thomas

Young in 1801, it provided crucial evidence for the theory of light as a wave and is fundamental to the field of wave optics.



Path difference: $r_2 - r_1 = d\sin\theta$

Condition for Bright Fringes at point P (Constructive Interference): $d\sin\theta_{bright} = m\lambda \quad (m = 0, \pm1, \pm2, \ldots)$

For $m = 0 \Rightarrow d\sin\theta = 0 \Rightarrow \theta = 0$ is called zero maximum fringe (central maximum).

First maximum fringe corresponding to $m = \pm1$

Condition for Dark Fringes at point P (Destructive Interference):

$$d\sin\theta_{Dark} = \left(m + \frac{1}{2}\right)\lambda \qquad (m = 0, \pm1, \pm2, \ldots)$$

Relationship between small angles and distance between screens: $\tan\theta = \dfrac{y}{L}$

Hence, the distance of the fringe on the screen is $y = L\tan\theta$

❍ **Bright and Dark Fringes in Interference Patterns**

Condition for Bright Fringes: $y_{bright} = L\tan\theta_{bright}$

For small angles, $\tan\theta \approx \sin\theta$ where θ is in radians.

The condition for bright fringes is given by: $d\sin\theta_{bright} = m\lambda \Rightarrow \sin\theta_{bright} = m\lambda/d$

Hence, the position of bright fringes is $y_{bright} = L\dfrac{m\lambda}{d} \quad (m = 0, \pm1, \pm2, \ldots)$

Condition for Dark Fringes: $y_{Dark} = L\tan\theta_{Dark}$

For small angles, $\tan\theta \approx \sin\theta$, where θ is in radians.

The condition for dark fringes is given by: $d\sin\theta_{Dark} = \left(m+\frac{1}{2}\right)\lambda \Rightarrow \sin\theta_{Dark} = \frac{\left(m+\frac{1}{2}\right)\lambda}{d}$

Hence, the position of dark fringes is $y_{Dark} = L\frac{\left(m+\frac{1}{2}\right)\lambda}{d} \quad (m = 0, \pm1, \pm2, \ldots)$

III. Diffraction

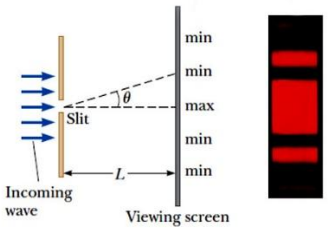
❍ **Diffraction** is a phenomenon that occurs when waves encounter an obstacle or pass through an opening and bend around the edges, causing them to spread out or change direction. This behavior is most noticeable when the size of the obstacle or opening is comparable to the wavelength of the waves.

❍ **Mechanical diffraction** refers to the diffraction of mechanical waves, such as sound waves or water waves, when they encounter obstacles or pass through openings. Similar to the diffraction of light or other types of waves, mechanical diffraction involves the bending and spreading of waves around objects or through narrow gaps $d \approx \lambda$.

❍ **Sound diffraction** is the phenomenon where sound waves bend around obstacles or pass through openings, causing them to spread out and create interference patterns. This behavior is similar to diffraction observed with other types of waves, such as light or water waves.

❍ **Light diffraction** is the bending and spreading of light waves when they encounter an obstacle or pass through a narrow aperture. This phenomenon is similar to diffraction observed with other types of waves, such as sound or water waves.

Condition for Minima (Dark Fringes): $a\sin\theta = m\lambda \quad (m = \pm1, \pm2, \pm3, \ldots)$



Where, a is the width of the slit and m is the order of the minimum fringe.

Condition for Minima (Dark Fringes): $y = L\tan\theta$

For small angles, $\tan\theta \approx \sin\theta$ where θ is in radians.

Using the condition for minima: $\sin\theta = \frac{m\lambda}{a} \Rightarrow y = L\frac{m\lambda}{a} \quad (m = \pm1, \pm2, \pm3, \ldots)$

Hence, the position of dark fringe on the screen is $y = L\frac{m\lambda}{a} \quad (m = \pm1, \pm2, \pm3, \ldots)$

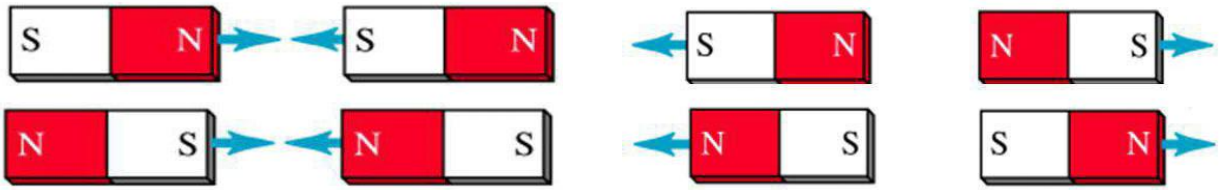
Chpater3 **Electricity and Magnetism**
Lesson1 **Magnetic Field**

I. Magnet and Magnetic Field

❍ A **magnet** is an object or material that produces a magnetic field, which exerts a force on other materials that are either magnetic themselves or susceptible to magnetism. Magnets attract or repel other magnets and can attract ferromagnetic materials like iron, nickel, cobalt, and some alloys. There are 2 types of magnets such as permanent magnet and temporary magnet.

❍ A magnet always has **two poles**: a **north pole** (N) and a **south pole** (S).

❍ Law of magnet: Likes poles or same poles repel each other and unlike poles or opposite poles attract each other.

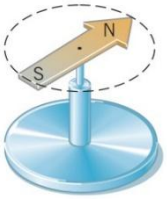


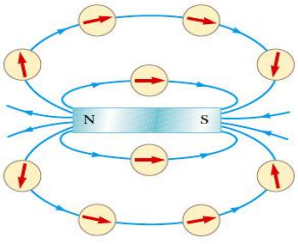
II. Characteristics of Magnetic Fields:

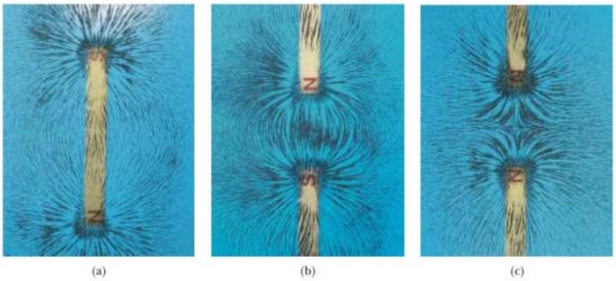
❍ Magnetic field is the space around a bar magnet, electric current passed through magnetic field lines and earth and the needle experienced by a magnetic force.

❍ The size represented by magnetic field at any points is called magnetic field $\vec{B}$.

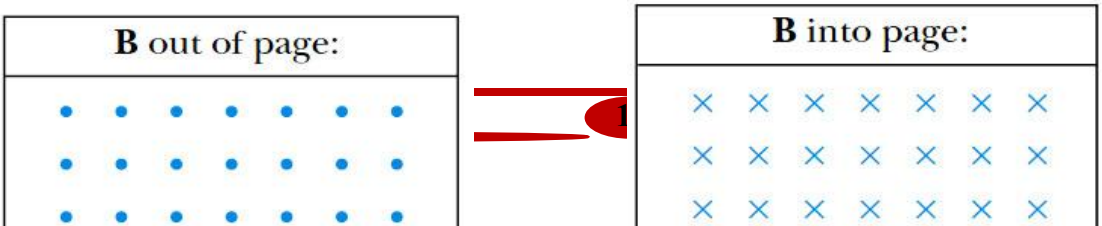
❍ The magnetic field lines are the line that all point of magnetic field lines touched magnetic field $\vec{B}$. Magnetic spectrum is the spectrum of magnetic field.







❍ A **uniform magnetic field** is a magnetic field in which the magnetic field strength and direction are the same at every point in space. In a uniform magnetic field, the magnetic field lines are parallel, equally spaced, and point in the same direction.



❍ **Earth's Magnetic Field**

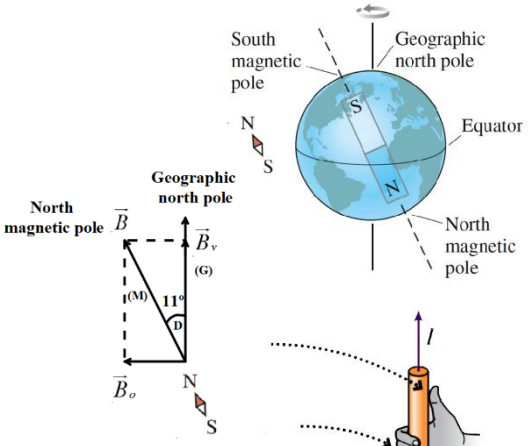
The magnetic longitudinal plane (M) is a vertical plane that magnetic induction $\vec{B}$ located.

The longitudinal plane (G) is the vertical plane that intersects the Earth north and south's poles.

Magnetic declination D is the angle created by longitudinal plane.

The Earth's magnetic inductance can be divided into two components: Horizontal inductance $\vec{B}_0$ and vertical inductance $\vec{B}_v$.

In vector form: $\vec{B} = \vec{B}_o + \vec{B}_v$ where $B_o = 2 \times 10^5$ T.



III. Magnetic field due to a straight current-carrying wire

❍ The characteristic of magnetic fields at point M have

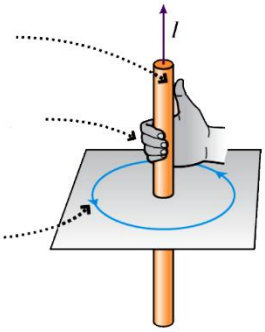
Direction is perpendicular to the plan of current carrying-wire.

The direction of magnetic field is found by using right-hand rule

❶ Point your *right* thumb in I the direction of the current.

❷ Curl your fingers around the wire to indicate a circle.

❸ Your fingers point in the direction of the magnetic field lines around the wire.



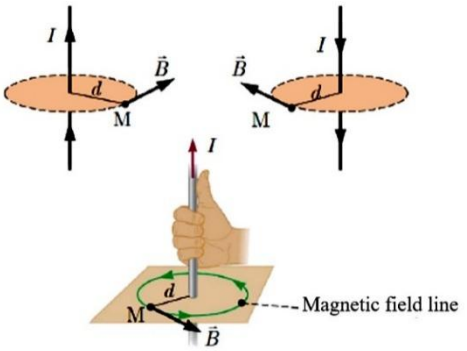
The magnitude of magnetic field $\vec{B}$ is $B = \dfrac{\mu_0 I}{2\pi d}$

Where, $\mu_0 = 4\pi \times 10^{-7} SI$ is the vacuum permittivity.

I is the electric current measured in (A).

d is the distance from point M to the wire measured in (m).

B is the magnetic field measured in (T).



IV. Magnetic field due to a circular loop

❍ The characteristic of magnetic fields $\vec{B}$ at point O have

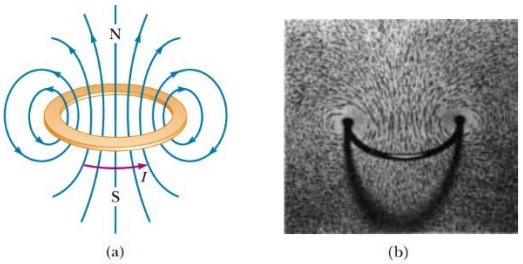
Direction is perpendicular to the loop of wire.

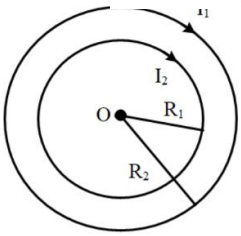
Direction of magnetic field is found by using right-hand grip rule

❶ Wrap your 4 fingers around a circular loop of wire

❷ Outstretch your right thumb in the direction of magnetic field.

The magnitude of magnetic field $\vec{B}$ is $B = \dfrac{\mu_0 I}{2R}$



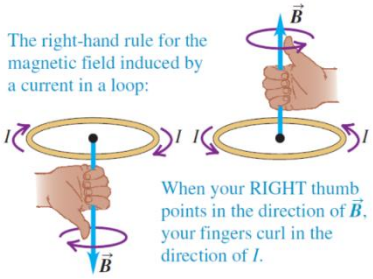


Where, $\mu_0 = 4\pi \times 10^{-7}$ SI is the vacuum permittivity.

R is the radius of a circular loop measured in (m).

I is the electric current measured in (A).

B is the magnetic field measured in (T).



For a circular with N-turns of wire: $B = \dfrac{\mu_0 NI}{2R}$

Where, $R = \dfrac{R_1 + R_2}{2}$ is the average radius measured in (m).

V. Magnetic field due to Solenoid

❍ Solenoid is a loop of wire wrapped around a core produces magnetic field when electric current passed through it, and a solenoid has long length compared to its radius $(l \geq 5R)$.

❍ The characteristic of vector magnetic induction $\vec{B}$ at point O inside Solenoid have

❶ Direction is parallel the axis of solenoid.

❷ Direction is determined by using right-hand grip rule (Same as a circular loop of wire)

❸ The magnitude of magnetic field $\vec{B}$ is $B = \mu_0 nI$

If n is the number of turns per meter, the magnetic field can be written as $B = \mu_0 \dfrac{NI}{l}$

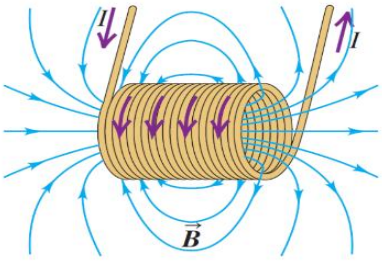
Where, $\mu_0 = 4\pi \times 10^{-7}\, SI$ is the vacuum permittivity.

I is the electric current measured in (A).

B is the magnetic field measured in (T).

l is the length of Solenoid measured in (m).

N is the total number of turns of wire.



❍ The relationship between number of turns, diameter and length of a Solenoid is given by

$$N = \frac{l}{d} \Rightarrow n = \frac{l/d}{l} = \frac{1}{d}$$

Where, l is the length of solenoid measured in (m).

d is the diameter of wire measured in (m).

The relationship between length of a wire, total number of turns and diameter of wire is given by

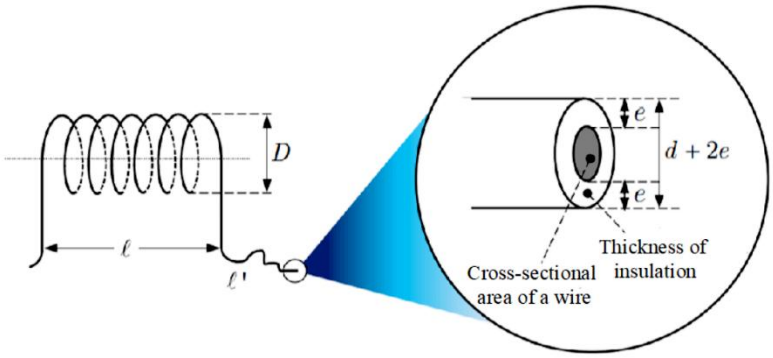
$$l' = \pi DN$$

Where, l' is the length of a wire measured in (m).

N is the total number of turns of wire.

D is the diameter of a Solenoid measured in (m).

❍ **Noted:** Key Steps to Calculate the Total Number of Turns and resistance of a solenoid:



- Since the wire has insulation around it, the effective diameter of the wire, including the insulation, is:

$$d_{effective} = d + 2e$$

- In a single layer of the solenoid (assuming the turns are tightly packed), the number of turns per unit length along the core (in one layer) is given by:

$$N_{per\ layer} = \frac{l}{d_{effective}}$$

- Since there are x layers of wire, and each layer has approximately the same number of turns, the total number of turns N is:

$$N = \frac{l}{d_{effective}}x = \frac{l}{d+2e}x$$

- Hence, the total number of turns N is $N = \frac{l}{d+2e}x$

The resistance of a wire wrapped around a core of a solenoid is given by

$$R = \rho\frac{l'}{A} \quad \therefore\ l' = N\left(2\pi r\right) = \pi DN \quad \therefore A = \frac{\pi d^2}{4}$$

Where, x is the number of layers of the solenoid.

e is the thick of insulation.

r is the radius of a wire measured in (m).

A is the area of a wire measured in (m^2).

d is the diameter of wire measured in (m).

VI.Method to Determine the Earth's Magnetic Field ($\vec{B}_H$), Electric Current (I), and Angle of Deflection (α)

***** Objective: Find B_H; I; α**

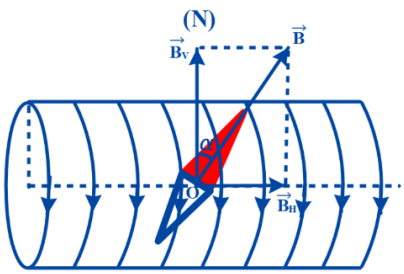


Fig.a

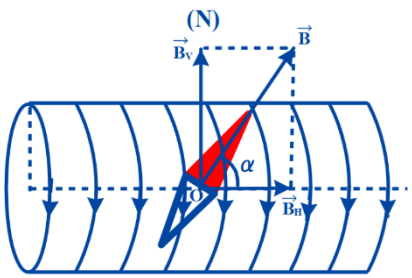


Fig.b

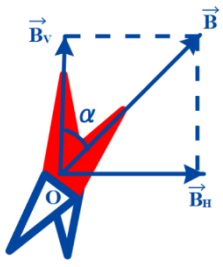


Fig.c

* At point ***O***, the needle experiences two **magnetic fields**: $\vec{B}_V$ (Earth's magnetic field) and $\vec{B}_H$ (magnetic field created by the solenoid).

The **net magnetic field** is: $\vec{B} = \vec{B}_V + \vec{B}_H$

(Figure1)* In **ΔOBvB**, the **angle of deflection α** is related to the fields by:

$$\tan\alpha = \frac{B_H}{B_V}$$

- Earth's magnetic field: $B_V = \dfrac{B_H}{\tan\alpha}$ $\therefore$ since $B_H = \mu_0 \dfrac{NI}{l}$
- Magnetic field created by the solenoid: $B_H = B_V \tan\alpha$

This implies: $\dfrac{\mu_0 NI}{l} = B_H \cdot \tan\alpha$

- The electric current: $I = \dfrac{B_H \tan\alpha l}{\mu_0 N}$

(Figure2)* In **ΔOBHB**, the **angle of deflection α** is related to the fields by:

$$\tan\alpha = \frac{B_V}{B_H}$$

- Earth's magnetic field: $B_V = B_H \tan\alpha$ $\therefore$ since $B_H = \mu_0 \dfrac{NI}{l}$
- Magnetic field created by the solenoid: $B_H = \dfrac{B_V}{\tan\alpha}$

This implies: $B_H = \dfrac{\mu_0 NI}{l} \cdot \tan\alpha$

- The electric current: $I = \dfrac{B_H l}{\mu_0 N \tan\alpha}$

Where, $B_V = 2\times10^{-5}\ \text{T}$ is the earth magnetic field.

B_H is the magnetic field of the solenoid.

B_2 is the U-shape magnetic field measured in (T).

B_1 is the magnetic field created by a solenoid measured in (T).

I is the electric current measured in (A).

N is the number of turns measured in (turns).

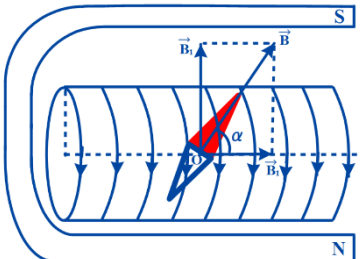
l is the length of the solenoid measured in (m).

VII. Method to Determine the U-shape magnet ($\vec{B}_2$), Electric Current (I), and Angle of Deflection (α)

***** Objective: Find B_2; I; α**

* At point ***O***, the needle experiences two **magnetic fields**: $\vec{B}_1$ magnetic field created by the solenoid) and $\vec{B}_2$ (magnetic field of a U-shape magnet).

The **net magnetic field** is: $\vec{B} = \vec{B}_1 + \vec{B}_2$

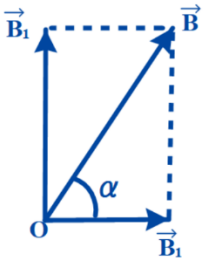


(Figure1)* In **ΔOB₁B**, the **angle of deflection α** is related to the fields by:

$$\tan\alpha = \frac{B_2}{B_1}$$

$$B_2 = B_1 \tan\alpha \quad \therefore \text{ since } B_1 = \mu_0 \frac{NI}{l}$$

$$B_2 = \mu_0 \frac{NI}{l} \tan\alpha$$



- Magnetic field of a U-shape magnet: $B_2 = \mu_0 \frac{NI}{l} \tan\alpha$
- The electric current: $I = \frac{B_2 l}{\mu_0 N \tan\alpha}$

❍ **Helmholtz coil:** A Helmholtz coil is formed using two identical circular magnetic coils positioned symmetrically along a common axis, with a separation $L = R$, where R is the radius of the coils. Electric current flows through both coils in the same direction. When the electric current flows through both coils, they create a uniform magnetic field. The direction of the magnetic field is determined using the right-hand rule: curl the four fingers of your right hand in the direction of the electric current I, and outstretch your right thumb in the direction of the magnetic field.

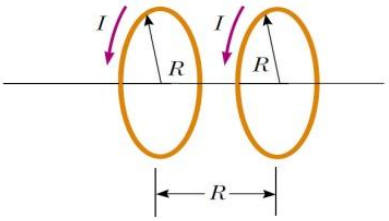
The magnetic field produced at the center of the Helmholtz coils is given by $B = \frac{8\mu_0 IN}{5\sqrt{5}R} = 0.7\mu_0 \frac{NI}{l}$

Where, $\mu_0 = 4\pi \times 10^{-7}$ SI is the vacuum permittivity.

I is the electric current through the coil measured in (A).

B is the magnetic field measured in (T).

R is the radius of the coils measured in (m).



VIII. Magnetic Force on a Current Carrying Conductor

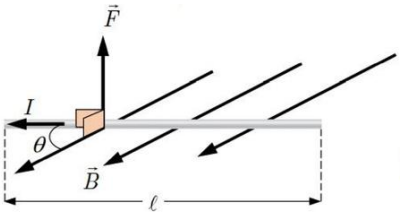
❍ A current carrying wire placed inside magnetic field, therefore the magnetic force will be exerted on a wire. The characteristic of magnetic force on a current carrying conductive have

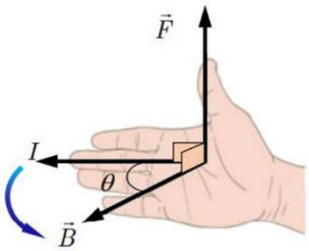
❶ Point of application on the current carrying conductor

❷ Direction is perpendicular to the plan determined by magnetic field $\vec{B}$ and current I.

❸ Direction is determined by using right-hand rule

- Point your fingers in the direction of the current.
- Curl them towards the magnetic field $\vec{B}$.
- Your thumb will point in the direction of the force $\vec{F}$.





The magnitude of the magnetic force is $F_m = IlB\sin\theta$ $\quad \theta(\vec{l}, \vec{B})$

Where, I is the electric current measured in (A).

B is the magnetic field measured in (T).

l is the length of a wire measured in (m).

IX. The Magnetic Force Between Two Parallel Conductors

❍ Experiment:

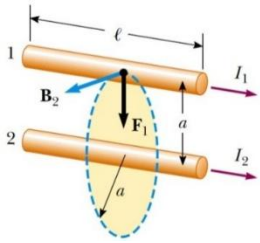
Two parallel conductors carry electric current in the same direction attract each other.

Two parallel conductors carry electric current in the opposite direction repel each other.

❍ Calculate the magnetic force experience on a part of parallel conductors.

Magnetic field $\vec{B}_1$ created by electric current I_1 at distance a from the second conductor is given by:

$$B_1 = \frac{\mu_0 I_1}{2\pi a}$$



- The force experience on second conductor is given by: $F_{12} = I_2 l B_1 = \dfrac{\mu_0 I_1 I_2 l}{2\pi a}$
- Similarly, the force experience on first conductor: $F_{21} = I_1 l B_2 = \dfrac{\mu_0 I_1 I_2 l}{2\pi a}$
- Hence, the magnetic force between two parallel conductors is $F_{12} = F_{21} = I_1 l B_2 = \dfrac{\mu_0 I_1 I_2 l}{2\pi a}$
- If two parallel conductors are unknown, the magnetic force per unit length can be written as

$$\frac{F_1}{l} = \frac{F_2}{l} = \frac{\mu_0 I_1 I_2}{2\pi a}$$

Where, $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{SI}$ is the vacuum permittivity.

l is the length of conductors measured in (m).

I_1 is the electric current in first conductor measured in (A).

I_2 is the electric current in second conductor measured in (A).

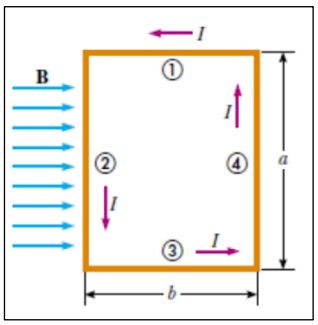
a is the distance two parallel conductors measured in (m).

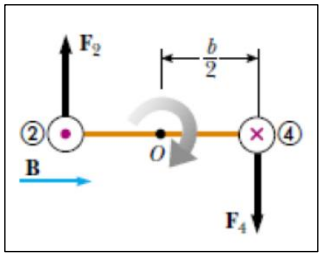
$F_{12} = F_{21}$ is the interaction magnetic force between two parallel conductors measured in (N).

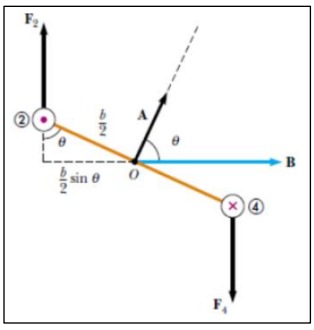
X. Torque on a Current Loop in a Uniform Magnetic Field

❍ The magnitude of the torque is directly proportional to the magnitude of the magnetic field.

❍ Consider a loop of wire carrying current suspended in a uniform magnetic field of flux density $\vec{B}$ as shown in the figure below:







There is no force act on sides ❶ and ❸ because these wires are parallel to the field.

However, magnetic forces do act on sides ❷ and ❹ because these sides are perpendicular to the filed.

The magnitude of these forces is $F_2 = F_4 = F = ILB\sin\theta = IaB\sin 90^\circ = IaB \quad \therefore L = a$

Magnetic forces act on sides ❷ and ❹ are opposite and parallel then two anti-parallel forces are separated by a small distance from a couple O.

$$\tau = F_1 \frac{b}{2}\sin\theta + F_4 \frac{b}{2}\sin\theta = IBa\frac{b}{2}\sin\theta + IBa\frac{b}{2}\sin\theta$$

$$\tau = IBA\sin\theta \quad \therefore A = ab = \text{area of a loop}$$

This is the expression of torque produced in a loop with single turn.
For a loop with N turns, the torque will be:

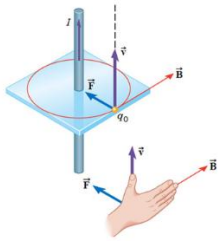
$$\tau = NIBA\sin\theta$$

This result shows that the torque has its maximum IBA when the field is perpendicular to the normal plane of the loop, and is zero when the field is parallel to the plane of the loop.

Hence, $\tau_{\max} = NIBA \quad \therefore \theta = 90^\circ$

XI. Magnetic Force on a Charged Particle

❍ Consider a charged object having charge q is projected a uniform magnetic field of flux density $\vec{B}$ with velocity $\vec{v}$. The magnetic force acting on the object can be expressed as:



$$F_m = q(\vec{v}\times\vec{B})$$

Direction is perpendicular to a plan of $\vec{v}$ and $\vec{B}$ and $\vec{B}$ $(\vec{F}_m \perp \vec{v}, \vec{F}_m \perp \vec{B})$

Direction

- If $q>0$ $\vec{F}_m$ is in the same direction from $\vec{v}\times\vec{B}$
- If $q<0$ $\vec{F}_m$ is in the opposite direction from $\vec{v}\times\vec{B}$

The magnitude of magnetic force is $F_m = qvB\sin\alpha \quad \alpha(\vec{v},\vec{B})$

- If $\vec{v}\uparrow\uparrow\vec{B}$ therefore, $\alpha = 0$ or $\alpha = \pi \Rightarrow F_m = 0$ or $\alpha = \pi \Rightarrow F_m = 0$ (Particle moves in straight path)
- If $\vec{v}\perp\vec{B}$ therefore, $\alpha = \pi/2 \Rightarrow F_m = qvB$ (Particle moves in a circular motion)
- If $\vec{v}$ and $\vec{B}$ created an angle θ therefore, $\alpha = \theta \Rightarrow F_m = qvB\sin\theta$ (Particle moves in helical path)

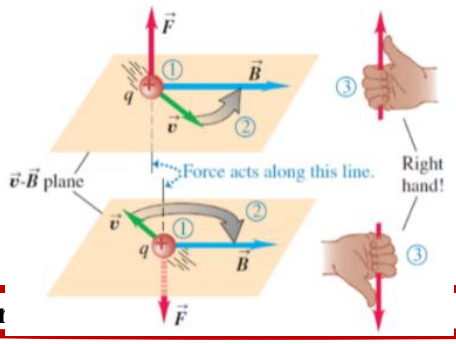
∗ **Right-hand rule** for the direction of magnetic force on a positive charge moving in a magnetic field:

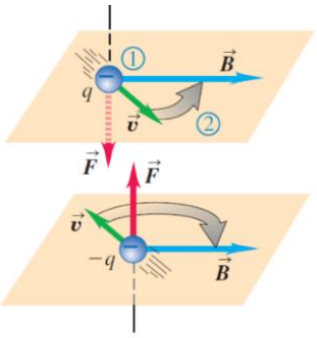
❶ Place the $\vec{v}$ and $\vec{B}$ vectors tail to tail.

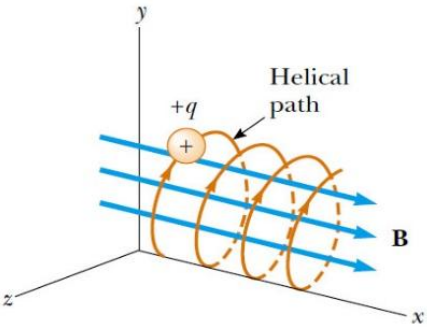
❷ Imagine turning $\vec{v}$ toward $\vec{B}$ in the $\vec{v}-\vec{B}$ plane (through the smaller angle).

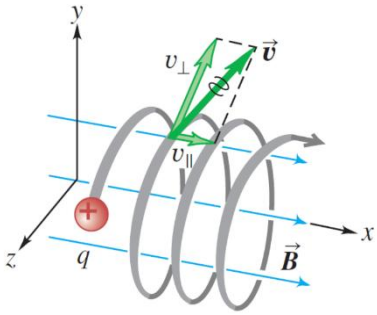
❸ The force acts along a line perpendicular to the $\vec{v}-\vec{B}$ plane. Curl the fingers of your *right hand* around this line in the same direction you rotated $\vec{v}$. Your thumb now points in the direction the force acts.

∗ **If the charge is negative,** the direction of the force is ***opposite*** to that given by Place the $\vec{v}$ and $\vec{B}$ vectors tail to tail. the right-hand rule.







❍ **Noted**: ation of circula  nagnetic field) and linear mo).

❍ **Velocity Decomposition**: The total velocity v of the particle can be broken into two components:

$v_{||} = v\cos\theta$, the velocity along the magnetic field.

$v_{\perp} = v\sin\theta$, the velocity perpendicular to the magnetic field.

❍ **Distance Traveled Along the Magnetic Field**: The distance x traveled along the magnetic field depends on the parallel velocity $v_\parallel$ and the time t for which the particle is in motion. Since $v_\parallel$ is constant, the distance x is: $x = v_{||}t = v\cos\theta t$

Where, x is the distance moved along the magnetic field axis (m).

v is the total particle's speed measured in $(\mathrm{m/s})$.

θ is the angle between the velocity and magnetic field.

t is the time for which the particle is in motion.

$v_{||}$ is the velocity along the magnetic field measured in $(\mathrm{m/s})$.

$v_{\perp}$ is the velocity perpendicular to the magnetic field measured in $(\mathrm{m/s})$.

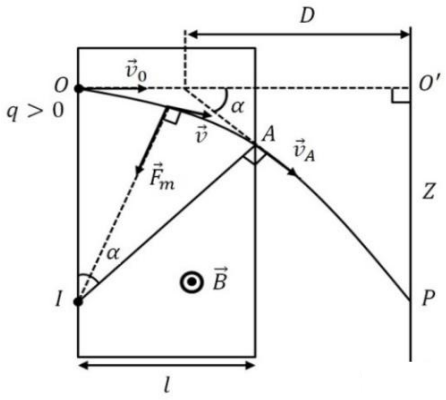
XII. Motion of a Charged Particle in a Uniform Magnetic Field

❍ The motion of a **charged particle in a uniform magnetic field** depends on the direction of the particle's velocity relative to the magnetic field. When a charged particle, such as an electron or proton, enters a magnetic field, it experiences a force called the **Lorentz force**, which is perpendicular to both its velocity and the magnetic field. This causes the particle to follow a curved or circular path, depending on the orientation of its velocity with respect to the field. The direction of magnetic force is inward to the center of a circular path.

Method#1

A positive charged particle is projected into uniform magnetic field at point O experienced by a magnetic force.

We have, $\vec{F}_m = q(v_0 \times \vec{B})$



Newton's Second Law: $\vec{F} = m\vec{a} = m(\vec{a}_T + \vec{a}_n) = \vec{F}_T + \vec{F}_N$

$$\vec{F}_m = \vec{F} = \vec{F}_T + \vec{F}_N$$

Since, $\vec{F}_m \perp \vec{v}_0 \Rightarrow \vec{F}_m = \vec{F}_N$ and $\vec{F}_T = 0$

$$\vec{F}_T = m\frac{dv}{dt} = 0 \Leftrightarrow \frac{dv}{dt} = 0$$

$$\Rightarrow v = v_0 = const$$

Method#2

Power: $P = \vec{F}_m \cdot \vec{v} = Fv\cos 90^\circ = 0 \quad (\vec{F}_m \perp \vec{v})$

Work done: $W = Pt = 0$

Work-Energy Theorem: $\Delta K = W$

We get, $\Delta K = K_f - K_i = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 = \frac{1}{2}m(v^2 - v_0^2)$

$\Rightarrow v = v_0 = const$

Since, $F_m = F_n$ (F_n is the centripetal force)

$$qvB = m\frac{v^2}{\rho} \Rightarrow \rho = \frac{mv_0}{qB} = const$$

$$R = \frac{mv_0}{qB} \quad \therefore \rho = R$$

Hence, the radius of circular path in uniform magnetic field is $R = \frac{mv_0}{qB}$

Where, R is the radius of a circular path measured in (m).

m is the mass of a particle measured in (kg).

q is the charge of a particle measured in (C).

v_0 is the velocity of a particle measured in $(\mathrm{m/s})$.

B is the uniform magnetic field measured in (T).

❍ **Deflection of a charged particle in a uniform magnetic field**

We have: $\alpha(rad) = \frac{OA}{R}$

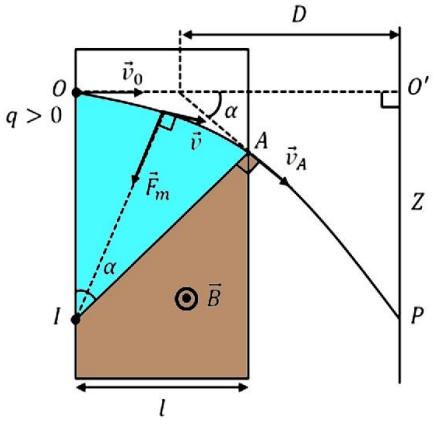
Since, α is small angle $\Rightarrow OA = l$ and $R = \frac{mv_0}{qB}$

Imply, $\alpha(rad) = \frac{qBl}{mv_0}$

Magnetic deflection: $\tan\alpha = \frac{Z}{D} \Rightarrow Z = D\tan\alpha$

Since, α is small angle therefore, $\tan\alpha = \alpha(rad)$

Hence, magnetic deflection is $Z = \frac{DqBl}{mv_0}$



XIII. Mass Spectrometer

❍ A **mass spectrometer** is a scientific instrument used to measure the **charge-to-mass ratio** (m/q) of ions. It is widely used for analyzing the composition of chemical and biological substances, identifying molecules, and determining isotopic composition. The data obtained from a mass spectrometer is often used to determine the molecular weight and structure of compounds.

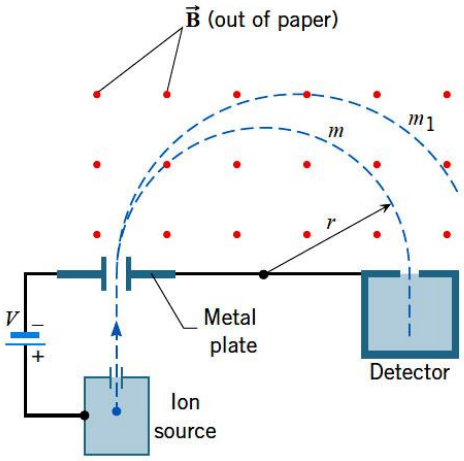
Work-Energy Theorem: $\Delta K = W \Leftrightarrow \frac{1}{2}mv^2 = |q|V$

The radius of the circular path in a magnetic field:

$$r = \frac{mv_0}{qB} \Rightarrow v^2 = \left(\frac{|q|BR}{m}\right)^2$$

We get, $\frac{1}{2}m\left(\frac{|q|Br}{m}\right)^2 = |q|V \Leftrightarrow \frac{|q|}{m} = \frac{2V}{(Br)^2}$

Hence, charge to mass ratio is $\frac{|q|}{m} = \frac{2V}{(Br)^2}$



Where, r is the radius of a circular path measured in (m).

m is the mass of a particle measured in (kg).

q is the charge of a particle measured in (C).

V is the voltage or potential difference measured in (V).

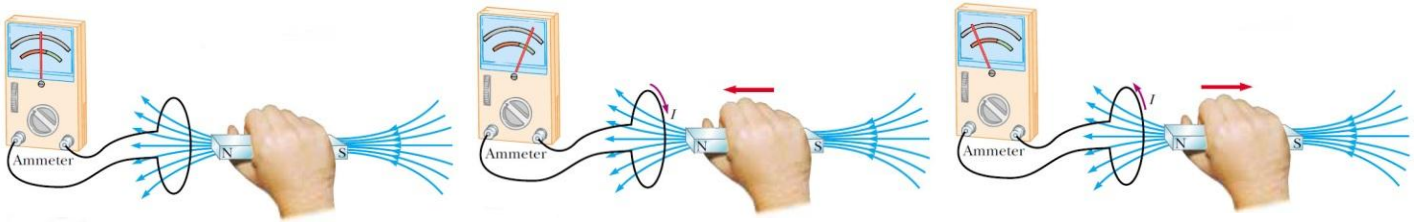
B is the uniform magnetic field measured in (T).

Chpater3 Electricity and Magnetism
Lesson2 Faraday's Law of Electromagnetic Induction

Joseph Henry (1797–1878) was an American scientist who made significant contributions to electromagnetism and is best known for his pioneering work in the development of the electromagnet and inductance. His research laid the foundation for the later invention of the telegraph and electric motor. Henry independently discovered electromagnetic induction before Michael Faraday, though Faraday is credited with the discovery. Henry also demonstrated the principle of self-induction and constructed one of the first electromagnets capable of lifting heavy loads. As the first Secretary of the Smithsonian Institution, Henry promoted scientific research and education in the United States.

I. Faraday's Law

❍ Experiment



When the magnet is brought to rest and held stationary relative to the loop figure, no deflection is observed.

When a magnet is moved toward the loop, the ammeter needle deflects in one direction, as shown in the figure.

When the magnet is moved away from the loop, the needle deflects in the opposite direction, as shown in figure. Finally, if the magnet is held stationary and the loop is moved either toward or away from it, the needle deflects.

These results are quite remarkable in view of the fact that a current is set up even though no batteries are present in the circuit. We call such a current the induced current which is produced by an induced emf. This phenomenon is called **electromagnetic induction**.

II. Magnetic Flux

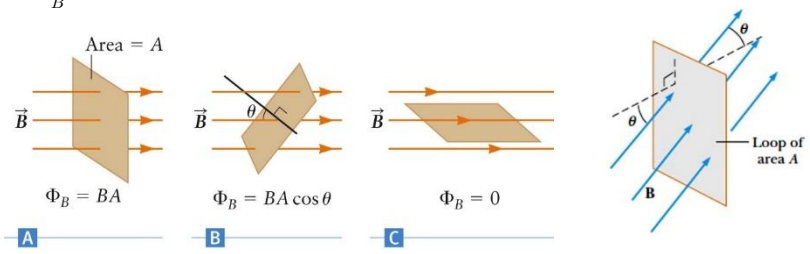
❍ **Magnetic flux** (Φ_B) is a measure of the total magnetic field passing through a given area.

$$\Phi_B = BA\cos\theta$$

If $\vec{B} \perp \vec{A}$ therefore, $\theta = 0 \Rightarrow \Phi_B = BA$

If $\vec{B} \uparrow\uparrow \vec{A}$ therefore, $\theta = 90^\circ \Rightarrow \Phi_B = 0$

If $\vec{B} \uparrow\downarrow \vec{A} \Rightarrow \theta = 180^\circ$, so $\Phi_B = -BA$



Where, B is the magnetic field measured in (T).

A is the area through which the magnetic field passes measured in (m^2).

θ is the angle between magnetic field $\vec{B}$ and normal (perpendicular) to the plane of the loop $\vec{A}$.

Φ_B is the magnetic flux measured in (Wb) or (Tm^2).

❍ **Faraday's Law:** Whenever the magnetic flux through a closed conducting loop change, an electromotive force (emf) is generated in the loop, resulting in induce current. Induced current is a temporary current which occurs when there is a change in magnetic flux within the electric circuit.

$$E = -N\frac{\Delta\Phi_B}{\Delta t} \text{ or } |E| = N\frac{|\Delta\Phi_B|}{\Delta t}$$

Where, $\Delta\Phi_B = \Phi_f - \Phi_i$ and $\Delta t = t_f - t_i$, and N is the total number loops.

Noted: The negative sign is due to the fact that the direction of induced current is such that it opposes the cause producing it.

Ohm's law: $I = \frac{V}{R} = \frac{E}{R}$

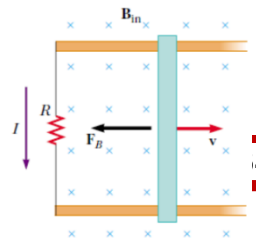
Hence, induced current is $I = \frac{V}{R} = \frac{E}{R}$

III. Motional Induction and Motional emf

❍ Whenever a metal rod of length l moves with velocity $\vec{v}$ is the magnetic field $\vec{B}$, the metal rod acts an electromotive force (emf) generator. The induced emf can be calculated using the formula

$$E = Blv\sin\alpha$$

Where, E is the electromotive force in the rod measured in (V).



l is the length of the metal rod measured in (m).

v is the velocity of the metal rod measured in $(\mathrm{m/s})$.

α is the angle between velocity $\vec{v}$ and magnetic field $\vec{B}$.

❍ The induced current occurs in the conductor is found by using right hand rule

Extend your right hand with your palm facing the direction of the magnetic field $\vec{B}$.

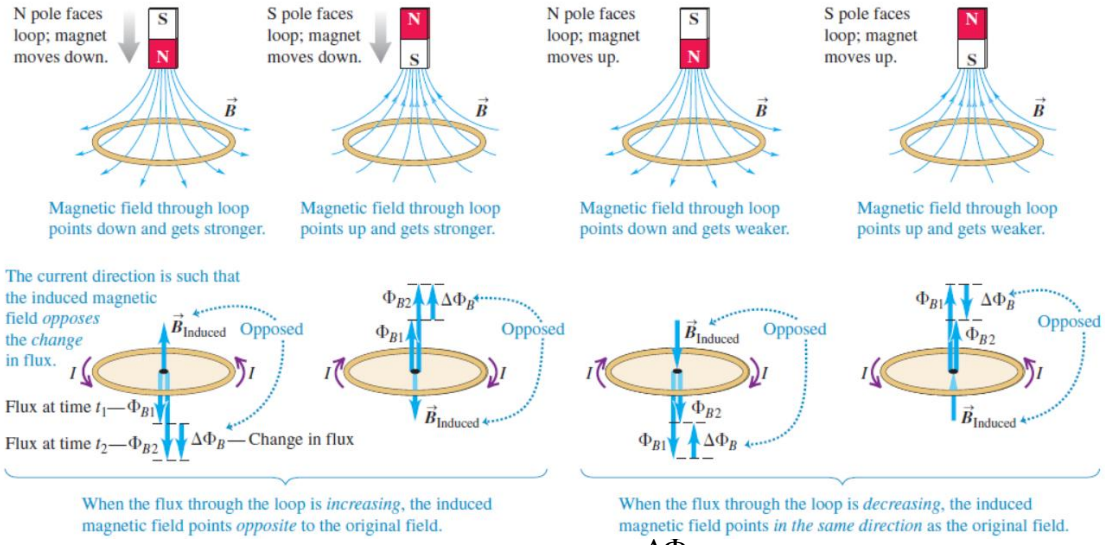
Point your thumb in the direction of the conductor's motion or velocity $\vec{v}$.

The direction your fingers curl represents the direction of the induced current in the conductor.

Lenz's Law

Lenz's law[1]: The induced current in a loop is in the direction that creates a magnetic field that opposes the change in magnetic flux through the area enclosed by the loop.

Lenz's law[2] states that the direction of induced current is such that it opposes the cause producing it.



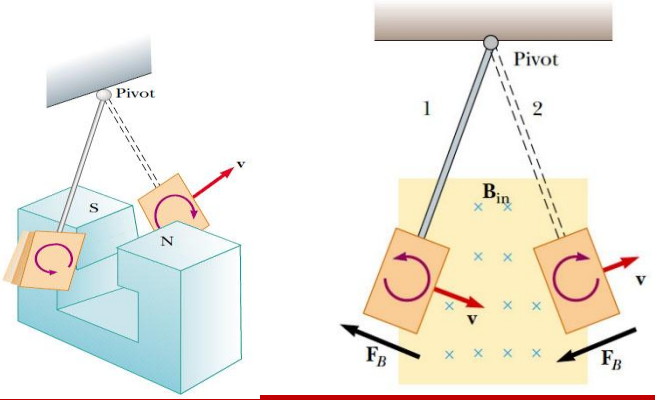
$$E = -N\frac{\Delta\Phi}{\Delta t}$$

When $\Delta t \to 0$, E approaches to at instance time is called instantaneous electromotive force.

$$e = \lim_{\Delta t \to 0}\left(-N\frac{\Delta\Phi}{\Delta t}\right) = -N\frac{d\Phi}{dt}$$

IV. Eddy Current

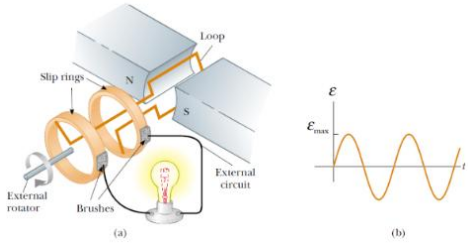
❍ Eddy currents are loops of electrical current that are induced within conductors by a changing magnetic field. These currents flow in closed loops within the conductor, and they are created due to Faraday's Law of Electromagnetic Induction.



V. Generator and electric motor

❍ Generator is an electrical device use to convert mechanical energy to electrical energy. Whenever the rectangular loop rotates, the magnetic flux through the loop changes. This change in magnetic flux can induce an electromotive force (emf) in the loop, leading to the generation of electric current.

$$e = -N\frac{d\Phi}{dt}$$



Magnetic flux through the loop at instant t is $\Phi = BA\cos\theta = BA\cos\omega t$

$$e = -N\frac{d}{dt}\ BA\cos\omega t$$

$$e = NBA\omega\sin\omega t$$

$$e = NBA\omega\sin\omega t = E_m\sin\omega t$$

Hence, the electromotive force at instant is $e = NBA\omega\sin\omega t$

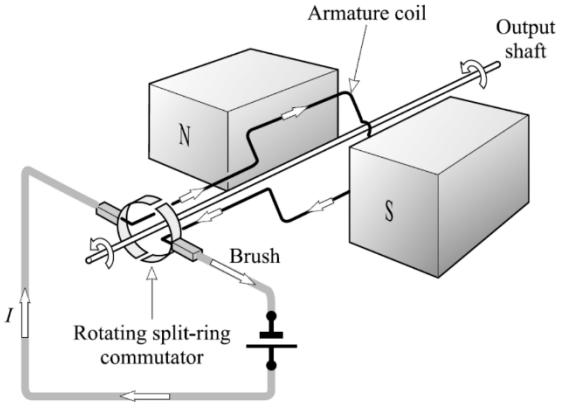
❍ **Electric motor** is an electrical device which converts from electrical energy to mechanical energy.

Armature Coil: Shown as a rotating coil located between two magnetic poles (represented by the letters N and S).

Magnetic Field: The presence of the north (N) and south (S) magnetic poles creates a magnetic field in which the armature coil rotates.

Brush: A brush makes contact with a **rotating split-ring commutator**, which reverses the current direction in the coil as it rotates, allowing for continuous rotation.

Output Shaft: The output shaft is connected to the armature, indicating that mechanical energy is produced.



The rotating armature coil acts as a generator, inducing a **back emf (electromotive force)** due to its rotation in the magnetic field.

This back emf opposes the applied voltage source that drives the motor, leading to a reduction in the net potential difference across the armature coil.

The net potential difference can be expressed as: $\text{Net p.d. across armature} = (\text{line voltage}) - (\text{back emf})$

The armature current can be defined as: $\text{Armature current} = \dfrac{(\text{line voltage}) - (\text{back emf})}{\text{armature resistance}}$

The mechanical power P developed within the armature of the motor is given by:

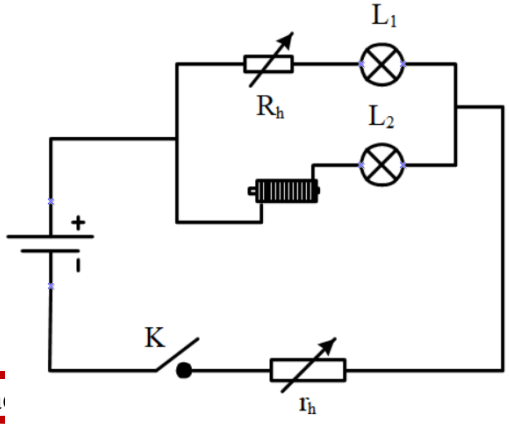
$$P = (\text{armature current})(\text{back emf})$$

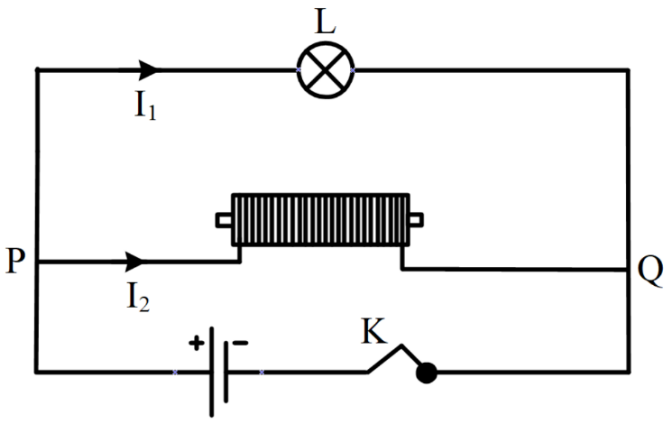
Chpater3 **Electricity and Magnetism**
Lesson3 **Self-Induction**

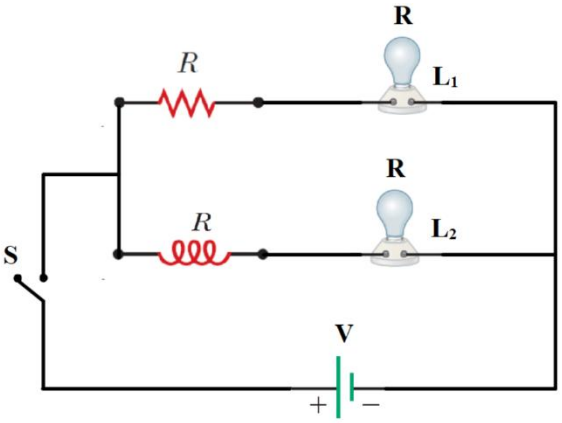
I.Self-Induction

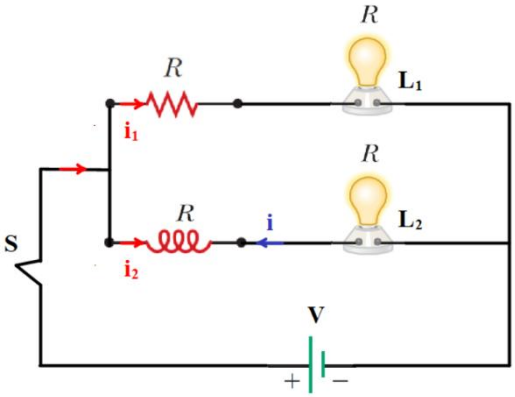
❍ Self-induction occurs when the current in a coil change over time, either increasing or decreasing. This change in current generates an electromotive force (emf) within the same circuit, effectively inducing a voltage in the coil itself.

❍ Experiment: The experiment of self-induction involves observing how a change in current through a coil induces an electromotive force (EMF) in the same coil due to its own changing magnetic field.









II.Inductance

❍ Inductance is a property of an electrical circuit or component that quantifies its ability to oppose changes in current. It is a measure of how effectively a coil or inductor can store energy in its magnetic field when current flows through it.

Total magnetic flux: $N\Phi \propto B$

And $B \propto I \Rightarrow N\Phi \propto I$

$$N\Phi = LI \Rightarrow L = \frac{N\Phi}{I}$$

Since, $\Phi = BA$

$$B = \frac{\mu_0 NI}{l}$$

Hence, the inductance of an electrical circuit is $L = \mu_0 \frac{N^2 A}{l}$

Where, $\mu_0 = 4\pi \times 10^{-7} SI$ is the vacuum permittivity.

N is the total number turns of Solenoid.

A is the area of a Solenoid measured in $\left(\mathrm{m}^2\right)$.

l is the inductance measured in $\left(\mathrm{H}\right)$.

Noted: A solenoid will have no inductance unless the wire is fold into two before being wrapped into the solenoid.

III.The induce electromotive force

❍ **Faraday's Law of Induction**: This law states that the induced emf in a closed circuit is proportional to the rate of change of the magnetic flux through the circuit.

$$e = -N\frac{d\Phi}{dt} = -\frac{d\left(N\Phi\right)}{dt} = -\frac{d\left(Li\right)}{dt}$$

$$e = -L\frac{di}{dt}$$

❍ **Negative Sign**: The negative sign in the equation represents Lenz's Law, which indicates that the induced EMF opposes the change in magnetic flux that produced it.

IV.Magnetic Energy

❍ Magnetic energy in inductance refers to the energy stored in the magnetic field created by a current flowing through an inductor or solenoid. This energy is stored as a result of the magnetic field generated by the coil's current.

Hence, the magnetic energy in inductance is $E_L = \frac{1}{2}Li^2$

Where, E_L is the magnetic energy measured in (J).

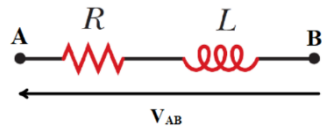
L is the inductance measured in (H).

i is the electric current measured in (A).

V.RL Circuit

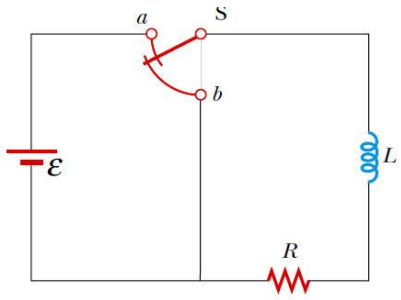
❍ An **RL circuit** is a type of electrical circuit that consists of a **resistor (R)** and an **inductor (L)** connected in series or parallel with a power source.

Relationship between electric current and voltage



$$V_{AB} = Ri + L\frac{di}{dt}$$

When the switch S attach to point a



$$E = Ri + L\frac{di}{dt} \Leftrightarrow r\left(\frac{E}{R} - i\right) = L\frac{di}{dt}$$

$$\Rightarrow \frac{R}{L}dt = \frac{1}{\left(\frac{E}{R} - i\right)}di$$

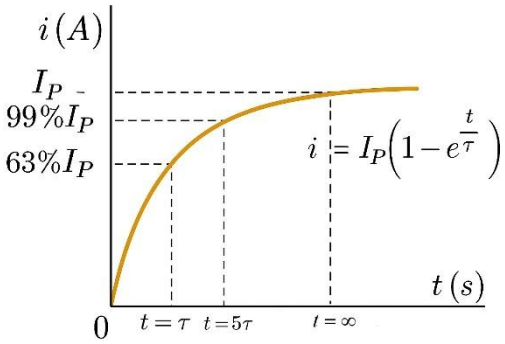
Sove the equation above which gives,

$$i = \frac{E}{R}\left(1 - e^{-\frac{t}{L/R}}\right)$$

Since, $I_P = E/R$, and $\tau = L/R$ (τ is called time constant)

Hence, the equation of growth of current is $i = I_P\ 1 - e^{-t/\tau}$

- If $t = \tau \Rightarrow i = I_P(1 - 0.37) = 0.63I_P$ or $i = 63\%I_P$ or $i = 63\%I_P$
- If $t = 5\tau \Rightarrow i = 0.99I_P$ or $i = 99\%I_P$ or $i = 99\%I_P$



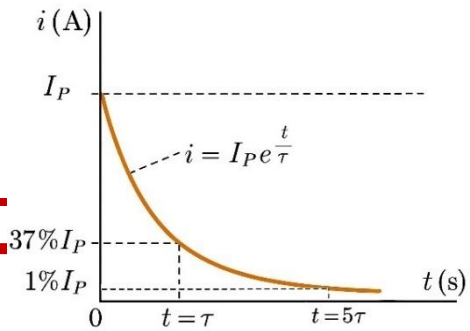
When the switch S attach to point b

$$0 = Ri + L\frac{di}{dt} \Rightarrow i = -\frac{L}{R}\frac{di}{dt}$$

$$\Leftrightarrow \frac{di}{dt} = -\frac{R}{L}dt$$

$$\Leftrightarrow \int_{I_P}^{i}\frac{di}{dt} = -\frac{L}{R}\int_0^t dt$$

$$\Rightarrow i = I_P e^{-\frac{t}{L/R}}$$



Hence, the equation of decay of current in an RL circuit is $i = I_P e^{-\frac{t}{\tau}}$

• If $t = \tau \Rightarrow i = 0.37 I_P = 37\% I_P$

VI. LC Circuit

❍ Energy Stored in the Magnetic Field

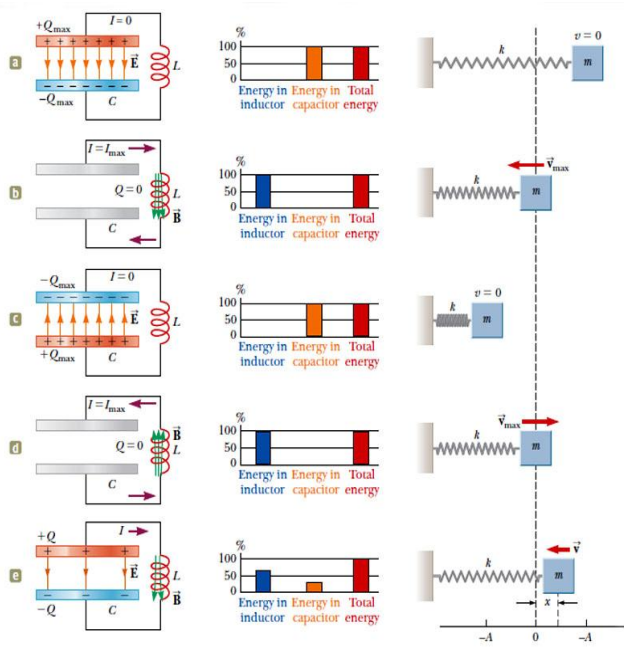
When switch S is connected to point A, the capacitor begins to charge until it reaches its maximum electric charge. At this point, the capacitor is fully charged, and the voltage across its plates is at its peak q_m.

Energy stored in a capacitor: $E_C = \frac{1}{2} CV^2 = \frac{1}{2} \frac{q_m^2}{C}$

Energy stored in an inductor: $E_L = \frac{1}{2} Li^2$

Hence, the energy stored in dipole LC is $E_{CL} = E_C + E_L = \frac{1}{2} \frac{q_n^2}{C}$

When the switch S attached to point b, the capacitor begins disc



VII. LC Circuit: Electromagnetic Oscillations

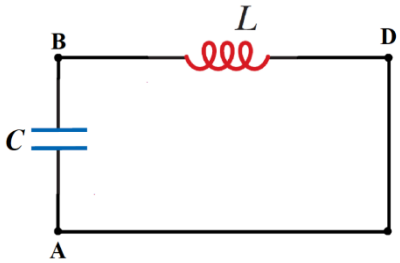
❍ Differential equation of voltage between terminal of a capacitor

$$V_{AA} = V_{AB} + V_{BD} + V_{DA}$$

Since, $V_{AA} = V_A - V_A = 0$, $V_{DA} = 0$, $(V_D = V_A)$, and $V_{AB} = V_C$ $V_{BD} = V_L = L\frac{di}{dt}$

We get, $V_C + L\frac{di}{dt} = 0$

Since, $q = CV_C$ and $i = \frac{di}{dt} = \frac{d^2(CV_C)}{dt^2} = C\frac{d^2V_C}{dt^2}$



Imply, $\frac{di}{dt} = \frac{d}{dt}\left(C\frac{dV_C}{dt}\right) = C\frac{d^2V_C}{dt^2}$

$$V_C + LC\frac{d^2V_C}{dt^2} = 0$$

$\ddot{V}_C + \frac{1}{LC}V_C = 0$ is the differential equation across voltage of a capacitor.

The voltage across the capacitor is given by solving differential equation. The solution of this equation

$$V_C = V_m\cos(\omega t + \varphi)$$

Verify answer: $\dot{V}_C = \frac{dV_C}{dt} = -V_m\omega\sin(\omega t + \varphi)$

$$\ddot{V}_C = \frac{d^2V_C}{dt} = -\omega^2V_m\cos(\omega t + \varphi) = -\omega^2V_C$$

We get, $-\omega^2V_C + \frac{1}{LC}V_C = 0$

$$-\omega^2 + \frac{1}{LC} = 0 \Rightarrow \omega^2 = \frac{1}{LC}$$

$$\omega = \frac{1}{\sqrt{LC}}$$

Hence, the period of LC oscillation is $T = \frac{2\pi}{\omega} = 2\pi\sqrt{LC}$

VIII. Changing in electric charge and current

❍ The expression of charge of a capacitor: $q = CV_C$

$$q = CV_m\cos\ \omega t + \varphi\ = q_m\cos\ \omega t + \varphi$$

Since, $i = \frac{dq}{dt}$

Hence, the equation of changing in charge and current is $i = -\omega q_m\sin\ \omega t + \varphi$

IX. Enger stored in LC Circuit

$$E_{CL} = E_C + E_L = \frac{1}{2}CV^2 + \frac{1}{2}Li^2$$

Hence, $E_{CL} = \frac{1}{2}CV_m^2 + \frac{1}{2}LI_m^2$

Chpater3 **Electricity and Magnetism**
Lesson4 **Alternating Current**

I. Production of AC Sinusoidal Electromotive Force

❍ The production of **AC (Alternating Current) sinusoidal electromotive force (emf)** is based on **Faraday's Law of Electromagnetic Induction**. The most common way to generate an AC sinusoidal emf is through the use of a **rotating coil** in a magnetic field, which is the principle behind AC generators (alternators).

Electromotive force of AC sinusoidal is given by equation: $e = -\frac{d}{dt}\left(NBA\cos\omega t\right)$

Hence, the AC sinusoidal electromotive force is $e = E_m \sin\omega t$

II. Definition of Alternating Current

❍ Alternating current is the current that change its direction two times in every period.

$$i = I_m \sin\omega t$$

Where, I_m is the amplitude or maximum electric current measured in $\left(\text{A}\right)$.

$\omega = \frac{2\pi}{T} = 2\pi f$ is angular frequency measured in $\left(\text{rad/s}\right)$.

RMS Voltage $V_{rms} = \frac{V_m}{\sqrt{2}}$

RMS Current $I_{rms} = \frac{I_m}{\sqrt{2}}$

III.General Rule

❍ In studying alternating current (AC), we assume that:

The electric current i at any instant has the same value at all points in the electric circuit.

The laws and formulas that apply to direct current (DC) can also be used in alternating current circuits.

❍ The effects of **alternating current (AC)** are chemical effect, heating effect, magnetic effect and optic effect.

Ohm's Law Application Phasor Diagram and Phase Difference

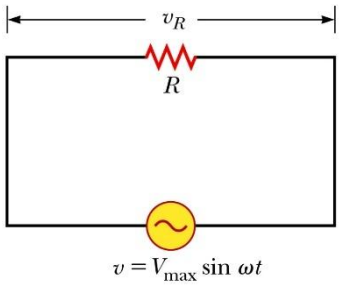
❍ **Resistor in AC circuit**

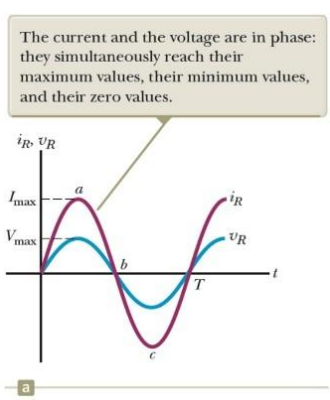
Instantaneous voltage across resistor: $v_R = V_{\max}\sin\omega t$

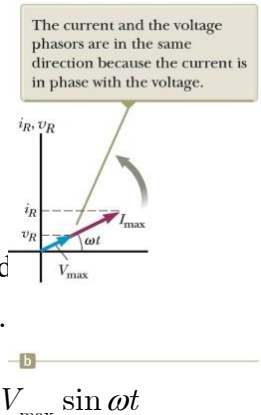
Ohm's Law: $i_R = \frac{v_R}{R} = \frac{V_{\max}}{R}\sin\omega t$

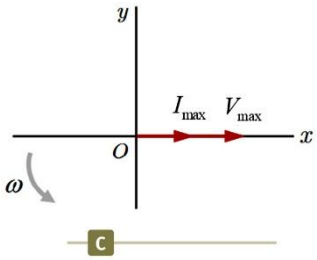
$$i_R = I_{\max}\sin\omega t \quad \left(I_{\max} = V_{\max}/R\right)$$

Hence, the instantaneous current in the resistor is: $i_R = I_{\max}\sin\omega t$









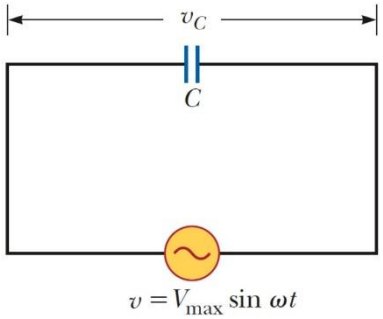
From plots nt i_R and ge v_R across a resistor as functions of time. The voltage.

❍ **Capaci**

Instantaneous voltage across capacitor: $v_C = V_{\max}\sin\omega t$

Instantaneous current of AC circuit: $i_c = \frac{dq}{dt} = \frac{d}{dt}\left(Cv_c\right) = C\frac{dv_c}{dt} = C\frac{d}{dt}\left(V_{\max}\sin\omega t\right)$

$$i_c = C\omega V_{\max}\cos\omega t$$

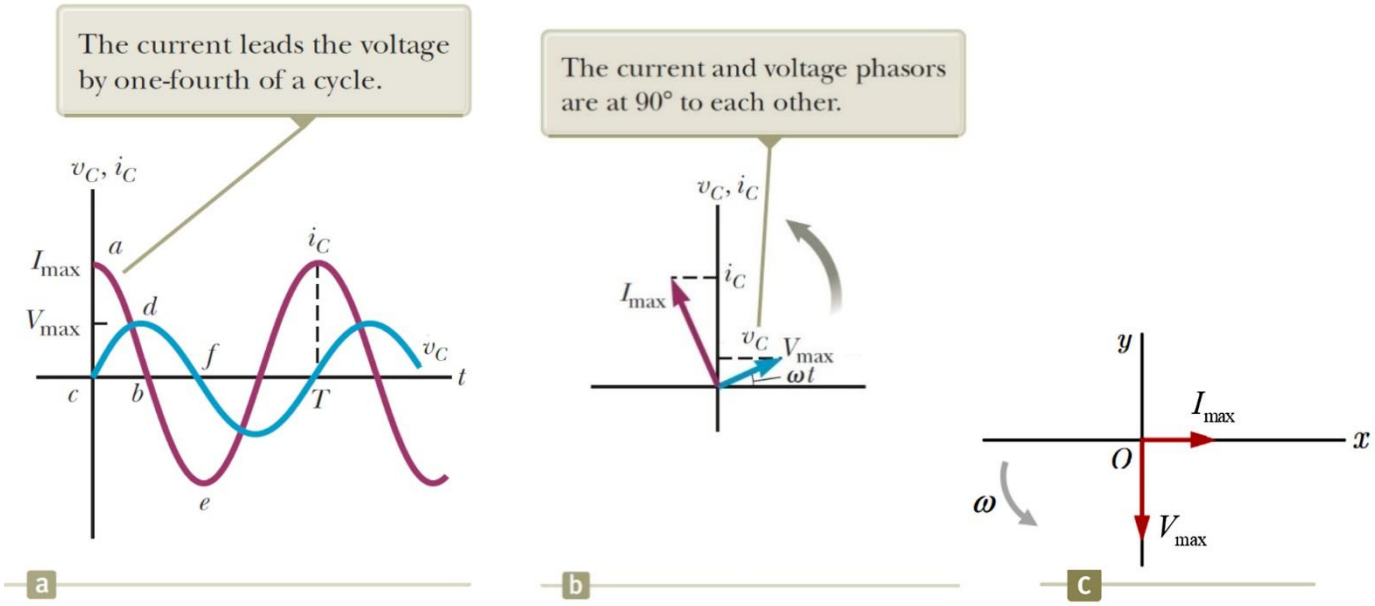


Since, $\cos\omega t = \sin\left(\omega t + \frac{\pi}{2}\right)$ $\quad i_c = C\omega V_{\max}\sin\left(\omega t + \frac{\pi}{2}\right) = I_{\max}\sin\left(\omega t + \frac{\pi}{2}\right)$

$$i_c = I_{\max}\sin\left(\omega t + \frac{\pi}{2}\right)$$

Therefore, $I_{\max} = C\omega V_{\max} \Leftrightarrow \frac{V_{\max}}{I_{\max}} = \frac{1}{C\omega} = Z_C$

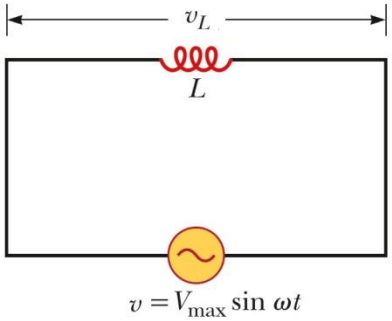
The capacitive reactance of the capacitor to the sinusoidal current is given by $Z_C = \frac{1}{C\omega}$



From plot of the instantaneous current i_C and instantaneous voltage V_C across a capacitor as functions of time. The voltage lags behind the current by 90°.

❍ **Inductor in AC circuit**

Instantaneous voltage across capacitor: $v_L = V_{\max}\sin\omega t$



Since, $v_L = L\frac{di}{dt} \Leftrightarrow L\frac{di}{dt} = V_{\max}\sin\omega t \Leftrightarrow di = \frac{V_{\max}}{L}\sin\omega t dt$

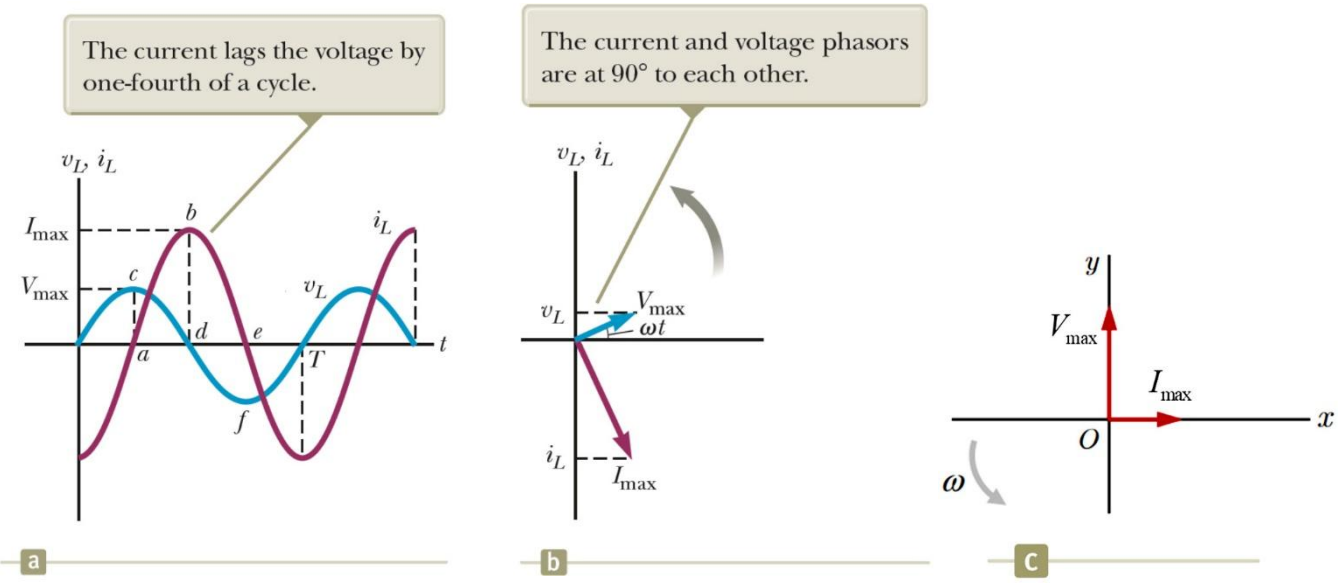
$$\int di_L = \frac{V_{\max}}{L}\int \sin\omega t\, dt \Rightarrow i_L = -\frac{V_{\max}}{L\omega}\cos\omega t$$

Since, $\cos\omega t = \sin\left(\frac{\pi}{2} - \omega t\right) = -\sin\left(\omega t - \frac{\pi}{2}\right)$

$$\Rightarrow i_L = \frac{V_{\max}}{L\omega}\sin\left(\omega t - \frac{\pi}{2}\right) = I_{\max}\sin\left(\omega t - \frac{\pi}{2}\right)$$

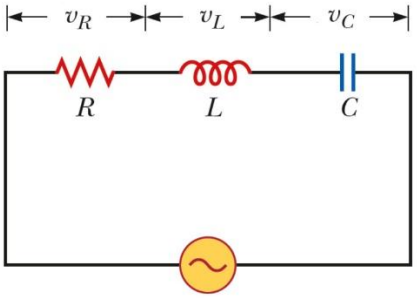
$$I_{\max} = \frac{V_{\max}}{L\omega} \Leftrightarrow \frac{V_{\max}}{I_{\max}} = L\omega = Z_L$$

The capacitive reactance of the inductor to the sinusoidal current is given by $Z_L = L\omega$



From plot of the instantaneous current i_L and instantaneous voltage v_L across an inductor as functions of time. The current lags behind the voltage by 90°.

V. RLC Series Circuit



Assume that current in AC circuit is $i = I_{\max} \sin \omega t$

Instantaneous voltage across resistor: $v_R = V_{\max} \sin \omega t$

Instantaneous voltage across inductor: $v_L = V_{\max} \sin\left(\omega t + \frac{\pi}{2}\right)$

Instantaneous voltage across capacitor: $v_c = V_{\max} \sin\left(\omega t - \frac{\pi}{2}\right)$

Total instantaneous voltage in AC circuit: $v = v_R + v_L + v_c$

Amplitude: $V_{\max}$

Vector amplitude of voltage: $\vec{V}_{\max} = \vec{V}_{R\max} + \vec{V}_{L\max} + \vec{V}_{C\max}$

$$V_{\max}^2 = V_{Rm}^2 + \left(V_{Lm} - V_{Cm}\right)^2$$

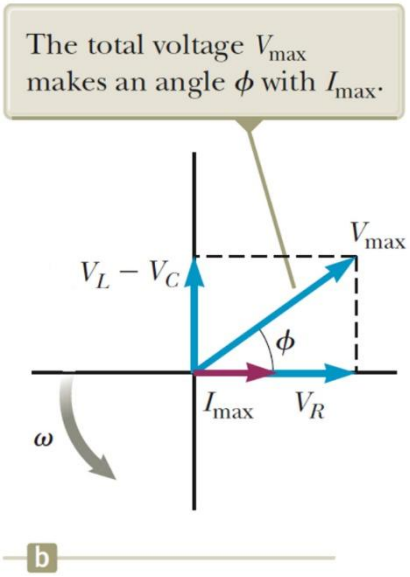
$$\left(ZI_{\max}\right)^2 = \left(RI_{\max}\right)^2 + \left(Z_L I_{\max} - Z_C I_{\max}\right)^2$$

$$Z^2 = R^2 + \left(Z_L - Z_C\right)^2$$

Hence, the **impedance** of RLC series circuit is

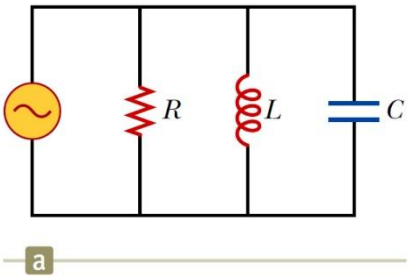
$$Z = \sqrt{R^2 + \left(Z_L - Z_C\right)^2}$$

The phase difference in RLC series circuit is given by Using the phasor diagram:



$$\tan\phi = \left(\frac{Z_L - Z_C}{R}\right) \Rightarrow \phi = \tan^{-1}\left(\frac{Z_L - Z_C}{R}\right)$$

VI.RLC Parallel Circuit

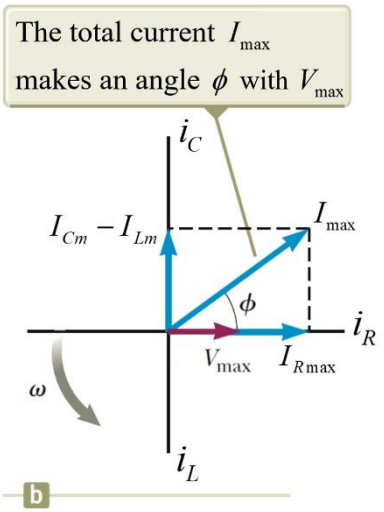


Assume that instantaneous voltage in AC circuit is $v = V_{\max}\sin\omega t$

Instantaneous current across resistor: $i_R = I_{R\max}\sin\omega t$

Instantaneous current across inductor: $i_L = I_{L\max}\sin\left(\omega t - \frac{\pi}{2}\right)$

Instantaneous current across capacitor: $i_c = I_{Cm}\sin\left(\omega t + \frac{\pi}{2}\right)$



Total current in RLC parallel circuit: $i = i_R + i_L + i_C$

$$i = I_{R\max}\sin\omega t + I_{L\max}\sin\left(\omega t - \frac{\pi}{2}\right) + I_{C\max}\sin\left(\omega t + \frac{\pi}{2}\right)$$

Amplitude $I_{\max}$

Vector amplitude of maximum current: $\vec{I}_{\max} = \vec{I}_{R\max} + \vec{I}_{L\max} + \vec{I}_{C\max}$

In magnitude: $I_{\max}^2 = I_{Rm}^2 + \left(I_{Cm} - I_{Lm}\right)^2$

$$\left(\frac{V_{\max}}{Z}\right)^2 = \left(\frac{V_{R\max}}{R}\right)^2 + \left(\frac{V_{C\max}}{Z_C} - \frac{V_{L\max}}{Z_L}\right)^2$$

$$\left(\frac{1}{Z}\right)^2 = \left(\frac{1}{R}\right)^2 + \left(\frac{1}{Z_C} - \frac{1}{Z_L}\right)^2$$

Hence, the **impedance** of RLC parallel circuit is $\frac{1}{Z} = \sqrt{\frac{1}{R^2} + \left(C\omega - \frac{1}{L\omega}\right)^2}$

The phase difference in RLC parallel circuit is given by Using the phasor diagram:

$$\tan\phi = \left(\frac{I_{C\max} - I_{L\max}}{I_{\max}}\right) = \frac{\frac{V_{\max}}{Z_C} - \frac{V_{\max}}{Z_L}}{\frac{V_{\max}}{R}} = \frac{\frac{1}{Z_C} - \frac{1}{Z_L}}{\frac{1}{R}}$$

Hence, the phase difference in RLC parallel circuit is $\phi = \tan^{-1}\left(\frac{\frac{1}{Z_C} - \frac{1}{Z_L}}{\frac{1}{R}}\right)$

VII. Resonance in a Series RLC Circuit

❍ Resonance in Series RLC circuit occur when $Z_L - Z_C = 0 \Leftrightarrow Z_L = Z_C \Leftrightarrow L\omega_0 = \frac{1}{C\omega_0}$

$$\omega_0^2 = \frac{1}{LC} \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

$$2\pi f_0 = \frac{1}{\sqrt{LC}} \Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}}$$

❍ Hence, this frequency corresponds to the natural frequency of oscillation of an LC circuit.

$$Z = \sqrt{R^2 + \left(Z_L - Z_C\right)^2} \Rightarrow R = Z \left(Z_L = Z_C\right)$$

$$\tan\phi = \left(\frac{Z_L - Z_C}{R}\right) \Rightarrow \phi = \tan^{-1}\left(\frac{Z_L - Z_C}{R}\right) = \tan^{-1}(0) \Rightarrow \phi = 0$$

VIII. Average Power and Power Factor in AC Circuit

❍ In an **AC circuit**, the **average power** refers to the average amount of electrical energy consumed or delivered by a circuit element over a complete cycle of the alternating current.

Average power: $P_{av} = V_{rms} I_{rms} \cos\phi$

Where, $\cos\phi$ is called power factor of AC circuit.

ϕ is the phase difference between current and voltage.

Apparatus power (total): $P_{av} = V_{rms} I_{rms}$

❍ **Average power in series RLC circuit**

For pure resistor:

Phase difference: $\phi = 0 \Rightarrow P_{av} = 0$

For pure inductor:

Current lag than voltage $\frac{\pi}{2} \Rightarrow \cos\phi = 0 \Rightarrow P_{av} = 0$

For pure capacitor:

Voltage lag than current $\frac{\pi}{2} \Rightarrow \cos\phi = 0 \Rightarrow P_{av} = 0$

For Series RLC circuit

$$\cos\phi = \frac{R}{Z}$$

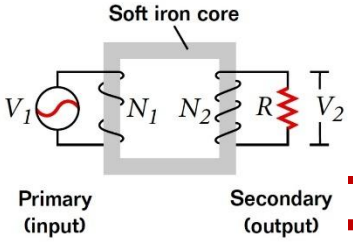
$$P_{av} = V_{rms} I_{rms} \cos\phi = V_{rms} I_{rms} \frac{R}{Z}$$

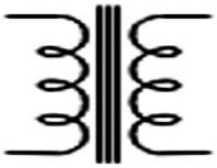
Since, $I_{rms} = V_{rms} / Z$

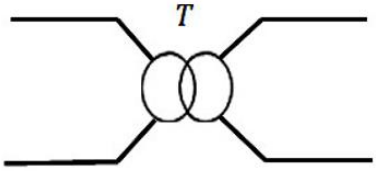
Hence, the average power is $P_{av} = R I_{rms}^2$

IX.The Transformer and Power Transmission

❍ Transformer is an electrical device used to transform power from one circuit to another without any direct connection to winding, also used to step-up and step-down output voltage.







❍ **The characteristic of a transformer**

The frequency at secondary coil is the same frequency at primary coil.

The relation between number of turns and voltages is given by

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} = K \text{ or } \frac{V_2}{V_1} = \frac{I_1}{I_2} \text{ or } \frac{V_2}{V_1} = \frac{I_1}{I_2}$$

Where, V_2 is the voltage at secondary coil measured in (V).

V_1 is the voltage at primary coil measured in (V).

N_2 is the number of turns at secondary coil.

N_1 is the number of turns at secondary coil.

I_1 is the current at primary coil measured in (A).

I_2 is the current at secondary coil measured in (A).

- If $K > 1$ therefore, it is called Step-Up transformer.
- If $K < 1$ therefore, it is called Step-Down transformer.

❍ **Joule Heating and Efficiency**

Power: $P_{e1} = P_{e2} + P_j$

Efficiency of transformers: $Rd = \frac{P_{e2}}{P_{e1}} \times 100\%$